



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON Central Office – Madhapur – Hyderabad

Sec: **Sr.Super-60& All**

Jee-Main

Date: 29-04-2021

Time: 09.00Am to 12.00

PTM-01

Max.Marks:300

IMPORTANT INSTRUCTIONS:

1. Immediately fill in the Admission number on this page of the Test Booklet with Blue/Black Ball Point Pen only.
2. The candidates should not write their Admission Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1, 2, 3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **two sections**.

(I) Section-I contains 20 **multiple choice** questions with **only one correct** option.

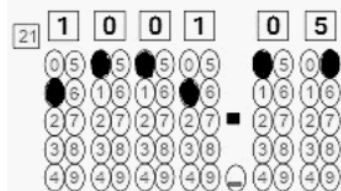
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

(II) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only.

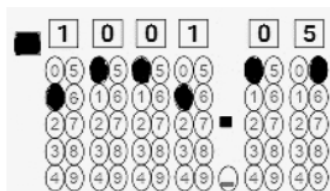
First 5 attempted questions will be considered if more than 5 questions attempted.

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted question 21, bubble on 21 as shown below.



Question Answered for marking



Question Cancelled for marking

Marking scheme : +4 for correct answer and 0 in all other cases.

6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capitals): _____

Admission Number:

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Candidate's Signature: _____

Invigilator's Signature: _____

29-04-2021_Sr.Super60 & All_Jee-Main_PT M-01_Test Syllabus

PHYSICS : Complete Modern Physics,Electrostatics(Excludingcapacitors)
Gravitation

CHEMISTRY : Thermodynamics, Thermochemistry, Hydrocarbons, (Excluding Benzene)Alkyl Halide, Aryl Halide,Qualitative analysis, Compounds of d-block metals, Zero group

MATHEMATICS :Indefinite Integration, definite Integration,AUC&D.E

PHYSICS

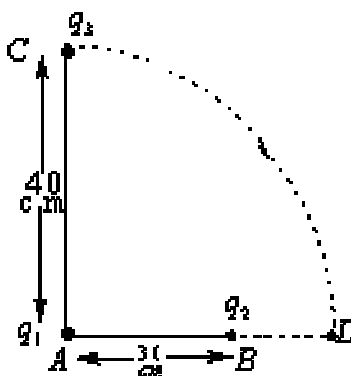
Max Marks: 100

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

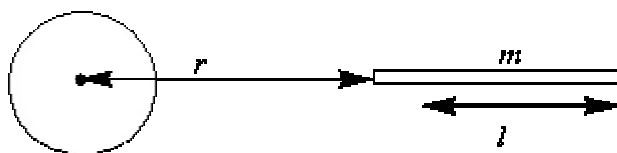
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- 1 Find the PE of three objects of masses 1 kg, 2 kg and 3 kg placed at the three vertices of an equilateral triangle of side 50 cm.
 1. -22 G 2. 55 G 3. 5 G 4. 60 J
- 2 If electrons are accelerated through a potential of V volt, the minimum wavelength of the continuous x-ray radiation is
 1. $\frac{hC}{Ve}$ 2. $\frac{Ve}{hC}$ 3. $\frac{Ve}{h}$ 4. $\frac{V}{hC}$
- 3 A metal sphere A of radius a is charged to potential V . What will be its potential if it is enclosed by a spherical conducting shell B of radius b and the two are connected by a wire
 1. $\frac{a}{b}V$ 2. $\frac{bV}{a}$ 3. $\frac{a^2V}{b}$ 4. $\frac{b^2V}{a}$
- 4 The radius of a planet is $1/4^{\text{th}}$ of R_e and its acc. due to gravity is $2g$. What would be the value of escape velocity on the planet, if escape velocity on earth is v_e .
 1. $\frac{v_e}{\sqrt{2}}$ 2. $v_e\sqrt{2}$ 3. $2v_e$ 4. $\frac{v_e}{2}$
- 5 The radius of the oxygen nucleus ($^{16}_8\text{O}$) is $2.8 \times 10^{-15} \text{ m}$. Find the radius of the lead nucleus ($^{205}_{62}\text{Pb}$).
 1. $8.55 \times 10^{-15} \text{ m}$ 2. $6.55 \times 10^{-15} \text{ m}$
 3. $6.55 \times 10^{-13} \text{ m}$ 4. $4.55 \times 10^{-12} \text{ m}$
- 6 Two charges q_1 and q_2 are placed 30 cm apart as shown in the Figure. Third charge q_3 is moved along the arc of a circle of radius 40 cm from C to D. The change in potential energy of the system is $\frac{q_3 k}{4\pi\epsilon_0}$ where k is



1. $8q_1$ 2. $6q_1$ 3. $8q_2$ 4. $6q_2$

- 7 The gravitational force of attraction between a uniform sphere of mass M and a uniform rod of length l and mass m oriented as shown is



1. $\frac{GMm}{r(r+l)}$ 2. $\frac{GM}{r^2}$ 3. $Mmr^2 + L$ 4. $(r^2 + L)mM$

- 8 The K_{α} X-ray emission line of tungsten occurs at $\lambda = 0.021 \text{ nm}$. The energy difference between K and L levels in this atoms is about:

1. 0.51 MeV 2. 1.2 MeV 3. 59 keV 4. 13.6 eV

- 9 From a point charge there is a fixed point A. At that point there is an electric field of 500 V/m and potential of 3000 V. Then the distance of A from the point charge is :

1. 6 m 2. 12 m 3. 36 m 4. 144 m

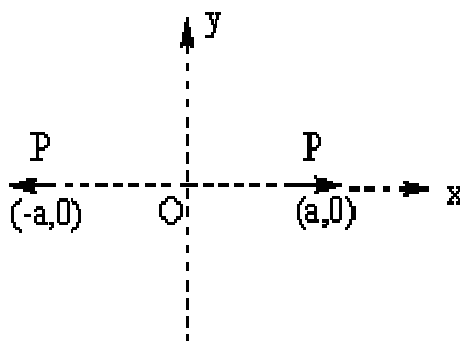
- 10 The orbiting frequency 'n' of a surface satellite is given by the relation

1. $2\pi\sqrt{\frac{R}{g}}$ 2. $\frac{1}{2\pi}\sqrt{\frac{g}{R}}$ 3. $2\pi\sqrt{2gR}$ 4. $2\pi\sqrt{\frac{g}{2R}}$

- 11 The photoelectric threshold wavelength for a certain metal is $4000 \text{ \AA}$. The maximum energy of the ejected photoelectrons by a radiation of $2000 \text{ \AA}$ is

1. 2.13 eV 2. 3.1 eV 3. 1.23 eV 4. 3.82 eV

- 12 If 'g' is acceleration due to gravity on the earth's surface. Then the gain in PE of an object of mass 'm' raised from the surface of earth to a height equal to double of the radius 'R' of the earth is
1. $\frac{mgR}{2}$ 2. $2mgR$ 3. mgR 4. $\frac{mgR}{4}$
- 13 At a given instant there are 25% undecayed radioactive nuclei in a sample. After 10 sec, the number of undecayed nuclei reduces to 12.5%. Then
1. Mean life of the nuclei is 14.43s
 2. The time in which the number of original undecayed nuclei would reduce to 6.25% is 40 sec.
 3. The ratio of mean life to half-life is 0.1443
 4. Suppose 12.5% of the radioactive nuclei remain undecayed at say $t=0$, then 100% would have remained 40 seconds earlier
- 14 A cube of side 10 cm encloses a charge of $0.1\mu C$ at its centre. Calculate the number of lines of force through each face of the cube.
1. 1.113×10^{11} 2. 1.13×10^4 3. 1.13×10^9 4. 1883
- 15 A man weight $60kg$ on the surface of the earth. The altitude where his weight will be $90kg$ is (R is the radius of the earth)
1. $\left(\sqrt{\frac{2}{3}} - 1\right)R$ 2. $\left(\sqrt{\frac{3}{2}} - 1\right)R$ 3. $\frac{\sqrt{3}}{2}R$ 4. $\sqrt{2}R$
- 16 A system of two electric dipoles, each of dipole moment having magnitude P are arranged in the configuration shown in figure. The electrostatic interaction energy of this system of two dipoles is



1. $\frac{5}{4} \frac{kp^2}{a^3}$ 2. $\frac{7kp^2}{a^3}$ 3. $\frac{kp^2}{4a^3}$ 4. None of these

- 17 Work function of a metal is 3.0 eV. It is illuminated by a light of wavelength $3 \times 10^{-7} \text{ m}$. Calculate the maximum energy of the electron.
 1. 4.14 eV 2. 2.14 eV 3. 3.16 eV 4. 1.13 eV
- 18 A planet is revolving around the sun in an elliptical orbit. Its closest distance from the sun is $r_{\min}$. The farthest distance from the sun is $r_{\max}$. If the orbital angular velocity of the planet when it is nearest to the sun is ω , then the orbital angular velocity at the point when it is at the farthest distance from the sun is
 1. $\sqrt{(r_{\min} / r_{\max})} \omega$ 2. $\sqrt{(r_{\max} / r_{\min})} \omega$
 3. $(r_{\max}^2 / r_{\min}^2) \omega$ 4. $(r_{\min}^2 / r_{\max}^2) \omega$
- 19 Two point charges $+8q$ and $-2q$ are located at $x = 0$ and $x = L$ respectively. The location of a point on the x axis at which the net electric field due to these two point charges is zero is
 1. $\frac{L}{4}$ 2. $2L$ 3. $4L$ 4. $8L$
- 20 If a satellite is moving around a planet of mass M in an elliptical orbit of semi major axis a . The orbital speed (V) of the satellite when it is at a distance r from the focus will be given by
 1. $V^2 = \frac{GM}{r}$ 2. $V^2 = \frac{2GM}{r}$
 3. $V^2 = GM \left[\frac{1}{r} - \frac{1}{a} \right]$ 4. $V^2 = GM \left[\frac{2}{r} - \frac{1}{a} \right]$

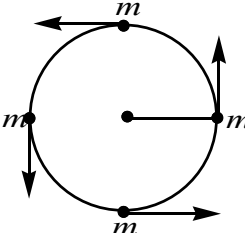
(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

- 21 Two identical copper spheres are separated by 1 m in vacuum. $___ \times 10^{13}$ electrons would have to be removed from one sphere and added to other so that they now attract each other with a force of 0.9 N.
- 22 The equivital potential energy of a satellite revolving around the earth in circular orbit is 4 MJ. Find the additional energy (in MJ) that should be given to the satellite so that it escapes from the gravitational field of the earth

- 23 Light of two different frequencies whose photons have energies 1 eV and 2.5 eV respectively, successively illuminates a metal of work function 0.5 eV. If the maximum kinetic energies of emitted electron is K_1 and K_2 respectively, and the ratio of $\frac{K_1}{K_2}$ is $\frac{p}{4}$, then the value of "p" is _____.
- 24 Four similar particles of mass ' m ' are orbiting in a circle of radius ' r ' in the same angular direction symmetrically because of their mutual gravitational attractive force. Velocity of a particle is given by $\sqrt{\frac{Gm}{r}\left(\frac{1}{\sqrt{2}} + \frac{1}{n}\right)}$. Then ' n ' is equal to
- 
- 25 A particle of mass $200 \text{ MeV}/c^2$ collides with a hydrogen atom at rest. Soon after the collision the particle comes to rest, and the atom recoils and goes to its first excited state. The initial kinetic energy of the particle (in eV) is $\frac{N}{4}$. The value of N is _____ (Given the mass of the hydrogen atom to be $1 \text{ GeV}/c^2$)
- 26 A potential difference of 3000 V is maintained between two large parallel plates 6 cm apart. The force on a proton midway between the plates is $x \times 10^{-15} \text{ N}$. What is the value of ' x ' ?
- 27 Gravitational acceleration on the surface of a planet is $\frac{\sqrt{6}}{11}g$, where g is the gravitational acceleration on the surface of the earth. The average mass density of the planet is $\frac{2}{3}$ times that of the earth. If the escape speed on the surface of the earth is taken to be 11 kms^{-1} . Find the escape speed on the surface of the planet in kms^{-1} .
- 28 If 20 J of work has to be done to move an electric charge of 4C from a point where potential is 10 volt to another point where potential is ' v ' volt. Find the value of ' v ' in volt
- 29 The wavelength K_α X-rays produced by an X-ray tube is $0.76 \text{ \AA}$. The atomic number of anticathode materials is (round off to nearest integer)
- 30 The charges $-q$, Q , $-q$ are placed along x-axis at $x = 0$, $x = a$ and $x = 2a$. If the potential energy of the system is 0 then Q/q is _____

CHEMISTRY**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- 31** A system absorbs 10 KJ of heat at constant volume and its temperature rises from 27°C to 37°C . The ΔE of reaction is
 1. 100 KJ 2. 10 KJ 3. 0 4. 1 KJ
- 32** The heat of atomization of $\text{PH}_{3(g)}$ is 228K.Cal mol^{-1} and that of $\text{P}_2\text{H}_{4(g)}$ is 355K.Cal mol^{-1} . The energy of the P-P bond is (in K.Cal);
 1. 102 2. 51 3. 26 4. 204
- 33** An isolated system is that in which :
 1. There is no exchange of energy with the surroundings
 2. There is exchange of mass and energy with the surroundings
 3. There is no exchange of mass and energy with the surroundings
 4. There is exchange of mass with the surroundings
- 34** A well stoppered thermos flask contains some ice cubes. This is an example of
 1. Closed system 2. Open system
 3. Isolated system 4. Non-thermodynamic system
- 35** ΔH° for a reaction $\text{F}_2 + 2\text{HCl} \rightarrow 2\text{HF} + \text{Cl}_2$ is given as -352.8 KJ . ΔH_f° for HF is -268.3KJ mol^{-1} , then ΔH_f° of HCl would be
 1. -22KJmol^{-1} 2. 880KJmol^{-1} 3. -91.9KJmol^{-1} 4. -183.8KJmol^{-1}
- 36** For the reaction $2\text{SO}_2(g) + \text{O}_2(g) \rightarrow 2\text{SO}_3(g)$, the entropy
 1. Increases 2. Decreases
 3. Remains unchanged 4. Change cannot be predicted
- 37** How many monochlorobutanes will be obtained on chlorination of n-butane?
 1. 5 2. 2 3. 3 4. 4
- 38** 2-chloro-2-methylpentane on reaction with sodium methoxide in methanol can form:

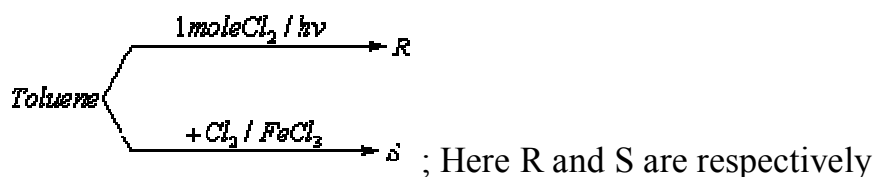
- $$\begin{array}{c} \text{CH}_3 \\ | \\ \text{C}_2\text{H}_5\text{CH}_2-\text{C}-\text{OCH}_3 \\ | \\ \text{CH}_3 \end{array}$$

$$\begin{array}{c} \text{C}_2\text{H}_5\text{CH}_2\text{C}=\text{CH}_2 \\ | \\ \text{CH}_3 \end{array}$$

$$\begin{array}{c} \text{C}_2\text{H}_5\text{CH}=\text{C}-\text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$$
- a) b) c)
1. (a) and (c) 2. (c) Only 3. (a) and (b) 4. All of these

- 39 Which of the following is not an isomer of 1-butyne?
1. But-2-yne
 2. Buta-1,3-diene
 3. But-2-ene
 4. Methyl cyclo propene
- 40 Which of the following is propargyl group?
1. $-CH_2 - C \equiv CH$
 2. $-C \equiv C - M_e$
 3. $-C \equiv CH$
 4. $-C \equiv C - CH_2 - CH_3$
- 41 Which of the following reaction conditions are used for the conversion of ethanol to ethylene?
1. *con* $H_2SO_4 / 70^\circ C$
 2. *dil* $H_2SO_4 / 140^\circ C$
 3. *dil* $H_2SO_4 / 100^\circ C$
 4. *conc* $H_2SO_4 / 170^\circ C$
- 42 Baeyer's reagent oxidizes ethylene to
1. Ethylene chlorohydrin
 2. Ethyl alcohol
 3. CO_2 and H_2O
 4. Ethane - 1, 2 - diol
- 43 The reaction, $(CH_3)_3C - OH \xrightarrow[H_2SO_4]{Heat} (CH_3)_2C = CH_2$, is an example of :
1. Sulphonation
 2. Decomposition
 3. Dehydration
 4. Alkylation
- 44 $CH_2 = CH_2 + HCl \xrightarrow{x} CH_3 - CH_2 - Cl$ what is 'x'?
1. Al_2O_3
 2. Anhydrous $AlCl_3$
 3. $NaCl$
 4. $MgCl_2$
- 45 SO_2 and HCl are by products in the reaction of ethyl alcohol with thionyl chloride. Which of the following is formed as main product.
1. $C_2H_5 - O - C_2H_5$
 2. C_2H_6
 3. $CH_3 - Cl$
 4. $C_2H_5 - Cl$

46



1. Benzyl chloride, p-chlorotoluene
2. Benzoyl chloride, p-chlorobenzene
3. p-Chlorobenzene, p-chlorobenzene
4. o-chlorobenzene, p-chlorobenzene

- 47 Schweitzer's reagent used for dissolving cellulose in the manufacture of artificial silk is
1. $CuSO_4 \cdot 5H_2O$
 2. CuI
 3. $Cu(NH_3)_4SO_4$
 4. $Cu(CH_3COO)_2 \cdot Cu(OH)_2$
- 48 Yellow Ammonium sulphide solution can be used for the separation of which of the following pair of species
1. CuS & PbS
 2. PbS & Bi_2S_3
 3. Bi_2S_3 & CuS
 4. CdS & As_2S_3
- 49 Number of lone pairs and bond pairs respectively that the central atom in xenon difluoride has
1. 2, 6
 2. 3, 2
 3. 2, 4
 4. 2, 2
- 50 The shape of XeF_4
1. Pentagonal bipyramidal
 2. Square planar
 3. Octahedral
 4. Distorted octahedral

(NUMERICAL VALUE TYPE)

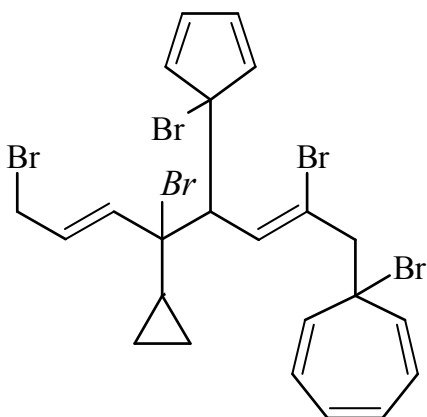
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Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

- 51 What is the proton affinity of PH_3 using Born Haber's cycle using Hess's law of constant heat summation ?
- Given : $\Delta H_f^\circ(PH_3) = -2 kJ$, Bond dissociation energy of $H_2 = +20 kJ$,
 Ionisation energy of $H = 10 kJ$, Bond dissociation energy of chlorine molecule = $+8 kJ$ Electron gain enthalpy of $Cl = -4 kJ$,
 Lattice enthalpy of $PH_4Cl = -28 kJ$
- $$PH_3 + \frac{1}{2}H_2 + \frac{1}{2}Cl_{2(g)} \rightarrow PH_4Cl_{(s)}; \Delta H_r^\circ = -8 kJ$$
- 52 $Me_3C-CH=CH-CHMe_2 \xrightarrow[Zn/H_2O]{O_3} ? \xrightarrow[\Delta]{conc.KOH} ?$ how many organic products are there in final product mixture?

53 How many moles of AgNO_3 will react with the given compound?



54 How many of the following cations can be precipitated with H_2S in acid medium only.

Pb^{2+} , Ag^+ , Hg^{2+} , Cu^{2+} , Cd^{2+} , Bi^{3+} , Sn^{2+} , As^{3+} , Sb^{3+} , Zn^{2+} , Mn^{2+} , Ni^{2+} , Fe^{2+} , Fe^{3+}

55 If the number of salts (or salt mixtures) given below that liberate colored gas/vapours with conc. H_2SO_4 is/are 'x' and that which liberate only colourless/gas vapours is/are 'y', then what is the value of $\left(1 - \frac{x}{y}\right)$.

$\text{KCl(s)} + \text{K}_2\text{Cr}_2\text{O}_7(\text{s}), \text{KNO}_2(\text{s}), \text{KI(s)}, \text{KBr(s)}, \text{KCl(s)}$

$\text{KBr(s)} + \text{MnO}_2(\text{s}), \text{KNO}_3, \text{KCl(s)} + \text{MnO}_2, \text{K}_2\text{SO}_3$

56 $\text{Fe} + 2\text{HCl} \rightarrow \text{X} + \text{H}_2$, the number of 'd' electrons of ' Fe^{+n} ' in 'X' is

57 The number of equivalent $\text{Cr}-\text{O}$ bonds in CrO_4^{2-} is ____

58 'X' is the 1st noble gas compound prepared in the laboratory. In which the oxidation state of 'Pt' is (only numerical value)

59 The total number of cyclic structural as well as stereo isomers possible for a compound with the molecular formula C_5H_{10} .

60 The total number of alkenes possible by dehydrobromination of 3-bromo-3-cyclopentylhexane using alcoholic KOH is ____

(SINGLE CORRECT ANSWER TYPE)

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61 $\int \frac{1 + \cos x}{(x + \sin x)^2} dx =$

1. $\frac{1}{x + \sin x} + c$

2. $-\frac{1}{x + \sin x} + c$

3. $\frac{2}{x + \sin x} + c$

4. $-\frac{2}{x + \sin x} + c$

62 If $\int \frac{x^{49} \tan^{-1}(x^{50})}{1 + x^{100}} dx = k [\tan^{-1}(x^{50})]^2 + c$ then k =

1. $\frac{1}{50}$

2. $-\frac{1}{50}$

3. $\frac{1}{100}$

4. $-\frac{1}{100}$

63 $\int \frac{\sin^6 x}{\cos^8 x} dx =$

1. $\tan(7x) + c$

2. $\frac{\tan^7 x}{7} + c$

3. $\frac{\tan 7x}{7}$

4. $\sec^7 x + c$

64 $\int \frac{dx}{(x+100)\sqrt{x+99}} = f(x) + c \Rightarrow f(x):$

1. $2(x+100)^{1/2}$

2. $3(x+100)^{1/2}$

3. $2 \tan^{-1}(\sqrt{x+99})$

4. $2 \tan^{-1}(\sqrt{x+100})$

65 $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx =$

1. $\sin x + \cos x + c$

2. $\tan x + \cot x + c$

3. $\sec x - \csc x + c$

4. $\sin x - \cos x + c$

66 $\int e^x (x^2 + 2x + 3) dx =$

1. $e^x (x^2 - 3) + c$

2. $e^x (x^2 + 3) - x + c$

3. $e^x (x^2 + 3) + c$

4. $e^x (x^2 - 3) - x + c$

67 If $\int_0^K \frac{\cos x}{1 + \sin^2 x} dx = \frac{\pi}{4}$, then K =

1. $\frac{\pi}{4}$

2. $\frac{\pi}{2}$

3. π

4. 2π

68 $\int_0^{\pi/2} \frac{1}{4\cos^2 x + 9\sin^2 x} dx =$

1. $\frac{\pi}{4}$

2. $\frac{\pi}{8}$

3. $\frac{\pi}{12}$

4. $\frac{\pi}{16}$

69 $\int_0^1 \frac{4x^3}{\sqrt{1-x^8}} dx =$

1. $\frac{\pi}{4}$

2. $\frac{\pi}{2}$

3. π

4. $\frac{\pi}{8}$

70 $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx =$

1. $\frac{\pi^2}{4}$

2. $\frac{\pi}{4}$

3. $\frac{\pi^2}{2}$

4. $\frac{\pi}{2}$

71 The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1 + e^x} dx$

1. 0

2. -1

3. 1

4. None of these

72 If $u_n = \int_0^{\pi/4} \tan^n x dx$, then $u_n + u_{n-2} =$

1. $\frac{1}{n-1}$

2. $\frac{1}{n+1}$

3. $\frac{1}{2n-1}$

4. $\frac{1}{2n-1}$

73 If $[x]$ denotes the greatest integer function, then the value of $\int_4^{10} \frac{[x^2]}{[x^2 - 28x + 196] + [x^2]} dx$

1. 0

2. 1

3. 3

4. None of these

74 $\lim_{x \rightarrow 0} \frac{\int_0^x \sin^2 t \cos t dt}{x^3} =$

1. 1

2. $\frac{1}{2}$

3. $\frac{1}{3}$

4. $\frac{2}{3}$

75 The area bounded by $y = e^x$, $y = e^{-x}$ and $x = 1$ is

1. $e + \frac{1}{e} + 2$

2. $e + \frac{1}{e} - 2$

3. $e - \frac{1}{e} + 2$

4. $e - \frac{1}{e} - 2$

76 The area bounded by the curves $y = x$, $y = x^3$ is

1. $\frac{1}{4}$

2. $\frac{1}{6}$

3. $\frac{1}{3}$

4. $\frac{1}{2}$

77 If the distance between the centres of two circles of unit radii is 1, the common area of the circles is

1. $\frac{2\pi}{3} - \sqrt{3}$ 2. $\frac{2\pi}{3} + \sqrt{3}$ 3. $\frac{2\pi}{3} - \frac{\sqrt{3}}{2}$ 4. $\frac{2\pi}{3} + \frac{\sqrt{3}}{2}$

78 The order and degree of the differential equation $y = x \frac{dy}{dx} - \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$ is

1. (2, 1) 2. (2, 2) 3. (1, 2) 4. (1, 1)

79 Form the differential equation by eliminating A, B from $y = Ae^{2x} + Be^{3x} + Ce^{-2x}$ is

1. $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 12y = 0$ 2. $\frac{d^3y}{dx^3} + 3\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 12y = 0$
 3. $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 12y = 0$ 4. $\frac{d^3y}{dx^3} + 3\frac{d^2y}{dx^2} - \frac{dy}{dx} + 12y = 0$

80 The solution of the differential equation $\frac{dy}{dx} = e^{y+x} + e^{y-x}$ is

1. $e^{-y} = e^{-x} - e^x + C$ 2. $e^{-y} = e^x - e^{-x} + C$
 3. $e^{-y} = e^x + e^{-x} + C$ 4. None of these

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

81 If $\int \sqrt[3]{\frac{\sin^n x}{\cos^{n+6} x}} dx = a(\tan x)^b + c$ then $ab =$

82 If $\int \frac{(x^4 - x)^{\frac{1}{4}}}{x^5} dx = \frac{A}{B} \left(1 - \frac{1}{x^3}\right)^{\frac{5}{4}} + c$ then $B - 2A =$ _____

83 The value of $\int_{-1}^7 \operatorname{sgn}(\{x\}) dx$, where $\{ \cdot \}$ denotes the fractional part function

84 If $L = \lim_{n \rightarrow \infty} \frac{(1^2 + 2^2 + 3^2 + \dots + n^2)(1^3 + 2^3 + 3^3 + \dots + n^3)}{1^6 + 2^6 + 3^6 + \dots + n^6}$ then the value of $12L =$

85 If $\int_0^1 \frac{x + \sqrt{1-x^2} \cdot \sin^{-1}(x)}{\sqrt{1-x^2}} dx = \frac{\pi k}{6}$ then $K =$ _____

86 $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sec^2 t dt}{x \sin x} =$

87 If $y = f(x)$ is the solution of the differential equation $x(y^3 - x)dy = y(x + y^3)dx$ and $f(1) = (-2)^{1/3}$ and $f^{-1}(-2) = k$. Then $|k| =$ _____

88 The area bounded by the curve $|x| + |y| = 1$ is

89 For the equation $ydx + y^2dy = xdy, x \in R, y > 0, y = y(x)$ and $y(1) = 1$. Then $y(-3)$ is given by

90 The sum of the order and degree of the differential equation $\frac{d^2y}{dx^2} = \left[y + \left(\frac{dy}{dx} \right)^6 \right]^{1/4}$ is _____.



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

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Max.Marks:300

Key Sheet PHYSICS

1	1	2	1	3	1	4	1	5	2
6	3	7	1	8	3	9	1	10	2
11	2	12	2	13	2	14	4	15	1
16	2	17	4	18	4	19	2	20	4
21	6.25	22	2.0	23	1.0	24	4.0	25	51.0
26	8.0	27	3.0	28	15.0	29	41.0	30	0.25

CHEMISTRY

31	2	32	2	33	3	34	3	35	3
36	2	37	2	38	4	39	3	40	1
41	4	42	4	43	3	44	2	45	4
46	1	47	3	48	4	49	2	50	2
51	2.0	52	4.0	53	3.0	54	9	55	-2.5
56	6.0	57	4.0	58	5.0	59	7.0	60	5.0

MATHEMATICS

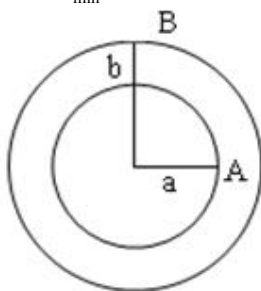
61	2	62	3	63	2	64	3	65	3
66	3	67	2	68	3	69	2	70	1
71	3	72	1	73	3	74	3	75	2
76	4	77	3	78	3	79	1	80	1
81	1.0	82	7.0	83	8.0	84	7.0	85	3.0
86	1.0	87	4.0	88	2.0	89	3.0	90	6.0

SOLUTIONS **PHYSICS**

1)
$$PE = \frac{-G}{r} (m_1 m_2 + m_2 m_3 + m_3 m_1)$$

$$= \frac{-G}{0.5} (1 \times 2 + 2 \times 3 + 1 \times 3) = \frac{-G(2+6+3)}{0.5} = \frac{-G(11)}{0.5} = -22G$$

2)
$$ev = \frac{hc}{\lambda_{\min}} \Rightarrow \lambda_{\min} = \frac{hc}{ev}$$



3)
$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{a} \quad V' = \frac{1}{4\pi\epsilon_0} \frac{q}{b} \quad \Rightarrow \frac{V'}{V} = \frac{a}{b}$$

4) The escape velocity on the earth is defined as $v_e = \sqrt{2g_e R_e}$
 Where R_e & g_e are the radius & acceleration due to gravity of earth.

Now for planet $g_p = 2g_e, R_p = R_e / 4$

So
$$v_p = \sqrt{2g_p R_p} = \sqrt{2 \times 2g_e \times R_e / 4} = \frac{v_e}{\sqrt{2}}$$

5)
$$R \propto A^{\frac{1}{3}}$$

$$\frac{R_1}{R_2} = \left(\frac{A_1}{A_2} \right)^{\frac{1}{3}}$$

$$\frac{2.8 \times 10^{-15}}{R_2} = \left(\frac{16}{205} \right)^{\frac{1}{3}} = \frac{1}{2.33}$$

$$R_2 = 2.33 \times 2.8 \times 10^{-15}$$

$$= 6.55 \times 10^{-15} \text{ m}$$

6)
$$c) PE = \frac{q_2 q_3}{4\pi\epsilon_0} \left[\frac{1}{0.1} - \frac{1}{0.5} \right] = \frac{8q_2 q_3}{4\pi\epsilon_0}$$

7)

$$\begin{aligned} df &= \frac{Gdm}{x^2} \quad dm = \frac{m}{L} dx \\ &= \frac{GM}{x^2} \frac{m}{L} dx \\ &= \frac{GMm}{L} \int_r^{L+r} \frac{dx}{x^2} \\ &= \frac{GMm}{L} \left[\frac{1}{L+r} - \frac{1}{r} \right] \\ &= \frac{-GMm}{L} \left[\frac{r-L-r}{r(L+r)} \right] \\ &= \frac{+GMm}{r(L+r)} \end{aligned}$$

8)

$$\Delta E = \frac{12400}{\lambda (\text{in } \text{\AA})} eV = \frac{12400}{0.21} = 59.047 \times 10^3 eV$$

9)

$$\begin{aligned} E &= \frac{V}{r} \\ \therefore r &= \frac{V}{E} = \frac{3000}{500} = 6m \end{aligned}$$

10)

$$T = 2\pi \sqrt{\frac{R}{g}} \Rightarrow n = \frac{1}{T}$$

11)

$$\begin{aligned} \frac{hc}{\lambda} &= \frac{hc}{\lambda_0} + kE \\ kE &= 12400 \left(\frac{1}{2000} - \frac{1}{4000} \right) \\ &= \frac{12400}{4000} = 3.1 eV \end{aligned}$$

12)

$$PE = \frac{-GMm}{R} - \frac{GMm}{R+2R} = GMm \left[\frac{1}{R} - \frac{1}{3R} \right] = \frac{2GMm}{R} = 2gmR$$

13)

It is clear that half-life of nuclei is 10 seconds.

Mean life of the nuclei is

$$T_{mean} = \frac{T_{1/2}}{0.693} = \frac{10}{0.693} = 14.43s$$

6.25% of original number means $\frac{1}{16}$ of the original

number = $\frac{1}{(2)^4}$ of original number.

Hence time taken is $4 \times T_{1/2} = 40 \text{ sec.}$

14)

$$\frac{q}{6\epsilon_0} = \frac{0.1 \times 10^{-6}}{6 \times 8.85 \times 10^{-12}} = 1883$$

15)

$$\frac{g^1}{g} = \left(\frac{R}{R+h} \right)^2 \Rightarrow \frac{90}{60} = \left(\frac{R}{R+h} \right)^2$$

16)

Electric field due to dipole at $(-a, 0)$ at the position of second dipole

will be $\vec{E} = k \times \frac{2p}{(2a)^3}(-\hat{i}) = \frac{kp}{4a^3}(-\hat{i})$

$\vec{P}$ for dipole at $(a, 0) = p\hat{i}$

$$U = -\vec{P} \cdot \vec{E} = \frac{kp^2}{4a^3}$$

17) $kE = \frac{12400}{3000} - 3$

$$= 4.13 - 3 = 1.13 \text{ eV}$$

18) $v_1 r_1 = v_2 r_2$ or $r_1^2 \omega_1 = r_2^2 \omega_2$ ($\because L = mrv = \text{constant}$)

Or $r_{\min}^2 \omega = r_{\max}^2 \omega'$

$$\therefore \omega' = \left(\frac{r_{\min}^2}{r_{\max}^2} \right) \omega$$

19) $\frac{-K2q}{(x-L)^2} + \frac{K8q}{x^2} = 0 \Rightarrow \frac{1}{(x-L)^2} = \frac{4}{x^2}$

Or $\frac{1}{x-L} = \frac{2}{x} \Rightarrow x = 2x - 2L$ or $x = 2L$

20) For the elliptical orbit of a satellite, the mechanical energy is given by

$$E = \frac{GMm}{2a} (\text{constant})$$

So, K.E + P.E = $-\frac{GMm}{2a}$

Here, K.E = $\frac{1}{2}mv^2$ and P.E = $-\frac{GMm}{r}$

Where at any position r , v is orbital velocity substituting the values we have

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = -\frac{GMm}{2a}$$

$$v^2 = GM \left(\frac{2}{r} - \frac{1}{a} \right)$$

21) $F = k \frac{(ne)(ne)}{r^2}$

$$0.9 = 9 \times 10^9 \frac{n^2 e^2}{(1)^2}$$

$$n = 6.25 \times 10^{13}$$

22) $PE = -4 \text{ MJ}, TE = -2 \text{ MJ}$

The additional energy required

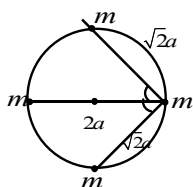
to escape = + 2 MJ

23) By using $E = W_0 + K_{\max} \Rightarrow K_{\max} = E - W_0$

Hence, $K_1 = 1 - 0.5 = 0.5$ and $K_2 = 2.5 - 0.5 = 2$

$$\frac{K_1}{K_2} = \frac{0.5}{2} = \frac{1}{4} = \frac{p}{4} \Rightarrow p = 1$$

24)



$$\sqrt{2} \frac{GM^2}{(\sqrt{2}a)^2} + \frac{GM^2}{(2a)^2} \frac{\sqrt{2}Gm^2}{2a^2} + \frac{Gm^2}{4a^2}$$

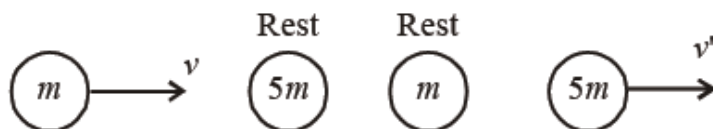
$$= \frac{Gm^2}{Ga^2} \left(\frac{1}{\sqrt{2}} + \frac{1}{4} \right)$$

$$V = \sqrt{\frac{Gm}{a} \left(\frac{1}{\sqrt{2}} + \frac{1}{4} \right)}$$

25)

Before collision

After collision



Particle

Hydrogen

Particle

Hydrogen

From linear momentum conservation, $L_i = L_f$

$$mV + 0 = 0 + 5mV' \Rightarrow V' = \frac{V}{5}$$

$$\text{Loss of KE} = KE_i - KE_f = \frac{1}{2}mv^2 - \frac{1}{2}(5m)\left(\frac{v}{5}\right)^2$$

$$= \frac{1}{2}mv^2 \left(1 - \frac{1}{5} \right) = \frac{4}{5} \left(\frac{mv^2}{2} \right)$$

$$= \frac{4}{5} KE_i = 10.2 \text{ eV}$$

[$\therefore$ Energy in first excited state of atom = 10.2 eV]

$$KE_i = 12.75 \text{ eV} = \frac{N}{4} \Rightarrow N = 51$$

The value of $N = 51$.

26)

$$E = \frac{3000}{6 \times 10^{-2}}$$

$$E = 500 \times 10^2 \text{ V} / m$$

$$F = Eq = 8 \times 10^{-15} \text{ N}$$

27)

$$\text{For earth : } g = \frac{GM}{R^2} = \frac{4\pi G\rho R}{3}$$

$$\text{For planet : } g' = \frac{4\pi G\rho'R'}{3}$$

$$\therefore \frac{g'}{g} = \frac{\rho'}{\rho} \times \frac{R'}{R}$$

$$\Rightarrow \frac{\sqrt{6}}{11} = \frac{2}{3} \times \frac{R'}{R} \Rightarrow \frac{R'}{R} = \frac{3\sqrt{6}}{22}$$

Escape velocity on earth is $v = \sqrt{2gR}$ and on the planet is $v' = \sqrt{2g'R'}$. Therefore,

$$\frac{v'}{v} = \sqrt{\frac{g'}{g} \times \frac{R'}{R}} = \sqrt{\frac{\sqrt{6}}{11} \times \frac{3\sqrt{6}}{22}} = \frac{3}{11}$$

$$\text{Thus } v' = \frac{3}{11} \times v = \frac{3}{11} \times 11 = 3 \text{ kms}^{-1}$$

28) 15.0

29) For K_{α} X-ray line ,

$$\frac{1}{\lambda} = R(Z-1)^2 \left[\frac{1}{1^2} - \frac{1}{2^2} \right] = R(z-1)^2 \left[1 - \frac{1}{4} \right]$$

$$\Rightarrow \frac{1}{\lambda} = \frac{3}{4} R(z-1)^2 \dots\dots(i)$$

30) $U_{12} = K \frac{-qQ}{d}, U_{13} = K \frac{q^2}{2d}$

$$U_{23} = K \frac{-qQ}{d}$$

$$\text{PE of the system } U = U_{12} + U_{13} + U_{23} = 0$$

$$\Rightarrow \frac{Q}{q} = \frac{1}{4} = 0.25.$$

CHEMISTRY

31) At constant volume work done = Zero

$$\therefore \Delta E = q = 10 \text{ KJ}$$

32) Conceptual

33) There is no exchange of mass and energy with the surroundings

34) Isolated system



$$\frac{1}{2} F_2 + HCl \rightarrow HF + \frac{1}{2} Cl_2 \quad \Delta H = \frac{-352.8}{2} = -176.4 \text{ KJ} \rightarrow 1$$

$$\frac{1}{2} F_2 + \frac{1}{2} H_2 \rightarrow HF \quad \Delta H = -268.3 \text{ KJ} \rightarrow 2$$

$$\frac{1}{2} H_2 + \frac{1}{2} Cl_2 \rightarrow HCl \quad \Delta H_f^0 = ?$$

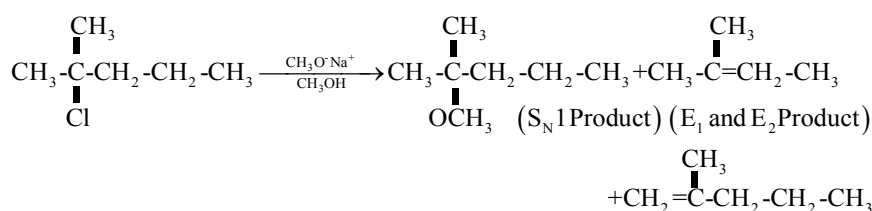
$$\text{Eq 2} - \text{Eq 1} = -268.3 - (-176.4)$$

$$= -91.9 \text{ KJ / mole}$$

36) Conceptual

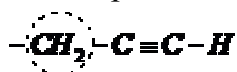
37) n-Butane, $\overset{1}{CH_3}\overset{2}{CH_2}\overset{2}{CH_2}\overset{1}{CH_3}$ has two different hydrogen atoms, marked by 1 and 2.

38)



39) Conceptual

40) Circled portion is called propargyl group



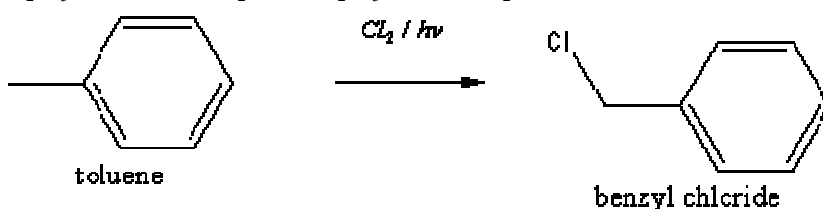
41) *conc H₂SO₄ / 170°C*

42) Ethane – 1, 2 – diol

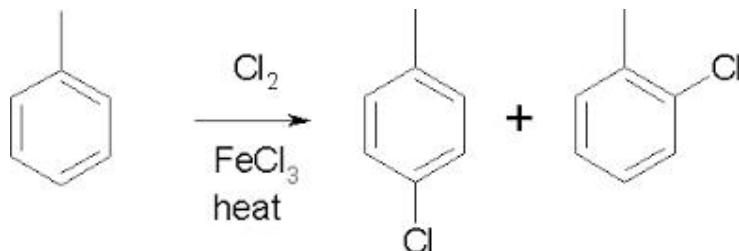
43) Conceptual

44) Because this is electrophilic addition reaction. Anhydrous AlCl_3 forms electrophilic (H^+) with HCl

45) $\text{C}_2\text{H}_5\text{---OH} + \text{SOCl}_2 \xrightarrow{\text{pyridine}} \text{C}_2\text{H}_5\text{---Cl} + \text{SO}_2 + \text{HCl}$

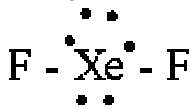


46)

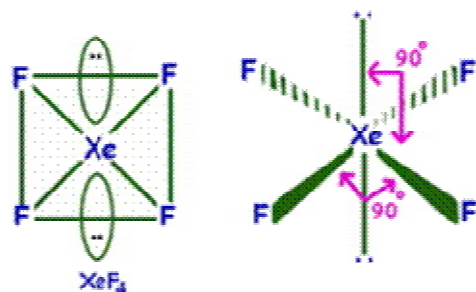


47) Conceptual

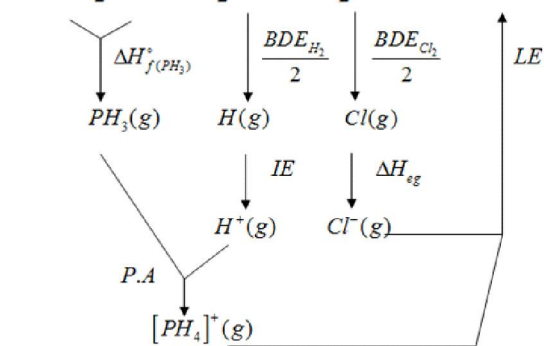
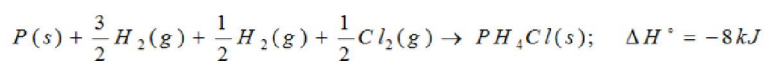
48) As_2S_3 is soluble in yellow ammonium sulphide but not CdS



49) has 3 – lone pairs ; 2 – bond pairs



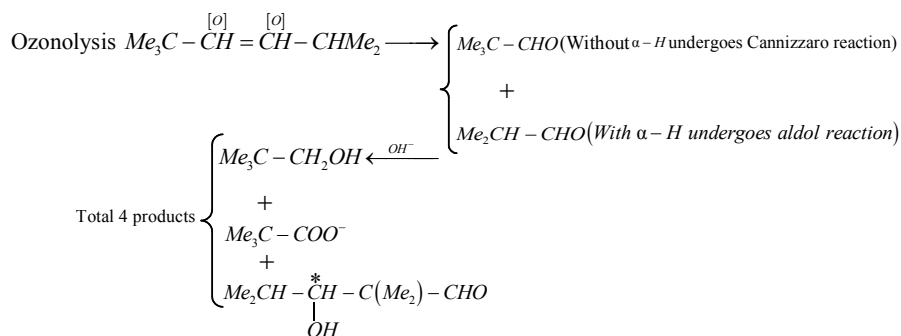
50)



$$-2 + PA + 10 + 10 + 4 - 4 - 28 = -8$$

$$PA = 2$$

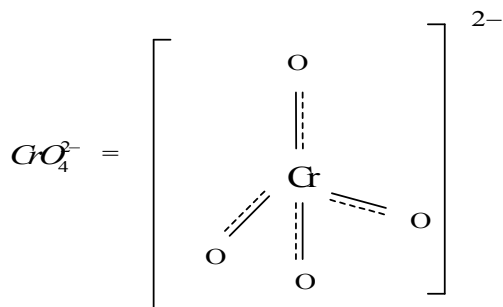
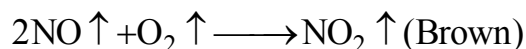
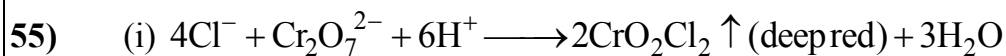
51)



52)

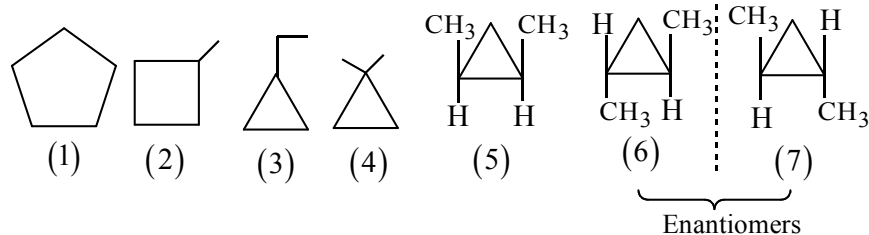
53) 3.0

54) All the Ist and IInd group cations of qualitative analysis precipitate as sulphides in acid medium.

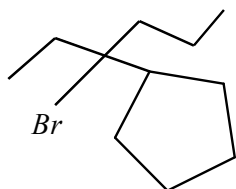


57)

58) 5.0

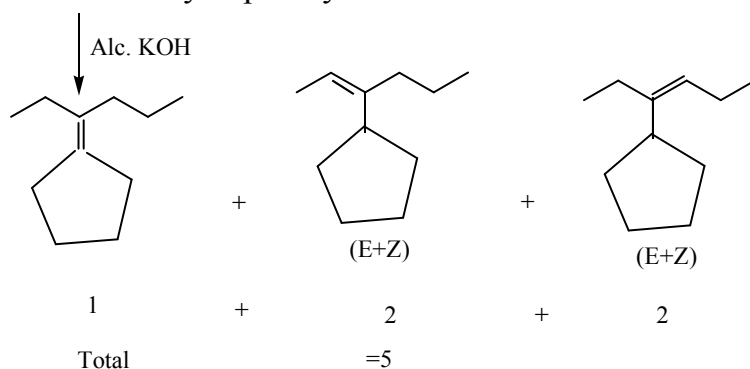


59)



60)

3-bromo-3-cyclopentylhexane



MATHEMATICS

61)

$$x + \sin x = t$$

$$\int \frac{1}{t^2} dt = -\frac{1}{t} + c$$

$$= \frac{-1}{x + \sin x} + c$$

62)

$$\tan^{-1} x^{50} = t \Rightarrow \frac{50x^{49}}{x^{100} + 1} dx = dt$$

$$\frac{1}{50} \int t dt = \frac{1}{5} \times \frac{t^2}{2} + c$$

$$\therefore k = \frac{1}{100}$$

63)

Conceptual

64)

$$2 \tan^{-1}(\sqrt{x+99})$$

65)

$$\int \left(\frac{\sin^3 x}{\sin^2 \cos^2 x} + \frac{\cos^3 x}{\sin^2 \cos^2 x} \right) dx$$

$$= \int (\sec x \tan x + \operatorname{cosec} x \cot x) dx$$

$$= \sec x - \operatorname{cosec} x + c$$

66)

Conceptual

67)

$$\left[\tan^{-1}(\sin x) \right]_0^k = \frac{\pi}{4}$$

$$\tan^{-1}(\sin k) = \frac{\pi}{4} \Rightarrow k = \frac{\pi}{2}$$

$$68) \int_0^{\pi/2} \frac{1}{2^2 \cos^2 x + 3^2 \sin^2 x} dx = \frac{\pi}{2ab} = \frac{\pi}{2 \times 2 \times 3} = \frac{\pi}{12}$$

$$69) \int_0^1 \frac{4x^3}{\sqrt{1-(x^4)^2}} dx = \left[\sin^{-1}(x^4) \right]_0^1 = \frac{\pi}{2}$$

$$70) \int_0^{\pi} \left(\frac{\sin x}{1 + \cos x} \right) dx = \frac{\pi + 0}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx$$

$$= \frac{\pi}{2} \left[-\tan^{-1}(\cos x) \right]_0^{\pi}$$

$$= -\frac{\pi}{2} \left(\frac{-\pi}{4} - \frac{\pi}{4} \right) = \frac{\pi^2}{4}$$

$$71) \int_{-\pi/2}^{\pi/2} \frac{\cos x}{1 + e^x} dx = \int_0^{\pi/2} \frac{\cos x}{1 + e^x} dx + \int_0^{\pi/2} \frac{\cos(-x)}{1 + e^{-x}} dx$$

$$= \int_0^{\pi/2} \cos x \left(\frac{1}{1 + e^x} + \frac{e^x}{1 + e^x} \right) dx = \left[\sin x \right]_0^{\pi/2} = 1$$

$$72) u_n = \int_0^{\pi/4} \tan^n x dx$$

$$= \int_0^{\pi/4} (\sec^2 x - 1) \tan^{n-2} x dx$$

$$= \int_0^{\pi/4} \sec^2 x \tan^{n-2} x dx - \int_a^{\pi/4} \tan^{n-2} x dx$$

$$= \left[\frac{\tan^{n-1} x}{n-1} \right]_0^{\pi/4} - u_{n-2}$$

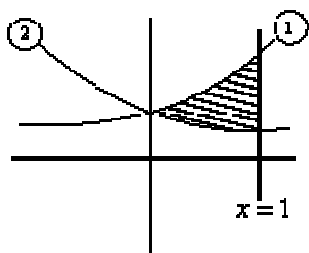
$$\Rightarrow u_n + u_{n-2} = \frac{1}{n-1}$$

$$73) 2I = \int_4^{10} \frac{[x^2]}{[(x-14)^2] + [x^2]} dx + \int_4^{10} \frac{[(14-x)^2]}{[(14-x-14)^2] + [(14-x)^2]} dx$$

$$= \int_4^{10} dx = 6 \quad \therefore I = 3.$$

$$74) \lim_{x \rightarrow 0} \frac{\int_0^x \sin^2 t \cos t dt}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{(\sin^2 x \cos x) - 0}{3x^2} = \frac{1}{3}$$



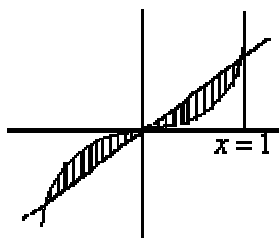
75)

$$y = e^x \rightarrow 1$$

$$y = e^{-x} \rightarrow 2$$

$$R.A = \int_0^1 (e^x - e^{-x}) dx$$

$$= e + \frac{1}{e} - 2$$



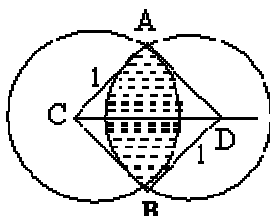
76)

$$y = x \rightarrow 1$$

$$y = x^3 \rightarrow 2$$

By solving we get $x = 0, x = 1$

$$R.A = 2 \int_0^1 (x - x^3) dx = \frac{1}{2}$$



77)

ACBD is rhombus.

$$\angle ACB = \angle ADB = \frac{2\pi}{3}$$

The desired area = Area of sector ACB + area of sector BAD – area of the rhombus

$$ACBD = 2 \cdot \frac{1}{2} \cdot \frac{2\pi}{3} - 2\Delta ACD = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$$

78)

$$\left(x \frac{dy}{dx} - y \right)^2 = 1 + \left(\frac{dy}{dx} \right)^2$$

Order = 1 degree = 2

79)

$$y_3 - s_1 y_2 + s_2 y_1 - s_3 y = 0$$

$$\Rightarrow y_3 - 3y_2 - 4y_1 + 12y = 0$$

80) Given differential equation is $\frac{dy}{dx} = e^y (e^x + e^{-x}) \Rightarrow e^{-y} dy = (e^x + e^{-x}) dx$, integrating, we get

$$\frac{e^{-y}}{-1} = e^x + \frac{e^{-x}}{-1} + C' \text{ or } e^{-y} = -e^x + e^{-x} - C' \text{ or } e^{-y} = e^{-x} - e^x + C$$

81) $\int \tan^{\frac{n}{3}} x \cdot \sec^2 x dx = \frac{(\tan x)^{\frac{n}{3}+1}}{\frac{n}{3}+1} + C$

$$= \frac{3}{n+3} (\tan x)^{\frac{n+3}{3}} + C \text{ ab} = 1$$

82) $\int \frac{(x^4)^{\frac{1}{4}} \left(1 - \frac{x}{x^4}\right)^{\frac{1}{4}}}{x^5} dx$

$$= \int \frac{x \left(1 - \frac{1}{x^3}\right)^{\frac{1}{4}}}{x^5} dx, \quad = \int \frac{\left(1 - \frac{1}{x^3}\right)^{\frac{1}{4}}}{x^4} dx$$

$$\text{let } 1 - \frac{1}{x^3} = t, \quad \frac{3}{x^4} dx = dt$$

$$\frac{1}{x^4} dx = \frac{1}{3} dt, \quad = \int t^{\frac{1}{4}} \cdot \frac{1}{3} dt$$

$$= \frac{1}{3} \cdot \frac{t^{\frac{5}{4}}}{\frac{5}{4}} + C, \quad = \frac{4}{15} t^{\frac{5}{4}} + C$$

$$= \frac{4}{15} \left(1 - \frac{1}{x^3}\right)^{\frac{5}{4}} + C$$

$$A = 4, B = 15$$

$$B - 2A = 7$$

83) CONCEPTUAL

84) $L = \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n r^2 \sum_{r=1}^n r^3}{\sum_{r=1}^n r^6}$

$$L = \lim_{n \rightarrow \infty} \frac{\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^2 \cdot \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^3}{\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^6} = \frac{7}{12}$$

$$12L = 7$$

85) $\int_0^1 \left[\frac{x}{\sqrt{1-x^2}} + \sin^{-1}(x) \right] dx = (x \sin^{-1} x)_0^1 = \frac{\pi}{2}$

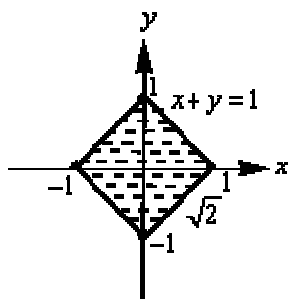
86) $\lim_{x \rightarrow 0} \frac{\tan x^2}{x \sin x}$

$$\lim_{x \rightarrow 0} \left(\frac{\tan x^2}{x^2} \times \frac{x}{\sin x} \right)$$

$$1 \times 1 = 1$$

87) 4.0

88) The region is a square of side $\sqrt{2}$
 $\therefore A = 2$



89) $ydx + y^2dy = xdy \Rightarrow \frac{ydx - xdy}{y^2} = -dy$

Integrating $\frac{x}{y} = -y + c$

$$y(1) = 1 \Rightarrow c = 2 \quad \Rightarrow \frac{x}{y} = -y + 2$$

$$x = -3 \Rightarrow -\frac{3}{y} = -y + 2 \Rightarrow y^2 - 2y - 3 = 0$$

$$\Rightarrow y = 3, -1, \quad y > 0 \Rightarrow y = 3$$

90) $\frac{d^2y}{dx^2} = \left[y + \left(\frac{dy}{dx} \right)^6 \right]^{1/4}$

$$\Rightarrow \left(\frac{d^2y}{dx^2} \right)^4 = \left[y + \left(\frac{dy}{dx} \right)^6 \right]$$

Hence, the order is 2 and degree is 4.



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON Central Office – Madhapur – Hyderabad

Sec: **Sr.Super-60& All**

Jee-Main

Date: 06-05-2021

Time: 09.00Am to 12.00

PTM-02

Max.Marks:300

IMPORTANT INSTRUCTIONS:

1. Immediately fill in the Admission number on this page of the Test Booklet with Blue/Black Ball Point Pen only.
2. The candidates should not write their Admission Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1, 2, 3 consisting of **Physics, Chemistry** and **Mathematics** having **30 questions** in each subject and each subject having **two sections**.

(I) Section-I contains 20 **multiple choice** questions with **only one correct** option.

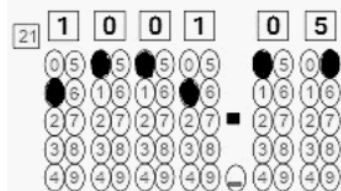
Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

(II) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only.

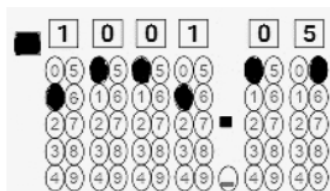
First 5 attempted questions will be considered if more than 5 questions attempted.

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted question 21, bubble on 21 as shown below.



Question Answered for marking



Question Cancelled for marking

Marking scheme : +4 for correct answer and 0 in all other cases.

6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capitals): _____

Admission Number:

--	--	--	--	--	--	--	--	--	--

Candidate's Signature: _____

Invigilator's Signature: _____

06-05-2021_Sr.Super60 & All_Jee-Main_PTMM-02_Test Syllabus

PHYSICS : Capacitors, Current electricity, Electromagnetism (Excluding EMI, AC)

CHEMISTRY : Chemical Equilibrium, Ionic Equilibrium, Grignard Reagent
GOC (Electronic effects; Reaction intermediates), Acidity-Basicity, Metallurgy
Coordination Compounds

MATHEMATICS : Complex numbers, P&C B.T, Probability

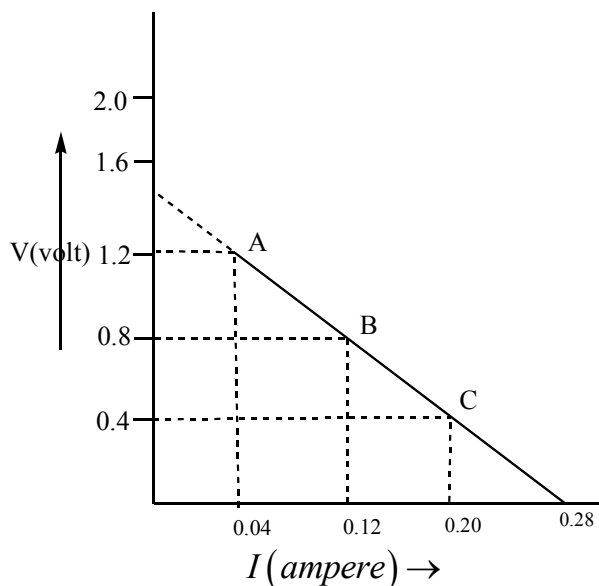
PHYSICS**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

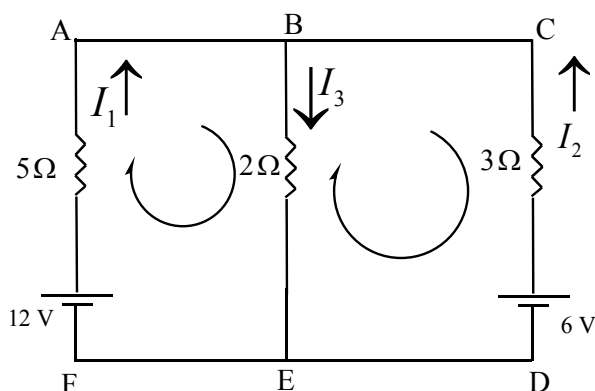
Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

01. A copper wire has a resistance of 10Ω an area of cross-section 1mm^2 . A potential difference of 10V exists across the wire. Calculate the drift speed of electrons if the number of electrons per cubic metre in copper is 8×10^{28} electrons.
 1) 0.078mms^{-1} 2) 0.087mms^{-1} 3) 0.088mms^{-1} 4) 0.068mms^{-1}
02. A cylindrical wire is stretched to increase its length by 10%. Calculate the percentage increase in resistance.
 1) 22% 2) 23% 3) 21% 4) 20%
03. At what temperature would the resistance of a copper conductor be double its resistance at 0°C ? (given $\alpha_{Cu} = 3.9 \times 10^{-3} \text{ }^\circ\text{C}^{-1}$)
 1) 256°C 2) 258°C 3) 260°C 4) 248°C
04. An electric toaster uses nichrome (an alloy of nickel and chromium) for its heating element. When negligibly small current passes through it, its resistance at room temperature (27.0°C) is found to be 75.3Ω . When the toaster is connected to a 230V supply, the current settles after a few seconds to a steady, the current value of 2.68A. What is the temperature of the nichrome element? The temperature coefficient of resistance of nichrome averaged over the temperature range involved is $1.70 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$
 1) 845°C 2) 851°C 3) 850°C 4) 847°C
05. The resistance of the platinum wire of a platinum resistance thermometer at the ice point is 5Ω . And at steam point is 5.23Ω . When the thermometer is inserted in a hot bath, the resistance of the platinum wire is 5.795Ω . Calculate the temperature of the bath.
 1) 346.65°C 2) 345.65°C 3) 346.65°C 4) 346.56°C
06. The reading on a high resistance voltmeter when a cell is connected across it is 2.2V. When the terminals of the cell are also connected to a resistance of 5Ω , the voltmeter reading drops to 1.8V. Find the internal resistance of the cell.
 1) 1.2Ω 2) 1.1Ω 3) 1.4Ω 4) 1.3Ω

07. Potential difference across terminals of a cell were measured (in volt) against different currents (in ampere) flowing through the cell. A graph drawn which was a straight line ABC, as shown in the fig. Determine from the graph internal resistance of the cell

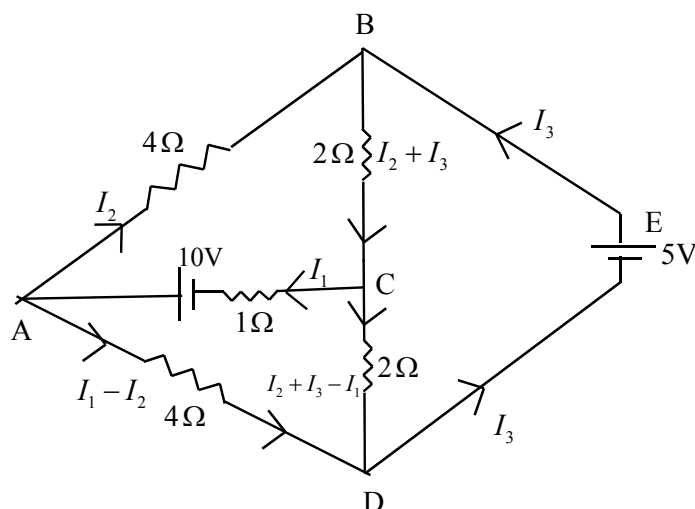


- 1) 3Ω 2) 4Ω 3) 5Ω 4) 1Ω
08. A cell of emf 1.1 V and internal resistance 0.5Ω is connected to a wire of resistance 0.5Ω . Another cell of the same emf is connected in series but the current remains the same. Find the internal resistance of the second cell.
- 1) 1Ω 2) 2Ω 3) 3Ω 4) 4Ω
09. Using Kirchhoff's laws in the electrical network shown in the figure. Calculate the values of I_1



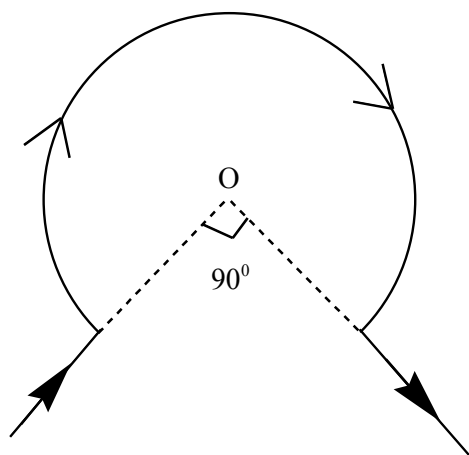
- 1) $\frac{18}{31}A$ 2) $\frac{66}{31}A$ 3) $\frac{48}{31}A$ 4) $\frac{33}{31}A$

10. Determine the current I_1 in given circuit



- 1) 2.5 A 2) $\frac{5}{8} A$ 3) $1\frac{7}{8} A$ 4) 3.25 A

11. The wire shown in fig. carries a current of 10A. Determine the magnitude of the magnetic field at the centre O. Given radius of the bent coil is 3 cm.

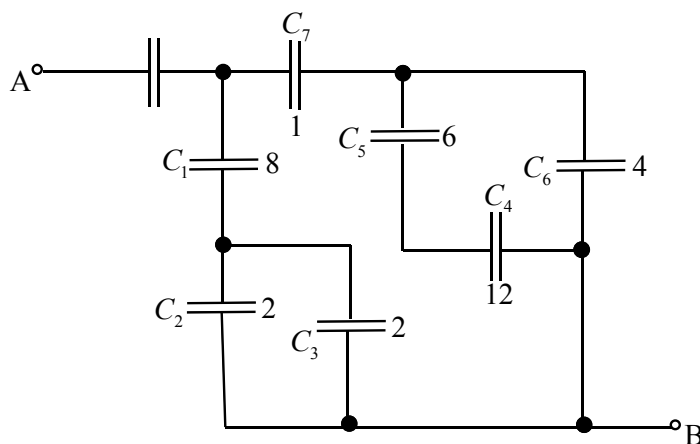


- 1) $1.56 \times 10^{-3} T$ 2) $1.57 \times 10^{-3} T$ 3) $1.50 \times 10^{-3} T$ 4) $1.60 \times 10^{-3} T$

12. The radius of the first orbit of hydrogen atom is $0.5 \text{ \AA}$. The electron moves in an orbit with a uniform speed of $2.2 \times 10^6 \text{ ms}^{-1}$. What is the magnetic field produced at the centre of the nucleus due to the motion of this electron? Use $\mu_0 / 4\pi = 10^{-7} \text{ NA}^{-2}$ and electric charge $= 1.6 \times 10^{-19} \text{ C}$

- 1) 13.01 T 2) 15.07 T 3) 16.07 T 4) 14.07 T

13. An electron is moving at 10^6 ms^{-1} in a direction parallel to a current of 5A, flowing through an infinitely long straight wire, separated by a perpendicular distance of 10cm in air. Calculate the magnitude of the force experienced by the electron.
 1) $1.6 \times 10^{-18} \text{ N}$ 2) $1.8 \times 10^{-18} \text{ N}$ 3) $1.9 \times 10^{-18} \text{ N}$ 4) $1.7 \times 10^{-18} \text{ N}$
14. Keeping the voltage of the charging source constant, what would be the percentage change in the energy stored in a parallel plate capacitor if the separation between its plates were to be decreased by 10%?
 1) 12.12 % 2) 11.11 % 3) 13.13 % 4) 14.14 %
15. An electron beam passes through a magnetic field of $2 \times 10^{-3} \text{ Wbm}^{-2}$ and an electric field of $3.4 \times 10^4 \text{ Vm}^{-1}$, both acting simultaneously. If the path of the electron remains un-deviated, calculate the speed of electrons. If the electric field is removed what will be the radius of the circular path? Mass of an electron = $9.1 \times 10^{-31} \text{ kg}$
 1) $4.7 \times 10^{-8} \text{ m}$ 2) $4.9 \times 10^{-8} \text{ m}$ 3) $4.8 \times 10^{-8} \text{ m}$ 4) $4.5 \times 10^{-8} \text{ m}$
16. Two capacitors of capacitances $3 \mu\text{F}$ & $6 \mu\text{F}$ are charged to potential of 2V and 5V respectively. These two charged capacitors are connected in series. Find the potential across C_1
 1) 11 V 2) 10 V 3) 12 V 4) 13 V
17. Three capacitors of equal capacitance, when connected in series have net capacitance C_1 and when connected in parallel have net capacitance C_2 . What the value is of $\frac{C_1}{C_2}$ is
 1) 1:3 2) 3:1 3) 1:9 4) 9:1
18. From the network shown in fig. Find the value of the capacitance C if the equivalent capacitance between points A and B is to be $1 \mu\text{F}$. All the capacitances are in μF



- 1) $\frac{33}{23} \mu\text{F}$ 2) $\frac{32}{23} \mu\text{F}$ 3) $\frac{31}{23} \mu\text{F}$ 4) $\frac{30}{23} \mu\text{F}$

19. An electric bulb rated for 500 watt at 100V is used in a circuit having a 200V supply. The resistance R that must be put in series with the bulbs, so that the bulb delivers 500 watt is
 1) 20Ω 2) 25Ω 3) 30Ω 4) 40Ω
20. A 1 Kw heater is meant to operate at 200V. How many units of electrical energy will it consume in the month of April. If it operates 10hr daily at specified voltage
 1) 300 J 2) 3000 J 3) 300 Kwh 4) 300 erg

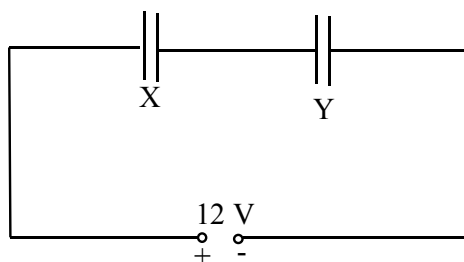
(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

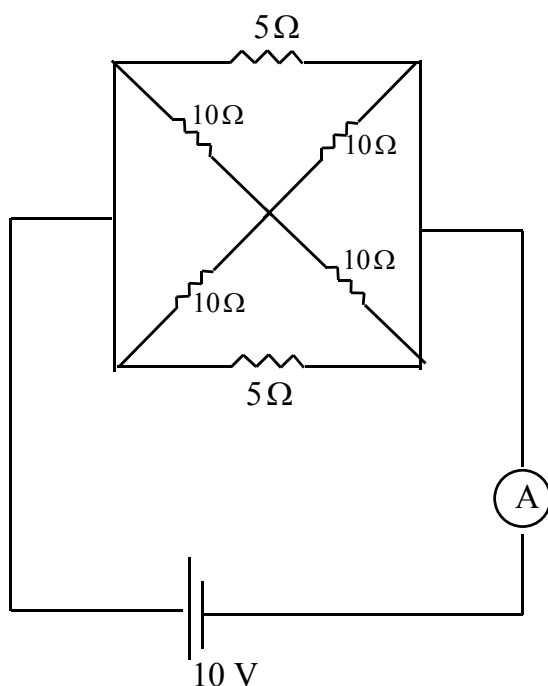
Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

21. A wire of resistance 4Ω is stretched to 4 times its length and is cut into 4 equal parts of which 3 are connected in parallel and one in series. What is the new effective resistance in ohms?
22. Two parallel plate capacitors, X and Y have same area of plates and same separation between them. X has air between the plates while Y contains a dielectric medium of $\epsilon_r = 4$. Calculate the potential difference between the x and Y in Volt?

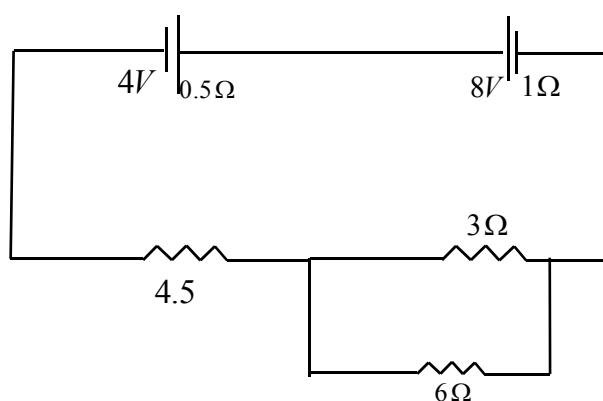


23. A 600 pF capacitor is charged by 200V supply. It is then disconnected from the supply and is connected to another uncharged 600pF capacitor. How much electrostatic energy is lost in μJ
24. A 900pF capacitor is charged by a 100V battery. The capacitor is disconnected & connected to another identical capacitor in parallel. The energy stored in the system is $\text{_____} \times 10^{-6} J$

25. Calculate the current in the Ammeter in Amperes in circuit given

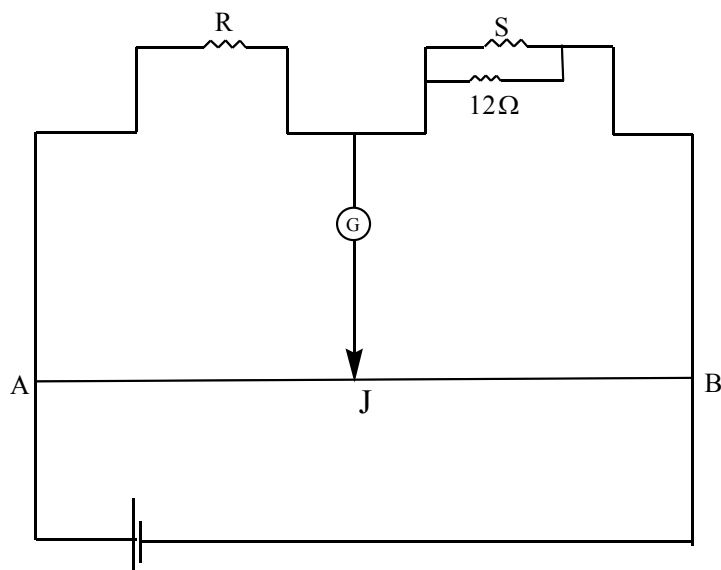


26. A p.d of 100V is applied to the ends of a copper wire one metre long. Find approximately the drift velocity of electrons in ms^{-1} . Given conductivity of copper = $5.81 \times 10^7 \Omega^{-1}m^{-1}$ and number density of electrons $n = 8.5 \times 10^{28} m^{-3}$
27. In the circuit shown find the current in 6Ω resistance in Ampere

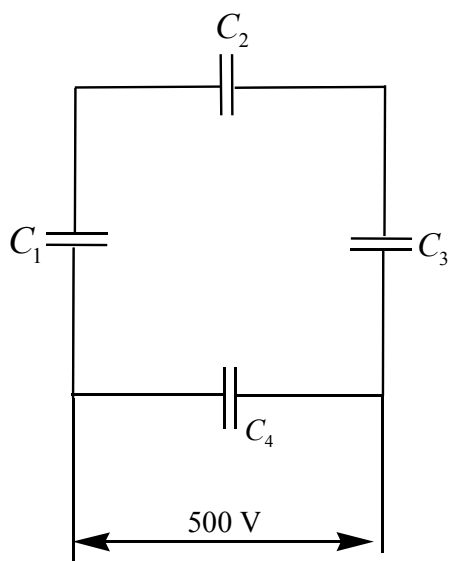


28. A 24V battery of internal resistance of 4Ω is connected to a variable resistor. At what value of the current in Ampere drawn from the battery is rate of heat produced in the resistor is maximum?

29. In a meter bridge the null point is found at a distance of 33.7 cm from A. If now a resistance of 12Ω is connected in parallel with S, the null point occurs at 51.9 cm. Find the value of S in ohm.



30. Four capacitors each of $10\mu F$ are connected to 500V supply. Find the p.d across ' C_3 '



(SINGLE CORRECT ANSWER TYPE)

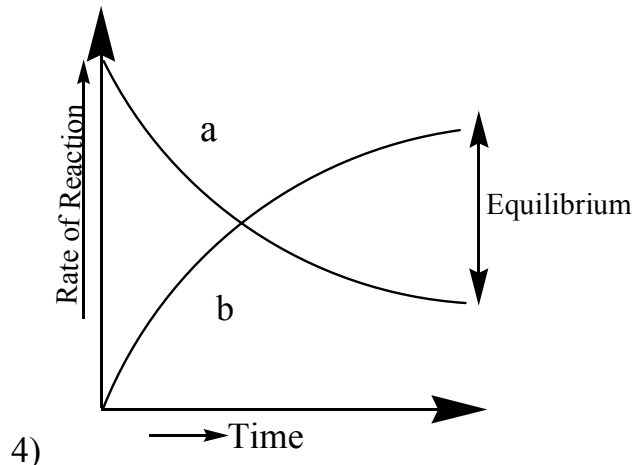
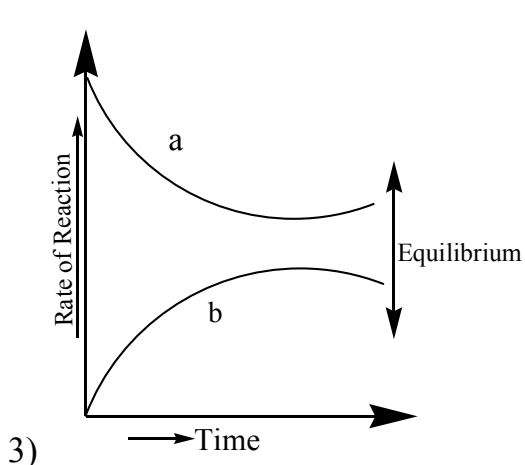
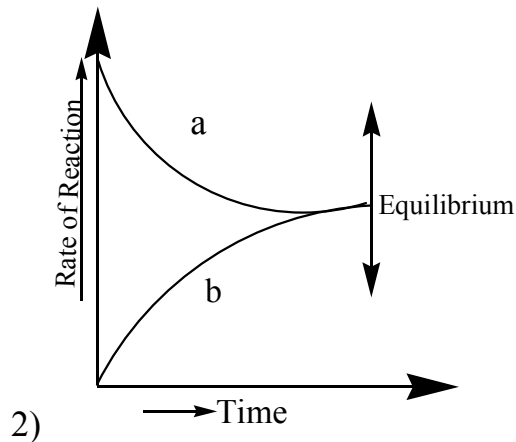
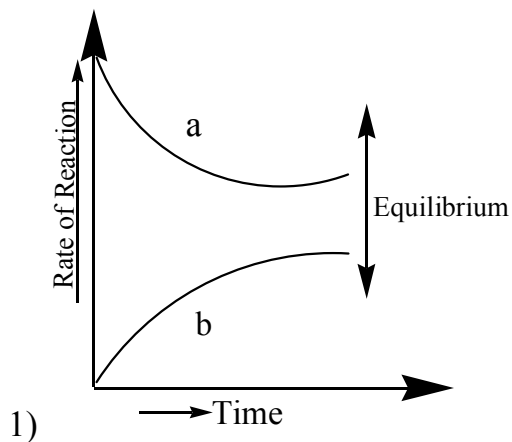
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

31. for the reaction $Fe_2N(s) + \frac{3}{2}H_2(g) \rightleftharpoons 2Fe(s) + NH_3(g)$

- 1) $K_c = K_p(RT)$ 2) $K_c = K_p(RT)^{-\frac{1}{2}}$ 3) $K_c = K_p(RT)^{\frac{1}{2}}$ 4) $K_c = K_p(RT)^{\frac{3}{2}}$

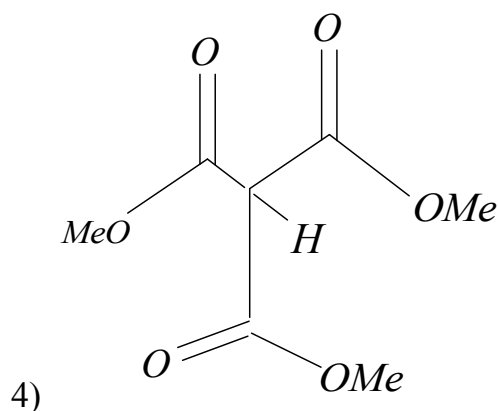
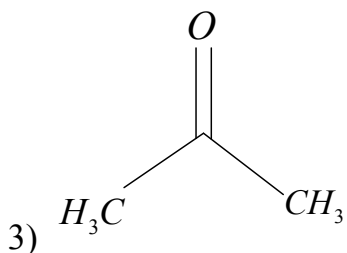
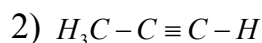
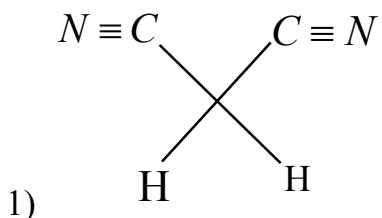
32. For the equilibrium $A \rightleftharpoons B$ the variation of the rate of the forward (a) and reverse (b) reaction with time is given by



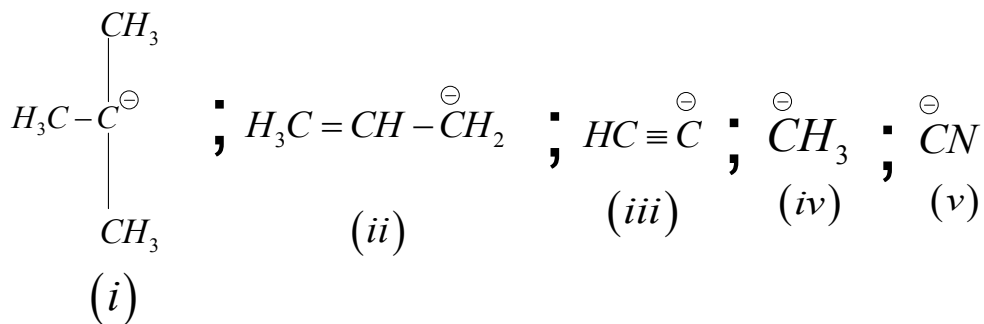
33. In chemical equilibrium reaction $A + 2B \rightleftharpoons 2C + D$, the initial concentration of B was 1.5 times of the concentration of A, but the equilibrium concentrations of A and B were found to be equal. The equilibrium constant (K) for the a foresaid chemical reaction is:

- 1) 4 2) 16 3) $\frac{1}{4}$ 4) 1

34. Following four solutions are prepared by mixing different volumes of NaOH and HCl of different concentrations, pH of which one of them will be equal to 1?
- 1) $55\text{ mL } \frac{M}{10} \text{ HCl} + 45\text{ mL } \frac{M}{10} \text{ NaOH}$ 2) $75\text{ mL } \frac{M}{5} \text{ HCl} + 25\text{ mL } \frac{M}{5} \text{ NaOH}$
 3) $100\text{ mL } \frac{M}{10} \text{ HCl} + 100\text{ mL } \frac{M}{10} \text{ NaOH}$ 4) $60\text{ mL } \frac{M}{10} \text{ HCl} + 40\text{ mL } \frac{M}{10} \text{ NaOH}$
35. The pK_a of a weak acid, HA, is 4.80. The pK_b of a weak base, BOH, is 4.78. The pH of an aqueous solution of the corresponding salt, BA, Will be
 1) 9.58 2) 4.79 3) 7.01 4) 9.22
36. An acidic buffer is obtained on mixing:
 1) 100 mL of 0.1M CH_3COOH and 100 mL of 0.1M NaOH
 2) 100 mL of 0.1M HCl and 200 mL of 0.1M NaCl
 3) 100 mL of 0.1M CH_3COOH and 200 mL of 0.1M NaOH
 4) 100 mL of 0.1M HCl and 200 mL of 0.1M CH_3COONa
37. If solubility product of $\text{Zr}_3(\text{PO}_4)_4$ is denoted by K_{sp} and its molar solubility is denoted by S, then which of the following relation between S and K_{sp} is correct?
 1) $S = \left(\frac{K_{sp}}{144}\right)^{1/6}$ 2) $S = \left(\frac{K_{sp}}{6912}\right)^{1/7}$ 3) $S = \left(\frac{K_{sp}}{929}\right)^{1/9}$ 4) $S = \left(\frac{K_{sp}}{216}\right)^{1/7}$
38. Which one of the following compounds possesses the most acidic hydrogen?



39. The increasing order of basicity for the following intermediates is (from weak to strong)



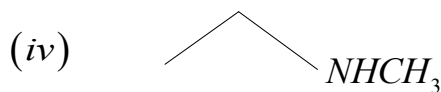
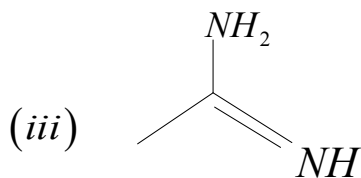
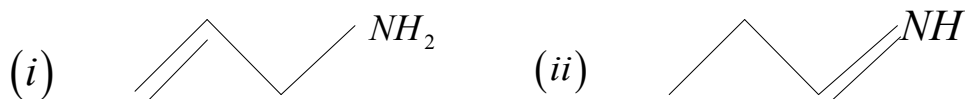
1) (iii) < (i) < (ii) < (iv) < (v)

2) (v) < (i) < (iv) < (ii) < (iii)

3) (v) < (iii) < (ii) < (iv) < (i)

4) (iii) < (iv) < (ii) < (i) < (v)

40. The increasing order of basicity of the following compounds is



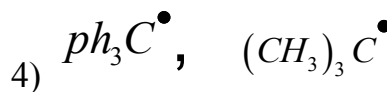
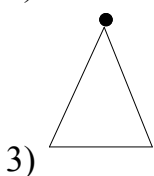
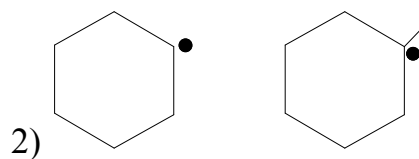
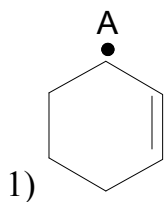
1) (i) < (iii) < (ii) < (iv)

2) (ii) < (i) < (iii) < (iv)

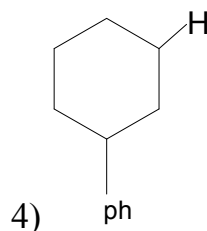
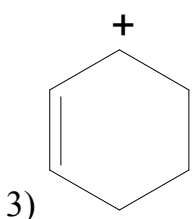
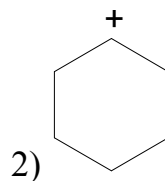
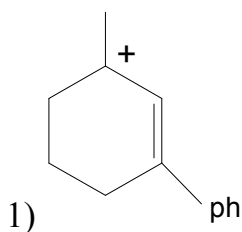
3) (ii) < (i) < (iv) < (iii)

4) (iv) < (ii) < (i) < (iii)

41. In which of the following pairs A is more stable than B?



42. Which of the following is most stable?



43. Consider the acidity of the carboxylic acids:

(i) $phCOOH$; (ii) $o-NO_2C_6H_4COOH$; (iii) $p-NO_2C_6H_4COOH$; (iv) $m-NO_2C_6H_4COOH$

Which of the following order is correct?

1) $ii > iv > i > iii$ 2) $ii > iv > iii > i$ 3) $i > ii > iii > iv$ 4) $ii > iii > iv > i$

44. Match the ores(column A) with the metals(column B)

	COLUMN-A (Ores)		COLUMN-B (Metals)
(I)	Siderite	(a)	Zinc
(II)	Kaolinite	(b)	Copper
(III)	Malachite	(c)	Iron
(IV)	Calamine	(d)	Aluminium

1) (I) $\rightarrow$ a; (II) $\rightarrow$ b; (III) $\rightarrow$ c; (IV) $\rightarrow$ d

2) (I) $\rightarrow$ c; (II) $\rightarrow$ d; (III) $\rightarrow$ b; (IV) $\rightarrow$ a

3) (I) $\rightarrow$ c; (II) $\rightarrow$ d; (III) $\rightarrow$ a; (IV) $\rightarrow$ b

4) (I) $\rightarrow$ b; (II) $\rightarrow$ c; (III) $\rightarrow$ d; (IV) $\rightarrow$ a

45. For a reaction, $4M(s) + nO_2(g) \rightarrow 2M_2O_n(s)$, the free energy change is plotted as a function of temperature. The temperature below which the oxide is stable could be inferred from the plot as the point at which:

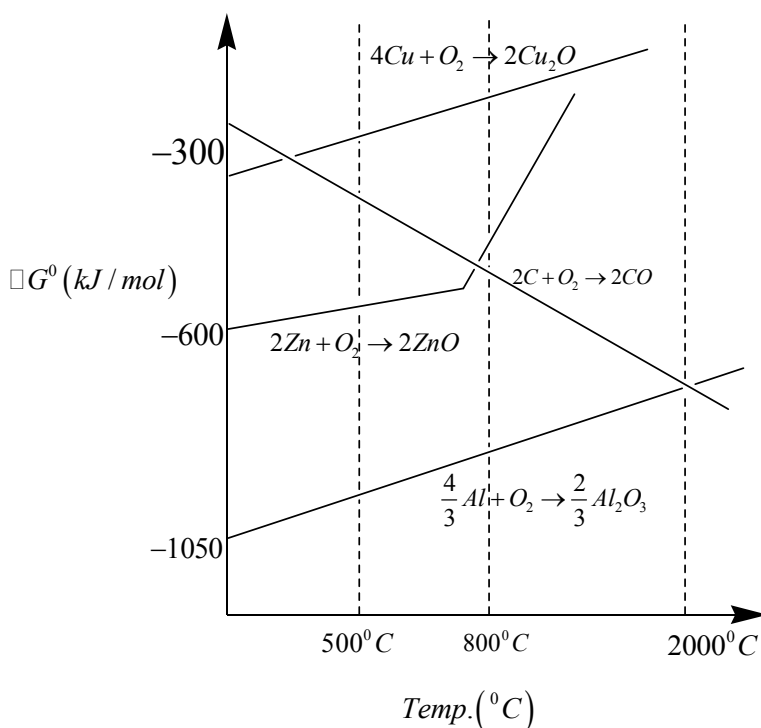
1) The slope change from negative to positive

2) The free energy change shows a change from negative to positive value

3) The slope changes from positive to negative

4) The slope changes from positive to zero

46. An Ellingham diagram provides information about:
- 1) The conditions of pH and potential under which a species is thermodynamically stable.
 - 2) The temperature dependence of the standard Gibbs energies of formation of some metal oxides.
 - 3) The pressure dependence of the standard electrode potentials of reduction reactions involved in the extraction of metals
 - 4) The kinetics of the reduction process
47. The reaction that does NOT define calcinations is:
- 1) $Fe_2O_3 \cdot xH_2O \xrightarrow{\square} Fe_2O_3 + xH_2O$
 - 2) $2Cu_2S + 3O_2 \xrightarrow{\square} 2Cu_2O + 2SO_2$
 - 3) $ZnCO_3 \xrightarrow{\square} ZnO + CO_2$
 - 4) $CaCO_3 \cdot MgCO_3 \xrightarrow{\square} CaO + MgO + 2CO_2$
48. The electrolytes usually used in the electroplating of gold and silver, respectively, are
- 1) $[Au(CN)_2]^-$ & $[Ag(CN)_2]^-$
 - 2) $[Au(CN)_2]^-$ & $[AgCl_2]^-$
 - 3) $[Au(OH)_4]^-$ & $[Ag(OH)_2]^-$
 - 4) $[Au(NH_3)_2]^+$ & $[Ag(CN)_2]^-$
49. The correct statement regarding the given Ellingham diagram is



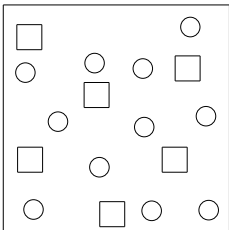
- 1) At 1400°C , Al can be used for the extraction of Zn from ZnO
 - 2) At 500°C , coke can be used for the extraction of Zn from ZnO
 - 3) Coke cannot be used for the extraction of Cu from Cu_2O
 - 4) At 800°C , Cu can be used for extraction of Zn from ZnO
50. Complex A has a composition of $\text{H}_{12}\text{O}_6\text{Cl}_3\text{Cr}$. If the complex on treatment with *conc.* H_2SO_4 loses 13.5% of its original mass, the correct molecular formula of A is:
[Given: atomic mass of Cr=25 amu and Cl=35 amu]
- 1) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
 - 2) $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3].3\text{H}_2\text{O}$
 - 3) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2.\text{H}_2\text{O}$
 - 4) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl}.2\text{H}_2\text{O}$

(NUMERICAL VALUE TYPE)

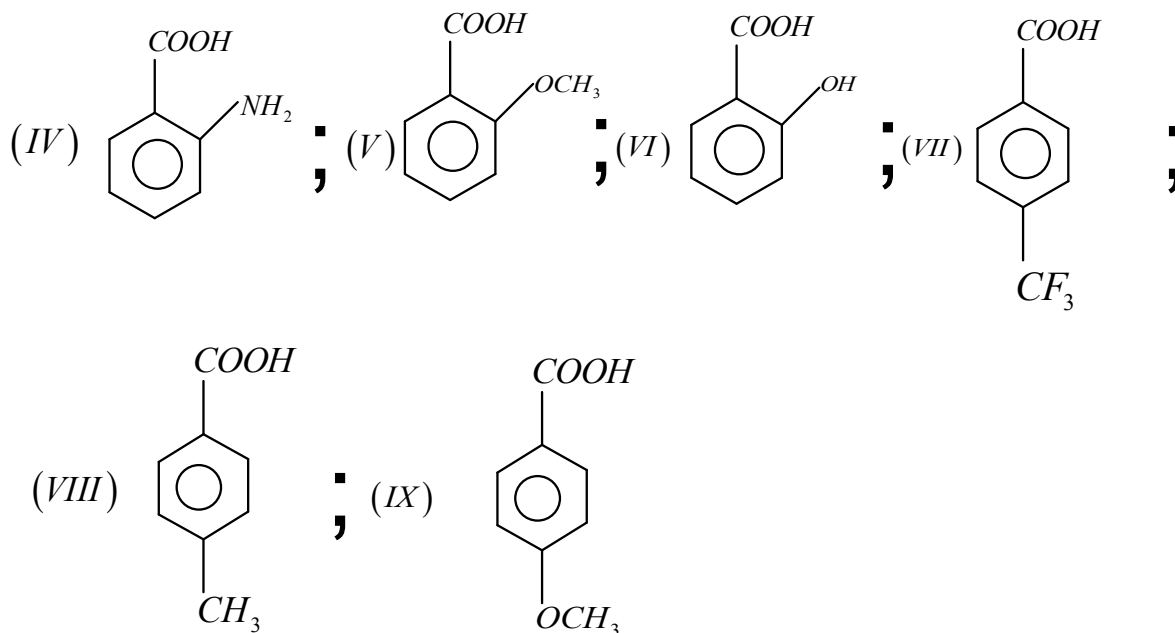
This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

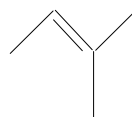
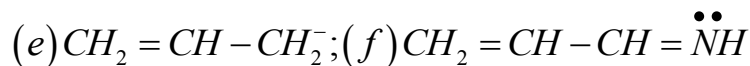
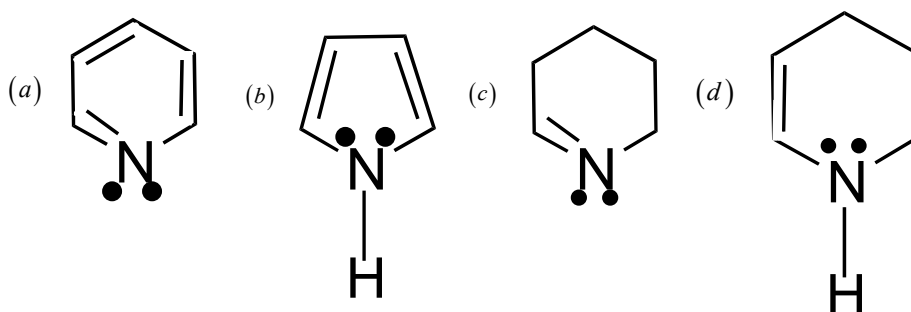
Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

51. For the reaction $\text{X} + \text{Y} \rightleftharpoons 2\text{Z}$, 1.0 mol of X, 1.5 mol of Y and 0.5 mol of Z were taken in a 1L vessel and allowed to react at equilibrium, the concentration of Z was 1.0 mol L^{-1} . The equilibrium constant of the reaction is $\frac{x}{15}$. The value of x is ____
52. In the figure shown below reactant A (represented by square) is in equilibrium with product B (represented by circle). The equilibrium constant is
- 
53. Two solutions, A and B, each of 100mL was made by dissolving 4g of NaOH and 9.8g of H_2SO_4 in water, respectively. The pH of the resultant solution obtained from mixing 40L of solution A and 10L of solution. B is ____
54. The total number of coordination sites in ethylenediaminetetraacetate (EDTA^{4-}) is ____
55. Considering that $\Delta_0 > p$, the magnetic moment (in BM) of $[\text{Ru}(\text{H}_2\text{O})_6]^{2+}$ would be ____

56. The volume (in mL) of 0.125 M $AgNO_3$ required to quantitatively precipitate chloride ions in 0.3 g of $[Co(NH_3)_6]Cl_3$ is__
57. How many of the following acids are more acidic than benzoic acid?
 (I) $HCOOH$; (II) $CH_3 - C \equiv C - COOH$; (III) $CF_3 - COOH$



58. How many of following functional groups show -M effect.
 1) $\overset{\oplus}{N}H_3$; 2) $-COOH$; 3) $\overset{\oplus}{P}H_3$; 4) $-CH(CH_3)_2$; 5) $-CN$; 6) $-SO_3H$
59. How many of the following lone-pair indicated is not involved in resonance:



60. Number of hyper conjugate structure possible for

MATHS**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. $\sin x + i \cos 2x$ & $\cos x - i \sin 2x$ are conjugate to each other for
 1) $x = n\pi$ 2) $x = \left(n + \frac{1}{2}\right)\frac{\pi}{2}$ 3) $x=0$ 4) No value of x
62. The value of $(z+3)(\bar{z}+3)$ is equivalent to
 1) $|z+3|^2$ 2) $|z-3|$ 3) z^2+3 4) None of these
63. If Z is a complex number, then
 1) $|z^2| > |z|$ 2) $|z^2| = |z|^2$ 3) $|z^2| < |z|^2$ 4) $|z^2| \geq |z|^2$
64. If $f(z) = \frac{7-z}{1-z^2}$, where $z = 1+2i$, then $|f(z)|$ is equal to
 1) $\frac{|z|}{2}$ 2) $|z|$ 3) $2|z|$ 4) None of these
65. A box contains two white, three black and four red balls. In how many ways can three balls be drawn from the box, if atleast one black ball is to be included in the drawn?
 1) 46 2) 64 3) 36 4) 63
66. The sum of the digits in unit place of all the numbers formed with the help of 3, 4, 5 and 6 taken all at a time is
 1) 432 2) 108 3) 36 4) 18
67. The number of triangles that are formed by choosing the vertices from a set of 12 points, seven of which lie on the same line is
 1) 105 2) 15 3) 175 4) 185
68. The number of 5 digit telephone numbers having atleast one of their digits repeated is
 1) 90000 2) 10000 3) 30240 4) 69760
69. If the integers $r>1$, $n>2$ and coefficient of $(3r)^{th}$ & $(r+2)^{th}$ terms in the binomial expansion of $(1+x)^{2n}$ are equals then,
 1) $n=2r$ 2) $n=3r$ 3) $n=2r+1$ 4) None of these
70. The coefficient of x^n in the expansion of $(1+x)^{2n}$ & $(1+x)^{2n-1}$ are in the ratio
 1) 1:2 2) 1:3 3) 3:1 4) 2:1

71. If the middle term of $\left(\frac{1}{x} + x \sin x\right)^{10}$ is equal to $7\frac{7}{8}$, then the value of x is
 1) $2n\pi + \frac{\pi}{6}$ 2) $n\pi + \frac{\pi}{6}$ 3) $n\pi + (-1)^n \frac{\pi}{6}$ 4) $n\pi + (-1)^n \frac{\pi}{3}$
72. Three numbers are chosen from 1 to 20. Find the probability that they are not consecutive
 1) $\frac{186}{190}$ 2) $\frac{187}{190}$ 3) $\frac{188}{190}$ 4) $\frac{18}{{}^{20}C_3}$
73. If seven persons are to be seated in a row, then the probability that two particular persons sit next to each other is
 1) $\frac{1}{3}$ 2) $\frac{1}{6}$ 3) $\frac{2}{7}$ 4) $\frac{1}{2}$
74. If M and N are any two events, the probability that atleast one of them occurs is
 1) $P(M) + P(N) - 2P(M \cap N)$ 2) $P(M) + P(N) - P(M \cap N)$
 3) $P(M) + P(N) + P(M \cap N)$ 4) $P(M) + P(N) + 2P(M \cap N)$
75. If 25^{15} is divided by 13, then the remainder is
 1) 14 2) 12 3) 15 4) 17
76. In a foot ball championship, 153 matches were played every two teams played one match with each other. The number of teams, participating in the championship is
 1) 20 2) 18 3) 16 4) 22
77. A bag contains 5 black and 6 red balls, determine the number of ways in which 2 black and 3 red balls can be selected from the lot
 1) 100 2) 300 3) 200 4) 400
78. The real value of θ for which the expression $\frac{1+i\cos\theta}{1-2i\cos\theta}$ is a real number is
 1) $n\pi + \frac{\pi}{4}$ 2) $n\pi + (-1)^n \frac{\pi}{4}$ 3) $2n\pi \pm \frac{\pi}{2}$ 4) None of these
79. If A and B are mutually exclusive events, then
 1) $P(A) \leq P(\bar{B})$ 2) $P(A) \geq P(\bar{B})$ 3) $P(A) < P(\bar{B})$ 4) None of these
80. The probability that atleast one of the events A and B occurs is 0.6. If A and B occur simultaneously with probability 0.2, then $P(\bar{A}) + P(\bar{B})$ is equal to
 1) 0.4 2) 0.8 3) 1.2 4) 1.6

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

81. Find the number of positive integers greater than 6000 and less than 7000 which are divisible by 5, provided that no digit is to be repeated.
82. If a convex polygon has 44 diagonals, then find the number of its sides
83. In a non-leap year, the probability of having 53 Tuesday or 53 Wednesday is
84. The probability of happening of an event A is 0.5 and that of B is 0.3. If A and B are mutually exclusive events, then the probability of neither A nor B is
85. Find the value of $\left| (1+i) \frac{(2+i)}{(3+i)} \right|$
86. Find the number of different words that can be formed from the letters of the word 'TRIANGLE' so that no vowels are together is n then $\frac{n}{10}$ is
87. If a five-digit number divisible by 3 is to be formed using the numbers 0, 1, 2, 3, 4 and 5 without repetitions then the total number of ways this can be done is
88. If the seventh term from the beginning and the end in the expansion of $\left(\sqrt[3]{2} + \frac{1}{\sqrt[3]{3}} \right)^n$ are equal, then n equal to
89. The sum of last two digits of the numbers 3^{400} is
90. The probability that the home team will win an upcoming football game is 0.77, the probability that it will the game is 0.08 and the probability that it will lose the game is



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PTM-02

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Key Sheet

PHYSICS

1	1	2	3	3	1	4	4	5	2
6	2	7	3	8	1	9	3	10	1
11	2	12	4	13	1	14	2	15	3
16	3	17	3	18	2	19	1	20	3
21	21.33	22	7.2	23	6	24	2.25	25	5
26	40	27	0.16	28	3	29	13	30	166.6

CHEMISTRY

31	3	32	2	33	1	34	2	35	3
36	4	37	2	38	4	39	3	40	3
41	4	42	1	43	4	44	2	45	2
46	2	47	2	48	1	49	1	50	4
51	16	52	2	53	10.60	54	6	55	0
56	26.92	57	6	58	5	59	3	60	9

MATHEMATICS

61	4	62	1	63	2	64	1	65	2
66	2	67	4	68	4	69	1	70	4
71	3	72	2	73	3	74	2	75	2
76	2	77	3	78	3	79	1	80	3
81	112	82	11	83	0.14	84	0.2	85	1
86	1440	87	216	88	12	89	1	90	0.15

SOLUTIONS

PHYSICS

01. Here $R = 10\Omega$, $A = 1\text{mm}^2 = 10^{-6}\text{m}^2$, $V = 10\text{V}$, $n = 8 \times 10^{28} \text{ electron / m}^3$

Now, $NI = enAv_d \quad \therefore \frac{V}{R} = emAv_d$

Or $V_d = \frac{V}{enAR}$

$$= \frac{10}{1.6 \times 10^{-19} \times 8 \times 10^{28} \times 20^{-6} \times 10}$$

$$= 0.078 \times 10^{-3} \text{ms}^{-1} \Rightarrow 0.078 \text{mms}^{-1}$$

02. New length $l' = l + 10\% \text{ of } l$

$$= 1 + 0.1l = 1.1l$$

Or $\frac{l'}{l} = 1.1$

$$Al = A'l'$$

OR $\frac{A}{A'} = \frac{l'}{l}$

$$\therefore \frac{R'}{R} = \frac{l'}{l} \times \frac{A}{A'} = \left(\frac{l'}{l}\right)^2 = (1.1)^2 = 1.21$$

The percentage increase in resistance, $\frac{R' - R}{R} \times 100 = \left(\frac{R'}{R} - 1\right) \times 100 \Rightarrow (1.21 - 1) \times 100 = 21\%$

03. $\alpha = \frac{R_2 - R_1}{R_1(t_2 - t_1)} = \frac{2R_0 - R_0}{R_0(t - 0)} = \frac{1}{t}$

$$T = \frac{1}{\alpha} = \frac{1}{3.9 \times 10^{-3}} = 256^\circ\text{C}$$

thus the resistance of copper conductor becomes double at 256°C

04. Here $R_1 = 75.3\Omega$, $t_1 = 27^\circ\text{C}$, $R_2 = \frac{230}{2.68} = 85.8\Omega$, $t_2 = ?$

$$t_2 - t_1 = \frac{R_2 - R_1}{R_1\alpha} = \frac{85.8 - 75.3}{75.3 \times 1.70 \times 10^{-4}} = 820^\circ\text{C}$$

Steady temperature,

$$t_2 = 820 + t_1 = 820 + 27 = 847^\circ\text{C}$$

05. Here $R_0 = 5\Omega$, $R_{100} = 5.23\Omega$, $R_t = 5.795\Omega$

As $R_t = R_0(1 + \alpha t)$

$$\therefore R_t - R_0 = R_0\alpha t \rightarrow (1) \text{ \& } R_{100} - R_0\alpha \times 100 \rightarrow (2)$$

On dividing (1) & (2), we get

$$\frac{R_t - R_0}{R_{100} - R_0} = \frac{t}{100}$$

Or $t = \frac{R_t - R_0}{R_{100} - R_0} \times 100$

$$= \frac{5.795 - 5}{5.23 - 5} = \frac{0.795}{0.23} \times 100 \Rightarrow 345.65^\circ\text{C}$$

06. $\varepsilon = 2.2\text{V}$, $R = 5\Omega$, $V = 1.8\text{V}$

- Internal resistance, $r = R \left(\frac{\varepsilon - V}{V} \right) = 5 \left(\frac{2.2 - 1.8}{1.8} \right) \Omega = 1.1 \Omega$
07. The potential difference corresponding to zero current equals the emf of the cell
EMF of the cell, $\varepsilon = 1.4V$
Maximum current is drawn from the cell when the terminal potential difference is zero
 $\therefore I_{\max} = 0.28A$
Internal resistance,
 $r = \frac{\varepsilon}{I_{\max}} \Rightarrow \frac{1.4V}{0.28A} = 5\Omega$
08. First case:
Total emf, $\varepsilon = 1.1V$
Total resistance, $R = 0.5 + 0.5 = 1\Omega$
 $\therefore \text{Current}, I = \frac{\varepsilon}{R} = \frac{1.1}{1} = 1.1A$
Second case:
Total emf, $\varepsilon = 1 + 1.1 = 2.2V$
Total resistance $R = 0.5 + 0.5 + r = (1 + r)\Omega$
Where r is the internal resistance of the second cell is
 $\therefore \text{Current}, I = \frac{2.2}{1 + r} = 1.1 \text{ or } r = 1\Omega$
09. Applying kirchhoff's first law at junction B, $I_1 + I_2 = I_3 \rightarrow (1)$
Applying kirchhoff's second law to loops ABEFA and BCDEB, we get
 $2I_1 + 5I_1 = 12 \rightarrow (2), -2I_3 - 3I_2 = -6 \rightarrow (3)$
Solving equations (1),(2),(3) we get
 $I_1 = \frac{48}{31}A$
10. Let I_1, I_2 & I_3 be the currents . Kirchhoff's second rule for the closed loop ADCA gives
 $10 - 4(I_1 - I_2) + 2(I_2 + I_3 - I_1) - I_1 = 0$
Or $7I_1 - 6I_2 - 2I_3 = 10 \rightarrow (1)$
For the closed loop ABCA, we get
 $10 - 4I_2 - 2(I_2 + I_3) - I_1 = 0$
Or $I_1 + 6I_2 + 2I_3 = 10 \rightarrow (2)$
For the closed loop BCDEB, we get
 $5 - 2(I_2 + I_3) - 2(I_2 + I_3 - I_1) = 0$
Or $2I_1 - 4I_2 - 4I_3 = -5 \rightarrow (3)$
On solving equations (1)(2)(3) we get
 $I_1 = 2.5A$
11. As $\theta(\text{rad}) = \frac{\text{Arc}}{\text{Radius}}$

$$\therefore \frac{3\pi}{2} = \frac{l}{r} \text{ or } l = \frac{3\pi r}{2}$$

According to Biot-savart law, magnetic field at the centre O is

$$B = \frac{\mu_0 I}{4\pi r^2} = \frac{\mu_0}{4\pi} \cdot \frac{I}{r^2} \cdot \frac{3\pi r}{2} = \frac{\mu_0}{4\pi} \cdot \frac{3}{2} \cdot \frac{\pi l}{r}$$

$$= \frac{4\pi \times 10^{-7}}{4\pi} \cdot \frac{3}{2} \cdot \frac{22}{7} \times \frac{10}{3 \times 10^{-2}} \Rightarrow 1.57 \times 10^{-3} T$$

12. $r = 0.5 \times 10^{-10} m, v = 2.2 \times 10^6 ms^{-1}$

Period of revolution of electron,

$$T = \frac{2\pi r}{v} = \frac{2\pi \times 0.5 \times 10^{-10}}{2.2 \times 10^6} \Rightarrow \frac{1}{7} \times 10^{-15} s$$

Equivalent current,

$$I = \frac{\text{charge}}{\text{Time}} = \frac{e}{T} = \frac{1.6 \times 10^{-19} \times 7}{10^{-15}} \Rightarrow 1.12 \times 10^{-3} A$$

Magnetic field produced at the centre of the nucleus

$$B = \frac{\mu_0 I}{2r} = \frac{4\pi \times 10^{-7} \times 1.12 \times 10^{-3}}{2 \times 0.5 \times 10^{-10}} = 14.07 T$$

13. Magnetic field of the straight wire carrying a current of 2A, at a distance of 10cm or 0.1m from it is

$$B = \frac{\mu_0 I}{2\pi r} = \frac{4\pi \times 10^{-7} \times 5}{2\pi \times 0.1} = 10^{-5} T$$

This field acts perpendicular to the direction of the electron. So magnetic force on the electron is

$$F = qvB \sin 90^\circ$$

$$= 1.6 \times 10^{-19} \times 10^6 \times 10^{-5} \times 1 \Rightarrow 1.6 \times 10^{-18} N$$

14. $U = \frac{1}{2} CV^2 = \frac{1}{2} \frac{\epsilon_0 A}{d} V^2$

When the separation between the plates is decreases by 10%, the energy stored becomes

$$U' = \frac{1}{2} \frac{\epsilon_0 A}{\frac{90}{100} d} V^2 = \frac{10}{9} \cdot \frac{1}{2} \frac{\epsilon_0 A}{d} V^2 = \frac{10}{9} U$$

Percentage change in the stored energy,

$$\frac{U' - U}{U} \times 100 = \left(\frac{U'}{U} - 1 \right) \times 100 = \left(\frac{10}{9} - 1 \right) \times 100 = \frac{100}{9} = 11.11\%$$

15. $B = 2 \times 10^{-3} Wbm^{-2}, E = 3.4 \times 10^4 Vm^{-1}$

Magnetic force on the electron = Electric force on the electron

$$\text{Or } evB = eE$$

Velocity of electrons

$$v = \frac{E}{B} = \frac{3.4 \times 10^4}{2 \times 10^{-3}} ms^{-1} = 1.7 \times 10^7 ms^{-1}$$

When electric field has been removed,

Force exerted by the magnetic field on an electron = Centripetal force on an electron

$$\text{i.e., } evB = \frac{mv^2}{R}$$

$$\text{Or } R = \frac{mv}{eB} = \frac{9.1 \times 10^{-31} \times 1.7 \times 10^7}{1.6 \times 10^{-19} \times 2 \times 10^{-3}} \Rightarrow 4.8 \times 10^{-8} m$$

16. Total charge on the two capacitors = $C_1 V_1 + C_2 V_2 = (3 \times 2 + 6 \times 5) \mu C = 36 \mu C$

In series combination, charge capacitor, $q = 36 \mu C$

Potential on $3 \mu F$ capacitor

$$= \frac{q}{C_1} = \frac{36 \mu C}{3 \mu F} = 12 V$$

17. Let C = capacitance of each capacitor ‘

For series combination,

$$\frac{1}{C_1} = \frac{1}{C} + \frac{1}{C} + \frac{1}{C} = \frac{3}{C} \text{ or } C_1 = \frac{C}{3}$$

For parallel combination,

$$C_2 = C + C + C = 3C \therefore \frac{C_1}{C_2} = \frac{C}{3} \cdot \frac{1}{3C} = \frac{1}{9}$$

18. Conceptual

19. $R_{\text{bulb}} = \frac{V_0^2}{P_0} = \frac{100 \times 100}{500} = 20 \Omega$

Pd across bulb is 100V

Pd across R must below 100V

$$R = 20 \Omega$$

20. Actual power consumed $P = \left(\frac{V_A}{V_C} \right)^2 \times w$

$$\therefore E = \frac{p \times t}{1000} = \frac{1000 \times 30 \times 10}{1000} = 300 \text{ kwh}$$

21. $\pi r_1^2 l_1 = \pi r_2^2 l_1$

$$\Rightarrow \frac{r_2}{r_1} = \frac{1}{2}$$

$$\text{New resistance of wire} = \frac{\rho 4 l_1 \times 4}{\pi r_1^2} = 16 \times 4 = 64 \Omega$$

22. Conceptual

23. $du = U_i - U_f = \sum C_1 v_1^2 - \sum C v_c^2$ where $V_c = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$

24. $T\varepsilon = 2 \left(\sum q^1 V^1 \right) = q^1 v^1$

$$= q^1 \frac{q^1}{c} = \frac{1}{4} \frac{q^2}{c}$$

$$= \frac{1}{2} \cdot \frac{1}{2} q \cdot v$$

$$= \frac{1}{2} \cdot \frac{1}{2} \times 900 \times 10^{-12} \times 100 \times 100$$

$$= \frac{9}{4} \times 10^{-6} = 2.25 \times 10^{-6} J$$

$$25. \quad \frac{1}{R} = \frac{1}{15} + \frac{1}{10} + \frac{1}{5} = \frac{1}{2}$$

$$i - \frac{V}{R} = \frac{10}{2} = 5A$$

$$26. \quad \varepsilon = \frac{V}{l} = \frac{100}{1} = 100Vm^{-1}$$

$$j = \sigma \varepsilon = enV_d$$

$$V_d = \frac{\sigma \varepsilon}{en} = 0.43ms^{-1}$$

$$27. \quad Reff = 10.5 + 1 + 4.5 + 2 = 8\Omega$$

$$i = \frac{4}{8} = \frac{1}{2}A$$

$$i_{6\Omega} = \frac{1}{3} + \frac{1}{2} = \frac{1}{6}A$$

$$28. \quad i \text{ is maximum, when } R = r = 2\Omega$$

$$\therefore i = \frac{\varepsilon}{Reff} = \frac{24}{4+4} = 3A$$

$$29. \quad \frac{R}{S} = \frac{33.7}{66.3}$$

$$\frac{R}{S_{eq}} = \frac{51.9}{48.1}$$

$$\frac{R(S+12)}{12.S} = \frac{51.9}{48.1}$$

$$\frac{33.7}{66.3} \times \frac{S+12}{12} = \frac{51.9}{48.1}$$

$$S+12 = \frac{51.9}{48.1} \times \frac{12 \times 66.3}{33.7}$$

$$S = 13\Omega$$

$$30. \quad q \left\{ \frac{1}{C} + \frac{1}{C} + \frac{1}{C} \right\} = 500$$

$$q = \frac{500}{3} \times 10\mu c$$

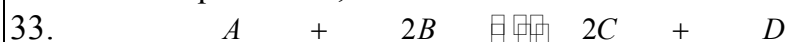
$$V = \frac{q}{C} = \frac{500}{3} V = 166.6V$$

CHEMISTRY

$$31. \quad K_p = K_c (RT)^{\Delta n_g} = K_c (RT)^{1-3/2} = K_c (RT)^{-1/2}$$

$$\Rightarrow K_c = K_p (RT)^{1/2}$$

32. At Equilibrium, rate of forward reaction = Rate of backward reaction.



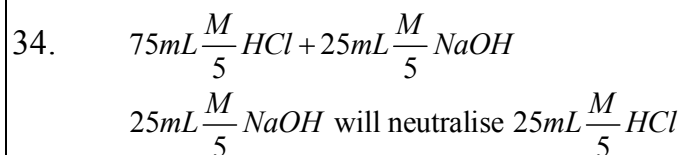
$$t_{eq} \quad 2-x \quad 3-2x \quad 2x \quad x$$

$$\text{Given, } 3-2x = 2-x$$

$$\Rightarrow x = 1$$

$$\therefore [C] = 2, [D] = 1, [A] = 1, [B] = 1$$

$$\therefore K_c = \left\{ \frac{2^2 \times 1}{1^2 \times 1} \right\} = 4$$



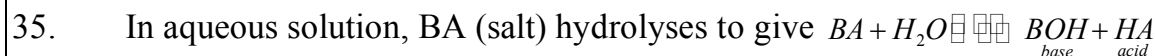
$$75 - 25 = 50\text{mL} \frac{M}{5} \text{HCl} \text{ will remain}$$

Total volume will be $75 + 25 = 100\text{mL}$

$$50\text{mL} \frac{M}{5} \text{HCl} \text{ is diluted to } 100\text{mL}$$

$$[H^+] = [HCl] = \frac{M}{5} \times \frac{50}{100} = \frac{M}{10}$$

$$pH = -\log_{10} [H^+] = -\log_{10} \frac{M}{10} = 1$$



Now pH is given by

$$pH = \frac{1}{2} pK_w + \frac{1}{2} pK_a - \frac{1}{2} pK_b$$

Substituting given values, we get

$$pH = \frac{1}{2} (14 + 4.80 - 4.78) = 7.01$$



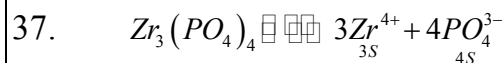
Millimoles	10	20	0	0
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at start

Millimoles	0	10	10	10
------------	---	----	----	----

after reaction

Buffer solution contains CH_3COONa (10 moles) and CH_3COOH (10 milimol) which is a acidic buffer



$$K_{SP} = [Zr^{4+}]^3 [PO_4^{3-}]^4 = (3S)^3 (4S)^4$$

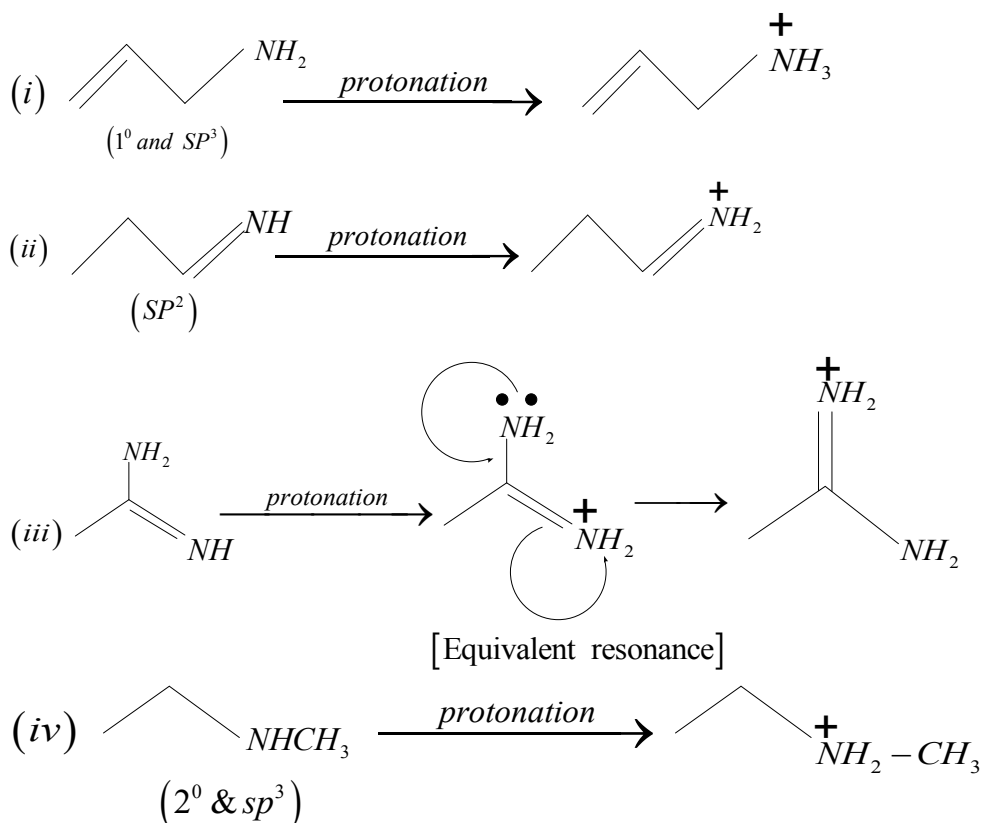
$$K_{SP} = 6912S^7$$

$$S = \left(\frac{K_{SP}}{6912} \right)^{1/7}$$

38. Acidic strength $\propto -I, -M$ effect. Due to strong-I and $-M$ effect of three $-COOCH_3$ groups, it has most acidic Hydrogen.

39. Basicity order can be determined by the cumulative effect of the factors on the electron density of concerned atom.

40.



Hence, correct order of basicity will be: (ii) < (i) < (iv) < (iii)

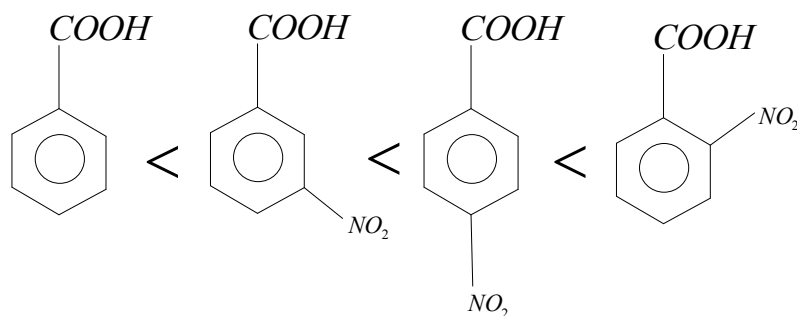
41. $ph_3C^\bullet$ is more stable than $(CH_3)_3C^\bullet$ because resonance stabilisation effect $ph_3C^\bullet$ is more pronounced as compared to hyper conjugation stabilisation effect in $(CH_3)_3C^\bullet$, overall stability order among free radical is

Triphenylmethyl > benzyl > allyl > tertiary alkyl > secondary > primary > methyl > vinyl

42. 3° carbocations are most stable.

43. In carboxylic acids, presence of electron withdrawing substituent e.g. $-NO_2$ disperses the negative charge of the anion and stabilizes it and hence increase acidity of the parent acid.

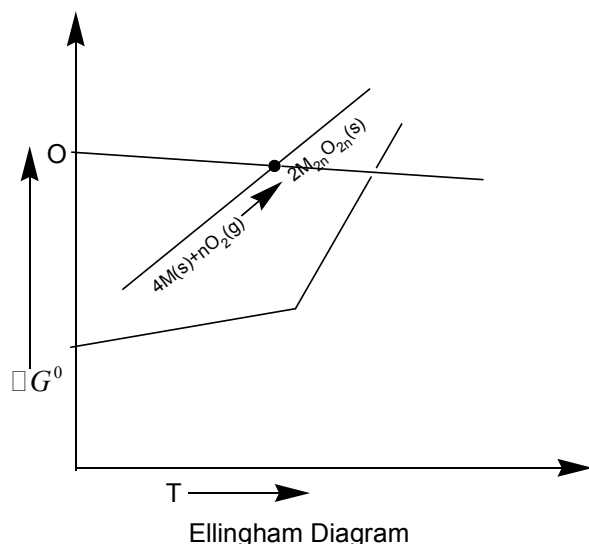
Further o-isomer will have higher acidity than corresponding m and p-isomers due to ortho and high inductive effect of $-NO_2$ group. Since nitro group at p-position has more pronounced electron withdrawing than $-NO_2$ group at m-position, hence the correct order is



44. Siderite = $FeCO_3$
Calamine = $ZnCO_3$



45.

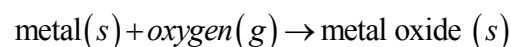


From graph it is evident that the temperature below which the oxide is stable, is the point at which free energy change shows a change from negative to positive

46. Ellingham diagram is the graph of ΔG° vs T of any metal/element oxide. Since

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

For most metal oxide formation.



$$\Delta H^\circ = -Ve; \Delta S^\circ = -Ve$$

So, graph will be straight line with -ve intercept and +ve slope.

47. Calcination involves heating of carbonates or hydrated oxides in absence of air.



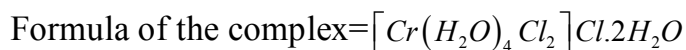
48. The electrolytes used in the electroplating of Au and Ag are $[\text{Au}(\text{CN})_2]^-$ & $[\text{Ag}(\text{CN})_2]^-$ respectively.

49. In the given Ellingham diagram, the metal which has a lower value of ΔG° (more negative) can reduce a metal oxide whose curve lies above it. So, Al can reduce ZnO at 1400°C

50. *conc.* H_2SO_4 acts as dehydrating agent. Molar mass of given complex = 266.5 g/mol

On treating with *conc.* H_2SO_4 the mass lost by the

$$\text{Complex} = \frac{13.5}{100} (266.5) \approx 36\text{g} = 2\text{moles of } \text{H}_2\text{O}$$



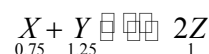
51.



$$t = 0 \quad 1\text{mol} \quad 1.5\text{mol} \quad 0.5\text{mol}$$

$$t = t_{eq} \quad 1-a \quad 1.5-a \quad 0.5+2a$$

$$0.5+2a=1; a=0.25$$



$$0.75 \quad 1.25 \quad 1$$

$$K_{eq} = \frac{[Z]^2}{[X][Y]} = \frac{1}{0.75 \times [1.25]} = \frac{x}{15}$$

$$\Rightarrow x = \frac{15}{(0.75 \times 1.25)} = 16$$

52. Equilibrium constant $K_c = \frac{[B]}{[A]} = \frac{11}{6} \approx 2$

53. $M_{H_2SO_4} = \frac{9.8}{98 \times 100} = 10^{-3} M$

$$M_{NaOH} = \frac{4}{40 \times 100} = 10^{-3} M$$

After neutralisation $[OH^-]$ can be calculated as

$$[OH^-] = \frac{(40 \times 10^{-3}) - (2 \times 10^{-3} \times 10)}{50} = \frac{20}{50} \times 10^{-3}$$

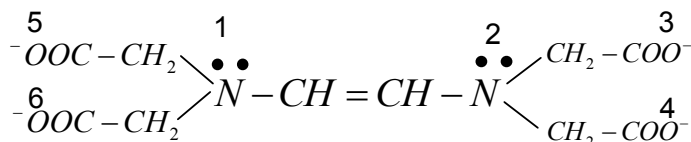
$$[OH^-] = \frac{2}{5} \times 10^{-3}$$

$$pOH = 3.397$$

$$pH = 14 - pOH$$

$$= 14 - 3.397 = 10.603$$

54.

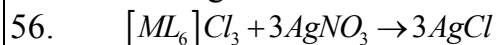


$EDTA^{4-}$ is a hexadentate ligand, since hexa means six and the ligand attaches six times.

55. $Ru^{2+} = 4d^6 = t_{2g}^6 e_g^0$ since $\Delta_0 > p$

No. of unpaired electrons = 0

$\therefore$ Magnetic moment = 0 B.M



$$0.3 g \quad V mL, 0.125 M$$

$$\text{Number of moles of the complex} = \frac{0.3}{267.46}$$

$$\text{Number of moles of } AgNO_3 = 0.125 \times V \times 10^{-3}$$

$$\frac{0.3}{267.46} \times 3 = 0.125 \times V \times 10^{-3}$$

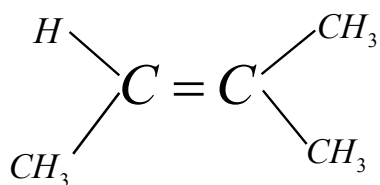
$$\text{Or, } V = \frac{0.3 \times 3 \times 1000}{267.46 \times 0.125} = 26.92 mL$$

57. I, III, IV, V, VI, VII are Acidic in nature.

58. Out of all $CH(CH_3)_2$ will not show M-effect Remaining 5 groups will show

59. Structures, b, d, e involved in Resonance

60.



$9\sigma_{C-H}$ are present so it will give 9 hyper conjugative structures.

MATHEMATICS

61. Let $z = \sin x + i \cos 2x$
 $\bar{z} = \sin x - i \cos 2x \longrightarrow (i)$
 Given that $\bar{z} = \sin x - i \sin 2x \longrightarrow (ii)$
 $\therefore \sin x - i \cos 2x = \cos x - i \sin 2x$
 $\Rightarrow \sin x = \cos x \& \cos 2x = \sin 2x$
 $\Rightarrow \tan x = 1 \& \tan 2x = 1$
 $\Rightarrow \tan x = \tan \frac{\pi}{4} \& \tan 2x = \tan \frac{\pi}{4}$
 $\Rightarrow x = n\pi + \frac{\pi}{4} \& 2x = n\pi + \frac{\pi}{4}$
 $\Rightarrow 2x - x = 0 \Rightarrow x = 0$
62. Given that $(z+3)(\bar{z}+3)$
 Let $z = x + iy$
 $\Rightarrow (z+3)(\bar{z}+3) = (x+iy+3)(x+3-iy)$
 $= (x+3)^2 - (iy)^2 = (x+3)^2 + y^2$
 $|x+3+iy|^2 = |z+3|^2$
63. If z is a complex number, then $z = x + iy$
 $|z| = |x+iy| \& |z|^2 = |x+iy|^2$
 $\Rightarrow |z|^2 = x^2 + y^2 \longrightarrow (i)$
 And $z^2 = (x+iy)^2 = x^2 + i^2 y^2 + i2xy$
 $z^2 = x^2 - y^2 + i2xy$
 $\Rightarrow |z^2| = \sqrt{(x^2 - y^2)^2 + (2xy)^2}$
 $\Rightarrow |z^2| = \sqrt{x^4 + y^4 - 2x^2 y^2 + 4x^2 y^2}$
 $\Rightarrow |z^2| = \sqrt{x^4 + y^4 + 2x^2 y^2} = \sqrt{(x^2 + y^2)^2}$
 $|z^2| = x^2 + y^2 \longrightarrow (ii)$
 From equations (i) & (ii)
 $|z|^2 = |z^2|$
64. Let $z = 1 + 2i$
 $\Rightarrow |z| = \sqrt{1+4} = \sqrt{5}$

$$\text{Now } f(z) = \frac{7-z}{1-z^2} = \frac{7-1-2i}{1-(1+2i)^2}$$

$$= \frac{6-2i}{1-1-4i^2-4i} = \frac{6-2i}{4-4i}$$

$$= \frac{(3-i)(2+2i)}{4-4i^2} = \frac{6+4i+2}{4+4}$$

$$= \frac{8+4i}{8} = 1 + \frac{1}{2}i$$

$$f(z) = 1 + \frac{1}{2}i$$

$$|f(z)| = \sqrt{1 + \frac{1}{4}} = \sqrt{\frac{4+1}{4}} = \frac{\sqrt{5}}{2} = \frac{|z|}{2}$$

65. There are 2 white, 3 black and 4 red balls. We have to draw 3 balls, out of these 9 balls in which atleast one black ball is included. Hence we can select the balls in the following ways

Black balls	1	2	3
Other than black	2	1	0

$$\text{Required number of selections} = {}^3C_1 \times {}^6C_2 + {}^3C_2 \times {}^6C_1 + {}^3C_3 \times {}^6C_0$$

$$= 3 \times 15 + 3 \times 6 + 1$$

$$= 45 + 18 + 1 = 64$$

66. If we fixed 3 at units place. Total possible number is 3! i.e 6
Sum of the digits in unit place of all these numbers = 3×3
Similarly, if we fixed 4, 5 and 6 at units place, in each case total possible numbers are 3!
Required sum of units of all such numbers

$$= (3+4+5+6) \times 3!$$

$$= 18 \times 3! = 18 \times 6 = 108$$

67. Total number of triangles formed from 12 points taking 3 at a time = ${}^{12}C_3$
But out of 12 points 7 are collinear so these 7 points constitute a straight line mean no. triangle is formed by joining these 7 points
Required no. of triangles = ${}^{12}C_3 - {}^7C_3 = 220 - 35 = 185$

68. If all the digits repeated, then number of 5 digits telephone numbers can be formed in 10^5 ways and if no digit repeated, then 5-digit telephone numbers can be formed in ${}^{10}P_5$ ways

Required number of ways

$$= 10^5 - {}^{10}P_5$$

$$= 100000 - \frac{10!}{5!}$$

$$= 100000 - 10 \times 9 \times 8 \times 7 \times 6$$

$$= 100000 - 30240 = 69760$$

69. Given that, $r > 1$, $n > 2$ and the coefficient of $(3r)^{th}$ & $(r+2)^{th}$ term are equal in the expansion of $(1+x)^{2n}$

$$\text{Then, } T_{3r} = T_{3r-1+1} = {}^{2n}C_{3r-1} x^{3r-1} \text{ \& } T_{r+2} = T_{r+1+1} = {}^{2n}C_{r+1} x^{r+1}$$

Given, ${}^{2n}C_{3r-1} = {}^{2n}C_{r+1}$ $\left[\because {}^nC_x = {}^nC_y \Rightarrow x + y = n \right]$

$$\Rightarrow 3r - 1 + r + 1 = 2n$$

$$\Rightarrow 4r = 2n \Rightarrow n = \frac{4r}{2}$$

$$\therefore n = 2r$$

70. Coefficient of x^n in the expansion of $(1+x)^{2n} = {}^{2n}C_n$ and coefficient of x^n in the expansion of $(1+x)^{2n-1} = {}^{2n-1}C_n$

$$\begin{aligned} \frac{{}^{2n}C_n}{{}^{2n-1}C_n} &= \frac{\frac{(2n)!}{n!n!}}{\frac{(2n-1)!}{n!(n-1)!}} \\ &= \frac{(2n)!n!(n-1)!}{n!n!(2n-1)!} \\ &= \frac{2n(2n-1)!n!(n-1)!}{n!n(n-1)!(2n-1)!} \\ &= \frac{2n}{1} = \frac{2}{1} \Rightarrow 2:1 \end{aligned}$$

71. Given expansion is $\left(\frac{1}{x} + x \sin x\right)^{10}$ since $n=10$ is even, so this expansion has only one middle term i.e, 6th term

$$\therefore T_6 = T_{5+1} = {}^{10}C_5 \left(\frac{1}{x}\right)^{10-5} (x \sin x)^5$$

$$\Rightarrow \frac{63}{8} = {}^{10}C_5 x^{-5} x^5 \sin^5 x$$

$$\Rightarrow \frac{63}{8} = \frac{10.9.8.7.6.5!}{5.4.3.2.1.5!} \sin^5 x$$

$$\Rightarrow \frac{63}{8} = 2.9.2.7. \sin^5 x$$

$$\Rightarrow \sin^5 x = \frac{1}{32}$$

$$\Rightarrow \sin^5 x = \left(\frac{1}{2}\right)^5$$

$$\Rightarrow \sin x = \frac{1}{2}$$

$$\therefore x = n\pi + (-1)^n \frac{\pi}{6}$$

72. Since the set of three consecutive numbers from 1 to 20 are 123, 234, 345, ..., 18, 19, 20 i.e, 18

$$P(\text{numbers are consecutive}) = \frac{18}{{}^{20}C_3} = \frac{18}{1140} = \frac{3}{190}$$

$$P(\text{three no's are not consecutive}) = 1 - \frac{3}{190} = \frac{187}{190}$$

73. If two persons sit next to each other, then consider these two persons as 1 group
Now, we have to arrange 6 persons. Number of arrangement = $2! \times 6!$

Total number of arrangement of 7 persons=7!

$$\text{Required probability} = \frac{2!6!}{7!} = \frac{2}{7}$$

74. If M and N are any two events

$$P(M \cup N) = P(M) + P(N) - P(M \cap N)$$

75. $25^{15} = (26-1)^{15}$

$$= {}^{15}C_0 26^{15} - {}^{15}C_1 26^{14} + \dots + {}^{15}C_{15}$$

$$= {}^{15}C_0 26^{15} - {}^{15}C_1 26^{14} + \dots + 1 - 13 + 13$$

$$= {}^{15}C_0 26^{15} - {}^{15}C_1 26^{14} + \dots + 13 + 12$$

It is clear that, when 25^{15} is divisible by 13, then remainder will be 12

76. Let the number of team participating in championship be n since, it is given that every two teams played one match with each other.

$$\text{Total match played} = {}^nC_2$$

According to the question,

$${}^nC_2 = 153$$

$$\frac{n(n-1)}{2} = 153$$

$$n^2 - n = 306$$

$$n^2 - n - 306 = 0$$

$$(n-18)(n+17) = 0$$

$$n = 18, -17$$

$$n = 18$$

77. It is given that bag contains 5 black and 6 red balls So, 2 black balls is selected from 5 black balls in 5C_2 ways 3 red balls are selected from 6 red balls in 6C_3 ways

Total number of ways in which 2 black and 3 red balls are selected = ${}^5C_2 \times {}^6C_3$

$$\Rightarrow 10 \times 20 = 200 \text{ ways}$$

78. Given expression = $\frac{1+i\cos\theta}{1-2i\cos\theta}$

$$= \frac{(1+i\cos\theta)(1+2i\cos\theta)}{(1-2i\cos\theta)(1+2i\cos\theta)}$$

$$= \frac{1+i\cos\theta+2i\cos\theta+2i^2\cos^2\theta}{1-4i^2\cos^2\theta}$$

$$= \frac{1+3i\cos\theta-2\cos^2\theta}{1+4\cos^2\theta}$$

For real value of θ , $\frac{3\cos\theta}{1+4\cos^2\theta} = 0$

$$\Rightarrow 3\cos\theta = 0$$

$$\Rightarrow \cos\theta = \cos\frac{\pi}{2}$$

$$\theta = 2n\pi \pm \frac{\pi}{2}$$

79. For mutually exclusive events,

$$P(A \cap B) = 0$$

$$\begin{aligned}
 P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\
 \Rightarrow P(A \cup B) &= P(A) + P(B) \\
 \Rightarrow P(A) + P(B) &\leq 1 \\
 \Rightarrow P(A) + 1 - P(\bar{B}) &\leq 1 \quad \left[\because P(B) = 1 - P(\bar{B}) \right] \\
 \Rightarrow P(A) &\leq P(\bar{B})
 \end{aligned}$$

80. Given $p(A \cup B) = 0.6$ & $P(A \cap B) = 0.2$

$$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow 0.6 = P(A) + P(B) - 0.2$$

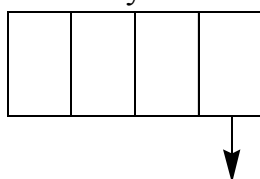
$$P(A) + P(B) = 0.8$$

$$\therefore P(\bar{A}) + P(\bar{B}) = 1 - P(A) + 1 - P(B)$$

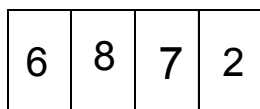
$$= 2 - [P(A) + P(B)]$$

$$= 2 - 0.8 = 1.2$$

81. We know that a number is divisible by 5, if at the units place of the number is 0 or 5. We have to form 4 digit numbers which is greater than 6000 and less than 7000. So unit digits can be filled in 2 ways



Since, repetition is not allowed, therefore tens place can be filled in 7 ways, Similarly hundreds place can be filled in 8. Hence thousand place can be filled in only 1 ways



Total number of integers = $1 \times 8 \times 7 \times 2 = 14 \times 8 = 112$

82. Let the convex polygon has n sides

$$\therefore \text{Number of diagonals} = {}^nC_2 - n$$

According to the question

$${}^nC_2 - n = 44$$

$$\frac{n!}{2!(n-2)!} - n = 44$$

$$\frac{n(n-1)}{2} - n = 44$$

$$n \left[\frac{(n-1)}{2} - 1 \right] = 44 \Rightarrow n \left(\frac{n-1-2}{2} \right) = 44$$

$$n(n-3) = 88$$

$$\Rightarrow n^2 - 3n - 88 = 0$$

$$\Rightarrow (n-11)(n+8) = 0$$

$$n = 11, -8$$

$$n = 11$$

83. In a non-leap year are 365 days which have 52 weeks and 1 day. If this day is a Tuesday or Wednesday then the year will have 53 Tuesday or 53 Wednesday.

Required probability = $1/7 = 0.14$

84. $P(\overline{A \cap B}) = P(\overline{A \cup B}) = 1 - P(A \cup B)$
 $\Rightarrow 1 - [P(A) + P(B)]$ [$\because$ Since A & B are mutually exclusive]
 $\Rightarrow 1 - (0.5 + 0.3)$
 $\Rightarrow 1 - 0.8 = 0.2$

85. Given that $\left| (1+i) \frac{(2+i)}{(3+i)} \right| = \left| \frac{(2+i+2i+i^2)}{(3+i)} \right| = \left| \frac{2+3i-1}{3+i} \right|$
 $= \left| \frac{1+3i}{3+i} \right| = \left| \frac{(1+3i)(3-i)}{(3+i)(3-i)} \right|$
 $= \left| \frac{3+9i-i-3i^2}{9-i^2} \right| = \left| \frac{3+8i+3}{9+1} \right|$
 $\left| \frac{6+8i}{10} \right| = \sqrt{\frac{6^2}{100} + \frac{8^2}{100}} \Rightarrow \sqrt{\frac{36+64}{100}} \Rightarrow \sqrt{\frac{100}{100}} = 1$

86. Number of letters in the word 'TRIANGLE' = 8 out of which 5 are consonants and 3 are vowels.

If vowels are not together, then we having following arrangement

V	C	V	C	V	C	V	C	V	C	V
---	---	---	---	---	---	---	---	---	---	---

Consonants can be arranged in $5! = 120$ way and vowels can occupy at 6 places.

The 3 vowels can be arranged at 6 places in 6P_3 ways = 120

87. We know that, a number is divisible by 3, when sum of digits the number must be divisible by 3. So, if we consider the digits 0, 1, 2, 4, 5 then $(0+1+2+4+5) = 12$

Sum is divisible by 3 there for five digit number Using the digit

$$0, 1, 2, 3, 5 = 4 \times 4 \times 3 \times 2 \times 1 = 96$$

4	4	3	2	1
---	---	---	---	---

And if we consider the digit 1, 2, 3, 4, 5 then $(1+2+3+4+5) = 15$ thus sum is also divisible by 3

So, five digits no can be formed using the digit 1, 2, 3, 4, 5, in $5!$ Ways

Total no. of ways = $96 + 5! = 96 + 120 = 216$

88. given expressions is $\left(\sqrt[3]{2} + \frac{1}{\sqrt[3]{3}} \right)^n$

$$T_7 = T_{6+1} = {}^nC_6 \left(\sqrt[3]{2} \right)^{n-6} \left(\frac{1}{\sqrt[3]{3}} \right)^6$$

Since T_7 from end is same as the T_7 from beginning of $\left(\sqrt[3]{2} + \frac{1}{\sqrt[3]{3}}\right)^n$ Then ,

$$T_7 = {}^nC_6 \left(\frac{1}{\sqrt[3]{3}}\right)^{n-6} \left(\sqrt[3]{2}\right)^6$$

$$\text{Given that } {}^nC_6 (2)^{\frac{n-6}{3}} (3)^{\frac{-6}{3}} = {}^nC_6 3^{\frac{-(n-6)}{3}} \cdot 2^{\frac{6}{3}}$$

$$\Rightarrow 2^{\frac{n-12}{3}} = \left(\frac{1}{3^{1/3}}\right)^{n-12}$$

Which is true, when $\frac{n-12}{3} = 0$

$$n-12=0 \Rightarrow n=12$$

89. Given that $3^{400} = 9^{200} = (10-1)^{200}$

$$\Rightarrow (10-1)^{200} = {}^{200}C_0 10^{200} - {}^{200}C_1 10^{199} + \dots + {}^{200}C_{199} 10^1 + {}^{200}C_{200}$$

$$\Rightarrow (10-1)^{200} = 100^{200} - 200 \times 10^{199} + \dots + 10 \times 200 + 1$$

So it is clear that the last two digits are 01.

90. $P(\text{lossing}) = 1 - (0.77 + 0.08) = 0.15$



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON Central Office – Madhapur – Hyderabad

Sec: **Sr.Super-60& All**

Jee-Main

Date: 13-05-2021

Time: 09.00Am to 12.00

PTM-03

Max.Marks:300

IMPORTANT INSTRUCTIONS:

1. Immediately fill in the Admission number on this page of the Test Booklet with Blue/Black Ball Point Pen only.
2. The candidates should not write their Admission Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1, 2, 3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **two sections**.

(I) Section-I contains 20 **multiple choice** questions with **only one correct** option.

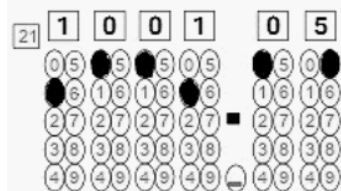
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

(II) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only.

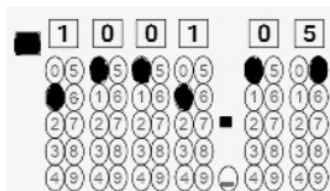
First 5 attempted questions will be considered if more than 5 questions attempted.

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted question 21, bubble on 21 as shown below.



Question Answered for marking



Question Cancelled for marking

Marking scheme : +4 for correct answer and 0 in all other cases.

6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capitals): _____

Admission Number:

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Candidate's Signature: _____ Invigilator's Signature: _____

13-05-2021_Sr.Super60 & All_Jee-Main_PTMM-03_Test Syllabus

PHYSICS : Complete EMI, AC, Work, Energy, Power, center of mass, Collisions

CHEMISTRY : Redox Electro chemistry, IUPAC nomenclature Isomerism including Tautomerism Alcohol-Phenol-Ether, Group-15 Transition elements

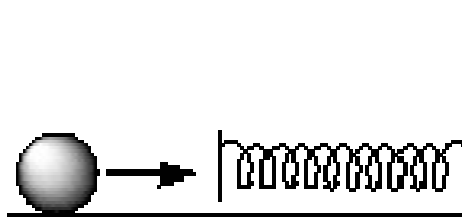
MATHEMATICS : Straight lines, Circles, Parabola, Ellipse, Hyperbola

PHYSICS**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

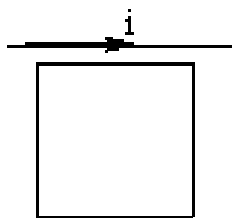
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

1. A bullet loses 40% of its initial KE after penetrating through a distance of 3 cm in a target. The further distance it travels before coming to rest is
 1) 2 cm 2) $\frac{8}{3}$ cm 3) 4.5 cm 4) 1 cm
2. The average resisting force that must act on a body mass 2kg to reduce its speed from 40ms^{-1} to 10ms^{-1} in 0.6s is
 1) 12.5 N 2) 25 N 3) 50 N 4) 100 N
3. A mass of 0.5kg moving with a speed of 1.5 m/s on a horizontal smooth surface, collides with a nearly weightless spring of force constant $k = 50\text{N/m}$. The maximum compression of the spring would be



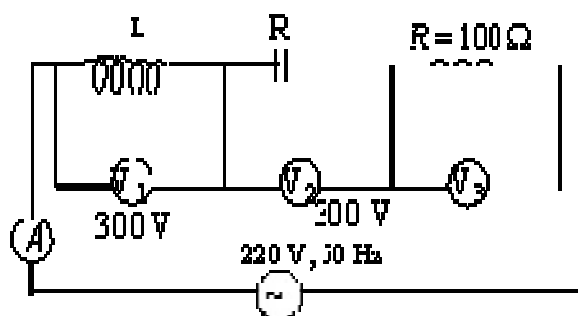
- 1) 0.5 m 2) 0.15 m 3) 0.12 m 4) 1.5 m
4. A motor of power P_0 is used to deliver water at a certain rate through a given horizontal pipe. To increase the rate of flow of water through the same pipe n times, the power of the motor is increased to P_1 . The ratio of P_1 to P_0 is
 1) $n:1$ 2) $n^2:1$ 3) $n^3:1$ 4) $n^4:1$
5. A force $\vec{F} = (5\vec{i} + 3\vec{j} + 2\vec{k})\text{N}$ is applied over a particle which displaces it from its origin to the point $\vec{r} = (2\vec{i} - \vec{j})\text{m}$. The work done on the particle in joules is
 1) +10 2) +7 3) -7 4) +13
6. The magnetic flux of $500\mu\text{Wb}$ per turn passing through a 200 turn coil is reversed in 20×10^{-3} seconds. The average e.m.f. induced in the coil in volt is
 1) 2.5 2) 5.0 3) 7.5 4) 10.0

7. Current i flows through a long wire. A square loop made of conducting wire held below the wire in the same vertical plane is released and allowed to fall under gravity. If acceleration of the loop is a , then :

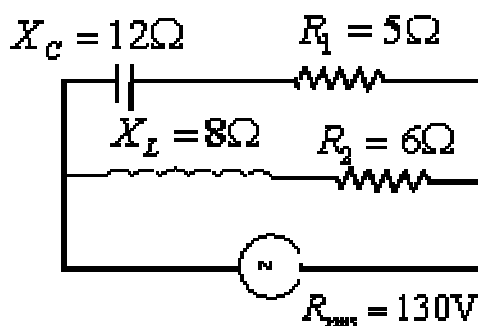


- 1) a current is induced in anticlockwise direction and $a > g$
 - 2) a current is induced in the clockwise direction and $a < g$
 - 3) a current is induced in anticlockwise direction and $a < g$
 - 4) no current is induced in the loop and $a = g$
8. An e.m.f of 5 volt is produced by a self inductance, when the current changes at a steady rate from 3A to 2A in 1 millisecond. The value of self inductance is
- 1) Zero
 - 2) 5H
 - 3) 5000H
 - 4) 5 mH
9. A coil having an area 2m^2 is placed in a magnetic field which changes from $1\text{Wb}/\text{m}^2$ to $4\text{Wb}/\text{m}^2$ in an interval of 2 seconds. The average e.m.f induced in the coil will be
- 1) 4V
 - 2) 3V
 - 3) 1.5V
 - 4) 2V
10. If the number of turns per unit length of a coil of a solenoid is doubled, the self inductance of the solenoid will
- 1) Remain unchanged
 - 2) Be halved
 - 3) Be doubled
 - 4) Become four times
11. A magnetic pole of strength 8Am is moved around a long straight wire carrying a current of 5A . The work done is
- 1) $8\pi \times 10^{-7} \text{J}$
 - 2) $16\pi \times 10^{-6} \text{J}$
 - 3) $2\pi \times 10^{-7} \text{J}$
 - 4) Zero
12. A coil of self-inductance L is placed in an external magnetic field (no current flows in the coil). The total magnetic flux linked with the coil is ϕ . The magnetic field energy stored in the coil is
- 1) Zero
 - 2) $\frac{\phi^2}{2L}$
 - 3) $\frac{\phi^2}{L}$
 - 4) $\frac{2\phi^2}{L}$
13. An inductor may store energy in
- 1) Its electric field
 - 2) Its coils
 - 3) Its magnetic field

- 4) Both in electric and magnetic fields
14. The coil a dynamo is rotating in a magnetic field. The developed induced e.m.f changes and the number of magnetic lines of force also changes. Which of the following condition is correct
- 1) Lines of force minimum but induced e.m.f. is zero
 - 2) Lines of force maximum but induced e.m.f. is zero
 - 3) Lines of force maximum but induced e.m.f. is not zero
 - 4) Lines of force maximum but induced e.m.f. is also maximum
15. In the circuit shown in fig. what will be the reading of the voltmeter V_3 and ammeter A ?



- 1) 800 V, 2 A
 - 2) 300 V, 2 A
 - 3) 220 V, 2.2 A
 - 4) 100 V, 2 A
16. What is the amount of power delivered by the ac source in the circuit shown (in watts)



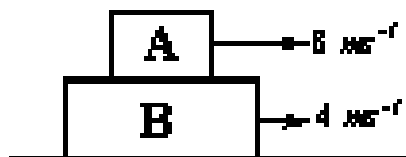
- 1) 500 watt
 - 2) 1014 watt
 - 3) 1514 watt
 - 4) 2013 watt
17. If power factor is $\frac{1}{2}$ in a series RL circuit, $R = 100\Omega$, then L is
- 1) $\frac{\sqrt{3}}{\pi} H$
 - 2) πH
 - 3) $\frac{\pi}{\sqrt{3}} H$
 - 4) None of these

- (NUMERICAL VALUE TYPE)**

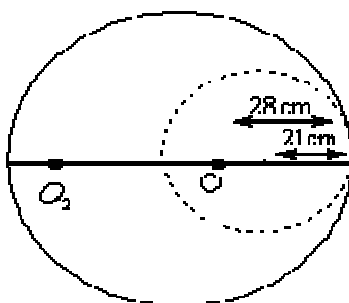
Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

-
- A diagram showing two rectangular blocks, labeled m and M , connected by a coiled spring. They are placed on a horizontal surface, indicated by a line with diagonal hatching below. A force F is applied to the left face of block m , pointing to the left. Another force F is applied to the right face of block M , pointing to the right.

23. Block A of mass 1kg is placed on a rough surface of block B of mass 3kg. Block B is placed on smooth horizontal surface. Blocks are given the velocities as shown in fig. Find the net work done by the frictional force in joules.

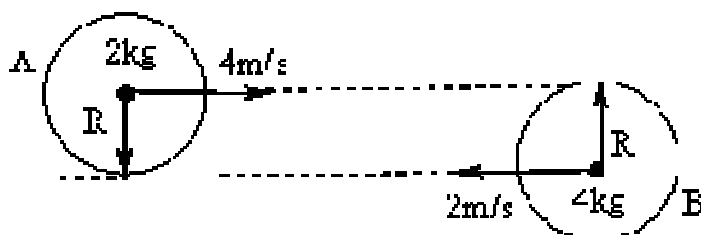


24. A conducting circular loop is placed in uniform magnetic field of $B = 0.02$ tesla with its plane perpendicular to the field. The radius of the loop starts shrinking at a constant rate of $\frac{1}{\pi} \text{ cm/sec}$. The induced e.m.f. in the loop at the instant when its radius is 2cm is $n \times 10^{-4} \text{ V}$ the value of n is
25. An ac ammeter is used to measure current in a circuit. When a given direct current passes through the circuit, the ac ammeter reads 3 A. When another alternating current passes through the circuit, the ac ammeter reads 4A. Then the reading of this ammeter (in A), if dc and ac flow through the circuit simultaneously is
26. A circular plate of uniform thickness has a diameter 56cm. A circular portion of diameter 42cm is removed from one edge as shown in the figure. The centre of mass of the remaining portion from the centre of plate will be

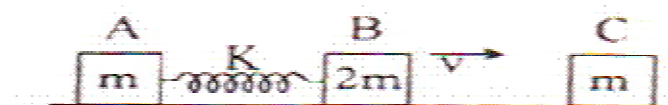


27. A dog weighing 5kg is standing on a flat boat so that it is 10m from the shore. The dog walks 4m on the boat towards the shore and then halts. The boat weighs 20kg and one can assume that there is no friction between it and the water. How far is the dog from the shore at the end of this time?(approximately)
28. A particle traveling horizontally with speed u collides and sticks with a particle of equal mass hanging at rest at the end of a light inextensible string of length $2l$. If the string rotates through an angle of 60° before its velocity becomes zero, if u is $\sqrt{xgl}$ then the value of x is _____

29. Two spheres are moving towards each other. Both have the same radius but their masses are 2kg and 4kg . If the velocities are 4ms^{-1} and 2ms^{-1} respectively and the coefficient of restitution is $\frac{1}{3}$ then common velocity along the line of impact is



30. Two blocks A and B of masses m and $2m$ respectively are connected by a spring of force constant k . The masses are moving to the right with uniform velocity v each, the heavier mass leading the lighter one. The spring in between them is of natural length during the motion. Block B collides with a third block C of mass m , at rest. The collision being completely inelastic. If the maximum compression of the spring is $\sqrt{\frac{mv^2}{nk}}$ find $\frac{n}{2}$



CHEMISTRY

Max Marks: 100

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

31. Which of the following statements is wrong ?
 - 1) Oxidation number of oxygen is +1 in peroxides
 - 2) Oxidation number of oxygen is +2 in oxygen difluorides
 - 3) Oxidation number of oxygen is $-\frac{1}{2}$ in superoxides
 - 4) Oxidation number of oxygen is -2 in most of its compounds.
32. Which one of the following is an example of disproportionation reaction ?
 - 1) $3Cl_{2(g)} + 6OH^-_{(aq)} \rightarrow ClO^-_{3(aq)} + 5Cl^-_{(aq)} + 3H_2O(l)$.
 - 2) $Ag^{+2}_{(aq)} + Ag(s) \rightarrow 2Ag^{-4}_{(aq)}$.
 - 3) $Zn(s) + CuSO_{4(aq)} \rightarrow Cu(s) + ZnSO_{4(aq)}$.
 - 4) $2KClO_{3(s)} \rightarrow 2KCl_{(s)} + 3O_{2(s)}$.
33. In a balanced equation $H_2SO_4 + xHI \rightarrow H_2S + yI_2 + zH_2O$ the value of x, y and z are
 - 1) $x = 3, y = 5, z = 2$
 - 2) $x = 4, y = 8, z = 5$
 - 3) $x = 8, y = 4, z = 4$
 - 4) $x = 5, y = 3, z = 1$
34. During electrolysis of $CuSO_4$ using pt - electrodes, the P^H of solution.
 - 1) increases
 - 2) decreases
 - 3) remains unchanged
 - 4) Cannot be predicted
35. The standard reduction electrode potential values of elements A, B, C are +0.68, - 2.50, - 0.50 V respectively . The order of their reducing power is
 - 1) $A > B > C$
 - 2) $A > C > B$
 - 3) $C > B > A$
 - 4) $B > C > A$
36. How many Coulomb of electricity will be consumed when 100 mA current passes through a solution of $AgNO_3$ for half an hour during electrolysis
 - 1) 108
 - 2) 180
 - 3) 1800
 - 4) 18000
37. If the same amount of electricity is passed through aqueous solution of $AgNO_3$ and $CuSO_4$ and the number of Ag and Cu atoms deposited are x and y respectively. Then
 - 1) $x = y$
 - 2) $x < y$
 - 3) $y = 2x$
 - 4) $x = 2y$

38. The correct decreasing order of priority for the functional groups of organic Compounds in the IUPAC system of nomenclature is
- $$\begin{array}{c} \text{O} \\ \parallel \\ \text{C} \end{array}$$
- 1) $-\text{COOH}, -\text{SO}_3\text{H}, -\text{C}(\text{O})-\text{NH}_2, -\text{CHO}$
 - 2) $-\text{SO}_3\text{H}, -\text{COOH}, -\text{CONH}_2, -\text{CHO}$
 - 3) $-\text{CHO}, -\text{COOH}, -\text{SO}_3\text{H}, -\text{CONH}_2$
 - 4) $-\text{CONH}_2, -\text{CHO}, -\text{SO}_3\text{H}, -\text{COOH}$
39. The functional group present in Anisole is
- 1) Ketone 2) Aldehyde 3) Amine 4) Ether
40. IUPAC name of $\text{HOOC}-\text{CH}_2-\text{CHO}$ is
- 1) Formyl ethanoic acid 2) 2 - Carboxy ethanal
 - 3) Prop - 3 - al - 1 - oic acid 4) prop - 1 - al - 3 - oic acid
41. Which of following pairs of Compounds are tautomers
- 1) propanol & propanone 2) Ethanol & Vinyl alcohol
 - 3) ethanol & allyl alcohol 4) Vinyl alcohol & ethanol
42. Carboxylic acids and esters are
- 1) Functional isomers 2) Keto - enol tautomers
 - 3) Geometrical isomers 4) Position isomers
43. In the esterification reaction, the Correct order of reactivity of alcohols is
- 1) $\text{CH}_3\text{OH} > \text{C}_2\text{H}_5\text{OH} > (\text{CH}_3)_2\text{CHOH}$.
 - 2) $\text{CH}_3\text{OH} > (\text{CH}_3)_2\text{CHOH} > \text{C}_2\text{H}_5\text{OH}$.
 - 3) $\text{CH}_3\text{CH}_2\text{OH} > (\text{CH}_3)_2\text{CHOH} > \text{CH}_3\text{OH}$.
 - 4) $(\text{CH}_3)_2\text{CHOH} > \text{CH}_3\text{CH}_2\text{OH} > \text{CH}_3\text{OH}$.
44. $\text{H}_5\text{C}_2-\text{O}-\text{C}_2\text{H}_5 + \text{HI}_{(\text{excess})} \rightarrow$
- 1) $\text{C}_2\text{H}_5\text{OH}$ and $\text{C}_2\text{H}_5\text{I}$ 2) 2 Moles of $\text{C}_2\text{H}_5\text{I}$
 - 3) 2 Moles of $\text{C}_2\text{H}_5\text{OH}$ 4) C_2H_4
45. The largest bond angle is in
- 1) NH_3 2) PH_3 3) AsH_3 4) H_2O
46. The oxyacid of phosphorous which has more non - ionisable hydrogens.
- 1) H_3PO_2 2) H_3PO_3 3) $\text{H}_4\text{P}_2\text{O}_7$ 4) $\text{H}_4\text{P}_2\text{O}_6$
47. In P_4O_{10} the number of the oxygen atoms bonded to each phosphorus atoms is
- 1) 2 2) 3 3) 4 4) 5
48. Which one of the following ions has same number of unpaired electrons as those present in V^{+3} ion
- 1) Fe^{+3} 2) Ni^{+2} 3) Mn^{+2} 4) Cr^{+3}

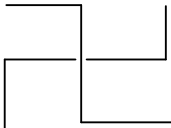
49. The ion having maximum magnetic moment is
 1) Co^{+3} 2) Cr^{+3} 3) Ni^{+2} 4) Cu^{+1}
50. Which one of the following is not oxidized by acidified $KMnO_4$
 1) Sodium oxalate 2) Potassium oxalate
 3) Ferrous sulphate 4) Sodium Sulphate

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place.
 (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

51. The number of moles of electrons are needed for the reduction of 20 ml of 0.5 M solution of $KMnO_4$ in acidic medium is $x \times 10^{-2} \text{ mole}^-$. What is x.
52. Specific Conductance of 0.1 M CH_3COOH at $25^\circ C$ is $3.9 \times 10^{-4} \text{ ohm}^{-1} \text{ cm}^{-1}$. If $\lambda^\infty(H_3O^+)$ and $\lambda^\infty(CH_3COO^-)$ at 25° are 349 and $41 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ respectively percentage of degree of ionization of CH_3COOH at the given concentration is _____.
53. For an electrochemical cell $Sn_{(s)} / Sn_{(aq, 1M)}^{+2} // Pb_{(aq, 1M)}^{+2} / Pb_{(s)}$ the ratio $\frac{[Sn^{+2}]}{[Pb^{+2}]}$ when this cell attains equilibrium is _____ ($E_0 \text{ Cell} = 0.01V$)
54. The number of 1° Carbons in the given compound is/are _____.

55. How many primary alcoholic isomers are possible for $C_5H_{11}OH$?
56. The Number of hydroxyl group in pyrophosphoric acid is _____.
57. How many bonds are equivalent in the dichromate dianion ?
58. When alkaline $KMnO_4$ is treated with KI , the iodide ions are oxidised to IO_3^- ions. The Number of moles $KMnO_4$ used for oxidise 2 moles of I^- ions are _____.
59. The total Number of compounds having at least one bridging oxo group among the molecules given below is _____. N_2O_5 , P_4O_6 , P_4O_{10} , N_2O_3
60. How many are secondary alcohols ? $C_6H_5CH(OH)CH_3$, $H_2C=CH-CH_2OH$, Benzyl alcohol, 3-methyl-2-pentanol, isopropylalcohol, isobutylalcohol.

MATHS**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. Given the system of straight lines $a(2x+y-3)+b(3x+2y-5)=0$, the line of the system situated farthest from the point $(4,-3)$ has the equation
- 1) $4x+11y-15=0$ 2) $7x+y-8=0$
 3) $4x+3y-7=0$ 4) $3x-4y+1=0$
62. In triangle ABC, if $y=x+1$ & $y=2x+3$ are altitude through A & angle bisector of B respectively. If vertex $C=(3,4)$ then equation of median through C is
- 1) $x+7y=25$ 2) $7x+2y=25$
 3) $7x+y=25$ 4) $2x+7y=25$
63. Locus of the image of the point $(2,3)$ in the line $(2x-3y+4)+\lambda(x-2y+3)=0$ ($\text{where } \lambda \in R$) is a circle of radius 'r' then r^4 is equal to
- 1) 4 2) 2 3) 8 4) 16
64. A circle has radius 3 units and its centre lies on the line $y=x-1$. Then the equation of this circle if it passes through point $(7,3)$, is:
- 1) $x^2+y^2-8x-6y+16=0$ 2) $x^2+y^2+8x+6y+16=0$
 3) $x^2+y^2-8x-6y-16=0$ 4) None of the above
65. If the circles $x^2+y^2=2$ and $x^2+y^2-4x-4y+\lambda=0$ have exactly three real common tangents then $\lambda=$
- 1) -10 2) 6 3) -6 4) 10
66. The equation of the circle which touches the circle $x^2+y^2-6x+6y+17=0$ externally and to which the lines $x^2-3xy-3x+9y=0$ are normals, is
- 1) $x^2+y^2-6x-2y-1=0$ 2) $x^2+y^2+6x+2y+1=0$
 3) $x^2+y^2-6x-6y+1=0$ 4) $x^2+y^2-6x-2y+1=0$
67. A straight line with slope 2 and y intercept 5 touches the circle $x^2+y^2+16x+12y+c=0$ at a point Q then the coordinates of Q are
- 1) $(-6,11)$ 2) $(-9,-13)$ 3) $(-10,-15)$ 4) $(-6,-7)$

68. If $x = 3$ is the chord of contact of the circle $x^2 + y^2 = 81$, then the equation of the corresponding pair of tangents is
- 1) $x^2 - 8y^2 + 54x + 729 = 0$ 2) $x^2 - 8y^2 - 54x + 729 = 0$
 3) $x^2 - 8y^2 - 54x - 729 = 0$ 4) $x^2 - 8y^2 = 729$
69. A chord through P cut the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ in A and B another chord through P in C and D, then
- 1) $PA \cdot PB < PC \cdot PD$ 2) $PA \cdot PB = PC \cdot PD$
 3) $PA \cdot PC = PB \cdot PD$ 4) $PA \cdot PB > PC \cdot PD$
70. Focus of the parabola is $(1, 1)$ and the tangent at the vertex has the equation $x + y = 1$ Then
- 1) Equation of the parabola is $(x - y)^2 = 2(x + y - 1)$
 2) Equation of the parabola is $(x - y)^2 = 4(x + y - 1)$
 3) Equation of the parabola is $(x - y)^2 = 2(x - y - 1)$
 4) Equation of the parabola is $(x + y)^2 = 4(x + y - 1)$
71. Find the length of the common chord of the parabola $y^2 = 4(x + 3)$ and the circle $x^2 + y^2 + 4x = 0$.
- 1) 4 2) 2 3) 6 4) 8
72. If one end of focal chord AB of parabola $y^2 = 8x$ is at $A\left(\frac{1}{2}, -2\right)$ then equation of normal through it at B is
- 1) $2x + y - 24 = 0$ 2) $x - 2y + 8 = 0$
 3) $4y = 2x - 7$ 4) $2x + 4y + 8 = 0$
73. The locus of the mid point of the line segment joining the focus to a moving point on the parabola $y^2 = 4ax$ is another parabola with directrix
- 1) $x = -a$ 2) $x = -a/2$ 3) $x = 0$ 4) $x = a/2$
74. An arch is in the shape of a parabola whose axis is vertically down words and measures 80 mts across its bottom on the ground. Its highest point is 24 mts. The measure of the horizontal beam across its cross section at a height of 18 mts from the centre is
- 1) 50 2) 40 3) 45 4) 60

75. If eccentricity of an ellipse is $\frac{1}{\sqrt{2}}$ and R_1, R_2 are the radii of auxiliary and director circles of the ellipse respectively, then
 1) $R_1^2 = 2R_2^2$ 2) $2R_2^2 = 3R_1^2$ 3) $2R_1^2 = R_2^2$ 4) $3R_1^2 = 4R_2^2$
76. Tangents are drawn from the point $P(2,3)$ to the circle $x^2 + y^2 - 8x + 6y + 1 = 0$ touch the circle at points A and B. The circumcircle of $\triangle PAB$ cuts the director circle of the ellipse $\frac{(x+5)^2}{9} + \frac{(y-3)^2}{b^2} = 1$ orthogonally, Then $b^2 =$
 1) 45 2) 50 3) 40 4) 54
77. If the ellipse $\frac{x^2}{4} + y^2 = 1$ meets the ellipse $x^2 + \frac{y^2}{a^2} = 1$ at four distinct points and $a = b^2 - 5b + 7$, then b does not lie in
 1) $[4,5]$ 2) $(-\infty, 2) \cup (3, \infty)$ 3) $(-\infty, 0)$ 4) $[2,3]$
78. The coordinates of a point on the hyperbola $\frac{x^2}{24} - \frac{y^2}{18} = 1$ which is the nearest to the line $3x + 2y + 1 = 0$ are
 1) $(6,3)$ 2) $(-6,-3)$ 3) $(6,-3)$ 4) $(-6,3)$
79. If focus is $(-1,2)$, directrix is $17y=4$ and length of latus rectum is $\frac{18}{5}$ then eccentricity of that conic, is
 1) $\frac{9}{5}$ 2) $\frac{18}{5}$ 3) $\sqrt{2}$ 4) $\frac{51}{50}$
80. The area (in square units) of the equilateral triangle formed by the tangent at $(\sqrt{3}, 0)$ to the hyperbola $x^2 - 3y^2 = 3$ with the pair of a asymptotes of the hyperbola is
 1) $\sqrt{2}$ 2) $\sqrt{3}$ 3) $\frac{1}{\sqrt{3}}$ 4) $2\sqrt{3}$

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

81. A variable line $x/a + y/b = 1$ moves in such a way that the harmonic mean of a and b is 8. Then the least area of triangle made by the line with the coordinate axes is _____ sq. units.

82. $A(-3,2), B(2,5)$ be two points. If the point $P(0,k)$ lies on y -axis such that $|PA - PB|$ is maximum. Then $\frac{k-1}{5} = \underline{\hspace{2cm}}$
83. Let the tangents drawn to the circle, $x^2 + y^2 = 16$ from the point $P(0,h)$ meet the x -axis at points A and B. If the area of $\triangle APB$ is minimum, then h is equal to
84. The number of distinct normals that can be drawn from $(-2,1)$ to the parabola $y^2 - 4x - 2y - 3 = 0$ is $\underline{\hspace{2cm}}$
85. The number of common tangents of the circle $x^2 + y^2 - 6y + 4 = 0$ and the parabola $y^2 = x$ is
86. If the shortest distance between the parabolas $2y^2 = 2x - 1, 2x^2 = 2y - 1$ is $\frac{1}{K\sqrt{2}}$. Then $K =$
87. If P is any point on ellipse $\frac{x^2}{4} + 4y^2 = 1$ whose foci are at A and B then, max. value of $|PA \cdot PB|$ is
88. Tangents are drawn to the ellipse $x^2 + 2y^2 = 2$, the locus of the mid points of the intercept made by the tangents between the coordinate axes is $\frac{1}{ax^2} + \frac{1}{by^2} = 1$ then $a + b$ is
89. The number of tangents to $x^2/9 - y^2/4 = 1$ through $(6,2)$ is
90. Let $S = (1, 2)$ and $S' = (5, 5)$ be the foci of a hyperbola. For any point P on the curve, it is given that $|S'P - SP| = 3$. If e' is the eccentricity of the conjugate hyperbola then e' is equal to $\underline{\hspace{2cm}}$



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Key Sheet PHYSICS

1	3	2	4	3	2	4	1	5	2
6	4	7	2	8	4	9	2	10	4
11	2	12	2	13	3	14	2	15	3
16	3	17	1	18	1	19	3	20	2
21	0.67	22	2	23	6	24	8	25	5.0
26	9	27	7	28	8	29	0	30	6

CHEMISTRY

31	1	32	1	33	3	34	2	35	4
36	2	37	4	38	1	39	4	40	3
41	4	42	1	43	1	44	2	45	1
46	1	47	3	48	2	49	1	50	4
51	5	52	1	53	2.15	54	4	55	4
56	4	57	6	58	4	59	3	60	3

MATHEMATICS

61	4	62	3	63	1	64	1	65	2
66	4	67	4	68	2	69	2	70	2
71	1	72	1	73	3	74	2	75	2
76	4	77	2	78	3	79	4	80	2
81	32.0	82	2.0	83	5.65	84	1.0	85	3
86	2	87	4.0	88	6	89	0	90	1.25

SOLUTIONS

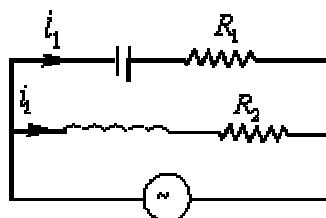
PHYSICS

1. $\frac{3}{x} = \frac{40}{100} \times \frac{100}{60} \Rightarrow x = \frac{9}{2} = 4.5 \text{ cm}$
2. $Ft = m(v - u), \quad F(0.6) = 2(40 - 10) \quad F = 100 \text{ N}$
3. $\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \Rightarrow mv^2 = kx^2 \text{ or } 0.5 \times (1.5)^2 = 50 \times x^2 \therefore x = 0.15 \text{ m}$
4. Power of motor initially = P_0
 Let, rate of flow of motor = (x)
 Since, power, $P_0 = \frac{\text{work}}{\text{time}} = \frac{mgy}{t} = mg\left(\frac{y}{t}\right),$
 $\frac{y}{t} = x = \text{rate of flow of water} = mgx \quad \dots\dots(i)$
 If rate of flow of water is increased by n times, i.e (nx) .
 Increased power $P_1 = \frac{mgy'}{t} = mg\left(\frac{y'}{t}\right) = mgn.x$
 The ratio of power $\frac{P_1}{P_0} = \frac{n}{1} \Rightarrow P_1 : P_0 = n : 1$
5. $W = \vec{F} \cdot \vec{x} = (5\hat{i} + 3\hat{j} + 2\hat{k}) \cdot (2\hat{i} - \hat{j}) = 10 - 3 = 7 \text{ joules}$
6. $\phi = 500 \times 10^{-6} \text{ Wb}, N = 200, t = 20 \times 10^{-3} \text{ s}$
 Given that coil is reversed
 $\therefore e = \frac{2N\phi}{t} = \frac{2 \times 200 \times 500 \times 10^{-6}}{20 \times 10^{-3}} = 10 \text{ V}$
7. a current is induced in the clockwise direction and $a < g$
8. $e = 5 \text{ V}, i = 3; i_2 = 2; t = 1 \text{ mill sec}$
 $t = 1 \times 10^{-3}$
 $e = -L \frac{di}{dt} \Rightarrow L = \frac{-e}{\frac{di}{dt}} \quad L = \frac{-5}{\left(\frac{2-3}{10^{-3}}\right)} = 5 \text{ mH}$
9. $A = 2 \text{ m}^2, B_1 = 1 \text{ wb} / \text{m}^2, B_2 = 4 \text{ wb} / \text{m}^2, t = 2 \text{ s}$
 $e = \frac{-A(B_2 - B_1)}{t} = \frac{2(4-1)}{2} = 3 \text{ V}$
10. $L \propto n^2$, where n is number of turns per unit length
11. Work done $W = \mu_0 m \times i \times n$
 $= 4\pi \times 10^{-7} \times 8 \times 5 \times 1$
 $= 160\pi \times 10^{-7} = 16\pi \times 10^{-6} \text{ J}$
12. $\frac{\phi^2}{2L}$
13. Its magnetic field
14. Lines of force maximum but induced e.m.f. is zero
15. Voltage across L and C cancels out.
 So, voltage across k is 220 V .

Again, $I_0 = \frac{220}{100} A = 2.2 A$

16. $i_{1rms} = \frac{E_{rms}}{\sqrt{X_C^2 + R_1^2}} = \frac{130}{13} = 10 A$

$i_{2rms} = \frac{E_{rms}}{\sqrt{X_L^2 + R_2^2}} = 13 A$



Power dissipated

$= i_{1rms}^2 R_1 + i_{2rms}^2 R_2 = 10^2 \times 5 + 13^2 \times 6$

= power delivered by battery

$= 500 + 169 \times 6 = 1514 \text{ watt}.$

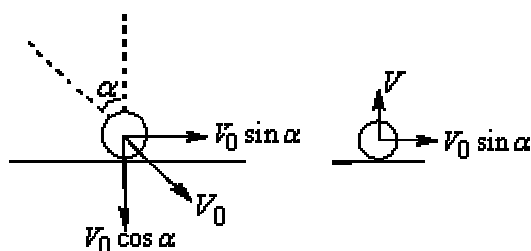
17. Power factor, $\cos \phi = \frac{1}{2} \quad \phi = 60^\circ$

$\tan \phi = \frac{X_L}{R} = \frac{\omega L}{R} = \tan 60^\circ = \sqrt{3}$

$L = \frac{\sqrt{3}R}{\omega} = \frac{\sqrt{3} \times 100}{2\pi \times 50} = \frac{\sqrt{3}}{\pi} H$

18. An ideal choke coil is a pure inductor. Hence inside it V leads I by 90° .

$P = V_{eq} I_{eq} \cos \theta = V_{eq} I_{eq} \cos 90^\circ = 0.$



19.

→ along common tangent direction

$V_0 \sin \alpha$ will remain unchanged

→ along common Normal direction

Relative speed separation =

e(relative speed approach) $v = ev_0 \cos \alpha$

20. Sol: $X_2 = \frac{m_1 d}{m_1 + m_2} = \frac{1^2 \times 1}{1^2 + 2^2} = \frac{1}{5} m$

21. (0.67)

Sol. ρ = mass per unit length

So, $\vec{F}_{ext} + \vec{F}_{Tn} = \frac{m d\vec{v}}{dt} \quad \rho g x - \left(\rho \frac{dx}{dt} \right) v = \frac{m dv}{dt} \quad \rho g x dt = \rho v dx + \rho x dv$

$g x dt = v dx + x dv$

Solve for v and x

22. Maximum elongation is given by

$$x_{\max} = \frac{2[F_1 m_2 + F_2 m_1]}{K(m_1 + m_2)}$$

Here $F_1 = F_2 = F$; $m_1 = m$ and $m_2 = M$

Put the values and solve to get $x_{\max} = \frac{2F}{K}$

23. A/c. L.C.M

$$1 \times 8 + 3 \times 4 = 4v \Rightarrow v = 5 \text{ ms}^{-1}$$

W.D by friction is equal to loss of K.E

$$\therefore W = \frac{1}{2}(1)(8)^2 + \frac{1}{2}(3)(4)^2 - \frac{1}{2}(4)(5)^2 = 32 + 24 - 50 = 6 \text{ J}$$

24. Let the radius be r at time t . the flux of the magnetic field at this instant is $\phi = \pi r^2 B$

Thus $\frac{d\phi}{dt} = 2\pi r B \frac{dr}{dt}$ The induced e.m.f. when $r=2\text{cm}$. $\mathcal{E} = 2\pi(2\text{cm})(0.02\text{T})(1.00\text{mm/sec}) = 2.5\mu\text{V}$

25. Net current $I = 3 + 4\sqrt{2} \sin \omega t$

$$\text{Now find } I_v = \sqrt{\frac{\int_0^T (3 + 4\sqrt{2} \sin \omega t)^2 dt}{T}} = 5 \text{ A}$$

26. Mass of the disc removed $= \frac{M}{2(28)^2} \times \pi(21)^2 = \frac{9M}{16}$

$$\text{Remaining mass} = \frac{7M}{16}$$

Using the same method as followed in the above question.

$$\frac{7M}{16} \times OO_2 = \frac{9M}{16} \times OO_1$$

$$\text{As, } OO_1 = (28 - 21) \text{ cm} = 7 \text{ cm}$$

$$\text{Hence, } OO_2 = \frac{9}{7} \times 7 \text{ cm} = 9 \text{ cm}$$

27. Given that initially the system is at rest, so

Now, as in the motion of dog no external force is applied to the system, hence

$$\vec{V}_{CM} = \text{const} \tan t = 0$$

$$\text{i.e., } \frac{m\vec{v}_1 + M\vec{v}_2}{m + M} = 0$$

$$\text{or } m\vec{v}_1 + M\vec{v}_2 = 0 \quad (\text{as } (m + M) = \text{finite})$$

$$\text{or } m \frac{\Delta \vec{r}_1}{\Delta t} + M \frac{\Delta \vec{r}_2}{\Delta t} = 0$$

$$\text{or } m\Delta \vec{r}_1 + M\Delta \vec{r}_2 = 0 \quad (\text{as } \Delta \vec{r} = \vec{d} = \text{displacement})$$

$$\text{or } md_1 - Md_2 = 0 \quad (\text{as } \vec{d}_2 \text{ is opposite to } \vec{d}_1)$$

$$\text{or } md_1 = Md_2 \quad \dots\dots\dots(i)$$

Now, when dog moves 4m towards shore relative to boat, the boat will shift a distance d_2 relative to shore opposite to the displacement of dog; so, the displacement of dog relative to shore (towards shore) will be.

$$d_1 = 4 - d_2 \quad \dots\dots\dots(ii)$$

$$(i.e., d_1 + d_2 = d_{rel} = 4)$$

Putting the value of d_2 from eqn. (ii) in eqn. (i)

$$md_1 = M(4 - d_1)$$

$$\text{or } d_1 = \frac{M \times 4}{m + M} = \frac{20 \times 4}{5 + 20} = 3.2m$$

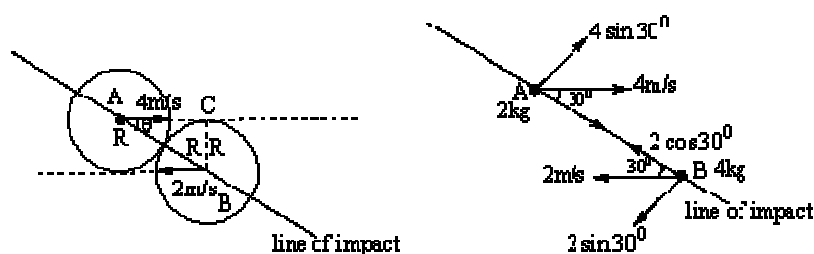
As initially the dog was 10m from the shore, so now he will be $(10 - 3.2) = 6.8$ m from the shore.

28. L.C.O M $\Rightarrow mu = (2m)V \Rightarrow v = \frac{4}{2}$

$$\text{L.C.O . Energy } \frac{1}{2}(2m)\frac{u^2}{4} = (2m)g(2\ell)(1 - \cos 60)$$

$$u = \sqrt{8g\ell} \text{ then } x = 8$$

29.



From triangle ABC

$$\sin \theta = \frac{BC}{AB}$$

Along the line of impact

$$2[4 \cos 30^\circ] - 4[2 \cos 30^\circ] = [2 + 4]V = \frac{R}{2R}$$

$$2\left[4 \frac{\sqrt{3}}{2}\right] - 4\left[2 \frac{\sqrt{3}}{2}\right] = 6V = \frac{1}{2}$$

$$0 = 6V \quad \theta = 30^\circ$$

$$V = 0$$

30. Immediately after collision, velocity of m is v but velocity of (B+C) is $\frac{2V}{3}$

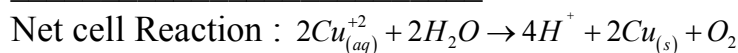
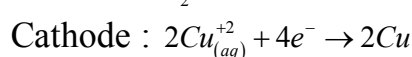
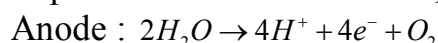
CHEMISTRY

31. Conceptual

32. A redox reaction in which one compound of intermediate oxidation state Converts to two Compounds, one of higher and one of lower oxidation states.

33. Conceptual

34. In presence of inert electrode, the cell reaction is



Due to increase in $[H^+]$, P^H decreases

35. More is reduction potential, more is the power to get itself reduced (or) lesser is reducing power.

36. Charge passed during electrolysis $= i \times t$
 $= (100 \times 10^{-3}) \times \frac{1}{2} \times 60 \times 60$
 $= 180 C$

37. $Ag^+ + e^- \rightarrow Ag$

$Cu^{+2} + 2e^- \rightarrow Cu$

Thus same quantity of electricity will deposit double the no. of Ag atoms than

Cu i.e $x = 2y$

38. Conceptual

39. Conceptual

40. Conceptual

41. Conceptual

42. Conceptual

43. Correct order of reactivity is $CH_3OH > CH_3CH_2OH > (CH_3)_2CHOH$

44. $H_5C_2 - O - C_2H_5 + HI_{(excess)} \rightarrow C_2H_5I + C_2H_5OH \xrightarrow{HI} C_2H_5I$

45. As electronegativity decreases down the group the electron density on Central atom of hydride decreases and thus repulsion between bond pair and lone pair of electrons decreases which decreases $H - M - H$ bond angle.

46. Conceptual

47. Conceptual

48. Conceptual

49. Conceptual

50. Conceptual

51. Moles of $KMnO_4 = m \times v_{(Lt)}$

$$= 0.5 \times 20 \times 10^{-3}$$

$$= 10^{-2} \text{ moles}$$

1 mol $KMnO_4$ required $= 5 \text{ mol } e^-$

$10^{-2} \text{ mol } KMnO_4$ required $= 5 \times 10^{-2} \text{ mol } e^-$

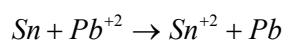
$$x = 5.$$

52. $\lambda_m(CH_3COOH) = \frac{1000 \times K}{0.1} = \frac{1000 \times 3.9 \times 10^{-4}}{0.1}$

$$= 3.9 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$\alpha_{CH_3COOH} = \frac{\lambda_m}{\lambda_m^\infty} = \frac{3.9}{41 + 349} = 1.0 \times 10^{-12} = 1$$

53. At equilibrium state $E_{cell} = 0$, $E_{cell}^0 = 0.01 \text{ V}$



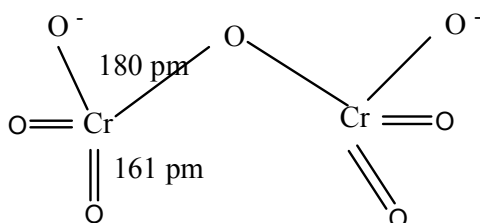
$$E = E_{cell}^0 - \frac{0.059}{n} \log \frac{[P]}{[R]}$$

$$0 = 0.01 - \frac{0.059}{2} \log \left[\frac{[Sn^{+2}]}{[Pb^{+2}]} \right]$$

$$\frac{1}{3} = \log \left[\frac{[Sn^{+2}]}{[Pb^{+2}]} \right]$$

$$\left[\frac{[Sn^{+2}]}{[Pb^{+2}]} \right] = 10^{1/3} = 2.15.$$

54. Conceptual
 55. Based on Structure
 56. Based on Structure
 57. $Cr_2O_7^{2-}$ has structure



58. $3MnO_4^- + H_2O + I^- \rightarrow 2MnO_2 + I_2 + IO_3^-$
 59. Based on Structure
 60. Conceptual

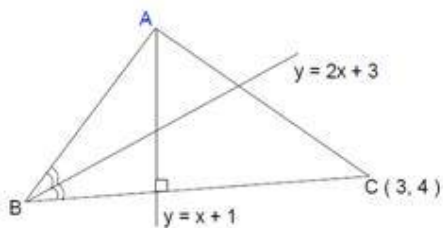
MATHEMATICS

61. The given system of lines passes through the point of intersection of the straight lines $2x + y - 3 = 0$ and $3x + 2y - 5 = 0$ [$L_1 + \lambda L_2 = 0$ form], which is (1,1).

The required line will also pass through this point, further, the line will be farthest from point (4,-3) if it is in direction

$$\therefore \text{the equation of the required line is } y - 1 = \frac{-1}{\frac{-3-1}{4-1}}(x - 1)$$

$$\Rightarrow 3x - 4y + 1 = 0$$



- 62.

Equation of BC is $x + y = 7$ so $B = \left(\frac{4}{3}, \frac{17}{3} \right)$

Let slope of AB is m so,

$$\left| \frac{m-2}{1+2m} \right| = \left| \frac{2+1}{1-2} \right|$$

So equation of AB is $3x + 21y - 123 = 0$

So $A = \left(\frac{17}{4}, \frac{21}{4}\right)$ mid-point of AB = $\left(\frac{67}{24}, \frac{131}{24}\right)$

Equation of median through C is $7x + y = 25$

63. point of intersection of lines $2x - 3y + 4 = 0$; $x - 2y + 3 = 0$ is $P(1, 2)$ let $B(h, k)$ be the image of $A(2, 3)$ w.r to given family of lines

$$AP = BP$$

$$AP^2 = BP^2$$

$$1 + 1 = (h - 1)^2 + (k - 2)^2$$

$$\text{Locus of } B(h, k) \text{ is } (x - 1)^2 + (y - 2)^2 = 2$$

This is a circle of radius $\sqrt{2} = r$

$$r^4 = (\sqrt{2})^4 = 4$$

64. Let its centre be (h, k) , then $h - k = 1$... (1)

Also, radius $a = 3$

$$\text{Equation is } (x - h)^2 + (y - k)^2 = 9$$

Also, it passes through $(7, 3)$

$$\text{i.e., } (7 - h)^2 + (3 - k)^2 = 9 \quad \dots (2)$$

We get h and k from Eqs. (1) and (2), solving simultaneously as $(4, 3)$. Equation is

$$x^2 + y^2 - 8x - 6y + 16 = 0$$

65. $C_1 C_2 = r_1 + r_2$

66. The given normals are $x - 3y = 0$, $x - 3 = 0$ which intersect at centre whose coordinates are $(3, 1)$. The given circle is $C_1(3, -3)$, $r_1 = 1$, C_2 is $(3, 1)$ and $r_2 = (?)$. If the two circles touch externally, then $C_1 C_2 = r_1 + r_2 \Rightarrow 4 = 1 + r_2 \Rightarrow r_2 = 3$

$$\therefore (x - 3)^2 + (y - 1)^2 = 3^2 \quad \text{or} \quad x^2 + y^2 - 6x - 2y + 1 = 0$$

67. Conceptual

68. Put $x = 3$ in equation of circle

$$\Rightarrow 9 + y^2 = 81 \quad y = \pm 6\sqrt{2}$$

Hence, equation as the pair of tangents is

$$(3x + 6\sqrt{2}y - 81)(3x - 6\sqrt{2}y - 81) = 0 \Rightarrow (x - 27)^2 - 8y^2 = 0$$

$$\Rightarrow x^2 - 54x - 8y^2 + 729 = 0$$

69. Given Circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

Since chord through P cut the circle at A & B then

$$PA \cdot PB = |S_{11}| \dots \dots \dots (1)$$

Since the chord through P and cuts the Circle at C and D

$$\text{Then } PC \cdot PD = |S_{11}| \dots \dots \dots (2)$$

From (1) & (2)



$$PA.PB = PC.PD = |S_{11}|$$

70. Directrix is $(x+y)=0$, parabola is $PM^2 = PN^2$

71. Solving given equations, we have $-x^2 - 4x = 4x + 12 \Rightarrow x^2 + 8x + 12 = 0$
 $\Rightarrow x = -2, (x = -6 \text{ is not possible})$

$$\text{From equation of parabola, } y^2 = 4(-2+3) = 4 \therefore y = \pm 2$$

So, curves intersect at $(-2, 2)$ and $(-2, -2)$.

So, length of common chord is 4

72. $\left(\frac{1}{2}, -2\right) \quad (at_1^2, 2at_1)$

$$2at_1 = -2 \quad t_1 = \frac{-1}{2}$$

$$t_1 t_2 = -1 \quad t_2 = 2$$

$$y + xt = 2at + at^2 \quad y + 2x = 8 + 16$$

$$y + 2x - 24 = 0$$

73. The focus of the parabola $y^2 = 4ax$ is $S(a, 0)$, let $P(at^2, 2at)$ be any point on the parabola then co-ordinates of the mid-point of SP are given by

$$x = \frac{a(t^2 + 1)}{2}, y = \frac{2at + 0}{2}$$

Eliminating t we get the locus of the mid-point as

$$y^2 = 2ax - a^2 \text{ or } y^2 = 2a(x - a/2) \quad (1)$$

$$\text{Which is a parabola of the form } Y^2 = 4AX \quad (2)$$

Where $Y = y, X = x - a/2$ and $A = a/2$

Equation of the directrix of (2) is $X = -A$

So equation the directrix of (1) is $x - a/2 = -a/2$

$$\Rightarrow x = 0$$

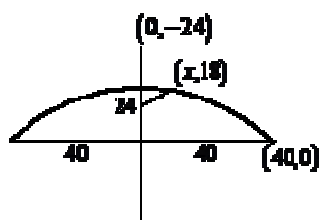
74. $(x-0)^2 = -4a(y-24)$

$$\Rightarrow 4a = \frac{200}{3}$$

$$\Rightarrow x^2 = \frac{-200}{3}(y-24)$$

$$\Rightarrow x = 20$$

$\therefore$ Measure of horizontal beam = 40



75. $e = \frac{1}{\sqrt{2}} \Rightarrow \frac{\sqrt{a^2 - b^2}}{a} = \frac{1}{\sqrt{2}} \Rightarrow a^2 = 2b^2 \quad \frac{R_1^2}{R_2^2} = \frac{a^2}{a^2 + b^2} = \frac{2}{3}$

76. Circumcircle of $\triangle PAB$ is $(x-2)(x-4) + (y-3)(y+3) = 0$

Director circle $(x+5)^2 + (y-3)^2 = 9 + b^2$ cut orthogonally $\Rightarrow b^2 = 54$

77. $a > 1$ (or) $b^2 - 5b + 7 > 1$ $b^2 - 5b + 6 > 0$ $b \in (-\infty, 2) \cup (3, \infty)$

78. $p(x, y)$ be the point on type nearest to the given $\frac{x_1^2}{24} - \frac{y_1^2}{18} = 1$

Tangent at p is parallel to the given line $\frac{\partial y}{\partial x \text{ at } p(x, y)} = \frac{-3}{2}$ $x = -2y$

79. $SZ = \frac{34-4}{17} = \frac{30}{17}$ and $\frac{l}{e} = \frac{9}{5e}$. For any conic $SZ = \frac{l}{e}$

80. Area of the triangle = ab

81. Given $\frac{2ab}{a+b} = 8 \Rightarrow a+b = \frac{ab}{4}$

Area of the triangle with coordinate axes

$= \frac{1}{2}|ab| \geq \frac{1}{2} \times 64^{32}$ $A.M \geq G.M$ $\frac{a+b}{2} \geq \sqrt{ab}$ $\frac{ab}{8} \geq \sqrt{ab}$ $\sqrt{ab} \geq 8$ $ab \geq 64$

82. Image of $A(-3, 2)$ is $A'(3, 2)$

$B = (2, 5)$ Equation of $A'B$ is $y - 5 = -\frac{3}{1}(x - 2)$

$3x + y = 11$ $P(0, k)$ is intersection point of y -axis

And $3x + y = 11 \therefore k = 11$ $\frac{k-1}{5} = 2$

83. Combined equation of pair of tangents $S_1^2 = S.S_{11}$

If these lines meets x-axis at A and B $AB = \frac{gh}{\sqrt{h^2 - 16}}$

Area of triangle $PAB = f(h) = \frac{4h^2}{\sqrt{h^2 - 16}}$

f is min $\Rightarrow f'(h) = 0 \Rightarrow h = 4\sqrt{2}$

84. grix the parabola is $y^2 - 2y + 1 = 4x + 4$

$\Rightarrow (y-1)^2 = 4(x+1)$

The axis $y-1=0$ also $(-2, 1)$ lies on the axis and it is extrios to the parabola , because $1^2 - 4(-2) - 2(1) - 3 = 1 + 8 - 2 - 3 > 0$

Rely one normal is passible.

85. Tangent to the parabola $y^2 = 4x$ is $y = mx + \frac{1}{4m}$. This touches the circle

$x^2 + (y-3)^2 = 5 \Rightarrow \sqrt{5} = \frac{|3 - \frac{1}{4m}|}{\sqrt{1+m^2}}$

$80m^4 - 64m^2 + 24m - 1 = 0$. Put $2m = t$. Then

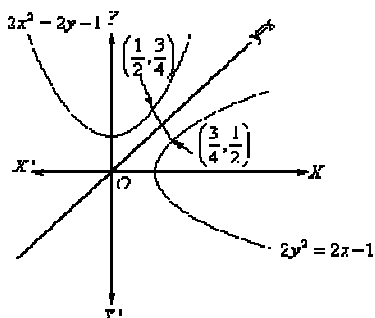
$\Rightarrow 5t^4 - 16t^2 + 12t - 1 = 0$. Clearly $t = 1$ is a root

$\Rightarrow 5t^3 + 5t^2 - 11t + 1 = 0$. Again $t = 1$ is a root

$$\Rightarrow 5t^2 + 10t - 1 = 0 \Rightarrow t = \frac{-10 \pm \sqrt{120}}{10} = \frac{-5 \pm \sqrt{30}}{5}$$

Hence 3 common tangents can be drawn

86.



Given parabolas $2y^2 = 2x - 1$ and $2x^2 = 2y - 1$ are symmetrical about the line $y = x$.

Also shortest distance occurs along the common normal which perpendicular to the line $y = x$. Differentiating $2y^2 = 2x - 1$ w.r.t. x , we get

$$2y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{2y} = 1 \Rightarrow y = \frac{1}{2}$$

Hence, points are as shown in the figure.

Then, the shortest distance

$$d = \sqrt{\frac{1}{16} + \frac{1}{16}} = \frac{1}{2\sqrt{2}}$$

87. $PA + PB = 4 \Rightarrow (PA \cdot PB)^{\frac{1}{2}} \leq 2 \Rightarrow |PA \cdot PB| \leq 4$

88. Let $P(\sqrt{2} \cos \theta, \sin \theta)$ be only pt on $\frac{x^2}{2} + \frac{y^2}{1} = 1$

Equation of tangent at P is

$$\frac{x\sqrt{2}}{2} \cos \theta + \frac{y}{1} \sin \theta = 1 \quad A(\sqrt{2} \sec \theta, 0) B(0, \csc \theta)$$

Midpoint of intercept between the axes is

$$\left(\frac{\sqrt{2}}{2} \sec \theta, \frac{1}{2} \csc \theta \right) = (h, k) \Rightarrow \cos \theta = \frac{1}{\sqrt{2}h}, \sin \theta = \frac{1}{2k}$$

Now, $\cos^2 \theta + \sin^2 \theta = 1 \Rightarrow \frac{1}{2h^2} + \frac{1}{4k^2} = 1 \therefore$ locus of (h, k) is

$$\frac{1}{2x^2} + \frac{1}{4y^2} = 1, \therefore a + b = 6$$

89. $s_{11} = 36/9 - 4/4 - 1 = 4 - 2 = 2 > 0$. The number of tangents = 0

90. $s(1, 2) s'(5, 5) \Rightarrow SS' = 5 \Rightarrow 2ae = 5 \Rightarrow ae = \frac{5}{2}$

and $|s'p - sp| = 3 \Rightarrow 2a = 3 \Rightarrow a = \frac{3}{2} \therefore e = \frac{5}{3}$

If $\frac{1}{e^2} + \frac{1}{e'^2} = 1 \Rightarrow \frac{1}{e'^2} = 1 - \frac{9}{25} = \frac{16}{25} \Rightarrow e' = \frac{5}{4} = 1.25$



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON Central Office – Madhapur – Hyderabad

Sec: **Sr.Super-60& All**

Jee-Main

Date: 20-05-2021

Time: 09.00Am to 12.00

PTM-04

Max.Marks:300

IMPORTANT INSTRUCTIONS:

1. Immediately fill in the Admission number on this page of the Test Booklet with Blue/Black Ball Point Pen only.
2. The candidates should not write their Admission Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1, 2, 3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **two sections**.

(I) Section-I contains 20 **multiple choice** questions with **only one correct** option.

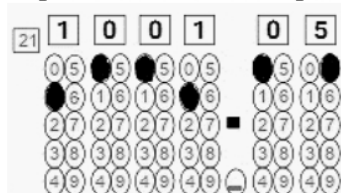
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

(II) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only.

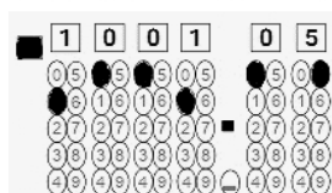
First 5 attempted questions will be considered if more than 5 questions attempted.

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted question 21, bubble on 21 as shown below.



Question Answered for marking



Question Cancelled for marking

Marking scheme : +4 for correct answer and 0 in all other cases.

6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capitals): _____

Admission Number:

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Candidate's Signature: _____

Invigilator's Signature: _____

20-05-2021_Sr.Super60 & All_Jee-Main_PT**M-04**_Test Syllabus

PHYSICS : Rotational Mechanics, NLM, friction, Circular motion, Kinematics

CHEMISTRY : Ideal Gas, Real Gas, Aldehydes & Ketones, Group-17, Group-16

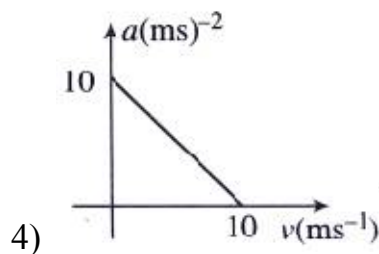
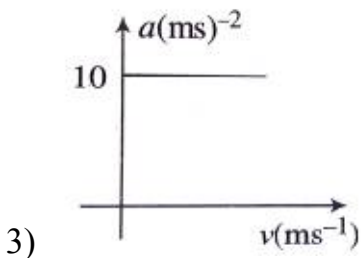
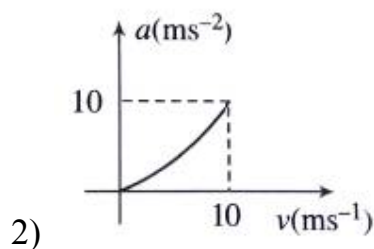
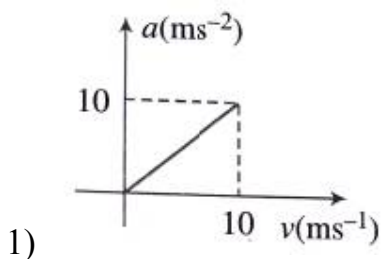
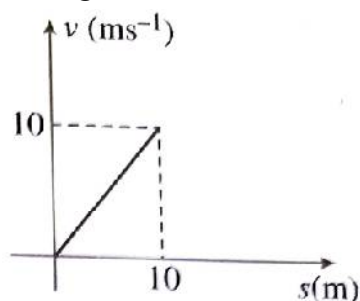
MATHEMATICS : Matrices & Determinants, Vectors & 3D, Functions

PHYSICS**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

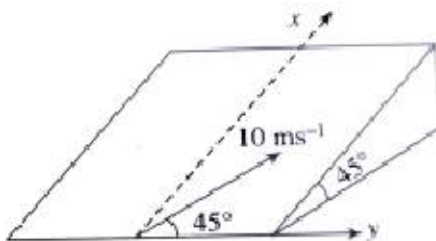
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

1. A person is throwing two balls in the air one after the other. He throws the second ball when the first ball is at the highest point. If he is throwing the balls every second, how high do they rise?
1) 5 m 2) 3.75 m 3) 2.50 m 4) 1.25 m
2. A juggler keeps on moving four balls in air throwing the balls after regular intervals. When one ball leaves his hand (speed = 20ms^{-1}), the position of other balls (height in meter) will be (take $g = 10\text{ms}^{-2}$)
1) 10,20,10 2) 15,20,15 3) 5,15,20 4) 5,10,20
3. Velocity versus displacement graph of a particle moving in a straight line is shown in figure. The corresponding acceleration versus velocity graph will be

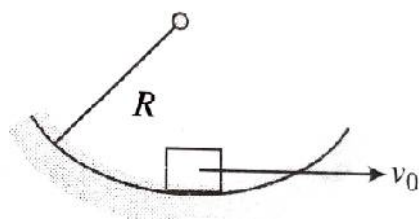


4. A man holds an umbrella at 30° with the vertical to keep himself dry. He, then, runs at a speed of 10ms^{-1} , and find the rain drops to be hitting vertically. Study the following statements and find the correct options.
- Velocity of rain w.r.t. Earth is 20ms^{-1}
 - Velocity of rain w.r.t. man is $10\sqrt{3}\text{ms}^{-1}$
 - Velocity of rain w.r.t. Earth is 30ms^{-1}
 - Velocity of rain w.r.t. man is $10\sqrt{2}\text{ms}^{-1}$
- Statements (i) and (ii) are correct
 - Statements (i) and (iii) are correct
 - Statements (iii) and (iv) are correct
 - Statements (ii) and (iv) are correct
5. Taxies leave station X for station Y every 10 min. Simultaneously, a taxi also leaves station Y for station X every 10 min. The taxies move at the same constant speed and go from X and Y or vice-versa in 2h. How many taxies coming from the other side will meet each taxi enroute from Y and X?
- 24
 - 23
 - 12
 - 11
6. A body is released from the top of a tower of height H m. After 2 s it is stopped and then instantaneously released. What will be its height after next 2 s?
- $(H-5)m$
 - $(H-10)m$
 - $(H-20)m$
 - $(H-40)m$
7. A projectile is fired from level ground at an angle θ above the horizontal. The elevation angle ϕ of the highest point as seen from the launch point is related to θ by the relation
- $\tan \phi = 2 \tan \theta$
 - $\tan \phi = \tan \theta$
 - $\tan \phi = \frac{1}{2} \tan \theta$
 - $\tan \phi = \frac{1}{4} \tan \theta$
8. The small marble is projected with a velocity of 10ms^{-1} in a direction 45° from the horizontal y-direction on the smooth inclined plane. Calculate the magnitude v of its velocity after 2 s.

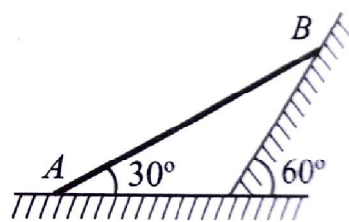


- $10\sqrt{2}\text{ms}^{-1}$
- 5ms^{-1}
- 10ms^{-1}
- $5\sqrt{2}\text{ms}^{-1}$

9. A lift is moving down with an acceleration a . A man in the lift drops a ball inside the lift. The acceleration of the ball as observed by the man in the lift, and a man standing stationary on the ground are, respectively.
- 1) a, g 2) $(g-a); g$ 3) a, a 4) g, g
10. A body of mass m is launched up on a rough inclined plane making an angle 45° with horizontal. If the time of ascent is half of the time of descent, the frictional coefficient between plane and body is
- 1) $\frac{2}{5}$ 2) $\frac{3}{5}$ 3) $\frac{3}{4}$ 4) $\frac{4}{5}$
11. A block of mass M is being pulled along rough horizontal surface. The coefficient of friction between the block and the surface is μ . If another block of mass $M/2$ is placed on the block and it is again pulled on the surface, the coefficient of friction between the block and the surface will be
- 1) μ 2) $\frac{3\mu}{2}$ 3) 2μ 4) $\frac{5\mu}{2}$
12. A particle is projected with a speed $v_0 = \sqrt{gR}$. The coefficient of friction between the particle and the hemi-spherical plane is $\mu = 0.5$. Then, the acceleration of the particle is

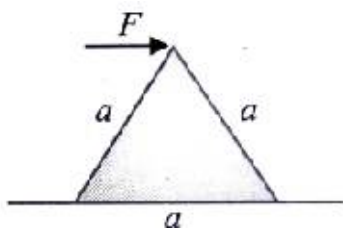


- 1) g 2) $\frac{\sqrt{5}g}{2}$ 3) $\sqrt{2}g$ 4) none of these
13. In the figure shown, the instantaneous speed of end A of the rod is v to the left. The angular velocity of the rod of length L must be

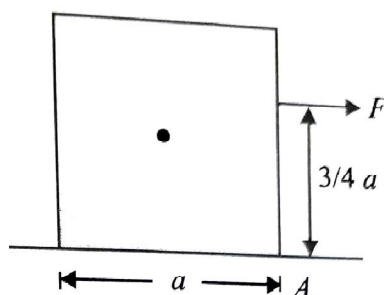


- 1) $\frac{v}{2L}$ 2) $\frac{v}{L}$ 3) $\frac{v\sqrt{3}}{2L}$ 4) none of these

14. An equilateral prism of mass m rests on a rough horizontal surface with coefficient of friction μ . A horizontal force F is applied on the prism as shown in figure. If the coefficient of friction is sufficiently high so that the prism does not slide before toppling, the minimum force required to topple the prism is

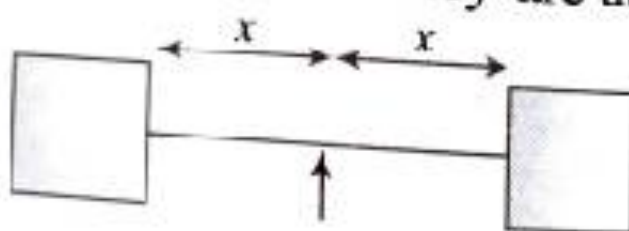


- 1) $\frac{mg}{\sqrt{3}}$ 2) $\frac{mg}{4}$ 3) $\frac{\mu mg}{\sqrt{3}}$ 4) $\frac{\mu mg}{4}$
15. A uniform disc of radius R lies in the $x-y$ plane, with its centre at origin. Its moment of inertia about z -axis is equal to its moment of inertia about line $y = x + c$. The value of c will be
- 1) $-\frac{R}{2}$ 2) $\pm \frac{R}{\sqrt{2}}$ 3) $+\frac{R}{4}$ 4) $-R$
16. A cube of side a and mass m is to be tilted at point A by applying a force F as shown in figure. The minimum force required is



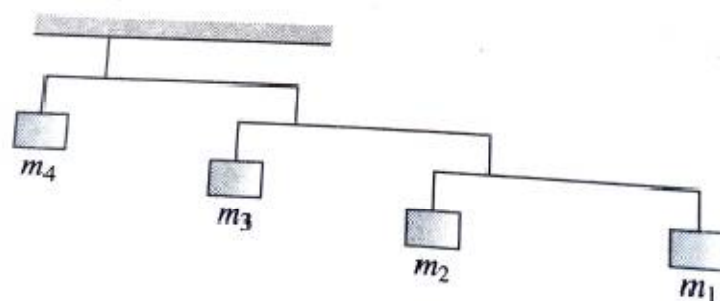
- 1) mg 2) $\frac{2}{3}mg$ 3) $\frac{3}{2}mg$ 4) $\frac{3}{4}mg$
17. The line of action of the resultant of two like parallel forces shifts by one-fourth of the distance between the forces when the two forces are interchanged. The ratio of the two forces is:
- 1) 1 : 2 2) 2 : 3 3) 3 : 4 4) 3 : 5
18. Two identical masses are connected to a horizontal thin massless rod as shown in the figure. When their distance from the pivot is x , a torque produces an angular acceleration α_1 . If the masses are now repositioned so that they are at distance $2x$

each from the pivot, the same torque will produce an angular acceleration α_2 such that



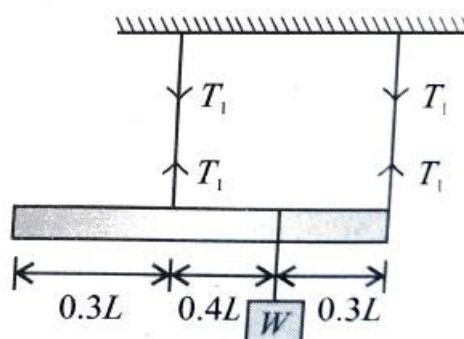
- 1) $\alpha_2 = 4\alpha_1$ 2) $\alpha_2 = \alpha_1$ 3) $\alpha_2 = \frac{\alpha_1}{2}$ 4) $\alpha_2 = \frac{\alpha_1}{4}$

19. Figure shows an arrangement of masses hanging from a ceiling. In equilibrium, each rod is horizontal, has negligible mass and extends three times as far to the right of the wire supporting it as to the left. If mass m_4 is 48 kg, then mass m_1 is equal to



- 1) 1 kg 2) 2 kg 3) 3 kg 4) 4 kg

20. In figure, the bar is uniform and weighing 500 N. How large must W be if T_1 and T_2 are to be equal?



- 1) 500 N 2) 300 N 3) 750 N 4) 1500 N

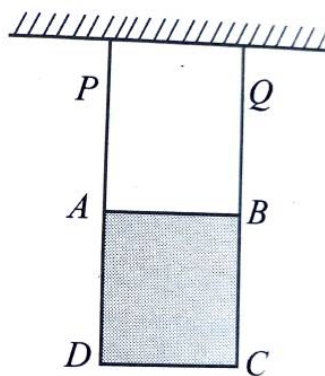
(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

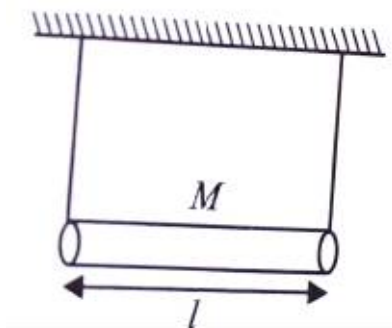
Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

21. A square plate ABCD of mass m and side l is suspended with the help of two ideal strings P and Q as shown. Determine the acceleration ($\text{in } m/s^2$) of corner A of the square just at the moment the string Q is cut. ($g = 10m/s^2$)

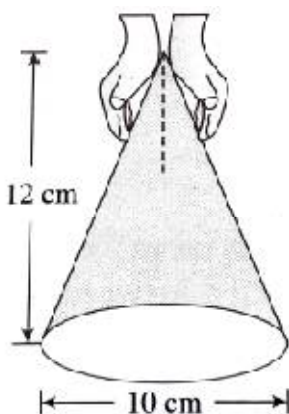


22. A uniform rod of length 1 m and mass 2 kg is suspended. Calculate tension T (in N) in the string at the instant when the right strings snaps ($g = 10m/s^2$)

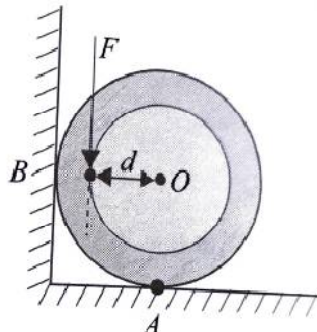


23. A balloon rises from rest on the ground with constant acceleration $1ms^{-2}$. A stone is dropped when the balloon has risen to a height of 39.2 m. Find the time taken by the stone to reach the ground.
24. A particle moves in a straight line. Its position (in m) as function of time is given by $x = (at^2 + b)$. What is the average velocity in the time interval $t = 3s$ to $t = 5s$ in ms^{-1} ?
(What a and b are constants and $a = 1ms^{-2}$, $b = 1m$)

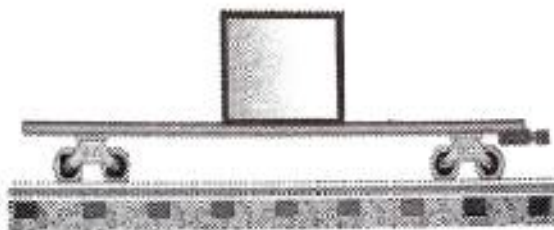
25. Two boats both having a mass of 150 kg including passengers in it are at rest. A sack of mass 50 kg makes 1st boat having total mass of 200 kg. It is thrown to the second boat with velocity whose horizontal component is 2ms^{-1} , relative to water. Calculate the distance (in m) between the boat 8.5 s. after the throw if the sack spent 0.5 s. in air. Neglect the resistance of air and water.
26. You are designing an elevator for a hospital. The force exerted on a passenger by the floor of the elevator is not to exceed 1.60 times the passenger's weight. The elevator accelerates upward with constant acceleration for a distance of 3.0 m and then starts to slow down. What is the maximum speed (in ms^{-1}) of the elevator?
27. A block is moving on an inclined plane making an angle 45° with the horizontal and the coefficient of friction is μ . The force required to just push it up the inclined plane is three times the force required to just prevent it from sliding down. If we define $N = 10\mu$, then N is
28. With two hands, you hold a cone motionless upside down, as shown in figure. The mass of the cone is ($m = 1\text{kg}$), and the coefficient of static friction between you fingers and the cone is ($\mu = 0.5$). If the minimum normal force you must apply with each hand in order to hold up the cone is found to be $13n\text{ N}$, find the value of 'n'. Consider only translational equilibrium.



29. A solid cylinder with $r = 0.1$ m and mass $M = 2$ kg is placed such that it is in contact with the vertical and a horizontal surface as shown in figure. The coefficient of static friction is $\mu = (1/3)$ for both the surfaces. Find the distance d (in CM) from the centre of the cylinder at which a force $F = 40$ N should be applied vertically so that the cylinder just starts rotating in anticlockwise direction.



30. A uniform cylinder rests on a cart as shown. The coefficient of static friction between the cylinder and the cart is 0.5. If the cylinder is 4 cm in diameter and 10 cm in height, then what is the minimum acceleration ($\text{in } m/s^2$) of the cart needed to cause the cylinder to tip over?

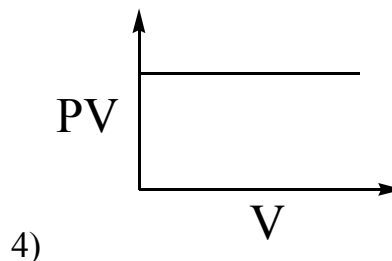
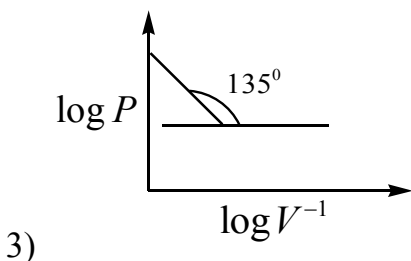
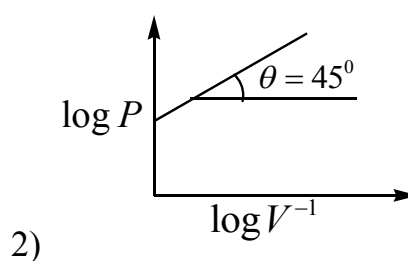
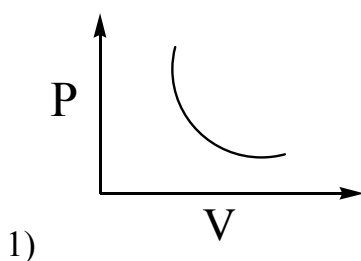


CHEMISTRY**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

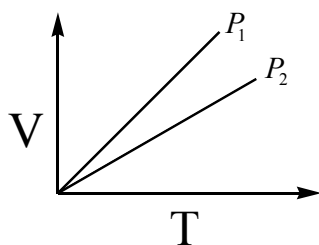
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

31. At 0°C , the density of a gaseous oxide at 2 bar is same as that of dinitrogen at 5 bar. What is the molecular mass of the oxide is
- 1) 30 2) 44 3) 70 4) 28
32. Pressure of 1 g of an ideal gas A at 27°C is found to be 2 atm. When 2 g of another ideal gas B is introduced in the same flask at same temperature, the pressure becomes 3 atm. Find the relation between their molecular masses
- 1) $M_A = 2M_B$ 2) $M_B = 4M_A$ 3) $M_B = 2M_A$ 4) $M_A = 4M_B$
33. In terms of charle's law, why -273°C is the lowest possible temperature
- 1) volume is directly proportional to temperature
 2) The lowest possible temperature is -273°C
 3) The pressure of gas is infinite at -273°C
 4) volume of the gas becomes zero at -273°C i.e the gas ceases to exist
34. Which of the following curve does not represents Boyle's law



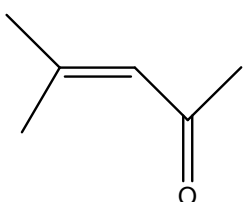
35. V versus T curves at constant P_1 and P_2 for an ideal gas are shown in figure. Which is correct?

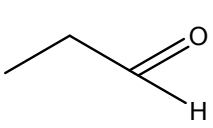
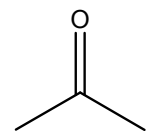
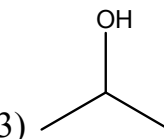
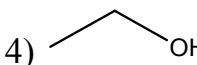


- 1) $P_1 > P_2$ 2) $P_1 < P_2$ 3) $P_1 = P_2$ 4) $P_1 = 2P_2$
36. 2.8 g of a gas at 1 atm and 273K occupies a volume of 2.24 L. The gas can't be
1) O_2 2) CO 3) N_2 4) C_2H_4
37. Consider the equation, $Z = \frac{PV}{RT}$. Which of the following statement is correct?
1) when $Z > 1$, real gases are easier to compress than the ideal gas
2) when $Z = 1$, real gases get compressed easily
3) when $Z > 1$, real gases are get compressed easily
4) If $Z = 1$, real gases are difficult to compress
38. In the contact process of manufacturing of H_2SO_4 , the catalyst used is
1) P_2O_5 2) V_2O_5 3) Rh 4) NO
39. Number of $S=O$ bonds in pyrosulphuric acid are
1) 2 2) 4 3) 5 4) 6
40. Which of the following is a laboratory method of preparation of O_2 ?
1) Heating of $KClO_3$ 2) From liquid air
3) By electrolysis of water 4) By Brin process
41. The reaction of chlorine with CO in the presence of sunlight gives
1) CO_2Cl_2 2) $COCl_2$ 3) $HOCl$ 4) $H_2Cl_2O_2$
42. When F_2 gas reacts with H_2O_2 , it forms
1) O_2 2) O_2F_2 3) HO_2F 4) both 2 & 3
43. Deacon process is used for the manufacture of
1) HNO_3 2) I_2 3) Cl_2 4) Br_2
44. When excess of Cl_2 reacts with ammonia products formed are
1) NH_3, N_2 2) NCl_3, HCl 3) NCl_3, N_2 4) NH_4Cl, N_2

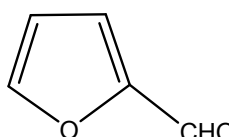
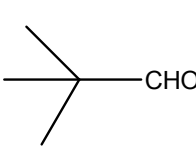
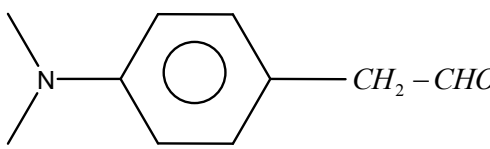
45. $ph-COOH \xrightarrow{SOCl_2} A \xrightarrow[pd-BaSO_4]{H_2} B$, Product 'B' is

- 1) $ph-CHO$
- 2) $ph-CH_2-OH$
- 3) $ph-CH_2-Cl$
- 4) $ph-CH=CH_2$

46. $X \xrightarrow[\Delta]{dil.KOH}$ , Structure of 'X' is

- 1) 
- 2) 
- 3) 
- 4) 

47. Which of the following aldehyde do not undergo a cannizzaro's reaction?

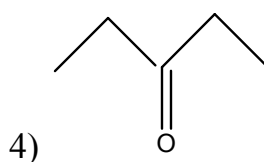
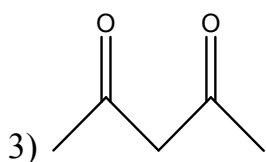
- 1) $ph-CHO$
- 2) 
- 3) 
- 4) 

48. Reagent that can distinguish a set of benzaldehyde and Formaldehyde is

- 1) $I_2 / NaOH$
- 2) Tollen's reagent
- 3) Fehling's Reagent
- 4) both 2 and 3

49. Which of the following will give -ve Tollen's but +ve iodoform test?

- 1) $HCHO$
- 2) CH_3-CHO



50. When nucleophile encounters a ketone, the site of attack is

- 1) The carbon atom of the carbonyl
- 2) The oxygen atom of the carbonyl
- 3) Both the carbon & oxygen atoms, with equal probability
- 4) No attack occurs

(NUMERICAL VALUE TYPE)

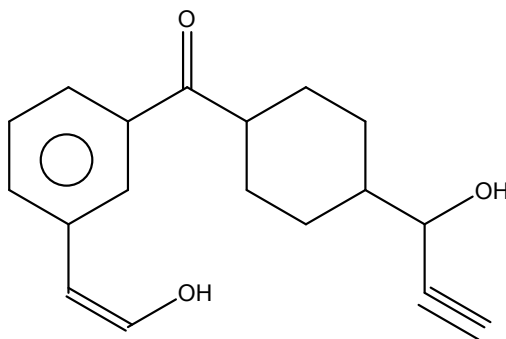
This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

51. $CH_3-\overset{\overset{O}{\parallel}}{C}-H \xrightarrow{HCN} A \xrightarrow{H_3O^+} B$. The number of oxygen atoms in the product B is

52. How many reagents (i to viii) are successively reacted with the following compound



i) $NaHCO_3$

ii) 2,4-DNP

iii) Na metal

iv) $AgNO_3 + OH^-$

v) Fehling solution

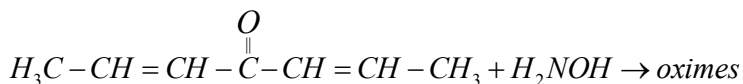
vi) $Cu_2Cl_2 + HCl$

vii) Br_2 / H_2O

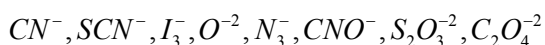
viii) $NaNO_2 + HCl$

53. In reaction of $C_6H_5COCH_3$ with $KOH + I_2$ to form iodoform, how many moles of KOH are consumed per mole of ketone

54. In the reaction given below, how many different oximes would be formed?



55. How many ions in the following behave like pseudohalide?



56. The number of mixed anhydrides among the following are $Cl_2O, ClO_2, Cl_2O_6, Cl_2O_7$

57. If an ideal gas at 100K is heated to 109K in a rigid container, the pressure increases by $x\%$ what is the value of x ?

58. A 1 L flask contains O_2 at $0^\circ C$ and $7.6 \times 10^{-10} \text{ mm of Hg}$. The number of molecules of O_2 is $x \times 10^{10}$, then x is ($N_A = 6.02 \times 10^{23}$)

59. 12 g N_2 , 4 g H_2 and 9 g O_2 are put into a 1 L container at $27^\circ C$. What is the total pressure?

60. At STP ($0^\circ C, 1 \text{ atm}$), the density of nitrogen monoxide is (in $g L^{-1}$)

MATHS**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. If matrix $A = [a_{ij}]_{3 \times 3}$, matrix $B = [b_{ij}]_{3 \times 3}$ where $a_{ij} + b_{ji} = 0$ and $b_{ij} - b_{ji} = 0$, then $A^4 \cdot B^3$ is
 1) skew-symmetric matrix 2) singular matrix
 3) symmetric matrix 4) zero matrix
62. If $A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$, then $\det(\text{Adj}(\text{Adj}A))$ is
 1) $(14)^4$ 2) $(14)^6$ 3) $(14)^9$ 4) $(14)^2$
63. If A is a skew-symmetric matrix of order 3, then the matrix A^4 is
 1) skew symmetric 2) symmetric 3) diagonal 4) null matrix
64. If $A = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, then A^8 equals
 1) 4 B 2) 128 B 3) -128 B 4) -64 B
65. The number of positive integral solutions of the equation $\begin{vmatrix} y^3+1 & y^2z & y^2x \\ yz^2 & z^3+1 & z^2x \\ yx^2 & x^2z & x^3+1 \end{vmatrix} = 11$ is
 1) 1 2) 2 3) 3 4) 4
66. If $a - 2b + c = 1$, then the value of $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix}$ is
 1) x 2) -x 3) -1 4) 1
67. If $p + q + r = 0$ and $\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix} = k \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$, then $K =$
 1) 0 2) abc 3) pqr 4) $a + b + c$
68. Suppose a Matrix A satisfies $A^2 - 5A + 7I = 0$. If $A^5 = aA + bI$, then the values of $2a + b$ is
 1) -87 2) -105 3) 1453 4) 1155
69. Let $\bar{a} = \bar{i} + \bar{j} + \bar{k}$, $\bar{b} = \bar{i} - \bar{j} + 2\bar{k}$ and $\bar{c} = x\bar{i} + (x-2)\bar{j} - \bar{k}$. If the vector $\bar{c}$ lies in the plane of $\bar{a}$ and $\bar{b}$, then x equals to
 1) 0 2) 1 3) -4 4) -2

70. If $\vec{a} + 2\vec{b} + 3\vec{c} = \vec{0}$, then $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$ is equal to
 1) $6(\vec{b} \times \vec{c})$ 2) $6(\vec{c} \times \vec{a})$ 3) $6(\vec{a} \times \vec{b})$ 4) $6(\vec{a} + \vec{b})$
71. A vector of magnitude $\sqrt{51}$ making equal angles with the vectors
 $\vec{a} = \frac{1}{3}(\vec{i} - 2\vec{j} + 2\vec{k})$, $\vec{b} = \frac{1}{5}(-4\vec{i} - 3\vec{k})$ and $\vec{c} = \vec{j}$ is
 1) $\pm(7\vec{i} + \vec{j} - \vec{k})$ 2) $\pm(5\vec{i} - \vec{j} - 5\vec{k})$
 3) $\pm(\vec{i} - \vec{j} + 7\vec{k})$ 4) $\pm(\vec{i} + 5\vec{j} - 5\vec{k})$
72. If $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, $\vec{b} = 4\vec{i} + 3\vec{j} + 4\vec{k}$, $\vec{c} = \vec{i} + \alpha\vec{j} + \beta\vec{k}$ are linearly dependent vectors and $|\vec{c}| = \sqrt{3}$ then
 1) $\alpha = 1, \beta = -1$ 2) $\alpha = 1, \beta = \pm 1$ 3) $\alpha = -1, \beta = \pm 1$ 4) $\alpha = \pm 1, \beta = 1$
73. If θ is the angle between $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ and the plane $2x - y + \sqrt{\lambda}z + 4 = 0$ and is such that $\sin \theta = 1/3$, the value of $\lambda =$
 1) $-\frac{4}{3}$ 2) $\frac{4}{3}$ 3) $-\frac{3}{5}$ 4) $\frac{5}{3}$
74. The image of the point $(-1, 3, 4)$ in the plane $x - 2y = 0$ is
 1) $(15, 11, 4)$ 2) $\left(-\frac{17}{3}, -\frac{19}{3}, 1\right)$ 3) $\left(\frac{9}{5}, -\frac{13}{5}, 4\right)$ 4) $\left(-\frac{17}{3}, -\frac{19}{3}, 4\right)$
75. An equation of the plane passing through the line of intersection of the planes $x + y + z = 6$ and $2x + 3y + 4z + 5 = 0$ and the point $(1, 1, 1)$ is
 1) $2x + 3y + 4z = 9$ 2) $x + y + z = 3$
 3) $x + 2y + 3z = 6$ 4) $20x + 23y + 26z = 69$
76. The volume of the tetrahedron included between the plane $3x + 4y - 5z - 60 = 0$ and the coordinate planes is
 1) 60 2) 600 3) 720 4) 400
77. The domain of the function $f(x) = \sin^{-1}\left(\log_2\left(\frac{x^2}{2}\right)\right)$ is
 1) $[-2, 2]$ 2) $[-2, -1]$ 3) $[1, 2]$ 4) $[-2, -1] \cup [1, 2]$
78. The range of expression $\frac{x^2 - x + 1}{x^2 + x + 1}$ is
 1) $\left[\frac{1}{3}, 3\right]$ 2) $\left[\frac{1}{3}, 1\right]$ 3) $[1, 3]$ 4) $\left(-\infty, \frac{1}{3}\right] \cup [3, \infty)$

79. Set A has 3 elements and set B has 4 elements. The number of injections that can be defined from A to B is
 1) 144 2) 12 3) 24 4) 64
80. If $0 < x < 1000$ and $\left[\frac{x}{2}\right] + \left[\frac{x}{3}\right] + \left[\frac{x}{5}\right] = \frac{31}{30}x$, where $[x]$ is the greatest integer less than or equal to x , the number of possible values of x is
 1) 34 2) 33 3) 32 4) 36

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

81. If $A = \begin{bmatrix} 5 & -6 \\ 1 & -1 \end{bmatrix}$ then determinant of $A^{1003} - 5A^{1002}$ is
82. If the integers a, b, c in order are in A.P., lying between 1 and 9 and a23, b53 and c83 are three-digit numbers, then the value of the determinant $\begin{vmatrix} 2 & 5 & 8 \\ a23 & b53 & c83 \\ a & b & c \end{vmatrix}$ is _____
83. If the adjoint of a 3×3 matrix P is $\begin{bmatrix} 1 & 2 & 1 \\ 4 & 1 & 1 \\ 4 & 7 & 3 \end{bmatrix}$, then the sum of squares of possible values of determinant of P is
84. If $f(x)$ satisfies the equation $\begin{vmatrix} f(x+1) & f(x+8) & f(x+1) \\ 1 & 2 & -5 \\ 2 & 3 & \lambda \end{vmatrix} = 0$ for all real x . If f is periodic with period 7, then the value of $|\lambda|$ is
85. Let $\vec{a}$ and $\vec{b}$ be vectors with $|\vec{a}|=2, |\vec{b}|=3$ and $\vec{a} \cdot \vec{b} = 0$. Define $\vec{c} = \vec{a} \times (\vec{a} \times (\vec{a} \times (\vec{a} \times \vec{b})))$. Then $|\vec{c}|$ equals to

86. If $\bar{a}, \bar{b}, \bar{c}$ are unit vectors such that $\bar{a}$ is perpendicular to plane of $\bar{b}$ and $\bar{c}$ and the angle between $\bar{b}$ & $\bar{c}$ is $\frac{\pi}{3}$ then $|\bar{a} + \bar{b} + \bar{c}|$ is
87. The equation of the plane passing through the intersection of the planes $2x - 5y + z = 3$ and $x + y + 4z = 5$ and parallel to the plane $x + 3y + 6z = 1$ is $x + 3y + 6z = k$, where k is
88. If equation of the plane through the line $\frac{x-1}{2} = \frac{y+2}{-3} = \frac{z}{5}$ and perpendicular to the plane $x - y + z + 2 = 0$ is $ax - by + cz + 4 = 0$, then the value of $10^3a + 10^2b + 10c$
89. If $f(x) = \cos[\pi^2]x + \cos[-\pi^2]x$ where $[x]$ stands for greatest integer function, then $f(2\pi) - f\left(\frac{\pi}{2}\right) =$
90. $f: R \rightarrow R$ is given by $f(x) = \frac{a^x}{a^x + \sqrt{a}} \quad \forall x \in R$, then $f\left(\frac{1}{11}\right) + f\left(\frac{2}{11}\right) + \dots + f\left(\frac{9}{11}\right) + f\left(\frac{10}{11}\right) =$



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Key Sheet PHYSICS

1	1	2	2	3	1	4	1	5	2
6	4	7	3	8	3	9	2	10	2
11	1	12	2	13	2	14	1	15	2
16	2	17	2	18	2	19	1	20	2
21	6.00	22	5.00	23	4.00	24	8.00	25	8.00
26	6.00	27	5.00	28	5.00	29	6.00	30	4.00

CHEMISTRY

31	3	32	2	33	4	34	3	35	2
36	1	37	3	38	2	39	2	40	1
41	2	42	1	43	3	44	2	45	1
46	2	47	4	48	3	49	3	50	1
51	3.00	52	6.00	53	4.00	54	6.00	55	4.00
56	2.00	57	9.00	58	268.00	59	66.74	60	1.34

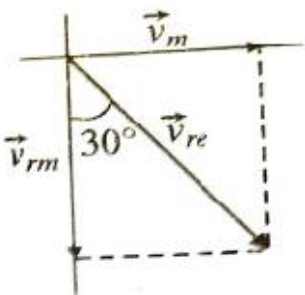
MATHEMATICS

61	2	62	1	63	2	64	2	65	3
66	3	67	3	68	1	69	4	70	1
71	2	72	4	73	4	74	3	75	4
76	2	77	4	78	1	79	3	80	2
81	6.00	82	0.00	83	8.00	84	4.00	85	48.00
86	2.00	87	7.00	88	1710.00	89	3.00	90	5.00

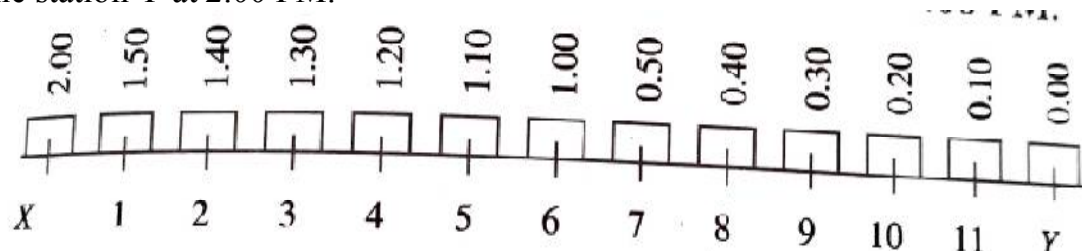
SOLUTIONS

PHYSICS

1. Time of ascent = 1 s $\Rightarrow \frac{u}{g} = 1 \Rightarrow u = 10 \text{ ms}^{-1}$
 Maximum height attained = $\Rightarrow \frac{u^2}{2g} = \frac{10^2}{2 \times 10} = 5 \text{ m}$
2. Time taken by the same ball to return to the hands of the juggler is $\frac{2u}{g} = \frac{2 \times 20}{10} = 4 \text{ s}$. So he is throwing the balls after 1 s each. Let at some instant he throws ball number 4. Before 1 s of throwing it, he throws ball 3. So the height of ball 3 is $h_3 = 20 \times 1 - \frac{1}{2} 10(1)^2 = 15 \text{ m}$
 Before 2 s, he throws ball 2. So the height of ball 2 is $h_2 = 20 \times 2 - \frac{1}{2} 10(2)^2 = 20 \text{ m}$
 Before 3 s, he throws ball 1. So the height of ball 1 is $h_1 = 20 \times 3 - \frac{1}{2} 10(3)^2 = 15 \text{ m}$
3. From graph $v = s$
 $\Rightarrow \frac{dv}{dt} = \frac{ds}{dt} \Rightarrow a = v$
 Acceleration versus velocity graph will be a straight line passing through origin with slope 1 s^{-1}
4. $\vec{v}_m$ = Velocity of man
 $\vec{v}_{re}$ = Velocity of rain w.r.t. earth
 $\vec{v}_{rm}$ = Velocity of rain w.r.t. man
 Velocity of man $|\vec{v}_m| = 10 \text{ ms}^{-1}$
 Using $\sin 30^\circ = \frac{v_m}{v_{re}}$
 $v_{re} = \frac{v_m}{\sin 30^\circ} = \frac{10}{1/2} = 20 \text{ ms}^{-1}$,
 $\cos 30^\circ = \frac{v_{rm}}{v_{re}}$
 $v_{rm} = v_{re} \cos 30^\circ = \frac{20 \times \sqrt{3}}{2} = 10\sqrt{3} \text{ ms}^{-1}$



shows the location of taxis going from X to Y at the instant 2.00 PM. The taxi which leaves the station X at 0.00 PM has just arrived at the station Y. Consider the taxi leaving the station Y at 2.00 PM.



It will meet all the 11 taxis marked 1 to 11 as well as 12 other taxis which would leave the station X from 2.00 PM to 3.50 PM. When it arrives at the station X at 4.00 PM, there will be one more taxi leaving that station. However, it will not be counted among the taxis crossed by taxi under consideration. That is, it will cross 23 taxis leaving the station X from 0.10 PM to 3.50 PM.

6. Distance covered by the object in first 2 s

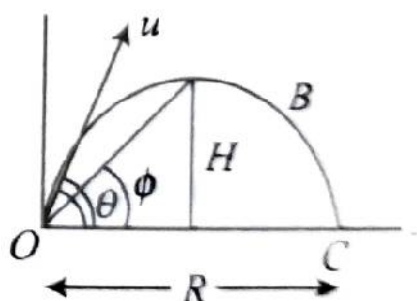
$$h_1 = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times 2^2 = 20\text{m}$$

Similarly, distance covered by the object in next 2 s will also be 20 m, hence the required height = $H - 20 - 20 = H - 40\text{m}$

7. From figure

$$\tan \phi = \frac{H}{R/2}$$

$$= \frac{u^2 \sin^2 \theta / 2g}{u^2 \sin 2\theta / 2g} = \frac{1}{2} \tan \theta$$



8. Given ($V = 10\text{ms}^{-1}$)

$$\text{After 2 s: } V_x = \frac{10}{\sqrt{2}} - \frac{10}{\sqrt{2}} \times 2 \Rightarrow V_x = -\frac{10}{\sqrt{2}}\text{ms}^{-1}$$

$$V_x = -\frac{10}{\sqrt{2}}\text{ms}^{-1} \text{ and } V_y = -\frac{10}{\sqrt{2}}\text{ms}^{-1}$$

$$V = \sqrt{\frac{100}{2} + \frac{100}{2}} = 10\sqrt{\frac{1}{2} + \frac{1}{2}} = 10\text{ms}^{-1}$$

9. $\vec{a}_{b,l} = \vec{a}_b - \vec{a}_l = (g - a) \downarrow \Rightarrow \vec{a}_b = g \downarrow$

10. $t_A = \sqrt{\frac{2s}{a_A}} \Rightarrow t_D = \sqrt{\frac{2s}{a_D}}, t_A = \frac{1}{2}t_D$

$$\sqrt{\frac{2s}{g \sin \theta + \mu g \cos \theta}} = \frac{1}{2} \sqrt{\frac{2s}{g \sin \theta - \mu g \cos \theta}}$$

$$4g \sin \theta - 4\mu g \cos \theta = g \sin \theta + \mu g \cos \theta$$

$$3g \sin \theta = 5\mu g \cos \theta \quad \text{or} \quad \mu = \frac{3}{5} \tan \theta = \frac{3}{5}$$

$$(\because \theta = 45^\circ)$$

11. Coefficient of friction is independent of the normal reaction. Hence, it will remain same.
 12. When the particle comes with the contact of hemisphere surface, it will slide and friction will be kinetic nature. This kinetic friction will provide tangential acceleration to the particle. Tangential acceleration of the particle,

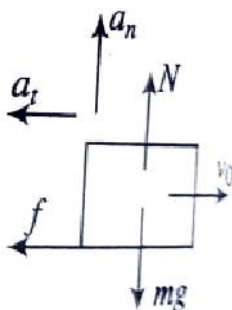
$$a_t = \frac{\mu mg}{m} = \mu g = \frac{g}{2} \leftarrow$$

Normal acceleration of the particle,

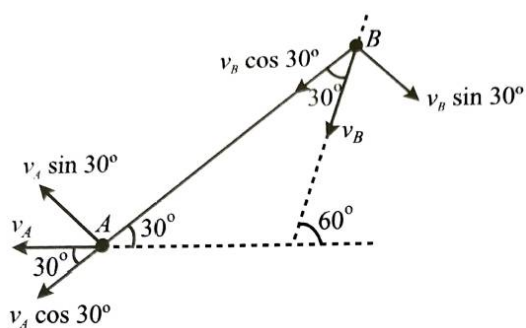
$$a_n = \frac{v^2}{R} = \frac{gR}{R} = g \uparrow$$

Net acceleration of the particle,

$$a_{net} = \sqrt{a_t^2 + a_n^2} = \sqrt{\frac{g^2}{4} + g^2} = \frac{\sqrt{5}g}{2} \uparrow$$



13.



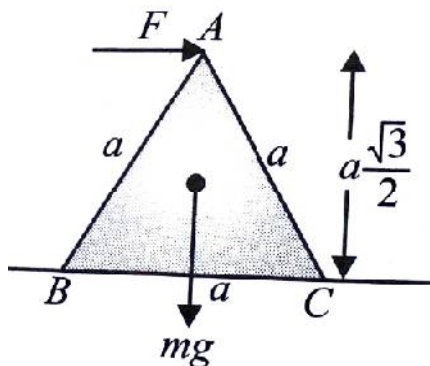
The velocities of the ends A and B along the length of rod should be the same.

$$\text{Hence, } v_A \cos 30^\circ = v_B \cos 30^\circ \Rightarrow v_A = v_B = v$$

Hence, the angular velocity of the rod is

$$\omega = \frac{(v_{AB})_{\perp}}{l} = \frac{2v \sin 30^\circ}{l} \Rightarrow \omega = \frac{v}{l}$$

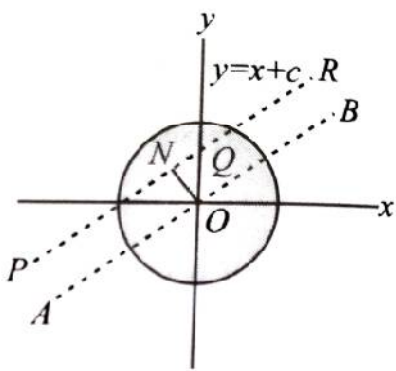
14. The tendency of rotating will be about point C. For the minimum force, the torque of F about C has to be equal to the torque of mg about C.



$$F \left(a \frac{\sqrt{3}}{2} \right) = mg \left(\frac{a}{2} \right)$$

$$\Rightarrow F = \frac{mg}{\sqrt{3}}$$

15.



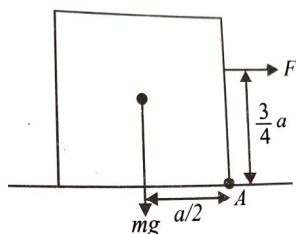
$$I_{PQR} = I_{AOB} + M (ON)^2$$

$$= \frac{1}{2} MR^2 + M \left(\frac{c}{\sqrt{2}} \right)^2$$

$$\text{But } \frac{1}{4} MR^2 = I_2$$

$$c = \pm \frac{R}{\sqrt{2}}$$

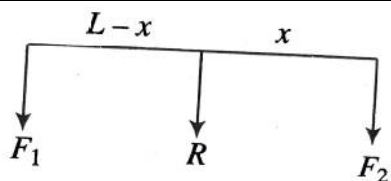
16. See the figure. For tilting about A the clockwise torque (due to F) should be greater than the anticlockwise torque about A.



$$mg \times \frac{a}{2} = \frac{F 3a}{4}; F = \frac{2mg}{3}$$

17. For $\tau = 0$

$$F_1 (L - x) = F_2 x$$



$$F_2 \left(\frac{3L}{4} - x \right) = F_1 \left(x + \frac{L}{4} \right) \frac{L-x}{x + (L/4)} = \frac{x}{(3L/4) - x}$$

$$\frac{3L}{4} = x + \frac{3x}{4} + \frac{x}{4} = 2x \Rightarrow 3L = 8x$$

So putting the value of x

$$F_1 \frac{5L}{8} = F_2 \frac{3L}{8} \Rightarrow F_1 : F_2 = 3 : 5$$

18. $\tau = (2mx^2)\alpha_1$

$$\tau = (2 \times m \times 4x^2)\alpha_2$$

Equating, $8mx^2\alpha_1$ or $\alpha_2 = \frac{\alpha_1}{4}$

19. $m_2 g \cdot 1 = m_1 g \cdot 3 \rightarrow m_2 = 3m_1$

$$\rightarrow 4m_1 g \cdot 3 = m_3 g \rightarrow m_3 = 12m_1$$

$$\rightarrow 16m_1 g \cdot 3 = m_4 g \rightarrow m_1 = \frac{48}{48} = 1\text{kg}$$

20. Taking torque about the attachment point for W, we get

$$-T_1(0.4L) + T_2(0.3L) + 500(0.2L) = 0$$

$$T = 1000\text{N}, \text{ where } T_1 = T_2 = T$$

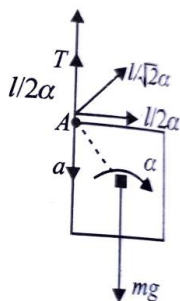
$$\sum F_y = 0 \Rightarrow 2T - W - 500 = 0$$

$$\Rightarrow W = 1500\text{N}$$

21. $T \frac{l}{2} = \frac{ml^2}{6} \times \alpha; mg - T = ma$

$$\alpha = \frac{3T}{ml} = \frac{3}{ml}(mg - ma)$$

Constraint relation: Acceleration of A in the vertical direction should be zero



$$\frac{l}{2} \alpha = a$$

$$\Rightarrow \alpha = \frac{2a}{l} = \frac{3(g-a)}{l}$$

$$5a = 3g \Rightarrow a = 3 \times \frac{g}{5} = 6\text{m/s}^2$$

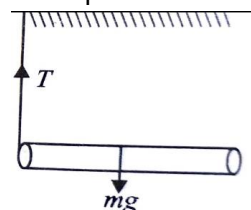
22. $Mg - T = Ma_y \dots\dots(i)$

$$T\left(\frac{L}{2}\right) = \frac{ML^2}{12}\alpha \quad \dots\dots(ii)$$

$$\text{And } a_y = \frac{L}{2}\alpha \quad \dots\dots(iii)$$

On solving, we get

$$T = \frac{Mg}{4} = 5N$$



$$23. \quad h = \frac{1}{2}at^2 \Rightarrow t = \sqrt{\frac{2h}{a}} = \sqrt{\frac{2 \times 40 \times 8}{10}} = 8s$$

Velocity after 8 s, $v = 0 + 1 \times 8 = 8ms^{-1}$

$$40 = -10t + \frac{1}{2}10t^2$$

$$10t^2 - 2 \times 10t - 2 \times 40 = 0$$

$$t^2 - 2t - 8 = 0$$

$$t - 4t + 2t - 8 = 0$$

$$t = 4s$$

$$24. \quad v_{av} = \frac{x_f - x_i}{t_f - t_i} = \frac{(1 \times 5^2 + 1) - (1 \times 3^2 + 1)}{5 - 3} = \frac{16}{2} = 8ms^{-1}$$

25. First boat

$$0 = 50 \times 2 + 150v$$

$$v_1 = -\frac{2}{3}ms^{-1}$$

$$\text{Second boat } 100 = 200v_2$$

$$v_2 = \frac{1}{2}ms^{-1}$$

$$T = \frac{2u_y}{g} = 0.5 \quad u_y = 2.5ms^{-1}$$

$$\text{Initial distance} = \frac{2 \times 2.5 \times 2}{10} = 1m$$

$$\text{Distance} = u_{rel} \times t = \left(\frac{1}{2} + \frac{2}{3}\right) \times 8$$

$$\text{Total distance} = 8m$$

26. Consider the forces on the person:

$$\sum F_y = ma_y$$

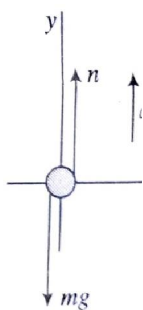
$$n - mg = ma$$

$$n = 1.6mg$$

$$\text{So, } a = 0.60g = 6ms^{-2}$$

$$v^2 = u^2 + 2as = 0^2 + 2 \times 6 \times 3$$

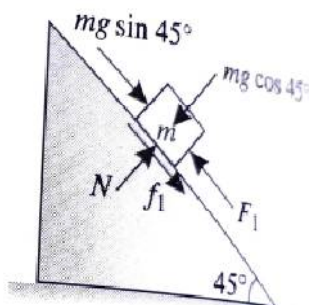
$$\Rightarrow v = 6ms^{-1}$$



27. Case I: For pushing the block up

$$F_1 = mg \sin 45^\circ + \mu mg \cos 45^\circ$$

$$\Rightarrow F_1 = \frac{mg}{\sqrt{2}} + \frac{\mu mg}{\sqrt{2}}$$



Case II: Force required to just prevent the block from sliding down.

$$F_2 = mg \sin 45^\circ - \mu mg \cos 45^\circ$$

$$\Rightarrow F_2 = \frac{mg}{\sqrt{2}} - \frac{\mu mg}{\sqrt{2}}$$

$$F_1 = 3F_2$$

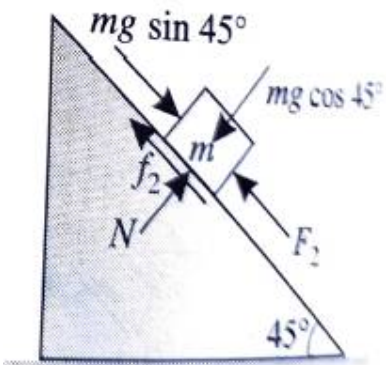
$$\Rightarrow 1 + \mu = 3 - 3\mu$$

$$4\mu = 2$$

$$\mu = \frac{1}{2}$$

$$\Rightarrow N = 10\mu$$

$$\Rightarrow N = 5$$



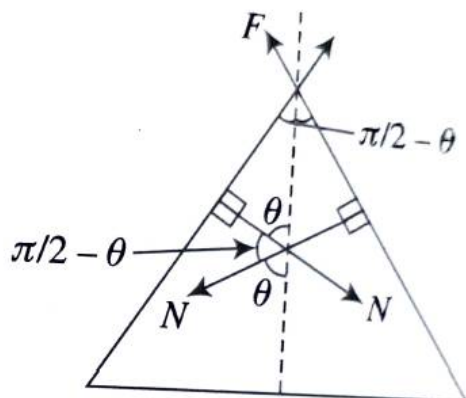
28. For equilibrium in vertical direction

$$2f \sin \theta - 2N \cos \theta - mg = 0 \quad \dots (i)$$

$$2\mu N \sin \theta - 2N \cos \theta = mg$$

$$\tan \theta = \left(\frac{\pi}{2} - \theta \right) = \frac{5}{12}$$

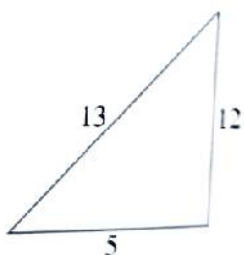
$$\cot \theta = \frac{5}{12}, \sin \theta = \frac{12}{13} \text{ \& \; } \cos \theta = \frac{4}{12}$$



$$2(-5)\left(\frac{12}{13}\right)N - 2N \cdot \frac{5}{13} = (1)(10)$$

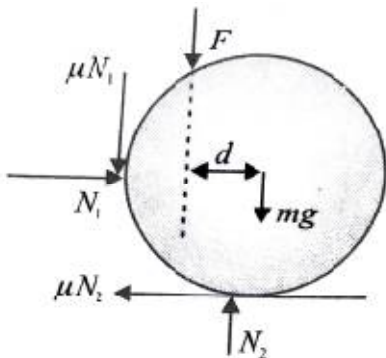
$$N\left(\frac{12}{13} - \frac{10}{13}\right) = (10) \Rightarrow N = \frac{10 \times 13}{2} = 65N$$

Hence $n = 5$



29. For equilibrium of the cylinder in the horizontal direction,

$$N_1 = \mu N_2 \quad \dots\dots\dots(i)$$



In the vertical direction,

$$N_2 + \mu N_1 = F + mg \quad \dots\dots\dots(ii)$$

Solving these two equations with $F = 40N, M = 2kg$ and $\mu = 1/3$, we get

$$N_1 = 18N \text{ and } N_2 = 54N$$

$$\text{Now, } Fd = \mu(N_1 + N_2)r$$

$$\therefore d = \frac{M(N_1 + N_2)r}{F}$$

$$= \frac{(1/3)(18+54)(0.1)}{40} = 0.06m$$

30. Maximum acceleration of cart so that cylinder does not slip: $a_m = \mu g = 0.5 \times 10 = 5ms^{-2}$

For tipping over: Let acceleration of cart is a.

Considering torque about A:

$$Ma \times 5 \geq mg \times 2$$

$$\Rightarrow a \geq 2g / 5 = 4ms^{-2}$$

CHEMISTRY

31. $d = \frac{PM}{RT}; P_1M_1 = P_2M_2$

32. $P \propto n$ at constant V and T

$$\frac{P_A}{P_B} = \frac{wt_A}{wt_B} \times \frac{M_B}{M_A}; P_A = 2atm, P_B = 1atm$$

33. Conceptual

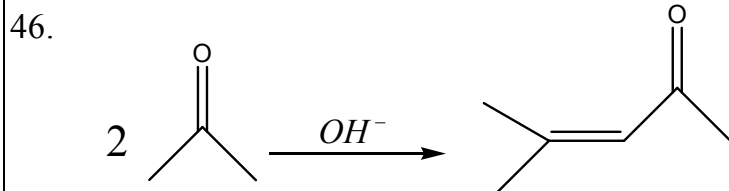
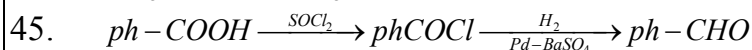
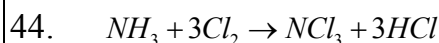
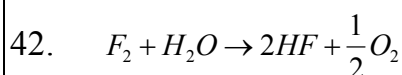
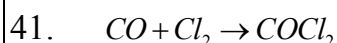
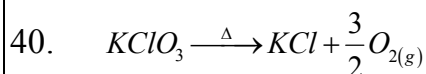
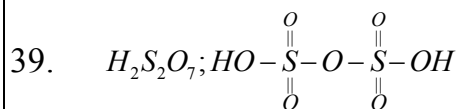
34. $P \propto \frac{1}{V}; PV = \text{constant}; \log P = \log V^{-1} + \log K$

35. at constant T, $P_1V_1 = P_2V_2; V_1 > V_2$ i.e. $P_1 < P_2$

36. $PV = nRT; PV = \frac{w}{M} RT; M = \frac{wRT}{PV} = 28$

37. Conceptual

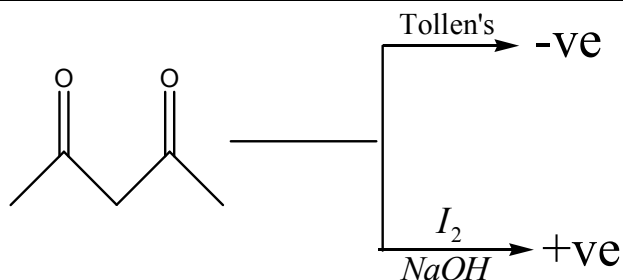
38. V_2O_5



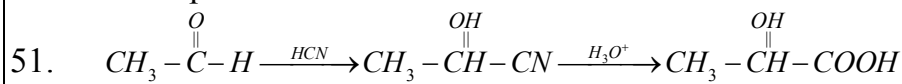
47. Aldehyde without α -Hydrogen undergoes cannizzaro's reaction

48. Conceptual

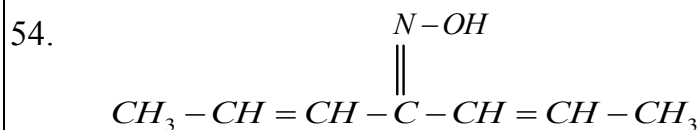
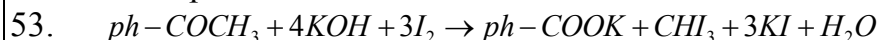
49.



50. Conceptual



52. Conceptual



55. CN^- , SCN^- , N_3^- , CNO^-

56. ClO_2 & Cl_2O_6

57. $\frac{P_i}{P_f} = \frac{T_i}{T_f} \Rightarrow P_f = \left(\frac{109}{100}\right) P_i$

$\Delta P = \frac{9P_i}{100}$, 9% increases

58. $n = \frac{PV}{RT} = 4.46 \times 10^{-12}$

Number of molecules = $nN_A = 4.46 \times 10^{-12} \times 6.02 \times 10^{23} = 268 \times 10^{10}$

59. $PV = nRT$

$P \times 1 = \left(\frac{12}{28} + \frac{4}{2} + \frac{9}{32}\right) \times 0.0821 \times 300$

$P = 66.74 \text{ atm}$

60. $d = \frac{PM}{RT}$

MATHEMATICS

61. Since matrix A is skew-symmetric,

$\therefore |A| = 0$

$\therefore |A^4 \cdot B^3| = 0$

62. $|A| = (1+2) - 2(-1-4) - (1-2) = 3 + 10 + 1 = 14$

$\therefore \det(A \text{adj}(A)) = |A|^4 = (14)^4$

63. We have $A^T = -A$

$(A^4)^T = (A.A.A.A)^T = A^T A^T A^T A^T$

$\Rightarrow (-A)(-A)(-A)(-A) = (-1)^4 A^4 = A^4$

64. We have $A = iB \Rightarrow (iB)^2 = i^2 B^2 = -B^2 = -\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = -2B$

$$\Rightarrow A^4 = (-2B)^2 = 4B^2 = 4(2B) = 8B$$

$$\Rightarrow (A^4)^2 = (8B)^2 \Rightarrow A^8 = 64B^2 = 128B$$

65. Multiply by y, z and x in rows 1, 2 and 3 respectively and then take common y, z and x from column 1, 2 and 3 respectively, then

$$\begin{vmatrix} y^3+1 & y^3 & y^3 \\ z^3 & z^3+1 & z^3 \\ x^3 & x^3 & x^3+1 \end{vmatrix} = 11 \Rightarrow \begin{vmatrix} 1 & 0 & y^3 \\ -1 & 1 & z^3 \\ 0 & -1 & x^3+1 \end{vmatrix} = 11 (C_1 \rightarrow C_1 - C_2 \text{ and } C_2 \rightarrow C_2 - C_3)$$

$$\Rightarrow 1(x^3+1+z^3) + y^3(1) = 11 \Rightarrow x^3 + y^3 + z^3 = 10$$

So solution are (1, 1, 2), (1, 2, 1) or (2, 1, 1)

66. $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix}$

$$a - 2b + c = 1$$

$$(a-b) + (c-b) = 1$$

Apply the operation, $R_1 \rightarrow R_1 - 2R_2 + R_3$ & $R_3 \rightarrow R_3 - R_2$, the determinant reduces to

$$\begin{vmatrix} 0 & 0 & 1 \\ x+2 & x+3 & x+b \\ 1 & 1 & c-b \end{vmatrix} = -1$$

67. $p+q+r=0 \Rightarrow p^3+q^3+r^3=3pqr$

$$\begin{vmatrix} pa & qb & rc \\ qc & rb & pb \\ rb & pc & qa \end{vmatrix} = pqr(a^3+b^3+c^3-3abc) = pqr \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix} \Rightarrow k = pqr$$

68. $A^3 = AA^2 = A(5A - 7I)$

$$= 5A^2 - 7A = 5(5A - 7I) - 7A = 18A - 35I$$

$$A^4 = AA^3 = A(18A - 35I) = 18(5A - 7I) - 35A$$

$$A^5 = 149A - 385I = 55A - 126I$$

$$A^5 = 149A - 385I$$

$$a = 149, b = -385$$

69. Since the three vectors are coplanar $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ x & x-2 & -1 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} 1 & 0 & 0 \\ 1 & -2 & 1 \\ x & -2 & -1-x \end{vmatrix} = 0$

$$\Rightarrow -2(-1-x) + 2 = 0 \Rightarrow x = -2$$

70. $\vec{a} + 2\vec{b} + 3\vec{c} = \vec{0} \Rightarrow \vec{a} \times \vec{b} + 3\vec{c} \times \vec{b} = \vec{0}$ i.e. $\vec{a} \times \vec{b} = 3\vec{b} \times \vec{c}$, $\vec{a} \times \vec{c} + 2\vec{b} \times \vec{c}$ i.e. $2\vec{b} \times \vec{c} = \vec{c} \times \vec{a}$
 $\therefore \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = 3\vec{b} \times \vec{c} + \vec{b} \times \vec{c} + 2\vec{b} \times \vec{c} = 6\vec{b} \times \vec{c}$

71. Let $\vec{d} = d_1\hat{i} + d_2\hat{j} + d_3\hat{k}$. Then $|\vec{d}|^2 = d_1^2 + d_2^2 + d_3^2 = 51$ (1)

$$\cos \theta = \frac{\vec{d} \cdot \vec{a}}{|\vec{d}| |\vec{a}|} = \frac{\vec{d} \cdot \vec{b}}{|\vec{d}| |\vec{b}|} = \frac{\vec{d} \cdot \vec{c}}{|\vec{d}| |\vec{c}|} = \frac{1}{3}(d_1 - 2d_2 + 2d_3) = \frac{1}{5}(-4d_1 - 3d_3) = d_2$$

$$\Rightarrow d_1 - 5d_2 + 2d_3 = 0, 4d_1 + 5d_2 + 3d_3 = 0$$

$$\Rightarrow \frac{d_1}{5} = \frac{d_2}{-1} = \frac{d_3}{-5} = \lambda \Rightarrow d_1 = 5\lambda, d_2 = -\lambda, d_3 = -5\lambda$$

$$\text{From (1) } \lambda = \pm 1 \quad \therefore \vec{d} = \pm(5\hat{i} - \hat{j} - 5\hat{k})$$

$$72. \begin{vmatrix} 1 & 1 & 1 \\ 4 & 3 & 4 \\ 1 & \alpha & \beta \end{vmatrix} = 0$$

$$1(3\beta - 4\alpha) - 1(4\beta - 4) + 4\alpha - 3 = 0$$

$$3\beta - 4\alpha - 4\beta + 4 + 4\alpha - 3 = 0$$

$$-\beta + 1 = 0 \Rightarrow \beta = 1$$

$$|\vec{c}| = \sqrt{3} \Rightarrow 1 + \alpha^2 + \beta^2 \Rightarrow \alpha^2 = 3 - 1 - 1 \Rightarrow \alpha = \pm 1$$

$$73. \sin \theta = \left| \frac{2 - 2 + 2\sqrt{\lambda}}{3\sqrt{5 + \lambda}} \right| = \frac{1}{3} \quad \lambda = \frac{5}{3}$$

$$74. \frac{h+1}{1} = \frac{k-3}{-2} = \frac{p-4}{0} = -2 \left(\frac{-1-6}{5} \right) \quad (h, k, p) = \left(\frac{9}{5}, \frac{-13}{5}, 4 \right)$$

75. Equation of any plane through the line of intersection of the given planes is

$$2x + 3y + 4z + 5 + \lambda(x + y + z - 6) = 0$$

It passes through (1,1,1) if $(2 + 3 + 4 + 5) + \lambda(1 + 1 + 1 - 6) = 0 \Rightarrow \lambda = 14/3$ and the required equation is therefore, $20x + 23y + 26z = 69$

$$76. \text{ Equation of the given plane can be written as } \frac{x}{20} + \frac{y}{15} + \frac{z}{-12} = 1$$

Which meets the coordinates axes in points $A(20, 0, 0)$, $B(0, 15, 0)$ and $C(0, 0, -12)$ and the coordinates of the origin are $(0, 0, 0)$

$\therefore$ the volume of the tetrahedron OABC is

$$\frac{1}{6} \begin{vmatrix} 0 & 0 & 0 & 1 \\ 20 & 0 & 0 & 1 \\ 0 & 15 & 0 & 1 \\ 0 & 0 & -12 & 1 \end{vmatrix} = \left| \frac{1}{6} \times 20 \times 15 \times (-12) \right| = 600$$

$$77. f(x) = \sin^{-1} \left(\log_2 \left(\frac{x^2}{2} \right) \right) \in R \Leftrightarrow \frac{1}{2} \leq \frac{x^2}{2} \leq 2 \Leftrightarrow 1 \leq x^2 \leq 4 \Leftrightarrow x \in [-2, -1] \cup [1, 2]$$

$$78. \text{ Let } y = \frac{x^2 - x + 1}{x^2 + 2x + 7} \Rightarrow yx^2 + yx + y = x^2 - x + 1 \Rightarrow (y-1)x^2 + (y+1)x + (y-1) = 0$$

$$x \in R \Rightarrow \text{Discriminant} \geq 0 \Rightarrow (y+1)^2 - 4(y-1)^2 \geq 0 \Rightarrow -3y^2 + 10y - 3 \geq 0$$

$$\Rightarrow 3y^2 - 10y + 3 \leq 0 \Rightarrow (3y-1)(y-3) \leq 0 \Rightarrow \frac{1}{3} \leq y \leq 3$$

$$\text{Range} = \left[\frac{1}{3}, 3 \right]$$

$$79. 3 \cdot {}^{n(B)}P_{n(A)} = {}^4P_3 = 4 \cdot 3 \cdot 2 = 24$$

80. $\therefore$ LHS is an integer $\therefore$ RHS must be an integer for which x is multiple of 30

$$\therefore x = 30, 60, 90, 120, \dots, 900$$

$\Rightarrow$ Number of possible values of x is 33

$$81. |A^{1003} - 5A^{1002}| = |A^{1002}(A - 5I)|$$

$$= |A^{1002}| |A - 5I| = |A|^{1002} |A - 5I| = 1 \times \begin{vmatrix} 0 & -6 \\ 1 & -6 \end{vmatrix} = 6$$

82. We have,

$$\begin{vmatrix} 2 & 5 & 8 \\ a23 & b53 & c83 \\ a & b & c \end{vmatrix} = \begin{vmatrix} 2 & 5 & 6 \\ 100a+20+3 & 100b+50+3 & 100c+80+3 \\ a & b & c \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 5 & 8 \\ 100a & 100b & 100c \\ a & b & c \end{vmatrix} + \begin{vmatrix} 2 & 5 & 8 \\ 20 & 50 & 80 \\ a & b & c \end{vmatrix} + \begin{vmatrix} 2 & 5 & 8 \\ 3 & 3 & 3 \\ a & b & c \end{vmatrix}$$

$$= 100 \begin{vmatrix} 2 & 5 & 8 \\ a & b & c \\ a & b & c \end{vmatrix} + 10 \begin{vmatrix} 2 & 5 & 8 \\ a & b & c \\ a & b & c \end{vmatrix} + 3 \begin{vmatrix} 2 & 5 & 8 \\ a & b & c \\ a & b & c \end{vmatrix}$$

$$= 0 + 0 + 3 \begin{vmatrix} 2 & 3 & 6 \\ 1 & 0 & 0 \\ a & b-a & c-a \end{vmatrix}$$

(Applying $C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$)

$$= -3[3(c-a) - 6(b-a)] = -9[c-a-2b+2a]$$

$$= -9(a-2b+c) = 0 \quad [Q \ a, b, c \text{ are in A.P., } \therefore 2b = a+c]$$

83. $Adj.P = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 1 & 1 \\ 4 & 7 & 3 \end{bmatrix} \quad |adj.P| = 4 \Rightarrow |P|^2 = 4 \Rightarrow |P| = \pm 2$

84. On solving we get

$$(2\lambda+15)f(x+1) - (\lambda+10)f(x+8) - f(x+1) = 0$$

$$(2\lambda+14)f(x+1) = (\lambda+10)f(x+8) \text{ Since } f \text{ is periodic with period 7}$$

$$\therefore f(x+1) = f(x+8) \Rightarrow 2\lambda+14 = \lambda+10 \Rightarrow |\lambda| = 4$$

85. $\vec{a} \times (\vec{a} \times \vec{b}) = (\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b} = -|\vec{a}|^2 \vec{b}$

Continuing in the same manner

$$\vec{C} = \vec{a} \times (\vec{a} \times (\vec{a} \times (\vec{a} \times \vec{b}))) = -|\vec{a}|^2 (-|\vec{a}|^2 \vec{b}) = |\vec{a}|^4 \vec{b}$$

$$|\vec{C}| = |\vec{a}|^4 |\vec{b}| = 2^4 \times 3 = 48$$

86. $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$ & $\vec{a} \cdot \vec{b} = 0$ & $\vec{a} \cdot \vec{c} = 0$

$$\vec{b} \cdot \vec{c} = |\vec{b}| |\vec{c}| \cos \frac{\pi}{3} = \frac{1}{2}$$

$$\therefore |\vec{a} + \vec{b} + \vec{c}|^2 = 3 + 2 \cdot 0 + 2 \cdot 0 + 1 = 4 \quad \therefore |\vec{a} + \vec{b} + \vec{c}| = 2$$

87. Equation of plane passing through the intersection of the planes

$$2x - 5y + z = 3 \text{ and } x + y + 4z = 5$$

$$(2x - 5y + z - 3) + \lambda(x + y + 4z - 5) = 0$$

$$\Rightarrow (2+\lambda)x + (-5+\lambda)y + (1+4\lambda)z - 3 - 5\lambda = 0 \quad \dots(i)$$

Which is parallel to the plane $x + 3y + 6z = 1$

$$\text{Then } \frac{2+\lambda}{1} = \frac{-5+\lambda}{3} = \frac{1+4\lambda}{6}$$

$$\therefore \lambda = \frac{-11}{2}$$

From eq. (i)

$$-\frac{7}{2}x - \frac{21}{2}y - 21z + \frac{49}{2} = 0$$

$$\therefore x + 3y + 6z = 7$$

Hence, $k = 7$

88. Let equation of a plane containing the line be $l(x-1) + m(y+2) + nz = 0$

$$\text{Then } 2l - 3m + 5n = 0 \text{ and } l - m + n = 0$$

$$\therefore \frac{l}{2} = \frac{m}{3} = \frac{n}{1}$$

$$\therefore \text{ the plane is } 2(x-1) + 3(y+2) + z = 0$$

$$\text{i.e. } 2x + 3y + z + 6 = 0$$

$$\therefore a = 2, b = -3, c = 1$$

$$\therefore 10^3 a + 10^2 b + 10c = 2000 - 300 + 10 = 1710$$

89. $f(x) = \cos 9x + \cos 10x$

$$f(2\pi) - f\left(\frac{\pi}{2}\right) = 1 + 1 - 0 + 1 = 3$$

90. $f(1-x) = 1 - f(x) \Rightarrow f(x) + f(1-x) = 1$

$$\Rightarrow f\left(\frac{1}{11}\right) + f\left(\frac{10}{11}\right) = 1$$



Sri Chaitanya IIT Academy, India

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A right Choice for the Real Aspirant
ICON CENTRAL OFFICE, MADHAPUR-HYD

20-05-2021_Sr.Super60 & All_Jee-Main_PTM-04_Final Key

S.NO	SUB	Q.NO	GIVEN KEY	FINALIZED KEY	EXPLANATION
1	PHY	12	2	3	$n \neq mg$
2	PHY	17	2	4	Calculation Error
3	PHY	18	2	4	Printing Mistake
4	PHY	20	2	4	Calculation Error
5	PHY	23	4	3.93 to 4.00	g value should be mentioned
6	PHY	25	8	10.67	Calculation Error
7	CHE	37	3	Delete	No option is correct
8	CHE	54	6	4	Getting only 4 possible options
9	CHE	58	268	2.68 to 2.69	Range must be given
10	MAT	61	2	3	From the given data $B = -A$

Question Paper Setter is **Total responsible** for the Key finalization:

Question Paper Setters & finalized		
MATHS	PHYSICS	CHEMISTRY
HYD_NKT IPL Mr.G MALLIKARJUN-9440223222	HYD_NKT IPL Mr.SYED JOHN-9849298991	HYD_NKT IPL Mr.G KONDAL REDDY-7013315798

NATIONAL IIT CO-ORDINATOR
MR. UMA SANKAR



Sri Chaitanya IIT Academy., India.

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A right Choice for the Real Aspirant

ICON Central Office – Madhapur – Hyderabad

Sec: **Sr.Super-60& All**

Jee-Main

Date: 27-05-2021

Time: 09.00Am to 12.00

PTM-05

Max.Marks:300

IMPORTANT INSTRUCTIONS:

1. Immediately fill in the Admission number on this page of the Test Booklet with Blue/Black Ball Point Pen only.
2. The candidates should not write their Admission Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1, 2, 3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **two sections**.

(I) Section-I contains 20 **multiple choice** questions with **only one correct** option.

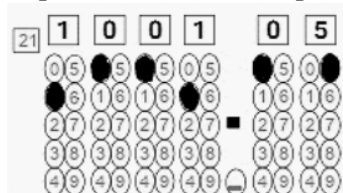
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

(II) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only.

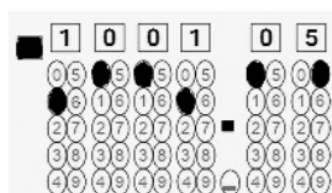
First 5 attempted questions will be considered if more than 5 questions attempted.

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted question 21, bubble on 21 as shown below.



Question Answered for marking



Question Cancelled for marking

Marking scheme : +4 for correct answer and 0 in all other cases.

6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capitals): _____

Admission Number:

--	--	--	--	--	--	--	--	--	--

Candidate's Signature: _____

Invigilator's Signature: _____

27-05-2021_Sr.Super60 & All_Jee-Main_PTM-05_Test Syllabus

PHYSICS : Heat and thermodynamics, Units, dimensions, Error analysis, Vernier calipers screw gauge, Waves and sound

CHEMISTRY : Liquid solutions Solid state, Benzene Carboxylic Acid & Derivatives, Environmental chemistry, Periodic properties

MATHEMATICS : MOD, AOD & LCD

PHYSICS**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

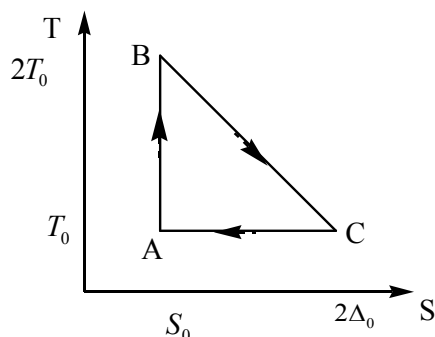
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- 2 kg of ice at -20°C is mixed with 5kg of water at 20°C in an insulating vessel having a negligible heat capacity. Calculate the final mass of water remaining in the container. It is given that the specific heats of water and ice are $1\text{Kcal/kg}^{\circ}\text{C}$ and $0.5\text{Kcal/kg}^{\circ}\text{C}$. While the latent heat of fusion of ice is 80Kcal/kg
 1) 4kg 2) 6kg 3) 5kg 4) 10kg
- Statement A: The temperature of a body be negative on the Kelvin scale
 Statement B: The temperature coefficient always positive
 1) Both A & B are true 2) Both A & B are false
 3) A is true and B is false 4) A is false and B is true
- When water is heated from 0°C to 10°C , its volume
 1) decreases continuously
 2) increases continuously
 3) first decreases and then increases
 4) first increases and then decreases
- We plot a graph, having temperature in $^{\circ}\text{C}$ on x axis and in $^{\circ}\text{F}$ on y axis. If the graph is straight line, then it
 1) passes through origin
 2) intercepts the positive x axis
 3) intercepts the positive y axis
 4) intercepts the negative of both x and y axis
- On increasing the temperature of a substance gradually, its colour becomes
 1) red 2) green 3) white 4) yellow
- In a given process of an ideal gas, $dw=0$ and $dQ<0$. Then for the gas
 1) the temperature will decrease
 2) the volume will increase
 3) the pressure will remain constant
 4) the temperature will increase

7. A monoatomic ideal gas, initially at temperature T_1 is enclosed in a cylinder fitted with a frictionless piston the gas is allowed to expand adiabatically to a temperature T_2 by releasing the piston suddenly. L_1 and L_2 are the lengths of the gas column before and after expansion respectively, then $\frac{T_1}{T_2}$ is

1) $\left(\frac{L_1}{L_2}\right)^{2/3}$ 2) $\frac{L_1}{L_2}$ 3) $\frac{L_2}{L_1}$ 4) $\left(\frac{L_2}{L_1}\right)^{2/3}$

8. The temperature entropy diagram of a reversible engine cycle is given in the figure. Its efficiency is



- 1) $\frac{1}{3}$ 2) $\frac{1}{2}$ 3) $\frac{2}{3}$ 4) $\frac{1}{4}$
9. The temperature of sink of Carnot engine is 27°C efficiency of engine is 25%. Then temperature of source is
- 1) 127°C 2) 227°C 3) 327°C 4) 27°C
10. A heat engine is 20% efficient. If the engine does 500J of work every second, how much heat does the engine exhaust every second?
- 1) 2500J 2) 2000J 3) 400J 4) 500J
11. In an adiabatic system which is true?
- 1) $p^\gamma T^{\gamma-1} = \text{constant}$ 2) $p^\gamma T^{1-\gamma} = \text{constant}$
 3) $p^\gamma T^\gamma = \text{constant}$ 4) $p^{1-\gamma} T^\gamma = \text{constant}$
12. How far does the sound travel in air when a tuning fork of frequency 256Hz makes 64 vibrations? (Velocity of sound in air = 320m/s)
- 1) 20m 2) 40m 3) 80m 4) 100m

13. Find the ratio of velocity of sound in hydrogen gas ($\gamma = 7/5$) to that in helium gas ($\gamma = 5/3$) at the same temperature. Given that molecular weight of hydrogen and helium are 2 and 4 respectively.
 1) 2.68 2) 10.68 3) 1.68 4) 4.68
14. The phase difference between the vibrations of two medium particles due to the transmission of a wave is $\frac{2\pi}{3}$. The distance between the particles is 15cm. Determine the wavelength of the wave.
 1) 15cm 2) 45cm 3) 60cm 4) 90cm
15. A tuning fork of frequency 300 Hz resonates with an air column closed at one end at 27°C . How many beats will be heard in the vibrations of the fork and the air column at 0°C ? (end correction is negligible)
 1) 14 2) 20 3) 28 4) 18
16. A bat is flying about in a cave, navigating via ultrasonic beeps. Assume that the sound emission frequency of the bat is 40 kHz. During one fast swoop directly towards a flat wall surface, the bat is moving at 0.03 times the speed of sound in air what frequency does the bat hear reflected off the wall?
 1) 32.47 kHz 2) 52.57 kHz 3) 12.47 kHz 4) 42.47 kHz
17. A quantity X is given by $\epsilon_0 L \frac{\Delta V}{\Delta t}$, where ϵ_0 is the permittivity of free space, L is a length, ΔV is a potential difference and Δt is a time interval. The dimensional formula for X is same as that of
 1) resistance 2) charge 3) voltage 4) current
18. Number of significant figures can reduce in
 1) addition 2) subtraction 3) multiplication 4) division
19. The dimensional formula for entropy is
 1) $[MLT^{-2}K^1]$ 2) $[ML^0T^{-2}]$
 3) $[ML^2T^{-2}K^{-1}]$ 4) $[ML^2T^{-2}K]$

20. A screw gauge having too equal divisions and a pitch of length 1mm is used to measure the diameter of wire of length 5.6cm. The main scale reading is 1mm and 47th circular division coincides with the main scale. Find the curved surface area of the wire in cm^2 to appropriate significant figures (use $\pi = 22/7$)
- 1) 2.6 2) 2.7 3) 2.8 4) 3.6

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

21. The heat generated in a circuit is given by $H = i^2 R t$ joule where I is current, R is resistance and t is time. If the percentage error in measuring I, R and t are 2%, 1% and 1% respectively. The maximum error in measuring heat will be $x\%$. Find the value of x
22. If x and a stand for distance, and the given equation $\int \frac{dx}{\sqrt{a^2 - x^n}} = \sin^{-1} \frac{x}{a}$ is dimensionally correct the value of n is
23. Two liquids A and B are at $32^\circ C$ and $24^\circ C$. When mixed in equal masses the temperature of the mixture is found to be $28^\circ C$. Their specific heats are in the ratio of $x : y$. Then find the value of $x + y$
24. A gas mixture consists of 2 moles of oxygen and 4 moles of Ar at temperature T. Neglecting all vibrational moles, the total internal energy of the system is PRT. What is the value of P?
25. Heat $Q = \frac{3}{2} RT$ is supplied to 4 moles of an ideal diatomic at temperature T, which remain constant the no of the gas are dissolved in to atom is ____
26. The value of $\gamma = \frac{C_p}{C_v}$ is $4/3$ for an adiabatic process of an ideal gas for which internal energy $U = k + n p v$. Find the value of n (k is a constant)

27. If 1 mole of a monoatomic ($\gamma = 5/3$) is mixed with 1 mole of a diatomic gas ($\gamma = 7/5$).
The values of r for the mixture is $\frac{x}{y}$. The value of $x + y$ is _____
28. When an ideal diatomic gas is heated at constant pressure, the fraction of heat energy supplied which increase the internal energy of the gas is $\frac{p}{q}$. The value of $p + q$ is _____
29. The frequency of a man's voice is 300Hz and its wavelength is 1 meter. If the wavelength of a child's voice is 1.5m, then the frequency of the child's voice is x Hz. The value of x is _____
30. Two identical wires are stretched by the same tension of 101N and each emits a note of frequency 202Hz. If the tension in one wire is increasing by 1N, then the beat frequency is y . The value of y is _____

CHEMISTRY**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

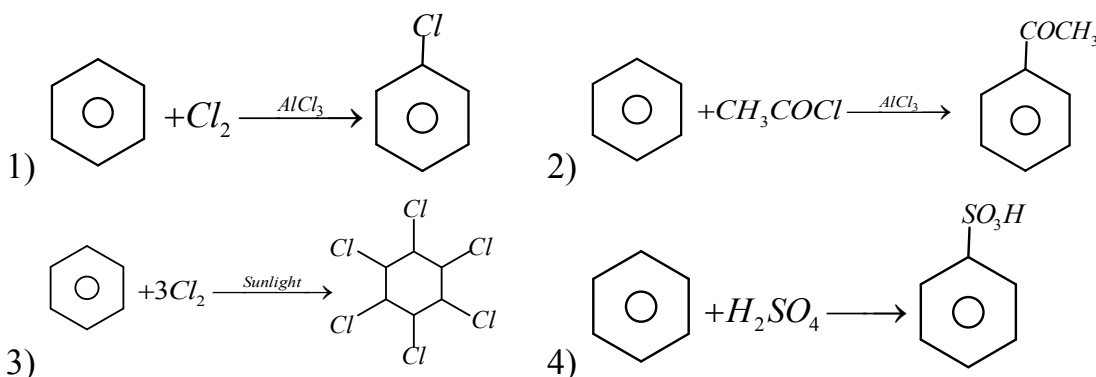
31. The formation of the oxide ion, $O_{(g)}^{-2}$ from oxygen atom requires first an exothermic and then an endothermic step as shown below:



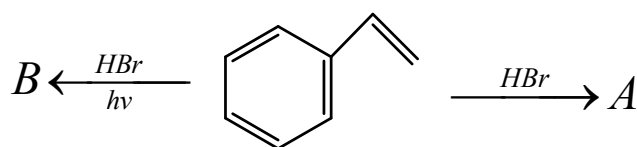
Thus, process of formation of O^{-2} in gas phase is unfavorable even though O^{-2} is iso electronic with neon. It is due to fact that

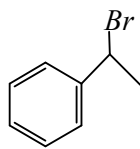
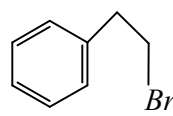
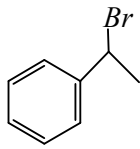
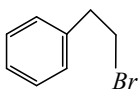
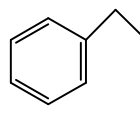
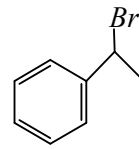
- 1) Oxygen is more electronegative
 - 2) Addition of electron in oxygen results in larger size of the ion
 - 3) electron repulsion outweighs the stability gained by achieving noble gas configuration
 - 4) O^{-} ion has comparatively smaller size than oxygen atom
32. IN which of the following arrangements, the order is not according to the property indicated against it?
- 1) $Al^{+3} < Mg^{+2} < Na^{+} < F^{-}$:Ionic size
 - 2) $B < C < N < O$:First ionization enthalpy
 - 3) $I < Br < F < Cl$:Electron gain enthalpy with Negative sign
 - 4) $Li < Na < K < Rb$: Metallic radius
33. First ionisation Enthalpies of group 2 elements are greater than that for group 1 and group 13 elements due to
- 1) Stable electronic configuration of group 2 elements
 - 2) Bigger atomic size of group 2 elements
 - 3) smaller effective nuclear charge of group 2 elements
 - 4) All of these

34. Assertion: Helium has the highest value of ionisation energy among all the elements known.
Reason: Helium has the highest value of electron affinity among all the elements known.
- 1) If both assertion and reason are true and reason is the correct explanation of assertion
 - 2) If both assertion and reason are true but reason is not the correct explanation of assertion
 - 3) If assertion is true but reason is false
 - 4) If both assertion and reason are false
35. Which of the following reactions is not an example of electrophilic substitution in benzene ring?



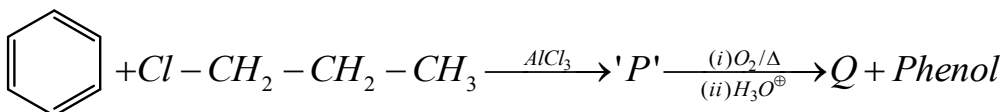
36. Observe the following reactions and predict the products A and B.



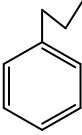
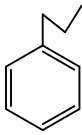
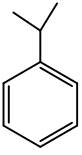
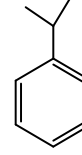
- 1) A and B both are  2) A and B both are 
- 3) A is  and B is  4) 'A' is  and 'B' is 

37. Which of the following steps is not correct in the mechanism of electrophilic substitution of benzene?
- 1) Generation of electrophile like X^+ , R^+ , NO_2^+ ,
 - 2) Attack of electrophile resulting in the formation of arenium ion in which one of the carbon is sp^3
 - 3) Addition of proton on between ring to give carbocation
 - 4) Removal of proton from sp^3 carbon atom to restore aromatic character

38.



The major products P and Q are

- | | |
|---|--|
| 1)  and $\text{CH}_3\text{CH}_2\text{CHO}$ | 2)  and CH_3COCH_3 |
| 3)  and CH_3COCH_3 | 4)  and $\text{CH}_3\text{CH}_2\text{CHO}$ |

39. Which of the following statements is not correct?
- 1) CO is the main air pollutant
 - 2) All pollutants are not wastes
 - 3) Lichens are pollution indicators
 - 4) water is polluted if B.O.D is less
40. Which of the following conditions in drinking water causes methemoglobinemia?
- 1) $> 50 \text{ ppm}$ of chloride
 - 2) $> 50 \text{ ppm}$ of nitrate
 - 3) $> 50 \text{ ppm}$ of lead
 - 4) $> 100 \text{ ppm}$ of sulphate
41. Which of the following conditions show the polluted environment?
- I) P^H of rain water is 5.6
 - II) Amount of carbon dioxide in the atmosphere is 0.03%
 - III) Bio-chemical oxygen demand is 10ppm
 - IV) Eutrophication
- 1) I and II only
 - 2) II and III only
 - 3) III and IV only
 - 4) I,II,III and IV

42. The maximum prescribed concentration of copper in drinking water is
 1) 0.05 ppm 2) 3 ppm 3) 5 ppm 4) 0.5 ppm
43. The edge lengths of the unit cells in terms of the radius of spheres constituting fcc, bcc and simple cubic unit cell are respectively.
 1) $2\sqrt{2}, \frac{4r}{\sqrt{3}}, 2r$ 2) $\frac{4r}{\sqrt{3}}, 2\sqrt{2}r, 2r$
 3) $2r, 2\sqrt{2}r, \frac{4r}{\sqrt{3}}$ 4) $2r, \frac{4r}{\sqrt{3}}, 2\sqrt{2}r$
44. The edge length of fcc unit cell is 508pm, if radius of cation is 110pm, the radius of anion is
 1) 110pm 2) 220pm 3) 285 pm 4) 144 pm
45. How many lithium atoms are present in a unit cell with edge with $3.5\text{\AA}$ and density 0.5gm cm^{-3} ?
 (Atomic mass of $Li = 6.94$)
 1) 2 2) 1 3) 4 4) 6
46. Assertion: Packing efficiency of body centred cubic structure is 68%
 Reason: 68% is the maximum packing efficiency any crystal can have
 1) If both assertion and reason are true and reason is the correct explanation of assertion.
 2) If both assertion and reason are true but reason is not the correct explanation of assertion
 3) If assertion is true but reason is false
 4) If both assertion and reason are false
47. According to Henry's law "The partial pressure of the Gas in vapour phase (P) is proportional to the mole fraction of the gas (χ) in the solution". For different Gases the correct statement about Henry's constant is
 1) Higher the value of k_H at a given pressure, higher is the solubility of the gas.
 2) Higher the value of k_H at a given pressure, lower is the solubility of the gas.
 3) k_H is not function of nature of gas
 4) k_H value for all gases is same at a given pressure.

48. Of the following 0.10m aqueous solutions, which one will exhibit the largest freezing point depression?
 1) KCl 2) $C_6H_{12}O_6$ 3) $Al_2(SO_4)_3$ 4) K_2SO_4
49. A solution containing 62gm ethylene glycol in 250gm water is cooled to $-10^\circ C$. If k_f for water is $1.86 k\ kg\ mol^{-1}$, the amount of water (in gm) separated as ice is
 1) 64 2) 32 3) 16 4) 48
50. Assertion: A solution of phenol and Aniline will show negative deviation from Raoult's law
 Reason: In case of negative deviation from Raoult's law, A-B forces are stronger than A-A and B-B forces.
 1) If both Assertion and reason are true and reason is correct explanation of assertion
 2) If both assertion and reason are true but reason is not correct explanation of assertion
 3) If assertion is true, but reason is false
 4) If both assertion and reason are false

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

51. If formic acid, acetic acid, propanoic acid and Benzoic acid are mixed with phosphorous and Bromine, then the number of products formed is _____
52. Among the following reagents how many can be used to convert carboxylic acid to the corresponding acid chloride?
 i) $NaCl$ ii) Cl_2
 iii) PCl_3 iv) PCl_5 v) $SOCl_2$
53. How many of the following solutions will show negative deviation from Raoult's law?
 i) Benzene + acetone ii) Toluene + CCl_4
 iii) Benzene + chloroform iv) Acetone + chloroform
 v) Acetone + ethanol vi) water + ethanol
 vii) water + HNO_3 viii) Methanol + CH_3COOH

54. Among the given compounds the number of compounds responsible for photo chemical smog is _____
 CF_2Cl_2 , ozone, sulphuric acid, Nitric oxide, Nitric acid, Hydro carbons, Formaldehyde, Acrolein, PAN, Carbonic acid
55. The number of primary pollutants present in the given list is _____
 $SO_2, NH_3, SO_3, O_3, HNO_3, NO, CO$
56. The α -form of a metal fcc structure with edge length of 450pm and β -form has bcc structure with edge length of 225pm. The ratio of density in α -form to β -form is 1 : x . What is the value of x ?
57. Among following how many are covalent solids CaO, LiF, CO_2 , graphite, diamond, Na, Brass, SiC, SiO_2, AlN, S_8
58. How many of the following compounds reacts slower than Benzene for electrophilic substitution reactions?
 i) Aniline ii) Toulene iii) Phenol
 iv) Nitro Benzene v) trichloro toluene
 vi) chloro Benzene vii) Methoxy Benzene
59. In how many below mentioned reactions, Benzene would be a major product?
 i) $C_6H_5OH \xrightarrow[\Delta]{Zn} ?$ ii) $C_6H_5Cl \xrightarrow{NaBH_4} ?$
 iii) $C_6H_5CHO \xrightarrow{LiAlH_4} ?$ iv) $C_6H_5SO_3H \xrightarrow[\text{pressure}]{\Delta, H_2O, H^+} ?$
 v) $C_6H_5N_2Cl \xrightarrow[SnCl_2 / NaOH]{H_2} ?$ vi) $C_2H_2 \xrightarrow[2000^\circ C]{Fe} ?$
60. What will be the minimum mass (in gm) of ethylene glycol that should be added to 1kg of water to prevent it from freezing at $-10^\circ C$?
 (molar mass of ethylene glycol = 62 gm/mol and k_f for water = $1.86 k \text{ kg mol}^{-1}$)

MATHS**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. For each $x \in R$, let $[x]$ be the greatest integer less than or equal to x . Then

$$\lim_{x \rightarrow 0^-} \frac{x([x] + |x|) \sin [x]}{|x|} \text{ is equal to}$$

- 1) 0 2) 1 3) $\sin 1$ 4) $-\sin 1$

62. If $\lim_{x \rightarrow \infty} \left(\sqrt{x^4 + ax^3 + 3x^2 + bx + 2} - \sqrt{x^4 + 2x^3 - cx^2 + 3x - d} \right) = 4$, then the value of a, b, c and d is

- 1) $a = 1, b = 2, c = 3$ and $d = 4$ 2) $a = 2, b = 3, c = 5$ and $d = 0$
 3) $a = 2, b = 4, c = 5$ and $d \in R$ 4) $a = 2, c = 5$ and $b, d \in R$

63. Let $f(x) = \begin{cases} x^2 - 1, & 0 < x < 2 \\ 2x + 3, & 2 \leq x < 3 \end{cases}$, the quadratic equation whose roots are $\lim_{x \rightarrow 2^-} f(x)$ and

$$\lim_{x \rightarrow 2^+} f(x) \text{ is}$$

- 1) $x^2 - 6x + 9 = 0$ 2) $x^2 - 7x + 8 = 0$
 3) $x^2 - 14x + 49 = 0$ 4) $x^2 - 10x + 21 = 0$

64. If $y = \log_a x + \log_x a + \log_x x + \log_a a$, then $\frac{dy}{dx}$ is equal to

- 1) $\frac{1}{x} + x \log a$ 2) $\frac{\log a}{x} + \frac{x}{\log a}$
 3) $\frac{1}{x \log a} + x \log a$ 4) $\frac{1}{x \log a} - \frac{\log a}{x(\log x)^2}$

65. If $y = \frac{\ln x}{x}$, then the value of $y''(e)$ is

- 1) 1 2) $-\frac{1}{e}$ 3) $-\frac{1}{e^2}$ 4) $-\frac{1}{e^3}$

66. If $f(x) = 2x$ and $g(x) = \frac{x^2}{2} + 1$, then which of the following can be a discontinuous function?

- 1) $f(x) + g(x)$ 2) $f(x) - g(x)$
 3) $f(x) \cdot g(x)$ 4) $\frac{g(x)}{f(x)}$

67. Let $a, b \in \mathbb{R}, (a \neq 0)$. If the function f defined as

$$f(x) = \begin{cases} \frac{2x^2}{a}, & 0 \leq x < 1 \\ a, & 1 \leq x < \sqrt{2} \\ \frac{2b^2 - 4b}{x^3}, & \sqrt{2} \leq x < \infty \end{cases}$$

is continuous in the interval $[0, \infty)$, then an ordered pair (a, b) is

- 1) $(-\sqrt{2}, 1 - \sqrt{3})$ 2) $(\sqrt{2}, -1 + \sqrt{3})$
 3) $(\sqrt{2}, 1 - \sqrt{3})$ 4) $(-\sqrt{2}, 1 + \sqrt{3})$

68. If $f(x)$ is a differentiable function in the interval $(0, \infty)$ such that $f(1) = 1$ and

$$\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1, \text{ for each } x > 0, \text{ then } f\left(\frac{3}{2}\right) \text{ is equal to}$$

- 1) $\frac{23}{18}$ 2) $\frac{13}{6}$ 3) $\frac{25}{9}$ 4) $\frac{31}{18}$

69. The set of points where the function f given by $f(x) = |2x - 1| \sin x$ is differentiable is

- 1) $\mathbb{R}$ 2) $\mathbb{R} - \left\{\frac{1}{2}\right\}$ 3) $(0, \infty)$ 4) None of these

70. The function $f(x) = e^{|x|}$ is

- 1) continuous everywhere but not differentiable at $x = 0$
 2) continuous and differentiable everywhere
 3) not continuous at $x = 0$
 4) None of these

71. A spherical iron ball 10cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of $50 \text{ cm}^3 / \text{min}$. When the thickness of ice is 15cm, then the rate at which the thickness of the ice decreases, is
- 1) $\frac{5}{6\pi} \text{ cm} / \text{min}$ 2) $\frac{1}{54\pi} \text{ cm} / \text{min}$
 3) $\frac{1}{18\pi} \text{ cm} / \text{min}$ 4) $\frac{1}{36\pi} \text{ cm} / \text{min}$
72. A water tank has the shape of an inverted right circular cone, whose semi-vertical angle is $\tan^{-1}\left(\frac{1}{2}\right)$. Water is poured into it at a constant rate of 5 cubic meter per minute. Then the rate (in m/min), at which the level of water is rising at the instant when the depth of water in the tank is 10 m, is
- 1) $1/5\pi$ 2) $1/15\pi$ 3) $2/\pi$ 4) $1/10\pi$
73. The tangent at the point $(2, -2)$ to the curve $x^2 y^2 - 2x = 4(1 - y)$ does not pass through the point
- 1) $(-2, -7)$ 2) $(-4, -9)$ 3) $\left(4, \frac{1}{3}\right)$ 4) $(8, 5)$
74. If the curves $y^2 = 6x, 9x^2 + by^2 = 16$ intersect each other at right angles, then the value of b is
- 1) $\frac{9}{2}$ 2) 6 3) $\frac{7}{2}$ 4) 4
75. The function $f(x) = x^3 + 6x^2 + (9 + 2k)x + 1$ is strictly increasing for all x , if
- 1) $k > \frac{3}{2}$ 2) $k \geq \frac{3}{2}$ 3) $k < \frac{3}{2}$ 4) $k \leq \frac{3}{2}$
76. The coordinates of the point on the parabola $y^2 = 8x$ which is at minimum distance from the circle $x^2 + (y + 6)^2 = 1$ are
- 1) $(2, -4)$ 2) $(18, -12)$ 3) $(2, 4)$ 4) none of these

77. If $f(x)$ is a non-zero polynomial of degree four, having local extreme points at $x = -1, 0, 1$, then the set $S = \{x \in R : f(x) = f(0)\}$ contains exactly
- 1) four rational numbers.
 - 2) four irrational numbers
 - 3) two irrational and two rational numbers
 - 4) two irrational and one rational number
78. Let $f(x) = e^x - x$ and $g(x) = x^2 - x, \forall x \in R$. Then the set of all $x \in R$, where the function $h(x) = (f \circ g)(x)$ is increasing, is
- 1) $\left[\frac{-1}{2}, 0\right] \cup [1, \infty)$
 - 2) $[0, \infty)$
 - 3) $\left[0, \frac{1}{2}\right] \cup [1, \infty)$
 - 4) $\left[-1, \frac{-1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$
79. Let $f(x) = \frac{x}{\sqrt{a^2 + x^2}} - \frac{d-x}{\sqrt{b^2 + (d-x)^2}}, x \in R$ where a, b and d are non-zero real constants. Then
- 1) f' is not a continuous function of x
 - 2) f is neither increasing nor decreasing function of x
 - 3) f is an increasing function of x
 - 4) f is a decreasing function of x
80. Let $f(x) = 5 - |x - 2|$ and $g(x) = |x + 1|, x \in R$. If $f(x)$ attains maximum value at α and $g(x)$ attains minimum value at β , then $\lim_{x \rightarrow -\alpha\beta} \frac{(x-1)(x^2 - 5x + 6)}{x^2 - 6x + 8}$ is equal to
- 1) $\frac{-1}{2}$
 - 2) $\frac{1}{2}$
 - 3) $\frac{-3}{2}$
 - 4) $\frac{3}{2}$

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

81. $\lim_{x \rightarrow 1} \frac{(\sqrt{x}-1)(2x-3)}{2x^2+x-3}$ is

82. If $f(x) = 1 - x + x^2 - x^3 \dots - x^{99} + x^{100}$, then $f'(1)$ is equal to

83. If $x^2 + 2x + 2y^2 = 1$, then $\frac{dy}{dx}$ at the point where $y = 1$ is equal to

84. If $f(x) = \begin{cases} \frac{8^x - 4^x - 2^x + 1^x}{x^2}, & x > 0 \\ e^x \sin x + \pi x + \lambda \ln 4, & x \leq 0 \end{cases}$ is continuous at $x = 0$, then the value of

$1000e^\lambda$ must be _____

85. Let $h(x)$ be differentiable for all x and let $f(x) = (kx + e^x)h(x)$, where k is some constant. If $h(0) = 5$, $h'(0) = -2$ and $f'(0) = 18$, then the value of k is equal to _____

86. If $u = x^2 + y^2$ and $x = s + 3t$, $y = 2s - t$, then $\frac{d^2u}{ds^2}$ is equal to

87. A ladder, 5 meter long, standing on a horizontal floor, leans against a vertical wall. If the top of the ladder slides downwards at the rate of 10cm/sec, then the rate at which the angle between the floor and the ladder is decreasing when lower end of ladder is 2 metres from the wall is _____ radian/sec

88. The distance, from the origin, of the normal to the curve, $x = 2 \cos t + 2t \sin t$, $y = 2 \sin t - 2t \cos t$ at $t = \pi / 4$ is

89. The graph of the function $y = f(x)$ has a unique tangent at the point $(a, 0)$ through which the graph passes. Then $\lim_{x \rightarrow a} \frac{\log_e \{1 + 6f(x)\}}{3f(x)}$ is _____

90. Maximum slope of the curve $y = -x^3 + 3x^2 + 9x - 27$ is



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Key Sheet

PHYSICS

1	2	2	2	3	3	4	3	5	4
6	1	7	4	8	1	9	1	10	2
11	4	12	3	13	3	14	2	15	1
16	4	17	4	18	2	19	3	20	1
21	6	22	2	23	2	24	11	25	3
26	3	27	5	28	12	29	200	30	1

CHEMISTRY

31	3	32	2	33	1	34	3	35	3
36	3	37	3	38	3	39	4	40	2
41	3	42	2	43	1	44	4	45	1
46	3	47	2	48	3	49	1	50	1
51	3	52	3	53	4	54	6	55	4
56	4	57	5	58	3	59	4	60	333

MATHEMATICS

61	4	62	4	63	4	64	4	65	4
66	4	67	3	68	4	69	2	70	1
71	3	72	1	73	1	74	1	75	1
76	1	77	4	78	3	79	3	80	2
81	-0.1	82	50	83	0	84	2000	85	3
86	10	87	0.05	88	2	89	2	90	12

SOLUTIONS

PHYSICS

1. Suppose m kg of ice melts into water. As the net heat changes is zero, so

$$mL + m_i S_i (\Delta T)_{ice} = M_w S_w (\Delta T)_w$$

$$m80 + 2 \times 0.5 \times (20) = 5 \times 1 \times (20)$$

$$m = 1\text{kg}$$
 Final mass of water remaining in the beaker is 6kg
2. 0 K is the minimum possible Temp on Kelvin scale and Temp coefficient is negative for semiconductors and insulators.
3. As water heated, its volume first decreases from 0°C to 4°C and then increases from 4°C to 10°C .
4. $F = \frac{9}{5}C + 160$
5. By Wien's law $\lambda_m \propto \frac{1}{T}$
 On increasing the temp, wavelength decreases. As yellow colour has minimum wavelength, the substance gradually becomes yellow.
6. $dQ = du + dw$
7. FQ adiabatic process

$$\left[\gamma = \frac{5}{3} \right], T v^{\gamma-1} = \text{constant}$$

$$T_1 v_1^{\gamma-1} = T_2 v_2^{\gamma-1}$$

$$\frac{T_1}{T_2} = \left[\frac{V_2}{V_1} \right]^{\gamma-1} = \left[\frac{AL_2}{AL_1} \right]^{\frac{5}{3}-1} = \left[\frac{L_2}{L_1} \right]^{2/3}$$
8. $\eta = \frac{\Delta w}{Q_{BC}} = \frac{\text{Area of } ABC}{\text{Area under } BC}$

$$\frac{\frac{1}{2} S_0 T_0}{S_0 T_0 + \frac{1}{2} S_0 T} = \frac{\frac{1}{2} S_0 T_0}{\frac{3}{2} S_0 T_0} = \frac{1}{3}$$
9. $\eta = 1 - \frac{T_2}{T_1}$

$$\frac{25}{100} = 1 - \frac{300}{T_1}$$

$$T_1 = 400\text{K} = 127^\circ\text{C}$$
10. $\eta = \frac{w}{Q_1} \Rightarrow 0.2 = \frac{500}{Q_1}$

$$Q_1 = 2500\text{J}$$

$$R = Q_1 - w = 2500 - 500 = 2000\text{J}$$
11. Conceptual
12. $\lambda = \frac{v}{f} = \frac{320}{256} = 1.25\text{m}$
 Distance travelled by the wave in 64 vibrations $= 1.25 \times 64 = 80\text{m}$

$$13. \quad V = \sqrt{\frac{\gamma RT}{M}}$$

$$\frac{V_H}{V_{He}} = \sqrt{\frac{\gamma_H M_{He}}{\gamma_{He} M_H}} = 1.68$$

$$14. \quad \Delta Q = \frac{2\pi}{\pi} \Delta x$$

$$\frac{2\pi}{3} = \frac{2\pi}{\pi} (15)$$

$$\lambda = 45$$

15. Frequency at $27^\circ C$

$$F_{27} = \frac{V_{27}}{4L} = 300 \text{ (given)}$$

$$\frac{V_0}{V_{27}} = \sqrt{\frac{273}{300}} = 0.954$$

$$V_0 = 0.954 V_{27}$$

Frequency at $27^\circ C$

$$F_0 = \frac{V_0}{4L} = \frac{0.954 \times V_{27}}{4L} = 0.954 \times 300 = 286$$

Beats produced per second $300 - 286 = 14$

16. bat approaches the stationary flat wall surface, the apparent frequency is $f' = f \left(\frac{V}{V - V_S} \right)$

The stationary flat surface (source) reflects the sound of frequency f' to the bat (observer) moving towards the flat surface. So apparent frequency is

$$f'' = \left[\frac{V + V_0}{V} \right] f'$$

$$= \left[\frac{V + V_0}{V - V_s} \right] f = 42.47 \text{ kHz}$$

$$17. \quad C = \frac{\Delta q}{\Delta V}$$

$$\frac{\epsilon_0 A}{L} = \frac{\Delta q}{\Delta V}$$

$$\epsilon_0 = \frac{\Delta q}{\Delta V} \cdot \frac{L}{A}$$

$$X = \epsilon_0 L \epsilon_0 \frac{\Delta V}{\Delta t}$$

$$= \frac{\Delta q}{\Delta V} \cdot \frac{L}{A} \cdot L \frac{\Delta V}{\Delta t}$$

$$X = \frac{\Delta q}{\Delta t}$$

18. To find difference between 20.17 – 20.15 we get, 0.02 (one significant figure only), which is less than the significant figures in the number used

19. Conceptual

$$20. \quad L.C = 0.01 \text{ mm}$$

$$D = MSR + CS [L.C]$$

$$= 1\text{mm} + 47(0.01) = 1.47\text{mm}$$

$$\text{Surface area} = \pi DL = \frac{22}{7} \times 0.147 \times 5.6 = 2.5872 = 2.6\text{cm}^2$$

$$21. \quad \frac{\Delta H}{H} = \frac{2\Delta I}{I} + \frac{\Delta R}{R} + \frac{\Delta t}{t} = 2 \times 2\% + 1\% + 1\% = 6\%$$

$$22. \quad \text{Here } x^n \text{ has the same dimensions of } a^2 \text{ thus } n = 2$$

$$23. \quad ms_1(\Delta T) = ms_2\Delta T_2$$

$$ms_1(4) = ms_2(4)$$

$$\frac{s_1}{s_2} = 1:1$$

$$24. \quad U = n \frac{F}{2} RT = 2 \times \frac{5}{2} RT + 4 \times \frac{3}{2} RT = 11RT$$

$$25. \quad 2x \left(\frac{3RT}{2} \right) + (4-x) \left(\frac{5}{2} RT \right) - 4 \left(\frac{5}{2} RT \right) = \frac{3}{2} RT$$

$$x = 3 \text{ moles}$$

$$26. \quad pv^\gamma = \text{constant}$$

$$pv^{4/3} = \text{constant}$$

$$Lnv^4 + \ln p^3 = \text{constant}$$

$$4 \frac{dv}{v} = -3 \frac{dp}{p}$$

$$4p(dv) + 3vdp = 0$$

$$3(pdv + vdp) + pdv = 0$$

$$\text{Adiabatic process } dQ = 0$$

$$du + dw = 0$$

$$du + pdv = 0$$

$$du = 3(pdv + vdp)$$

$$U = k + 3pv$$

$$n = 3$$

$$27. \quad \gamma = \frac{n_1 C_{v_1} + n_2 C_{v_2}}{n_1 + n_2} = \frac{3}{2}$$

$$28. \quad f = \frac{\Delta U}{\Delta Q} = \frac{n C_v \Delta T}{n C_p \Delta T}$$

$$= \frac{C_v}{C_p} = \frac{1}{\gamma}$$

$$f = \frac{5}{7}$$

$$29. \quad f_1 \lambda_1 = f_2 \lambda_2$$

$$30. \quad f \propto \sqrt{T}$$

$$f = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

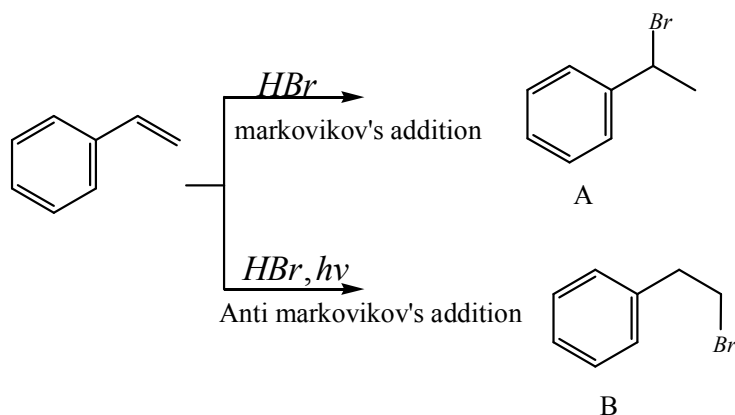
$$\frac{\Delta f}{f} = \frac{1}{2} \frac{\Delta T}{T}$$

$$\Delta f = \frac{200}{2} \times \frac{1}{101} = 1$$

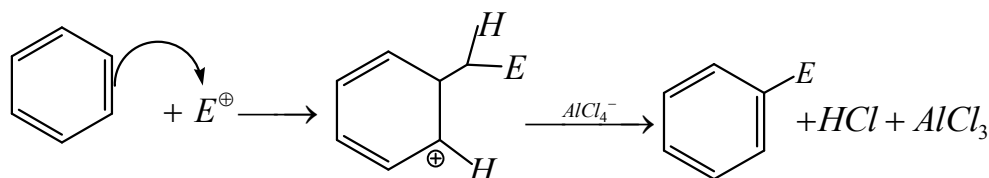
CHEMISTRY

31. Due to small size of 'O' atom, electron repulsion out weight the stability gained by achieving noble gas configuration.
32. 1) All of these are isoelectronic ions with decreasing nuclear charge and increasing ionic size
 $Al^{+3} (z=13) < Mg^{+2} (z=12) < Na^{+} (z=11) < F^{-} (z=9)$
 2) IE_1 of these elements are in the order: $B < C < O < N$
 IE_1 of 'N' ($1s^2 2s^2 2p^3$) is more than O ($1s^2 2s^2 2p^4$) due to extra stable half-filled 2p-subshell in the valence shell thus, the given order is not correct
 3) Electron gain enthalpies of these elements are
 $I(-295 kJ mol^{-1}) < Br(-325 kJ mol^{-1}) < F(-328 kJ mol^{-1}) < Cl(-349 kJ mol^{-1})$
 4) Metallic radius increases down the group in 1st group
33. Due to stable electronic configuration
34. Helium contains fully filled $1s^2$ orbital which has more penetrating effect and is very close to the nucleus and is very close to the nucleus and hence has higher value of ionisation energy.
35. Cl_2 is added to benzene ring in presence of sunlight to give benzene hexa chloride which is a free radical reaction.

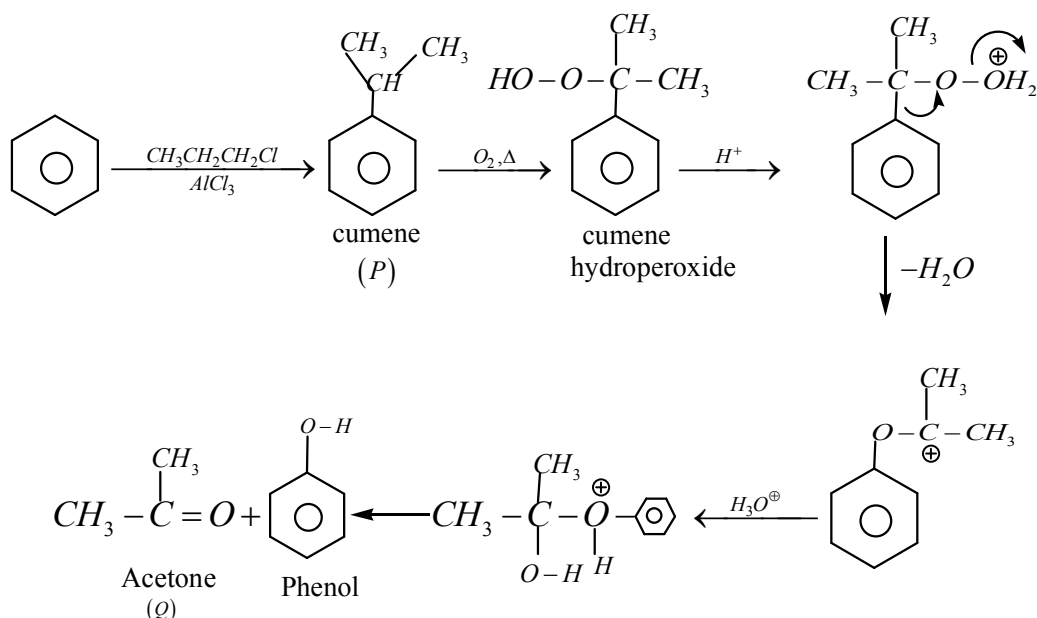
36.



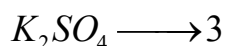
37. Arenium ion is formed as intermediate by the attack of electrophile not by addition of proton. Proton substituted by electrophile.



38. It is cumene hydroperoxide rearrangement reaction



39. Highly polluted water has BOD value of more than 17 ppm
40. > 50 ppm of nitrate concentration causes meth methemoglobinemia
41. BOD value more than 5ppm shows water is polluted the process in which nutrient enriched water bodies support a dense plant population, which kills aquatic life by depriving them of oxygen and results in subsequent loss of bio-diversity is known as eutrophication
42. Conceptual
43. Conceptual
44. For fcc $a = (r_c^+ + r_a^-)$
- $$508 = 2(110 + r_a^-)$$
- $$r_a^- = \frac{508}{2} - 110 = 144 \text{ pm}$$
45. $d = \frac{z \times m}{N_A \times a^3}$
- $$z = \frac{a^3 \times N_A \times d}{m} = \frac{(3.5 \times 10^{-8})^3 \times 0.53 \times 6.023 \times 10^{23}}{6.94} = 1.97 \approx 2$$
46. fcc has 74% packing efficiency
47. $P = k_H \cdot \chi$. Higher the value of k_H at a given pressure, lower is the solubility of gas in the liquid
48. $\Delta T_f = i \times k_f \times m$
- $\Delta T_f \propto i$ (Van't hoff factor)
- Salt i
- $\text{KCl} \longrightarrow 2$
- $\text{C}_6\text{H}_{12}\text{O}_6 \longrightarrow 1$
- $\text{Al}_2(\text{SO}_4)_3 \longrightarrow 5$



Hence, I is maximum i.e 5 for $Al_2(SO_4)_3$

$$49. \Delta T_f = k_f \times \frac{\omega_2}{m_2} \times \frac{1000}{\omega_1}$$

Where ω_1 = mass of solvent (H_2O)

ω_2 = mass of solute (ethylene glycol)

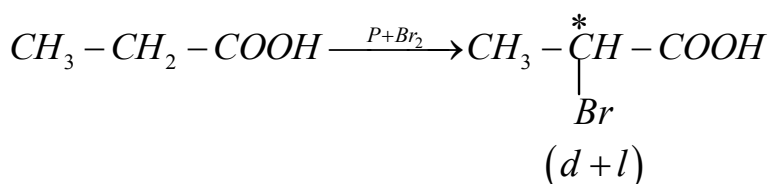
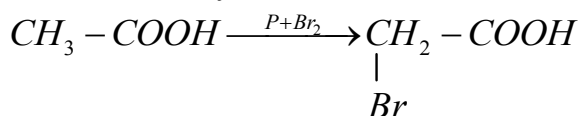
m_2 = molar mass of solute

$$\omega_1 = \frac{k_f \times \omega_2 \times 1000}{\Delta T_f \times m_2} = \frac{1.86 \times 62 \times 1000}{10 \times 62} = 186 \text{ gm}$$

$\therefore$ Amount of water separated as ice = $(250 - 186) = 64 \text{ gm}$

50. In negative deviation A-B forces are stronger than A-A and B-B forces and phenol-aniline will show negative deviation.

51. Formic acid and benzoic acid do not contain $\alpha - H$ atoms so they can not given Hell-volhard zelinsky reaction.



52. PCl_3, PCl_5 and $SOCl_2$ can be used for this conversion

53. (iii, iv, vii & viii)

54. Photo chemical smog contains neither smoke nor fog. It is a mixture of compounds like O_3, NO_2, PAN , Hydro carbons, aldehydes, ketones, acrolein, and CO

55. SO_2, NH_3, NO and CO are primary pollutants whereas SO_3, O_3 and HNO_3 are secondary pollutants

56. Z_{eff} for fcc = 4

Z_{eff} for bcc = 2

$$d = \frac{z \times M}{N_0 \cdot a^3}$$

$$\frac{d_\alpha}{d_\beta} = \frac{4}{(450)^3} \div \frac{2}{(225)^3}$$

$$= \left(\frac{225}{450} \right)^3 \times \frac{4}{2} = \frac{1}{4}$$

57. Graphite, diamond, SiC, SiO_2 , AlN

58. $-NO_2$, $-CCl_3$ and $-Cl$ are deactivating groups. Hence compounds mentioned in (iv), (v) and (vi) will react slower than Benzene
59. Reaction (i), (iv), (v) and (vi) will give Benzene as a major product
60. $\Delta T_f = k_f \times m$
 $10 = 1.86 \times \frac{\omega}{62 \times 1}$
 $\omega = 333 \text{ gm}$

MATHEMATICS

61. We have, $\lim_{x \rightarrow 0^-} \frac{x([x] + |x|) \sin[x]}{|x|}$
 $\lim_{x \rightarrow 0^-} \frac{x(-1-x) \sin(-1)}{-x} \left[\because \lim_{x \rightarrow 0^-} [x] = -1 \text{ and } \lim_{x \rightarrow 0^-} |x| = -x \right]$
 $= -\sin 1$
62. Given that,
 $4 = \lim_{x \rightarrow \infty} \left(\sqrt{x^4 + ax^3 + 3x^2 + bx + 2} - \sqrt{x^4 + 2x^3 - cx^2 + 3x - d} \right)$
 $= \lim_{x \rightarrow \infty} \frac{(a-2)x^3 + (3+c)x^2 + (b-3)x + 2 + d}{\sqrt{x^4 + ax^3 + 3x^2 + bx + 2} + \sqrt{x^4 + 2x^3 - cx^2 + 3x - d}}$
 Since, the limits is finite, the degree of the numerator must be at the most 2 $\Rightarrow a-2=0$, i.e., $a=2$
 Hence,
 $4 = \lim_{x \rightarrow \infty} \frac{(3+c) + \frac{b-3}{x} + \frac{2+d}{x^2}}{\sqrt{1 + \frac{a}{x} + \frac{3}{x^2} + \frac{b}{x^3} + \frac{2}{x^4}} + \sqrt{1 + \frac{2}{x} - \frac{c}{x^2} + \frac{3}{x^3} - \frac{d}{x^4}}}$
 $\Rightarrow 4 = \frac{3+c}{2} \Rightarrow c=5$
 Hence, $a=2, c=5$ and b, d are any real numbers.
63. $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (x^2 - 1) = 4 - 1 = 3$
 $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} 2x + 3 = 4 + 3 = 7$
 $\therefore$ Required quadratic equation is
 $x^2 - (3+7)x + (3 \times 7) = 0$
 $\Rightarrow x^2 - 10x + 21 = 0$
64. $\because y = \log_a x + \frac{\log a}{\log x} + 1 + 1 \quad \{\because \log_x x = 1\}$
 $\Rightarrow \frac{dy}{dx} = \frac{1}{x \log a} - \frac{\log a}{x(\log x)^2}$

65. Given, $y = \frac{\ln x}{x}$

Differentiating w.r.t. x , we get

$$y' = \frac{(1 - \ln x)}{x^2} \Rightarrow y'' = \frac{x^2 \left(-\frac{1}{x} \right) - (1 - \ln x) 2x}{x^4}$$

$$\therefore y''(e) = \frac{-e - 0}{e^4} = -\frac{1}{e^3}$$

66. $f(x) = 2x$ and $g(x) = \frac{x^2}{2} + 1$ both are continuous function.

So the function $\frac{g(x)}{f(x)}$ can be discontinuous.

If, $f(x) = 0 \Rightarrow 2x = 0 \Rightarrow x = 0$

Hence, at $x = 0$, $\frac{g(x)}{f(x)}$ is discontinuous.

67. Since $f(x)$ is continuous at $x = 1$

$$\therefore \frac{2}{a} = a \Rightarrow a = \pm 2$$

Also $f(x)$ is continuous at $x = \sqrt{2}$

$$\therefore a = \frac{2b^2 - 4b}{2\sqrt{2}}$$

When $a = \sqrt{2}$, we get $2 = b^2 - 2b \Rightarrow b^2 - 2b - 2 = 0$

$$\Rightarrow b = \frac{2 \pm \sqrt{4 + 4 \cdot 2}}{2} = 1 \pm \sqrt{3} \quad \therefore (a, b) \equiv (\sqrt{2}, 1 \pm \sqrt{3})$$

When $a = -\sqrt{2}$, we get

$$-2 = b^2 - 2b \Rightarrow b^2 - 2b + 2 = 0$$

$$\Rightarrow b = \frac{2 \pm \sqrt{4 - 8}}{2} = 1 \pm i \text{ (Neglected)}$$

68. Let $L = \lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$

Applying L' Hospital rule, we get

$$L = \lim_{t \rightarrow x} \frac{2t f(x) - x^2 f'(t)}{1} = 1$$

$$\text{or } 2x f(x) - x^2 f'(x) = 1 \Rightarrow f'(x) - \frac{2}{x} f(x) = \frac{-1}{x^2}$$

$$I.F. = e^{\int \frac{-2}{x} dx} = e^{-2 \log x} = \frac{1}{x^2}$$

Solution is $\left[f(x) \right] \frac{1}{x^2} = \int \frac{1}{x^2} \left(-\frac{1}{x^2} \right) dx \Rightarrow \frac{f(x)}{x^2} = \frac{1}{3x^3} + C$

We have, $f(1) = 1$

$$\Rightarrow 1 = \frac{1}{3} + C \Rightarrow C = \frac{2}{3} \quad \therefore C = \frac{2}{3} \quad \therefore f(x) = \frac{2}{3}x^2 + \frac{1}{3x}$$

$$\therefore f\left(\frac{3}{2}\right) = \frac{2}{3} \times \left(\frac{3}{2}\right)^2 + \frac{1}{3} \times \frac{2}{3} = \frac{31}{18}$$

69. $f(x) = |2x - 1| \sin x$

$$f(x) = \begin{cases} -(2x-1)\sin x, & x < 1/2 \\ (2x-1)\sin x, & x \geq 1/2 \end{cases}$$

$$Lf'\left(\frac{1}{2}\right) = \lim_{h \rightarrow 0^-} \frac{f(1/2+h) - f(1/2)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{-[2(1/2+h)-1]\sin(1/2+h) - 0}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{-2h \sin(1/2+h)}{h} = -2 \sin(1/2)$$

$$Rf'(1/2) = \lim_{h \rightarrow 0^+} \frac{f(1/2+h) - f(1/2)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{[2(1/2+h)-1]\sin(1/2+h) - 0}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{2h \sin(1/2+h)}{h} = 2 \sin(1/2)$$

$$Lf'(1/2) \neq Rf'(1/2)$$

So, $f(x)$ is not differentiable at $x = 1/2$

Hence, $f(x)$ is differentiable $\forall x \in \mathbb{R} - \left\{ \frac{1}{2} \right\}$

70. $f(x) = e^{|x|}; f(x) = \begin{cases} e^{-x}, & x < 0 \\ e^x, & x \geq 0 \end{cases}$

$\therefore e^x$ is continuous for all $x > 0$ and e^{-x} is continuous for all $x < 0$.

Now,

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} e^{-x} = e^0 = 1$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} e^x = e^0 = 1$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0} f(x) = f(0)$$

Hence, $f(x)$ is continuous everywhere.

$$Lf'(0) = \lim_{h \rightarrow 0^-} \frac{f(0-h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{e^{-h} - 1}{h} = \lim_{h \rightarrow 0^-} \frac{\left(1 - h + \frac{h^2}{2!} - \frac{h^3}{3!} + \dots\right) - 1}{h}$$

$$= \lim_{h \rightarrow 0^-} \left(-1 + \frac{h}{2!} - \frac{h^2}{3!} + \dots\right) = -1$$

$$Rf'(0) = \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{e^h - 1}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{\left(1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \dots\right) - 1}{h} = \lim_{h \rightarrow 0^+} \left(1 + \frac{h}{2!} + \frac{h^2}{3!} + \dots\right) = 1$$

$$L(f'(0)) \neq R(f'(0))$$

So $f(x)$ is not differentiable at $x = 0$.

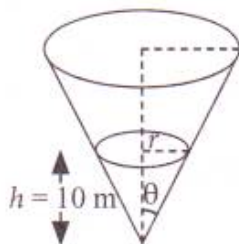
71. Given that, $\frac{dV}{dt} = 50 \text{ cm}^3 / \text{min}$

$$\Rightarrow \frac{d}{dt} \left(\frac{4}{3} \pi r^3 \right) = 50$$

$$\Rightarrow 3r^2 \frac{dr}{dt} = \frac{150}{4\pi} \Rightarrow \frac{dr}{dt} = \frac{50}{4\pi r^2}$$

$$\Rightarrow \left(\frac{dr}{dt} \right)_{r=15} = \frac{50}{4\pi \times 225} = \frac{1}{18\pi} \text{ cm} / \text{min}$$

72. Let r be the radius of the cone when level of water is 10m.



Given, $\frac{dV}{dt} = 5 \text{ m}^3 / \text{min}$ and $\tan^{-1} \frac{1}{2} = \theta$

$$\Rightarrow \tan \theta = \frac{1}{2} = \frac{r}{h} \therefore r = \frac{h}{2}$$

Now, $V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi \frac{h^3}{4}$

$$\therefore \frac{dV}{dt} = \frac{\pi}{12} \cdot 3h^2 \left(\frac{dh}{dt} \right)$$

$$\Rightarrow 5 = \frac{\pi}{4}(100) \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{1}{5\pi} m / \min$$

73. We have, $x^2 y^2 - 2x = 4(1 - y)$

$$\Rightarrow x^2 y^2 - 2x = 4 - 4y$$

Differentiating w.r.t. x , we get

$$2xy^2 + 2y.x^2 \frac{dy}{dx} - 2 = -4 \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx}(2yx^2 + 4) = 2 - 2xy^2 \Rightarrow \frac{dy}{dx} = \frac{2 - 2xy^2}{2yx^2 + 4}$$

$$\text{Now, } \left(\frac{dy}{dx} \right)_{(2, -2)} = \frac{2 - (2 \times 2 \times (-2)^2)}{(2(-2).2^2) + 4} = \frac{-14}{-12} = \frac{7}{6}$$

$$\therefore \text{Slope of tangent to the curve} = \frac{7}{6}$$

Equation of tangent passes through $(2, -2)$ is

$$y + 2 = \frac{7}{6}(x - 2) \Rightarrow 7x - 6y = 26$$

$\therefore$ Equation of tangent does not pass through $(-2, -7)$

74. Let the curves $y^2 = 6x$ and $9x^2 + by^2 = 16$ intersect at (α, β) .

$$\beta^2 = 6\alpha \text{ and } 9\alpha^2 + b\beta^2 = 16$$

$$\left[\frac{dy}{dx} \right]_{(\alpha, \beta)} \text{ for curve } y^2 = 6x \text{ is } \frac{dy}{dx} = \frac{3}{y} = \frac{3}{\beta}$$

$$\left[\frac{dy}{dx} \right]_{(\alpha, \beta)} \text{ for curve } 9x^2 + by^2 = 16 \text{ is}$$

$$\frac{dy}{dx} = -\frac{9x}{by} = -\frac{9\alpha}{b\beta}$$

As the curves are orthogonal, we have

$$\left(\frac{3}{\beta} \right) \left(-\frac{9\alpha}{b\beta} \right) = -1$$

$$\text{As } \beta^2 = 6\alpha, \text{ we get } 27\alpha = b(6\alpha)$$

$$\Rightarrow b = \frac{27}{6} = \frac{9}{2} \text{ (as } \alpha \neq 0 \text{)}$$

For $b = 0$ the intersection is non orthogonal. So we can rule out $b = 0$ in the beginning only to conclude $\alpha \neq 0$ in the end.

75. Hence, $f(x) = x^3 + 6x^2 + (9 + 2k)x + 1$

$$\Rightarrow f'(x) = 3x^2 + 12x + 9 + 2k$$

Now f is strictly increasing for all $x \in R$

if $f'(x) > \forall x \in R$

$$\Rightarrow 3x^2 + 12x + (9 + 2k) > 0 \text{ for all } x \in R$$

$$\Rightarrow 12^2 - 4.3(9 + 2k) < 0 \Rightarrow 12 - 9 - 2k < 0 \Rightarrow 3 - 2k < 0$$

$$\Rightarrow k > \frac{3}{2}$$

76. A point on the parabola is at a minimum distance from the circle if and only if it is at a minimum distance from the centre of the circle. A point on the parabola $y^2 = 8x$ is of the type $P(2t^2, 4t)$. Centre C of circle $x^2 + (y + 6)^2 = 1$ is $(0, -6)$.

$$\therefore CP^2 = 4t^4 + (4t + 6)^2 = 4(t^4 + 4t^2 + 12t + 9)$$

$$\Rightarrow \frac{d}{dt}(CP^2) = 4(4t^3 + 8t + 12) = 16(t + 1)(t^2 - t + 3)$$

$$\text{Also } \frac{d^2}{dt^2}(CP^2) = 48t^2 + 32$$

$$\text{Now, } \frac{d}{dt}(CP^2) = 0 \Rightarrow t = -1 \text{ (real value)}$$

$$\text{and } \left[\frac{d^2}{dt^2}(CP^2) \right]_{t=-1} = 80 > 0$$

$\therefore$ Required point is $(2, -4)$

77. Let $f'(x) = \lambda(x + 1)(x - 0)(x - 1) = \lambda(x^3 - x)$

$$\Rightarrow f(x) = \lambda \left(\frac{x^4}{4} - \frac{x^2}{2} \right) + C$$

$$\text{Given, } S = \{x \in R : f(x) = f(0)\}$$

$$\text{Now, } f(x) = f(0)$$

$$\Rightarrow \lambda \left(\frac{x^4}{4} - \frac{x^2}{2} \right) + C = C \Rightarrow \frac{x^4}{4} - \frac{x^2}{2} = 0 \Rightarrow x = 0, \pm\sqrt{2}$$

Hence, S has one rational and two irrational numbers.

78. We have, $f(x) = e^x - x$ and $g(x) = x^2 - x, \forall x \in R$

$$\therefore f'(x) = e^x - 1 \text{ and } g'(x) = 2x - 1$$

$$\text{Now, } h(x) = f(g(x)) \text{ [Given]}$$

$$\Rightarrow h'(x) = f'(g(x))g'(x) = (e^{g(x)} - 1)(2x - 1)$$

$$= (e^{x^2 - x} - 1)(2x - 1)$$

For $h(x)$ to be increasing, $h'(x) \geq 0$

Case I: $e^{x^2-x} - 1 \geq 0, 2x - 1 \geq 0 \Rightarrow x \in [1, \infty)$ _____ (i)

Case II: $e^{x^2-x} - 1 \leq 0, 2x - 1 \leq 0 \Rightarrow x \in \left[0, \frac{1}{2}\right]$ _____ (ii)

Hence, from (i) and (ii), $x \in \left[0, \frac{1}{2}\right] \cup [1, \infty)$

79. Hence, $f(x) = \frac{x}{\sqrt{a^2 + x^2}} - \frac{d-x}{\sqrt{b^2 + (d-x)^2}}$
 $\Rightarrow f'(x) = \frac{a^2}{(a^2 + x^2)^{3/2}} + \frac{b^2}{(b^2 + (d-x)^2)^{3/2}} > 0 \forall x \in R$

$\therefore f(x)$ is an increasing function of x .

80. We have, $f(x) = 5 - |x - 2|$ and $g(x) = |x + 1|$
 $\therefore f(x)$ attains its maxima at $x = 2$ i.e., $\alpha = 2$
 $g(x)$ attains its minima at $x = -1$ i.e., $\beta = -1$

So, the given expression is $\lim_{x \rightarrow 2} \frac{(x-1)(x^2 - 5x + 6)}{x^2 - 6x + 8}$
 $= \lim_{x \rightarrow 2} \frac{(x-1)(x-2)(x-3)}{(x-2)(x-4)} = \lim_{x \rightarrow 2} \frac{(x-1)(x-3)}{(x-4)} = \frac{1 \times (-1)}{(-2)} = \frac{1}{2}$

81. We have, $\lim_{x \rightarrow 1} \frac{(\sqrt{x} - 1)(2x - 3)}{2x^2 + x - 3}$
 $= \lim_{x \rightarrow 1} \frac{(\sqrt{x} - 1)(2x - 3)}{(x - 1)(2x + 3)}$
 $= \lim_{x \rightarrow 1} \frac{(2x - 3)}{(2x + 3)} \times \lim_{x \rightarrow 1} \left(\frac{x^{\frac{1}{2}} - 1^{\frac{1}{2}}}{x - 1} \right) = \frac{-1}{5} \times \frac{1}{2} (1)^{\frac{1}{2}}$
 $= \frac{-1}{5} \times \frac{1}{2} = \frac{-1}{10}$

82. We have, $f(x) = 1 - x + x^2 - x^3 + \dots - x^{99} + x^{100}$ _____ (i)
Differentiating (i) w.r.t. x , we get
 $f'(x) = 0 - 1 + 2x - 3x^2 + \dots - 99x^{98} + 100x^{99}$
 $\Rightarrow f'(x) = -1 + 2x - 3x^2 + \dots - 99x^{98} + 100x^{99}$
 $\therefore f'(1) = -1 + 2 - 3 + \dots - 99 + 100$
 $= (-1 - 3 - 5 \dots - 99) + (2 + 4 + \dots + 100)$

$$\begin{aligned}
 &= -(1+3+5+\dots+99) + (2+4+\dots+100) \\
 &= -\frac{50}{2}\{2+(49\times 2)\} + \frac{50}{2}\{(2\times 2)+(49\times 2)\} \\
 &= -2500 + 2550 = 50
 \end{aligned}$$

83. We have, $x^2 + 2x + 2y^2 = 1$ _____ (i)

Differentiating both sides of (i) w.r.t. x , we get

$$2x + 2 + \left(2 \times 2y \frac{dy}{dx}\right) = 0$$

$$\Rightarrow 4y \frac{dy}{dx} = -(2x+2) \Rightarrow \frac{dy}{dx} = \frac{-(2x+2)}{4y}$$

$$\Rightarrow \frac{dy}{dx} \Big|_{at y=1} = \frac{-(2x+2)}{4} \text{ _____ (i)}$$

Now, at $y=1$, we have, $x^2 + 2x + 2 = 1 \Rightarrow x^2 + 2x + 1 = 0$

$$\Rightarrow (x+1)^2 = 0 \Rightarrow x = -1$$

$$\therefore \frac{dy}{dx} \Big|_{x=-1} = \frac{-(-2+2)}{4} = 0$$

84. $\therefore f(x)$ is continuous at $x=0$

$$\text{So, } f(0) = \lim_{h \rightarrow 0} f(0+h)$$

$$\Rightarrow \lambda \ln 4 = \lim_{h \rightarrow 0} \frac{8^h - 4^h - 2^h + 1^h}{h^2} = \lim_{h \rightarrow 0} \frac{(2^h - 1)(4^h - 1)}{h.h}$$

$$\Rightarrow \lambda \ln 4 = \ln 2 \cdot \ln 4$$

$$\therefore \lambda = \ln 2 \Rightarrow e^\lambda = 2$$

$$\therefore 1000e^\lambda = 1000 \times 2 = 2000$$

85. We have, $f(x) = (kx + e^x)h(x)$

$$f'(x) = (kx + e^x)h'(x) + h(x)(k + e^x)$$

$$\therefore f'(0) = h'(0) + h(0)(k+1)$$

$$\Rightarrow 18 = -2 + 5(k+1) \Rightarrow k = 3$$

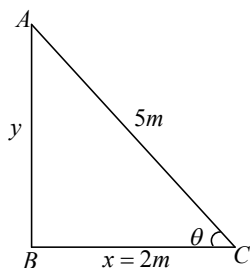
86. Given, $u = x^2 + y^2, x = s + 3t, y = 2s - t$

$$\Rightarrow \frac{dx}{ds} = 1, \frac{dy}{ds} = 2$$

$$\text{Now, } u = x^2 + y^2 \Rightarrow \frac{du}{ds} = 2x \frac{dx}{ds} + 2y \frac{dy}{ds} = 2x + 4y$$

$$\Rightarrow \frac{d^2u}{ds^2} = 2 \left(\frac{dx}{ds} \right) + 4 \left(\frac{dy}{ds} \right) \Rightarrow \frac{d^2u}{ds^2} = 2(1) + 4(2) = 10$$

87. Given, $\frac{dy}{dt} = 10 \text{ cm/s}$, $x = 2 \text{ m}$, $m = 200 \text{ cm}$ and Length of ladder = $5 \text{ m} = 500 \text{ cm}$



$\Rightarrow$ Let $\angle ACB = \theta$

Now $\sin \theta = \frac{y}{500}$

Differentiating w.r.t. t , we get

$$\cos \theta \frac{d\theta}{dt} = \frac{1}{500} \frac{dy}{dt}$$

$$\Rightarrow \frac{200}{500} \times \frac{d\theta}{dt} = \frac{1}{500} \times 10$$

$$\Rightarrow \frac{d\theta}{dt} = \frac{10}{200} \Rightarrow \frac{d\theta}{dt} = \frac{1}{20} \text{ rad/s}$$

88. Give that $x = 2 \cos t + 2t \sin t$

$$\Rightarrow \frac{dx}{dt} = -2 \sin t + 2[t \cos t + \sin t] = 2t \cos t$$

Also, $y = 2 \sin t - 2t \cos t$

$$\Rightarrow \frac{dy}{dt} = 2 \cos t - 2[-t \sin t + \cos t] = 2t \sin t$$

So, $\frac{dy}{dx} = \frac{2t \sin t}{2t \cos t}$

$$\Rightarrow \frac{dy}{dx} = \tan t \Rightarrow \left(\frac{dy}{dx} \right)_{t=\pi/4} = 1$$

So the slope of the normal is -1.

and at $t = \pi/4$, $x = \sqrt{2} + \frac{\pi}{2\sqrt{2}}$ and $y = \sqrt{2} - \frac{\pi}{2\sqrt{2}}$

$\therefore$ The equation of normal is

$$\left[y - \left(\sqrt{2} - \frac{\pi}{2\sqrt{2}} \right) \right] = -1 \left[x - \left(\sqrt{2} + \frac{\pi}{2\sqrt{2}} \right) \right]$$

$$\Rightarrow x + y = 2\sqrt{2}. \text{ So the distance from the origin is } 2.$$

89. We have, $y = f(x)$ and $f(a) = 0$

$$\therefore \lim_{x \rightarrow a} \frac{\log_e \{1 + 6f(x)\}}{3f(x)} = \lim_{x \rightarrow a} \frac{\log_e \{1 + 6f(x)\}}{6f(x)} \cdot 2 = 1 \cdot 2 = 2$$

90. $y = -x^2 + 3x^2 + 9x - 27$

$$\text{Slope, } m = \frac{dy}{dx} = -3x^2 + 6x + 9$$

$$\text{For maximum } \frac{dm}{dx} = -6x + 6$$

$$\text{Now, } \frac{dm}{dx} = 0 \Rightarrow -6x + 6 = 0 \Rightarrow x = 1$$

$$\therefore \frac{d^2m}{dx^2} = -6 < 0 \forall x$$

$\therefore x = 1$ is a point of local maximum.

$$\therefore \text{Maximum slope} = -3(1)^2 + 6(1) + 9 = 12$$



Sri Chaitanya IIT Academy., India.

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Jee-Main

Date: 03-06-2021

Time: 09.00Am to 12.00

PTM-06

Max.Marks:300

IMPORTANT INSTRUCTIONS:

1. Immediately fill in the Admission number on this page of the Test Booklet with Blue/Black Ball Point Pen only.
2. The candidates should not write their Admission Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1, 2, 3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **two sections**.

(I) Section-I contains 20 **multiple choice** questions with **only one correct** option.

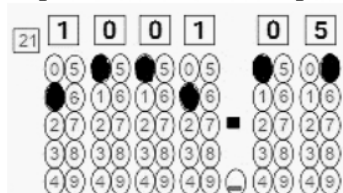
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

(II) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only.

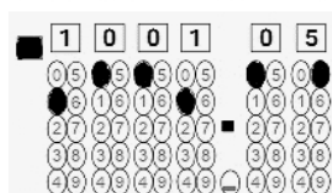
First 5 attempted questions will be considered if more than 5 questions attempted.

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted question 21, bubble on 21 as shown below.



Question Answered for marking



Question Cancelled for marking

Marking scheme : +4 for correct answer and 0 in all other cases.

6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capitals): _____

Admission Number:

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Candidate's Signature: _____

Invigilator's Signature: _____

03-06-2021_Sr.Super60 & All_Jee-Main_PT M-06_Test Syllabus

PHYSICS : Ray optics, Wave optics, SHM

CHEMISTRY : Atomic structure Chemical kinetics, Amines Diazonium salts Nitro Cyano Compounds Polymers, Chemical bonding, Group-IIIA

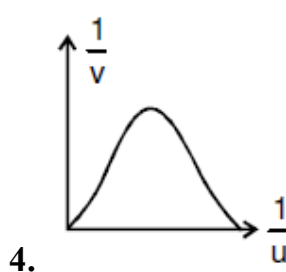
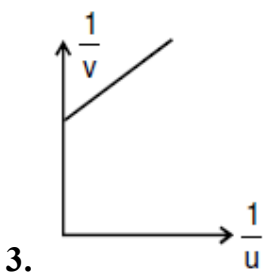
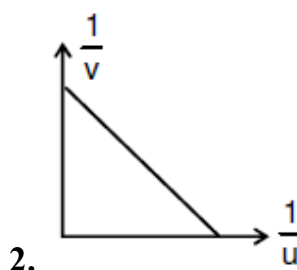
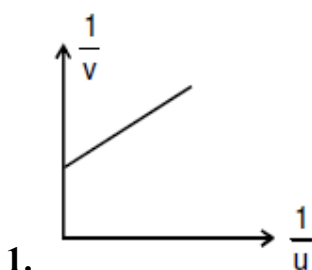
MATHEMATICS : Log, Q.E Trigonometric Function, Trigonometric, Equation, POT, I.T.F

(SINGLE CORRECT ANSWER TYPE)

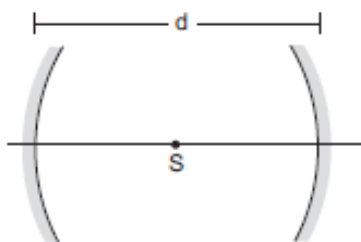
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

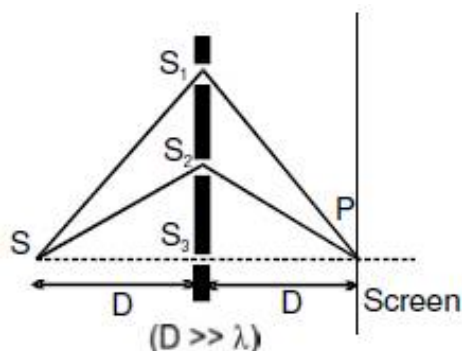
- Plane microwaves from a transmitter are directed normally towards a plane reflector. A detector moves along the normal to the reflector. Between positions of 14 successive maxima, the detector travels a distance 0.13 m. If the velocity of light is $3 \times 10^8 \text{ m/s}$ find the frequency of the transmitter
 1. $1.5 \times 10^{10} \text{ Hz}$ 2. 10^{10} Hz 3. $3 \times 10^{10} \text{ Hz}$ 4. $6 \times 10^{10} \text{ Hz}$
- beam of light consisting of two wavelengths $6300 \text{ \AA}$ and $\lambda \text{ \AA}$ is used to obtain interference fringes in a Young's double slit experiment. If 4th bright fringe of $6300 \text{ \AA}$ coincides with 5th dark fringe of $\lambda \text{ \AA}$, the value of λ (in $\text{\AA}$) is
 1. 5200 2. 4800 3. 6200 4. 5600
- A boy of height 1.5 m with his eye level at 1.4 m stands before a plane mirror of length 0.75 m fixed on the wall. The height of the lower edge of the mirror above the floor is 0.8 m. Then :
 1. the boy will see his full image 2. the boy cannot see his hair
 3. the boy cannot see his feet 4. the boy cannot see neither his hair nor his feet
- An object is placed at a distance u from a concave mirror and its real image is received on a screen placed at a distance of v from the mirror. If f is the focal length of the mirror, then the graph between $1/v$ versus $1/u$ is



5. A ray of light in a liquid of refractive index 1.4, approaches the boundary surface between the liquid and air at an angle of incidence whose sine is 0.8. Which of the following statements is correct about the behaviour of the light
1. It is impossible to predict the behaviour of the light ray on the basis of the information supplied
 2. The sine of the angle of refraction of the emergent ray will be less than 0.8
 3. The ray will be internally reflected
 4. The sine of the angle of refraction of the emergent ray will be greater than 0.8
6. A prism of refractive index $\sqrt{2}$ has refracting angle 60° . In order that a ray suffers minimum deviation it should be incident at an angle
1. 45°
 2. 90°
 3. 30°
 4. none
7. A point source S is placed midway between two converging mirrors having equal focal length f as shown in figure (18-E3). Find the values of d for which only one image is formed

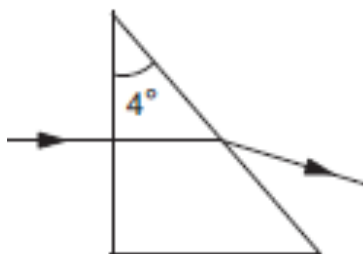


1. $3f, 5f$
 2. $2f, 4f$
 3. $f, 2f$
 4. $3f, f$
8. A monochromatic light source of wavelength λ is placed at S . Three slits S_1, S_2 and S_3 are equidistant from the source S and the point P on the screen. $S_1P - S_2P = \lambda / 6$ and $S_1P - S_3P = 2\lambda / 3$. If I be the intensity at P when only one slit is open, the intensity at P when all the three slits are open is

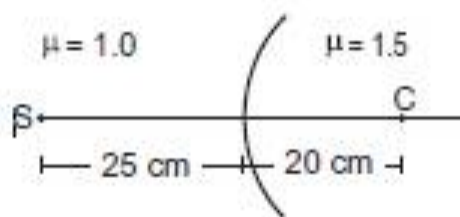


1. 31
2. 51
3. 81
4. zero

9. Find the angle of deviation suffered by the light ray shown in figure. The refractive index $\mu = 1.5$ for the prism material

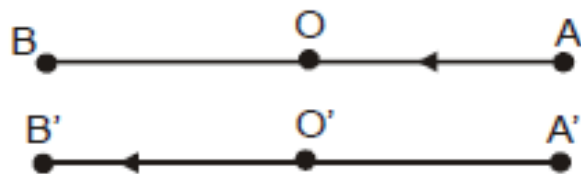


1. 2° 2. 4° 3. 6° 4. 8°
10. Locate the image formed by refraction in the situation shown in figure. (from the surface on the side of S)

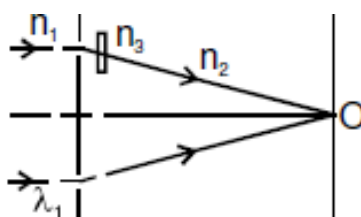


1. 40cm 2. 50cm 3. 200cm 4. 100cm
11. The time period of a particle in simple harmonic motion is equal to the time between consecutive appearance of the particle at a particular point in its motion. This point is
1. the mean position
 2. an extreme position
 3. between the mean position and the positive Extreme
 4. between the mean position and the negative extreme
12. The maximum acceleration of a particle in SHM is made two times keeping the maximum speed to be constant. It is possible when
1. amplitude of oscillation is doubled while frequency remains constant
 2. amplitude is doubled while frequency is halved
 3. frequency is doubled while amplitude is halved
 4. frequency is doubled while amplitude remains constant
13. The average acceleration in one time period in a simple harmonic motion is
1. $A\omega^2$ 2. $A\omega^2 / 2$ 3. $A\omega^2 / \sqrt{2}$ 4. zero

14. Two particles undergo SHM along parallel lines with the same time period (T) and equal amplitudes. At a particular instant, one particle is at its extreme position while the other is at its mean position. They move in the same direction. They will cross each other after a further time

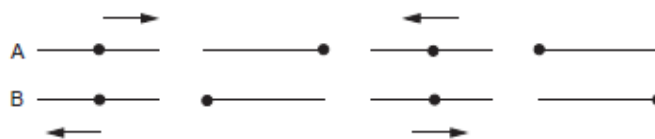


1. $T/8$ 2. $3T/8$ 3. $T/6$ 4. $4T/3$
15. In the figure shown in YDSE, a parallel beam of light is incident on the slit from a medium of refractive index n_1 . The wavelength of light in this medium is λ_1 . A transparent slab to thickness 't' and refractive index n_3 is put in front of one slit. The medium between the screen and the plane of the slits is n_2 . The phase difference between the light waves reaching point 'O' (symmetrical, relative to the slits) is :



1. $\frac{2\pi}{n_1\lambda_1}(n_3 - n_2)t$ 2. $\frac{2\pi}{\lambda_1}(n_3 - n_2)t$
3. $\frac{2\pi n_1}{n_2\lambda_1}\left(\frac{n_3}{n_2} - 1\right)t$ 4. $\frac{2\pi n_1}{\lambda_1}(n_3 - n_1)t$
16. Two particles A and B perform SHM along the same straight line with the same amplitude 'a', same frequency 'f' and same equilibrium position 'O'. The greatest distance between them is found to be $3a/2$. At some instant of time they have the same displacement from mean position. What is the displacement?
1. $a/2$ 2. $a\sqrt{7}/4$ 3. $\sqrt{3}a/2$ 4. $3a/4$

17. Figure represents two simple harmonic motions.



The parameter which has different values in the two motions is

1. amplitude 2. frequency 3. phase 4. maximum velocity.
18. It is necessary to coat a glass lens with a non-reflecting layer. If the wavelength of the light in the coating is λ , the best choice is a layer of material having an index of refraction between those of glass and air and a thickness of
1. $\lambda / 4$ 2. $\lambda / 2$ 3. $3\lambda / 8$ 4. λ
19. In a biprism experiment the distance of source from biprism is 1 m and the distance of screen from biprism is 4 meters. The angle of refraction of biprism is 2×10^{-3} radians. μ of biprism is 1.5 and the wavelength of light used is $6000 \text{ \AA}$. How many fringes will be seen on the screen?
1. 4 2. 5 3. 3 4. 6
20. The image of a small electric bulb fixed on the wall of a room is to be obtained on the opposite wall 3m away by means of a large convex lens. What is the maximum possible focal length of the lens required for the purpose
1. 1.75 2. 0.75 3. 2.75 4. 3.75

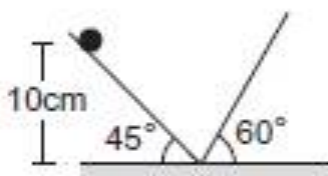
(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

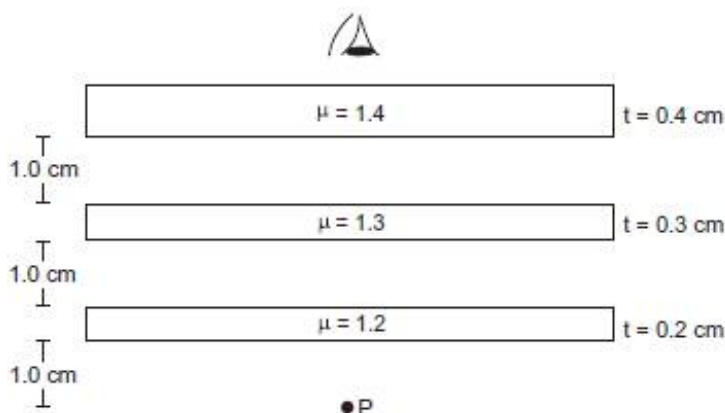
Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

21. A uniform meter stick is suspended through a small pin hole at the 10 cm mark. Find the time period of small oscillation about the point of suspension
22. Find the time period of the motion of the particle shown in figure. Neglect the small effect of the bend near the bottom



23. A point object is kept in front of a plane mirror. The plane mirror is doing SHM of amplitude 2cm. The plane mirror moves along the x-axis and x-axis is normal to the mirror. The amplitude of the mirror is such that the object is always in front of the mirror. The amplitude of SHM of the image is (in cm)
24. If a ray of light is incident on a plane mirror at an angle 60° from the mirror surface, then deviation produced by mirror is : (in degrees)
25. Locate the image of the point P as seen by the eye in the figure. (in cm above P)



26. In Young's double-slit experiment using the monochromatic light of wavelength λ , the intensity of light at a point on the screen where path difference is λ , is K units. The intensity of light at a point where path difference is $\frac{\lambda}{3}$ is given as $\frac{K}{x}$ then find 'x'
27. In Young's double slit experiment, the wavelength of red light is $7800 \text{ \AA}$ and that of blue light is $5200 \text{ \AA}$. The value of n for which n^{th} bright band due to red light coincides with $(n + 1)^{\text{th}}$ bright band due to blue light, is
28. In a double - slit experiment using the light of wavelength 600nm , the angular width of a fringe formed on a distance screen is 0.1° . The spacing between the two slits is $n \times 10^{-4}$. The find value of 'n' ?
29. A spring balance has a scale that reads from 0 to 50kg. The length of the scale is 20cm. A body suspended from this balance, when displaced and released, oscillates with a period 0.6s. What is the weight of the body (take $g = 9.8$)
30. A tank is filled with water to height of 12.5 cm. The apparent depth of a needle lying at the bottom of the tank is measured by a microscope to be 9.4cm. What is the refractive index of water ? If water is replaced by a liquid of refractive index 1.63 up to the same height, by what distance would the microscope have to be moved to focus on the needle again? (in cm)

CHEMISTRY**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

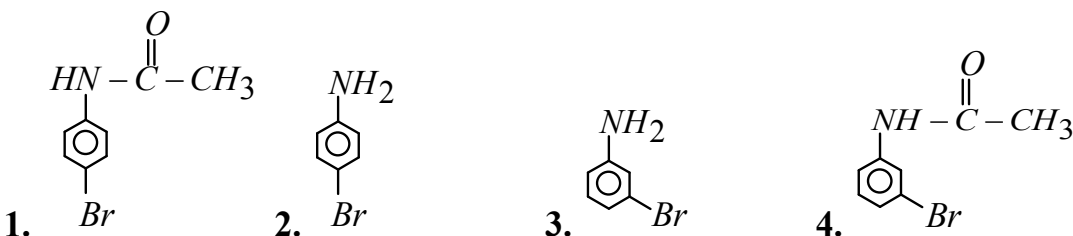
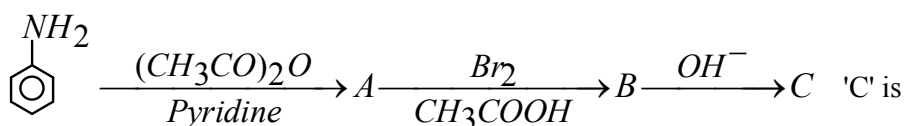
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

31. Which of the following is correct match about B, Al, Ga, In, Tl respectively
1. non-metal, typical metal, metal, metal & metal
 2. typical non-metal, metal, metal, metal & metal
 3. non-metal, metalloid, metalloid, metal & metal
 4. non-metal, metalloid, metalloid, metal & metal
32. Which of the following correct sequence of sum of first three ionization energies ($IE_1 + IE_2 + IE_3$)
1. $B > Al > Ga > In > Tl$
 2. $B > Tl > Ga > Al > In$
 3. $B > Ga > Tl > In > Al$
 4. $B > Ga > Tl > Al > In$
33. Which element is used to measure temperature of materials at high temperature
1. B
 2. Ga
 3. In
 4. Tl
34. Select correct statement about diborane (B_2H_6)
- | | |
|--------------------|------------------|
| Hybridization of B | Bond angle [HBH] |
| 1. sp^3 | 97° |
| 2. sp^2 | 120° |
| 3. sp^3 | 127° |
| 4. Both 1 and 3 | |
35. Correct sequence bond length is
1. $H_2 < F_2 < HF < O_2$
 2. $O_2 < H_2 < F_2 < HF$
 3. $H_2 < O_2 < HF < F_2$
 4. $H_2 < HF < O_2 < F_2$
36. Which quantum number gives the information about the spatial orientation of the orbital
1. Subsidiary quantum number
 2. Principal quantum number
 3. Magnetic quantum number
 4. Spin quantum number
37. Incorrect statement about Dalton's atomic theory is
1. Matter consists of indivisible atoms
 2. Atoms of different elements differ in mass
 3. All atoms of a given element have identical properties
 4. The theory could not explain the laws of chemical combination

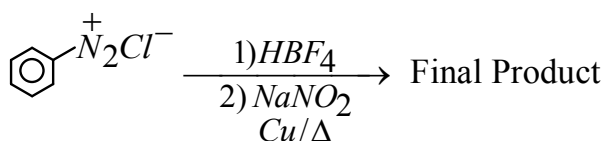
38. Maximum number of possible exchange pairs for d^5 configuration is
 1. 5 2. 8 3. 1 4. 10
39. Consider the following statement about photoelectric effect
 (a) the number of electrons ejected is proportional to the intensity of light
 (b) Kinetic energy of ejected electrons increase with the increase of wavelength of the light
 (c) work function of potassium is greater than lithium
 Incorrect statements are
 1. (a) & (b) only 2. (b) & (c) only
 3. (a) & (c) only 4. (a) (b) & (c)
40. For the reaction $2A \rightarrow B$ the concentration of reactant A changes from $0.04M$ to $0.02M$ in 2min. The average rate of reaction is
 1. $0.01Ms^{-1}$ 2. $1.66 \times 100^{-4} Ms^{-1}$
 3. $8.33 \times 10^{-5} Ms^{-1}$ 4. $5 \times 10^{-3} Ms^{-1}$
41. Which of the following statement is incorrect ?
 1. Trimolecular reactions involve simultaneous collision between three reacting species
 2. The probability that more than three molecules can collide is very small
 3. Reaction with the molecularity three are very common
 4. Reaction with molecularity three are slow to proceed
42. Addition of catalyst
 1. Catalyses the forward reaction only
 2. Catalyses the backward reaction only
 3. Catalyses the forward as well as backward reactions to the same extent
 4. Catalyses the forward as well as backward reactions to the different extent
43. Select the correct expression on the basis of collision theory
 1. Rate = $PZ_{AB}e^{-Ea/Rt}$ 2. Rate = $Ae^{-Ea/Rt}$
 3. Rate = $PZ_{AB}e^{Ea/Rt}$ 4. Rate = $Ae^{Ea/Rt}$
44. Which reaction can go for completion ?
 1. 0 order 2. 1st order 3. 2nd order 4. 3rd order

45. Select the correct statement
1. Order is applicable to elementary as well as complex reactions
 2. Order of a reaction is an experimental quantity
 3. molecularity is not valid for complex reaction
 4. All of these
46. Aniline undergoes condensation to form Schiff's base on reacting with
1. acetylchloride
 2. Ammonia
 3. Acetone
 4. Benzaldehyde
47. Aniline dissolves in HCl due to formation of
1. O-Chloroaniline
 2. Azodye
 3. Anilinium chloride
 4. Diazonium salt
48. Which of the following amine have less pK_b
1. N-ethylethanamine
 2. N, N-Dimethylethanamine
 3. N-Methyl methanamine
 4. Ethanamine

49.



50.

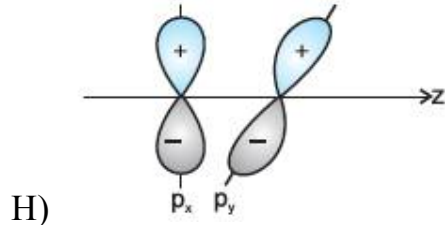
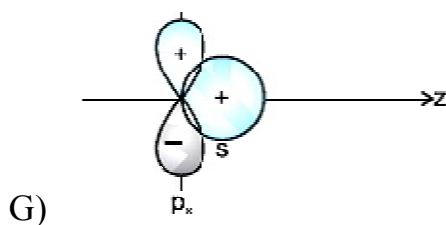
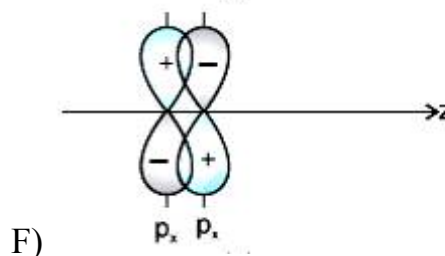
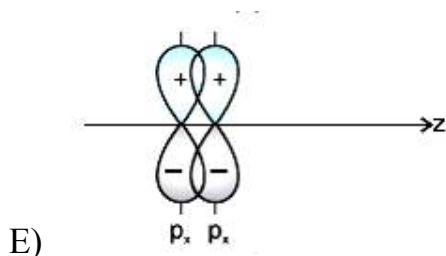
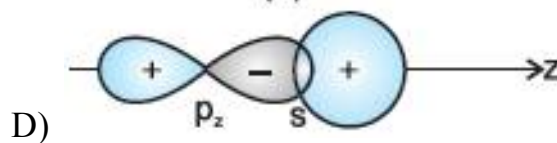
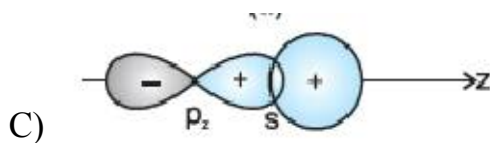
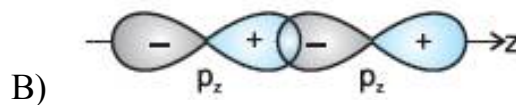
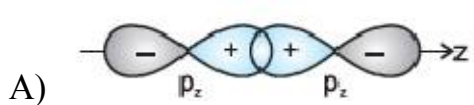
**(NUMERICAL VALUE TYPE)**

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Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

51. Volume of H_2 gas in lts enrolled at STP when 6 mole of B_2H_6 react with sufficient quantity of NH_3
52. Number of possible amines for $C_4H_{11}N$ reacts with Hinsberg's reagent which are soluble in $NaOH$
53. How many of the following statements are correct
 - i) Novolac is used in paints
 - ii) Novolac on heating with formaldehyde to form backlite
 - iii) backlite is used to make combs
 - iv) Buna-S is used to make floor tiles
 - v) PHBV under goes bacterial degradation
 - vi) Neoprene is the monomer in chloroprene
54. Number of zero overlaps among the following is/ are?



55. total no of corect statements is / are
- In d^3s hybridizations, the d-orbital involved is $d_{x^2-y^2}, d_{z^2}, d_{xy}$
 - In sp^3d hybridizations, the d-orbital involved is d_{z^2}
 - In sp^3d^2 hybridizations, the d-orbital involved is $d_{x^2-y^2}, d_{z^2}$
 - In sp^3d^2 hybridizations, the d-orbital involved is d_{xy}, d_{zx}
56. Half - life period of the decomposition of a compound is 20 minutes. If the initial concentration is doubled, the half - life period is reduced to 10 mins. What is the order of the reaction?
57. In a reaction $A_2 + 3B_2 \rightarrow 2C$, the order is 1.5. The ratio of rate of formation of C to that of disappearance of B_2 if given by x : y then x + y may be given as
58. A reaction $X_{2(g)} \rightarrow Z(g) + \frac{1}{2}Y(g)$ exhibits an increase in pressure from 150mm to 170 mm in 10 min. The rate of disappearance of X_2 in mm per min is
59. Half life of a reaction is 20 sec. If t_2 is second half life of reaction assuming it to be zero order and t_3 is third half life assuming it to be 2nd order reaction, then $\frac{t_3}{t_2} = ?$
60. For the first order reaction $t_{99.99\%} = xt_{90\%}$ then, find the value of x.

MATHS**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. Let $f(x) = x^2 - bx + c$, b is an odd positive integer $f(x) = 0$ have two prime numbers as roots and $b + c = 35$. Then global minimum value of $f(x)$ is
1. $-\frac{183}{4}$
 2. $\frac{173}{6}$
 3. $-\frac{81}{4}$
 4. data is not sufficient
62. If $f(x) = x^2 + bx + c$ and $f(2+t) = f(2-t)$ for all real numbers t , then which of the following is true?
1. $f(1) < f(2) < f(4)$
 2. $f(2) < f(1) < f(4)$
 3. $f(2) < f(4) < f(1)$
 4. $f(4) < f(2) < f(1)$
63. If $x^2 + (\log_3 2 + \log_2 3)x + \log_2 \sqrt{6} + \log_2 \sqrt{\frac{2}{3}} < 0$ then $x \in$
1. $(-\log_2 3, -\log_3 2)$
 2. $(\log_{1/2} 3, \log_{1/3} 2)$
 3. $\left(\log_2 \frac{1}{3}, \log_3 \frac{1}{2}\right)$
 4. All of these
64. If $(x^2 + x + 2)^2 - (a-3)(x^2 + x + 1)(x^2 + x + 2) + (a-4)(x^2 + x + 1)^2 = 0$ has at least one root, then the complete set of values of a is
1. $(-\infty, 5)$
 2. $(3, 6)$
 3. $\left(5, \frac{19}{2}\right)$
 4. $\left(5, \frac{19}{3}\right]$
65. If the equations $ax^2 + bx + c = 0$, $a, b, c \in R$ and $x^3 + 3x^2 + 3x + 2 = 0$ have two common roots, then $2a + 3b - 5$ is
1. -1
 2. 0
 3. 1
 4. 2
66. If α, β are the roots $4x^2 - 16x + \lambda = 0, \lambda \in R$, such that $1 < \alpha < 2$ and $2 < \beta < 3$, then the number of integral solutions of λ is
1. 6
 2. 5
 3. 3
 4. 2

67. The product of real values of x such that $(\log_x 2)^2 - (\log_2 x)^2 = \log_{2x} \left(\frac{2}{x} \right)$ is
1. $\sqrt{2}$ 2. $\frac{1}{\sqrt{2}}$ 3. $2^{\sqrt{2}}$ 4. $\frac{1}{2^{\sqrt{2}}}$
68. Let $S \equiv \{a \in \mathbb{N}, a \leq 100\}$. If the equation $\left[\tan^2 x \right] - \tan x - a = 0$ has real roots then number of elements in S is, [where $[.]$ greatest integer function]
1. 10 2. 8 3. 9 4. 0
69. The number of solutions of $|[x] - 2x| = 4$ where $[x]$ is the greatest integer $\leq x$ is
1. 2 2. 4 3. 1 4. Infinite
70. Sum of all the values of x satisfying the equation $\log_{17} \log_{11} (\sqrt{x+11} + \sqrt{x}) = 0$ is
1. 25 2. 36 3. 171 4. 0
71. The values of ' a ' for which the quadratic expression $ax^2 + (a-2)x - 2$ is negative for exactly two integral values of x , belongs to
1. $[-1, 1]$ 2. $[1, 2)$ 3. $[3, 4]$ 4. $[2, -1)$
72. If $2^{\sqrt{\sin^2 x - 2 \sin x + 5}} \frac{1}{4^{\sin^2 y}} \leq 1$, then the ordered pair (x, y) is equal to $(m, n \in \mathbb{I})$
1. $x = (4n+1)\frac{\pi}{2}, y = (2m+1)\frac{\pi}{2}$ 2. $x = 2n\pi, y = 2m\pi$
3. $x = (2n+1)\frac{\pi}{2}, y = (2m+1)\frac{\pi}{2}$ 4. $x = n\pi, y = m\pi$
73. If $\cos A + \cos B = 4 \sin^2 \frac{C}{2}$, then the sides of the triangle ABC are in
1. A.P 2. G.P 3. H.P 4. None of these
74. Let $f(x) = x^{11} + x^9 - x^7 + x^3 + 1$ and $f(\sin^{-1}(\sin 8)) = \alpha$ then $f(\tan^{-1}(\tan 8)) =$
1. α 2. $\alpha - 2$ 3. $\alpha + 2$ 4. $2 - \alpha$
75. If $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ then the value of $\tan^{-1}\left(\frac{\tan x}{4}\right) + \tan^{-1}\left(\frac{3 \sin 2x}{5 + 3 \cos 2x}\right) =$
1. $\frac{x}{2}$ 2. $2x$ 3. $3x$ 4. x

76. The equation $a \sin x + \cos 2x = 2a - 7$ possess 'a' solution If
 1. $a > 6$ 2. $2 \leq a \leq 6$ 3. $a < 2$ 4. $a \geq 7$
77. In $\triangle ABC$ if $2(bc^2 + ca^2 + ab^2) = b^2c + c^2a + a^2b + 3abc$ then the triangle is.
 (BC=a, CA=b, AB=c)
 1. right angled 2. equilateral 3. Scalene 4. isosceles
78. If $\tan^2 \pi(x+y) + \cot^2 \pi(x+y) = 1 + \sqrt{\frac{2x}{1+x^2}}$, $x, y \in R$ then the least positive value of y
 1. $\frac{1}{6}$ 2. $\frac{1}{4}$ 3. $\frac{3}{4}$ 4. $\frac{5}{4}$
79. The value of $\sum_{r=1}^{10} \tan^{-1}\left(\frac{3}{9r^2 + 3r - 1}\right)$ is
 1. $\cot^{-1}\left(\frac{13}{6}\right)$ 2. $\tan^{-1}\left(\frac{13}{6}\right)$ 3. $\pi - \cot^{-1}\left(\frac{6}{13}\right)$ 4. $\frac{\pi}{4}$
80. In $(0, 2\pi)$, the equation $\sec \theta + \operatorname{cosec} \theta = C$ has x_1 solutions if $C^2 < 8$ and has x_2 solutions if $C^2 > 8$, the ordered pair (x_1, x_2) is
 1. (4,2) 2. (2,4) 3. (3,3) 4. (3,5)

(NUMERICAL VALUE TYPE)

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Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

81. If a, b, c are the sides of a triangle, then the minimum value of $\frac{2a}{b+c-a} + \frac{2b}{c+a-b} + \frac{2c}{a+b-c}$ is
82. If $A_1, A_2, A_3, \dots, A_n$ are vertices of a regular polygon of 'n' sides and $\frac{1}{A_1 A_2} = \frac{1}{A_1 A_3} + \frac{1}{A_1 A_4}$ then n is
83. In $\triangle ABC$, $a \geq b \geq c$ and if $\frac{a^3 + b^3 + c^3}{\sin^3 A + \sin^3 B + \sin^3 C} = 8$ then the maximum value of 'a' is

84. In $\triangle ABC$, angles A, B, C are in A.P then $\lim_{A \rightarrow C} \frac{\sqrt{12 - 16 \sin A \sin C}}{|A - C|}$ is
85. Number of real values of 'x' satisfying the equation $\frac{(\sec^{-1} x)^3 + (\operatorname{cosec}^{-1} x)^3}{(\tan^{-1} x + \cot^{-1} x)^3} = 7$ is
86. The maximum value of the expression $\left| \sqrt{\sin^2 x + 2a^2} - \sqrt{2a^2 - 1 - \cos^2 x} \right|$, where a and x are real numbers is
87. The number of solutions of $\sin x + \sin y = \sin(x + y)$ and $|x| + |y| = 1$, $x > 0$ is
88. Number of solutions of the equation $\tan x + \sec x = 2 \cos x$ lying in the interval $[0, 2\pi]$ is
89. If $x = \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7}$ and $y = \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7}$ then $x^2 + y^2 =$
90. The number of points inside or on the circle $x^2 + y^2 = 4$ satisfying $\tan^4 x + \cot^4 x + 1 = 3 \sin^2 y$ is



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A right Choice for the Real Aspirant

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Date:03-06-2021

Time:09.00Am to 12.00

PTM-06

Max.Marks:300

Key Sheet

PHYSICS

1	1	2	4	3	3	4	2	5	3
6	1	7	2	8	1	9	1	10	4
11	2	12	3	13	4	14	2	15	1
16	2	17	3	18	1	19	2	20	2
21	1.55	22	0.73	23	4	24	120	25	0.2
26	4	27	2	28	3.44	29	219 to 219.17	30	1.73

CHEMISTRY

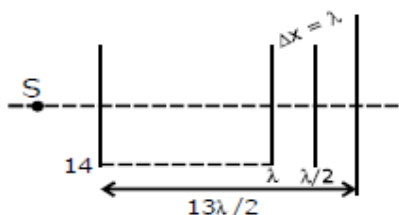
31	2	32	4	33	1	34	1	35	4
36	3	37	4	38	4	39	2	40	4
41	3	42	3	43	1	44	1	45	4
46	4	47	2	48	1	49	2	50	3
51	537.6	52	4	53	5	54	2	55	2
56	2	57	5	58	4	59	8	60	4

MATHEMATICS

61	3	62	2	63	4	64	4	65	2
66	3	67	4	68	3	69	2	70	1
71	2	72	1	73	1	74	4	75	4
76	2	77	2	78	2	79	1	80	2
81	6.00	82	7.00	83	2.00	84	2.00	85	1
86	1.41	87	2	88	2.00	89	2.00	90	4.00

SOLUTIONS PHYSICS

01.



$$\frac{13\lambda}{2} = 0.13$$

$$\Rightarrow \lambda = \frac{2}{100} m$$

$$\therefore f = \frac{v}{\lambda}$$

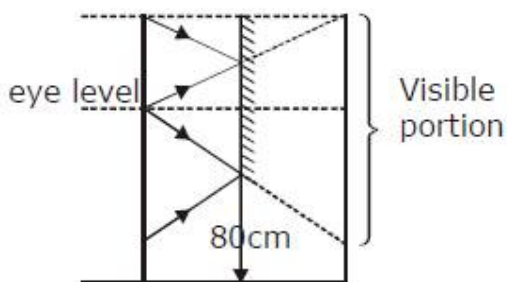
$$\Rightarrow f = \frac{3 \times 10^8 \times 100}{2} = 1.5 \times 10^{10} \text{ Hz}$$

02.

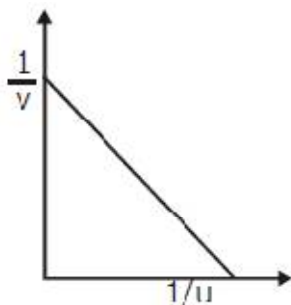
$$4 \times 6300 = (4.5)\lambda$$

$$\lambda = \frac{4 \times 6300}{9} \text{ nm} = 5600 \text{ nm}$$

03.

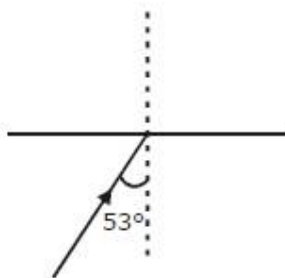


04.



$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

05.



$$\sin C = \frac{1}{1.4}$$

$$C = 45.58$$

06. For minimum deviation

$$r_1 = r_2 = r \Rightarrow 2r = A$$

$$r = \frac{A}{2} = 30^\circ$$

Now from Snell's law

$$1 \sin i = \sqrt{2} \sin 30^\circ$$

$$i = 45^\circ$$

07.

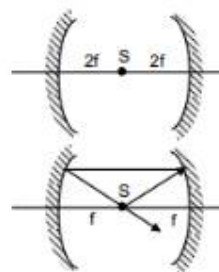
Both the mirrors have equal focal length f .

They will produce one image under two conditions.

Case I : When the source is at distance ' $2f$ ' from each mirror i.e. the source is at centre of curvature of the mirrors, the image will be produced at the same point S . So, $d = 2f + 2f = 4f$.

Case II : When the source S is at distance ' f ' from each mirror, the rays from the source after reflecting from one mirror will become parallel and so these parallel rays after the reflection from the other mirror the object itself. So, only one image is formed.

Here, $d = f + f = 2f$.



08.

$$S_1P - S_2P = \lambda/6$$

$$\therefore SS_1P - SS_2P = \lambda/3 \quad \dots\dots(1)$$

$$SS_1P - SS_3P = 4\lambda/3 \quad \dots\dots(2)$$

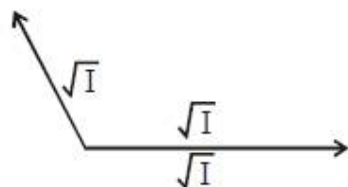
$$\Rightarrow \Delta\phi = \frac{2\pi}{\lambda} \times \frac{4\lambda}{3}$$

$$(2) - (1)$$

$$SS_2P - SS_3P = \lambda$$

$$\Rightarrow \Delta\phi = \frac{2\pi}{\lambda} \times \lambda = 2\pi$$

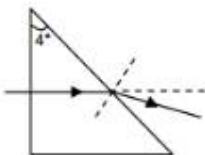
Take base SS_3P



$$I_{\text{net}} = (2\sqrt{I})^2 + (\sqrt{I}) + 2 \cdot 2\sqrt{I}\sqrt{I} \cos 120^\circ$$

$$I_{\text{net}} = 3I$$

09.



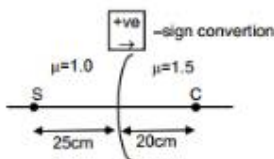
Given $\mu = 1.5$

And angle of prism = 4°

$$\therefore \mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin A/2} = \frac{(A + \delta_m)/2}{(A/2)} \quad (\text{for small angle } \sin \theta = \theta)$$

$$\Rightarrow \mu = \frac{A + \delta_m}{2} \Rightarrow 1.5 = \frac{4^\circ + \delta_m}{4^\circ} \Rightarrow \delta_m = 4^\circ \times (1.5) - 4^\circ = 2^\circ.$$

10.



$\mu_1 = 1, \mu_2 = 1.5, R = 20 \text{ cm}$ (Radius of curvature), $u = -25 \text{ cm}$

$$\therefore \frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R} \Rightarrow \frac{1.5}{v} = \frac{0.5}{20} - \frac{1}{25} = \frac{1}{40} - \frac{1}{25} = \frac{-3}{200}$$

$$\Rightarrow v = -200 \times 0.5 = -100 \text{ cm}.$$

So, the image is 100 cm from (P) the surface on the side of S.

11. Conceptual

12. Conceptual

13.

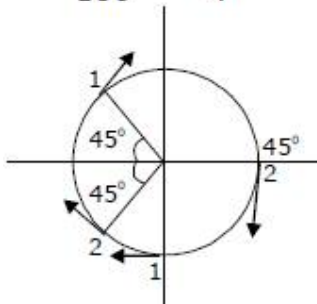
Total Accⁿ = 0

$$a_{av} = \frac{\text{Total change in Velocity in one T.P.}}{\text{Time Period}}$$

$$a_{av} = \frac{0}{T} = 0$$

14.

$$\frac{135 \times \pi}{180} = \frac{3\pi}{4}$$



$$\therefore t = \frac{3\pi}{4 \times 2\pi} T = \frac{3T}{8}$$

15.

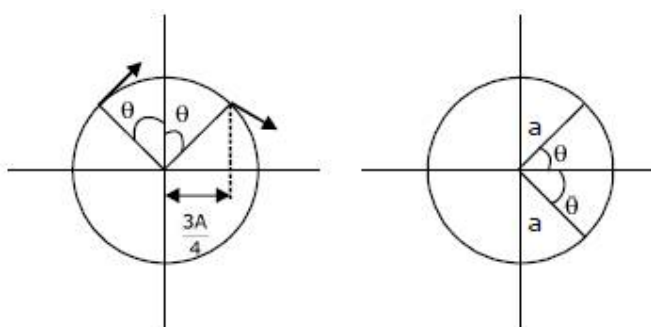
$$\left(\frac{n_3}{n_2} - 1 \right) t \times \frac{2\pi}{\lambda_2}$$

$$\therefore \frac{\lambda_1}{\lambda_2} = \frac{n_2}{n_1}$$

$$\Rightarrow \left(\frac{n_3}{n_2} - 1 \right) t \times \frac{2\pi n_2}{\lambda_1 n_1}$$

$$= \frac{2\pi}{\lambda_1 n_1} (n_3 - n_2) t$$

16.

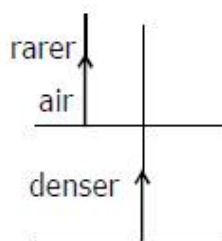


$$\Rightarrow \cos \theta = \frac{\sqrt{7}}{4}$$

$$\Rightarrow a \cos \theta = \frac{a\sqrt{7}}{4}$$

17. Conceptual

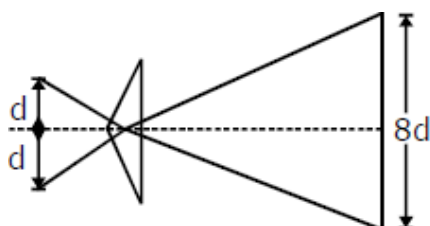
18.



$$2ut = \frac{\lambda}{2}$$

$$t = \frac{\lambda}{4}$$

19.



$$d = (\mu - 1) A \times 1$$

$$\text{no. of fringes} = \frac{8d^2 \cdot 2}{\lambda D}$$

$$= \frac{16d^2}{\lambda D} = \frac{16[(\mu - 1)A \cdot 1]^2}{6000 \times 10^{-10} \times 5}$$

$$= 5.33$$

20. Distance between the object the image $d = 3\text{m}$

Maximum focal length of the convex = f_{max}

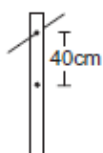
For real images, the maximum focal length is given as $f_{\text{max}} = \frac{d}{4} = \frac{3}{4} = 0.75\text{m}$

Hence for the required purpose, the maximum possible focal length of the convex lens is 0.75m

21. Let the mass of the stick be m . The moment of inertia of the stick about the axis of rotation through the point of suspension is

$$I = \frac{ml^2}{12} + md^2$$

where $l \square\square 1\text{ m}$ and $d \square\square 40\text{ cm}$.



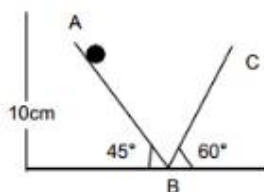
The separation between the centre of mass of the stick and the point of suspension is $d = 40\text{ cm}$. The time period of this physical pendulum is

$$T = 2\pi \sqrt{\frac{I}{mgd}}$$

$$= 2\pi \sqrt{\left(\frac{ml^2}{12} + md^2\right) / (mgd)}$$

$$= 2\pi \left[\sqrt{\left(\frac{1}{12} + 0.16\right) / 4} \right] \text{ s} = 1.55 \text{ s.}$$

- 22.



Let the time taken to travel AB and BC be t_1 and t_2 respectively

For part AB, $a_1 = g \sin 45^\circ$. $s_1 = \frac{0.1}{\sin 45^\circ} = 2\text{m}$

Let, v = velocity at B

$$\therefore v^2 - u^2 = 2a_1 s_1$$

$$\Rightarrow v^2 = 2 \times g \sin 45^\circ \times \frac{0.1}{\sin 45^\circ} = 2$$

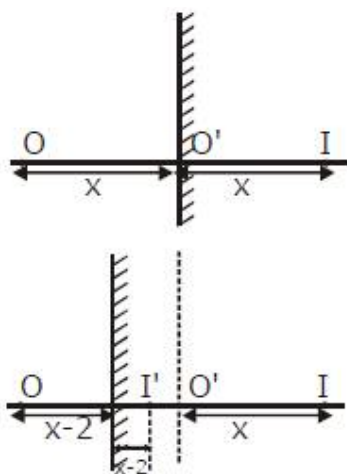
$$\Rightarrow v = \sqrt{2} \text{ m/s}$$

$$\therefore t_1 = \frac{v - u}{a_1} = \frac{\sqrt{2} - 0}{\frac{g}{\sqrt{2}}} = \frac{2}{g} = \frac{2}{10} = 0.2 \text{ sec}$$

Again for part BC, $a_2 = -g \sin 60^\circ$, $u = \sqrt{2}$, $v = 0$

$$\therefore t_2 = \frac{0 - \sqrt{2}}{-g \left(\frac{\sqrt{3}}{2} \right)} = \frac{2\sqrt{2}}{\sqrt{3}g} = \frac{2 \times (1.414)}{(1.732) \times 10} = 0.165 \text{ sec.}$$

23.



$$\begin{aligned} II' &= O'I + OO' - 2(OI') \\ &= x + x - 2(x - 2) \\ II' &= 4\text{cm} \end{aligned}$$

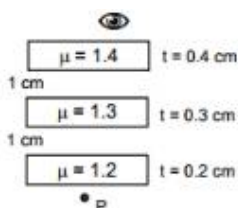
24. Deviation produced by plane mirror

$$\delta = 180 - 2i$$

$$\text{here } i = 90 - 90 = 30^\circ$$

$$\delta = 180 - 60 = 120^\circ$$

25.



The presence of air medium in between the sheets does not affect the shift.

The shift will be due to 3 sheets of different refractive index other than air.

$$\begin{aligned} &= \left(1 - \frac{1}{1.2}\right)(0.2) + \left(1 - \frac{1}{1.3}\right)(0.3) + \left(1 - \frac{1}{1.4}\right)(0.4) \\ &= 0.2 \text{ cm above point P.} \end{aligned}$$

26. Intensity of light = K units

Phase difference corresponding to λ is 2π and phase difference corresponding to $\lambda/3$ is $2\pi/3$.

Now using the formula of intensity,

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

$$\text{Let } I_1 = I_2 = I_0$$

In the first case, when path difference is λ

$$K = I_0 + I_0 + 2I_0 \cos 2\pi = 4I_0$$

In the second case, when path difference is $\lambda/3$

$$K' = I_0 + I_0 + 2I_0 \cos \frac{2\pi}{3} = I_0 + I_0 - 2I_0 \left(\frac{1}{2} \right) = I_0$$

Now

$$\frac{K'}{K} = \frac{I_0}{4I_0} = \frac{1}{4} = K' = \frac{K}{4}$$

27.

$$\frac{n\lambda_R D}{d} = (n+1) \frac{\lambda_B D}{d}$$

$$\Rightarrow n \cdot 7800 = (n+1) 5200$$

$$\Rightarrow n = 2$$

28. Wavelength of the light, $\lambda = 600 \text{ nm}$

The angular width of the fringe formed, $\theta = 0.1^\circ = 0.1\pi/180$

Spacing between the slits, $d = \lambda / \theta = (600 \times 10^{-9} \times 180) / (0.1 \times 3.14)$

$$= 108000 \times 10^{-9} / 0.314$$

$$d = 3.44 \times 10^{-4} \text{ m}$$

29.

Maximum mass that the scale can read, $M = 50 \text{ kg}$

Maximum displacement of the spring = Length of the scale, $l = 20$

cm = 0.2 m

Time period, $T = 0.6 \text{ s}$

Maximum force exerted on the spring, $F = Mg$

where,

g = acceleration due to gravity = 9.8 m/s^2

$$F = 50 \times 9.8 = 490$$

$$\therefore \text{Spring constant, } k = \frac{F}{l} = \frac{490}{0.2} = 2450 \text{ N/m}$$

Mass m , is suspended from the balance.

$$\text{Time period, } T = 2\pi \sqrt{\frac{m}{K}}$$

$$\begin{aligned}\therefore m &= \left(\frac{T}{2\pi}\right)^2 \times k \\ &= \left(\frac{0.6}{2 \times 3.14}\right)^2 \times 2450 \\ &= 22.36 \text{ kg}\end{aligned}$$

$\therefore$ Weight of the body = mg

$$= 22.36 \times 9.8$$

$$= 219.167 \text{ N}$$

Hence, the weight of the body is about 219 N.

30.

We know that $\eta = \frac{\text{real distance}}{\text{apparent distance}}$

$$\eta = \frac{12.5}{9.4} = 1.329$$

When the water is replaced by liquid of 1.63

$$\eta = \frac{12.5}{\text{apparent depth}}$$

$$1.63 = \frac{12.5}{x}$$

$$x = 7.67$$

So Distance by which microscope have to be moved is $9.4 - 7.67 = 1.73 \text{ cm}$

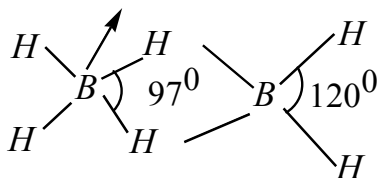
CHEMISTRY

31. As per NCERT Page - 309

32. $B - 6887, Al - 5137, Ga - 5520, In - 5082, \pi - 5437$

33. $B \rightarrow MP: 303K, BP: 2676K$ Due to its availability in liquid state is over wide range

34.



35. $H_2 - 74 \text{ pm}$

$F_2 - 144 \text{ pm}$

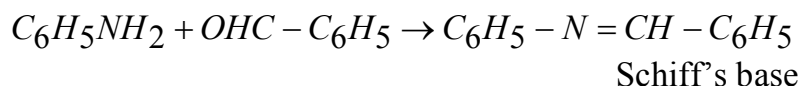
$O_2 - 121 \text{ pm}$

$HF - 92 \text{ pm}$

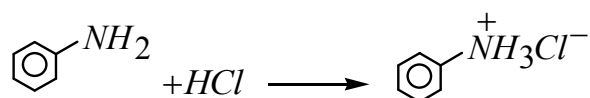
36. NCERT Reference : Structure of atom, XI, part-1 page no - 55

37. NCERT Reference : Structure of atom, XI, part-1 page no - 71

38. NCERT Reference : Structure of atom, XI, part-1 page no - 45
 39. NCERT Reference : Structure of atom, XI, part-1 page no - 41
 40. NCERT Reference : Chemical Kinetics, XII, page no - 100
 41. NCERT Reference : Chemical Kinetics, XII, page no - 104
 42. NCERT Reference : Chemical Kinetics, XII, page no - 116
 43. NCERT Reference : Chemical Kinetics, XII, page no - 117
 44. NCERT Reference : Chemical Kinetics, XII, page no - 105
 45. NCERT Reference : Chemical Kinetics, XII, page no - 104
 46.



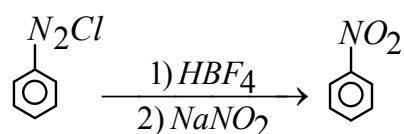
47.



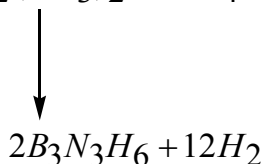
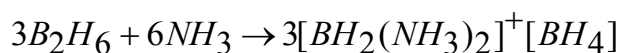
48. NCERT

49. 2

50.



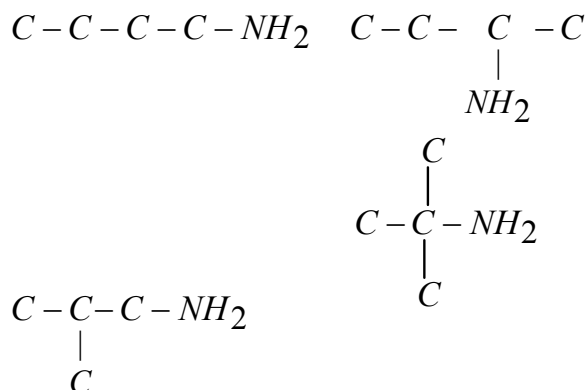
51.



3 moles B_2H_6 produce 12 mole of $\text{H}_2 = 12 \times 22.4$

6 moles B_2H_6 produce 24 mole of $\text{H}_2 = 24 \times 22.4$
 $= 537.6\text{ lts}$

52.



53. NCERT

54. conceptual NCERT

55. BC

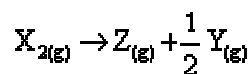
56. $t_{1/2} = 20 \text{ min}, t'_{1/2} = 10 \text{ min} \text{ \& } [A]_0' = 2[A]_0$

$$\therefore t_{1/2} \propto \frac{1}{[A]_0^{n-1}} \Rightarrow \frac{t'_{1/2}}{t_{1/2}} = \left[\frac{[A]_0'}{[A]_0} \right]^{n-1}$$

$$\text{or } \frac{20}{10} = \left[\frac{2 \times [A]_0}{[A]_0} \right]^{n-1} \rightarrow 2 = 2^{n-1} \rightarrow n-1=1 \Rightarrow n=2$$

57. $2+3=5$

58.



$$t = 0 \quad 150 \quad 0 \quad 0$$

$$t = 10 \text{ min} \quad 150-x \quad x \quad \frac{x}{2}$$

$$\text{Total pressure} = 150-x + x + \frac{x}{2}$$

$$170 = 150 + \frac{x}{2}$$

$$x = 40 \text{ mm}$$

$$\text{Rate of disappearance of } x_2 = \frac{40}{10} = 4 \text{ mm/min}$$

59. For zero order reaction $t_{1/2} \propto [a]$

For second order reaction $t_{1/2} \propto \frac{1}{[a]}$

$$[a] = \text{initial conc.}$$

$$\text{So, } t_2 = 10 \text{ sec and } t_3 = 80$$

$$\text{So, } t_3/t_2 = 8$$

60. $\frac{2.303}{k} \log \frac{100}{100-99.99} = x * \frac{2.303}{k} \log \frac{100}{100-90} \quad \text{Or, } \log 10^4 = x \log 10 \Rightarrow x = 4$

MATHEMATICS

61. Let α, β be roots of $x^2 - bx + c = 0$

$$\text{Then } \alpha + \beta = b$$

$$\Rightarrow \text{one of the roots is '2' (since } \alpha, \beta \text{ are primes and b is odd positive integer)}$$

$$\therefore f(2) = 0 \Rightarrow 2b - c = 4 \text{ and } b + c = 35$$

$$\therefore b = 13, c = 22$$

$$\text{Minimum value} = f\left(\frac{13}{2}\right) = -\frac{81}{4}$$

62. conceptual

63. $x^2 + (\log_3 2 + \log_2 3)x + \log_2 \left(\sqrt{6} \cdot \sqrt{\frac{2}{3}} \right) < 0$

$$\Rightarrow x^2 + (\log_3 2 + \log_2 3)x + 1 < 0$$

$$\Rightarrow (x + \log_3 2)(x + \log_2 3) < 0$$

$$\Rightarrow -\log_2 3 < x < -\log_3 2, [\because \log_3 2 < 1 \text{ \& } \log_2 3 > 1]$$

$$\therefore x \in (-\log_2 3, -\log_3 2) \text{ or } x \in \left(\log_2 \frac{1}{3}, \log_3 \frac{1}{2}\right)$$

$$\text{Or } x \in (\log_{1/2} 3, \log_{1/3} 2)$$

$\Rightarrow$ all options are correct

64. Let $x^2 + x + 1 = t \Rightarrow t \in \left[\frac{3}{4}, \infty\right)$

$$\Rightarrow t = \frac{1}{a-5} \Rightarrow \frac{1}{a-5} \geq \frac{3}{4}$$

65. $x^3 + 3x^2 + 3x + 2 = 0$
 $\Rightarrow (x^3 - 1) + 3(x^2 + x + 1) = 0 \Rightarrow (x^2 + x + 1)(x + 2) = 0$
 $\Rightarrow x = -2, \omega, \omega^2$

Common roots ω, ω^2 , $\therefore \omega, \omega^2$ are common roots of two equations $\therefore a = b = c = 1$

66. Conceptual

67. if $t = \log_2 x$, the given relation becomes

$$\frac{1}{t^2} - t^2 = \frac{1-t}{1+t} \Rightarrow (t-1)(t^4 + 2t^3 + t^2 + 2t + 1) = 0$$

$$\Rightarrow t = 1, t^2 + 2t + 1 + \frac{2}{t} + \frac{1}{t^2} = 0$$

$$\text{or } \left(t + \frac{1}{t}\right)^2 + 2\left(t + \frac{1}{t}\right) - 1 = 0$$

$$\therefore t + \frac{1}{t} = -(\sqrt{2} + 1) \in \mathbb{R} - (-2, 2)$$

The sum of its roots is $t_1 + t_2 = -(\sqrt{2} + 1)$

The product of the value of x is $2^{t+t_1+t_2} = 2^{-\sqrt{2}}$

68. The equation $[\tan^2 x] - \tan x - a = 0$ is true only if $\tan x$ is an integer. Since $[\tan^2 x]$ and a both are integers.

$$\Rightarrow \tan x = \frac{1 \pm \sqrt{4a+1}}{2} = \text{both value are integers}$$

$$\Rightarrow 4a+1 = (2k-1)^2 \Rightarrow k^2 - k = a \text{ and } a \leq 100$$

$\Rightarrow$ there are 9 values of a satisfied

69. If $x = n \in \mathbb{Z}$, $|n - 2n| = 4 \Rightarrow n = \pm 4$

If $x = n + K$ where $0 < K < 1$ then $|n - 2(n+k)| = 4$, it is possible if $K = \frac{1}{2}$

$$\Rightarrow |-n-1| = 4$$

$$\therefore n = 3, -5$$

70. $\log_{17} \log_{11}(\sqrt{x+11} + \sqrt{x}) = 0 \dots\dots\dots (1)$

Equation (1) is defined if $x \geq 0$.

We can rewrite (1) as $\log_{11}(\sqrt{x+11} + \sqrt{x}) = 17^0 = 1$

$$\Rightarrow \sqrt{x+11} + \sqrt{x} = 11^1 = 11$$

$$\Rightarrow \sqrt{x+11} = 11 - \sqrt{x}$$

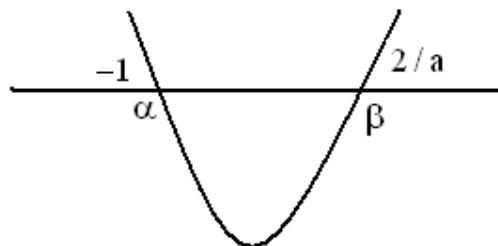
Squaring both sides we get $x+11 = 121 - 22\sqrt{x} + x$

$$\Rightarrow 22\sqrt{x} = 110 \Rightarrow \sqrt{x} = 5 \text{ or } x = 25$$

This clearly satisfies (1). Thus, sum of all the values satisfying (1) is 25

71. Let $f(x) = ax^2 + (a-2)x - 2$

$f(x)$ is negative for two integral values of x , so graph should be vertically upward parabola i.e., $a > 0$



Let two roots of $f(x) = 0$ are α and β then $\alpha, \beta = \frac{-(a-2) \pm (a+2)}{2a}$

$$\Rightarrow \alpha = -1, \beta = \frac{2}{a} \Rightarrow 1 < \beta \leq 2 \Rightarrow 1 < \frac{2}{a} \leq 2 \Rightarrow a \in [1, 2]$$

72. $\sin^2 x - 2\sin x + 5 = (\sin x - 1)^2 + 4 \geq 4$

$$\therefore 2^{\sqrt{\sin^2 x - 2\sin x + 5}} \geq 2^2 = 4$$

and $\sin^2 y \leq 1 \Rightarrow \frac{1}{4^{\sin^2 y}} \geq \frac{1}{4}$

$\therefore$ LHS ≥ 1 and according to question LHS ≤ 1 , so therefore, LHS = 1 for which

$$\therefore \sin^2 x - 2\sin x + 5 = 4 \text{ and } \operatorname{cosec}^2 y = 1, \sin^2 y = 1 \text{ or } \cos y = 0$$

$$(\sin x - 1)^2 = 0$$

$$y = 0$$

$$\sin x = 1 \Rightarrow x = (2n + 1)\frac{\pi}{2}$$

$$y = (2m + 1)\frac{\pi}{2}$$

73. $\cos A + \cos B = 4\sin^2\left(\frac{C}{2}\right)$

$$\Rightarrow 2\cos\frac{A+B}{2}\cos\frac{A-B}{2} = 4\sin^2\left(\frac{C}{2}\right)$$

$$\Rightarrow 2\sin\frac{C}{2}\cos\frac{A-B}{2} = 4\sin^2\frac{C}{2}$$

$$\Rightarrow 2\cos\frac{A-B}{2} = 4\sin\frac{C}{2}$$

$$\Rightarrow 2\cos\frac{A-B}{2}\sin\frac{A+B}{2} = 4\sin\frac{C}{2}\sin\frac{A+B}{2}$$

$$\Rightarrow \sin A + \sin B = 2.2\sin\frac{C}{2}\cos\frac{C}{2}$$

$$\Rightarrow \sin A + \sin B = 2\sin C \Rightarrow a + b = 2c.$$

$$\Rightarrow a, c, b \text{ are in A.P}$$

74. $\sin^{-1}(\sin 8) = 3\pi - 8$ $\tan^{-1}(\tan 8) = 8 - 3\pi$ but we have $f(x) + f(-x) = 2 \Rightarrow f(-x) = 2 - \alpha$

$$75. \quad \tan^{-1}\left(\frac{\tan x}{4}\right) + \tan^{-1}\left(\frac{3\left(\frac{2 \tan x}{1 + \tan^2 x}\right)}{5 + 3\left(\frac{1 - \tan^2 x}{1 + \tan^2 x}\right)}\right)$$

$$\tan^{-1}\left(\frac{\tan x}{4}\right) + \tan^{-1}\left(\frac{3 \tan x}{4 + \tan^2 x}\right) = \tan^{-1}\left(\frac{\frac{\tan x}{4} + \frac{3 \tan x}{4 + \tan^2 x}}{1 - \frac{3 \tan^2 x}{4(4 + \tan^2 x)}}\right)$$

$$= \tan^{-1}\left(\frac{(16 + \tan^2 x) \tan x}{16 + \tan^2 x}\right) = x$$

$$76. \quad a \sin x + \cos 2x = 2a - 7 \Rightarrow a \sin x + 1 - 2 \sin^2 x = 2a - 7$$

$$2 \sin^2 x - a \sin x + 2a - 8 = 0$$

$$\sin x = \frac{a \pm \sqrt{a^2 - 8(2a - 8)}}{4} = \frac{a \pm (a - 8)}{4} \Rightarrow \sin x = \frac{a - 4}{2}$$

$$-1 \leq \sin x \leq 1 \Rightarrow -1 \leq \frac{a - 4}{2} \leq 1 \Rightarrow -2 \leq a - 4 \leq 2$$

$$2 \leq a \leq 6$$

$$77. \quad (a - b)^2(a + c - b) + (b - c)^2(b + a - c) + (c - a)^2(c + b - a) = 0 \Rightarrow a = b = c$$

$$78. \quad LHS \in [2, \infty) \quad RHS \in [0, 2]$$

$$\therefore LHS = RHS = 2$$

$$\therefore x = 1 \text{ \& } \tan^2(\pi(1 + y)) = 1$$

$$\text{Leasty} = 1/4$$

$$79. \quad \sum_{r=1}^{10} \tan^{-1}\left(\frac{3}{1 + (3r - 1)(3r + 2)}\right) = \sum_{r=1}^{10} \tan^{-1}\left(\frac{3r + 2 - (3r - 1)}{1 + (3r + 2)(3r - 1)}\right) \quad \sum_{r=1}^{10} (\tan^{-1}(3r + 2) - \tan^{-1}(3r - 1))$$

$$\tan^{-1} 5 - \tan^{-1} 2 + \tan^{-1} 8 - \tan^{-1} 5 + \dots + \tan^{-1}(32) - \tan^{-1}(29)$$

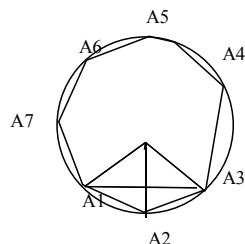
$$\tan^{-1}\left(\frac{32 - 2}{1 + 64}\right) = \tan^{-1}\left(\frac{6}{13}\right) = \cot^{-1}\left(\frac{13}{6}\right)$$

$$80. \quad \text{Draw the graph of } \sec \theta + \csc \theta$$

$$81. \quad \text{Required value} = \frac{a+b+c}{b+c-a} + \frac{a+b+c}{c+a-b} + \frac{a+b+c}{a+b-c} - 3 \text{ using A.M} \geq \text{H.M minimum value is}$$

$$9 - 3 = 6$$

$$82.$$



$$\frac{1}{2r \sin \frac{\pi}{n}} = \frac{1}{2r \sin \frac{2\pi}{n}} + \frac{1}{2r \sin \frac{3\pi}{n}} \quad \frac{1}{\sin \frac{\pi}{n}} = \frac{\sin \frac{3\pi}{n} + \sin \frac{2\pi}{n}}{\sin \frac{2\pi}{n} \sin \frac{3\pi}{n}}$$

$$\sin \frac{2\pi}{n} \sin \frac{3\pi}{n} = \sin \frac{\pi}{n} \sin \frac{3\pi}{n} + \sin \frac{\pi}{n} \sin \frac{2\pi}{n} \quad \cos \frac{\pi}{n} - \cos \frac{5\pi}{n} = \cos \frac{2\pi}{n} - \cos \frac{4\pi}{n} + \cos \frac{\pi}{n} - \cos \frac{3\pi}{n}$$

$$\cos \frac{5\pi}{n} - \cos \frac{3\pi}{n} = \cos \frac{4\pi}{n} - \cos \frac{2\pi}{n} \Rightarrow 2 \sin \frac{4\pi}{n} \sin \frac{\pi}{n} = -2 \sin \frac{3\pi}{n} \sin \frac{\pi}{n}$$

$$\frac{4\pi}{n} = \pi - \frac{3\pi}{n} \Rightarrow \frac{7\pi}{n} = \pi \Rightarrow \frac{7\pi}{n} = \pi \Rightarrow n = 7$$

83. $\frac{a^3 + b^3 + c^3}{a^3 + b^3 + c^3} \times 8R^3 = 8 \Rightarrow R = 1$ but $a \leq 2R \leq 2$

84. $2B = A + C$, $\lim_{A \rightarrow C} \frac{2\sqrt{3 - 2(\cos(A - C) - \cos(A + C))}}{|A - C|} B = 60^\circ$

$$= \lim_{A \rightarrow C} \frac{2\sqrt{2 \times 2 \sin^2 \frac{A - C}{2}}}{|A - C|} = \lim_{A \rightarrow C} 4 \left| \frac{\sin \frac{A - C}{2}}{\frac{A - C}{2}} \right| \times \frac{1}{2} = 2$$

85. $\left((\sec^{-1} x)^2 - (\sec^{-1} x)(\operatorname{cosec}^{-1} x) + (\operatorname{cosec}^{-1} x)^2 \right) = 7 \left(\frac{\pi}{2} \right)^2 - \frac{\pi^2}{4} - 3(\sec^{-1} x)(\operatorname{cosec}^{-1} x) = \frac{7\pi^2}{4}$

$$-3 \sec^{-1} x \left(\frac{\pi}{2} - \sec^{-1} x \right) = \frac{3\pi^2}{2} (\sec^{-1} x)^2 - \frac{\pi^2}{2} (\sec^{-1} x) - \frac{\pi^2}{2} = 0 \quad \sec^{-1} x = \frac{\frac{\pi}{2} \pm \sqrt{\frac{\pi^2}{4} + 2\pi^2}}{2} = \frac{\pi}{2} \pm \frac{3\pi}{2}$$

$$\sec^{-1} x = \pi \text{ or } \sec^{-1} x = -\frac{\pi}{2} \Rightarrow x = -1$$

86. $\cos^2 x = 2a^2 - 1$
 $\sin^2 x = 1 - \cos^2 x$
 $= 1 - 2a^2 + 1$
 $= 2(1 - a^2)$
 $2a^2 + \sin^2 x = 2$
 $\max \text{ value} = \sqrt{2} - 0$
 $= \sqrt{2}$

87. $|x| + |y| = 1$ is square and $\sin\left(\frac{x+y}{2}\right) = 0 \Rightarrow x = -y$

Given $x > 0 \Rightarrow$ no. of solutions = 2

88. The given equation can be written as

$$\frac{1 + \sin x}{\cos x} = 2 \cos x$$

$$\Rightarrow 1 + \sin x = 2 \cos^2 x = 2(1 - \sin^2 x)$$

$$\Rightarrow 2 \sin^2 x + \sin x - 1 = 0$$

$$\Rightarrow \sin x = -1 \text{ or } 1/2$$

Now $\sin x = -1 \Rightarrow \tan x$ and $\sec x$ not defined. $\sin x = 1/2 \Rightarrow x = \pi/6$ or

$$5\pi/6$$

The required number of solution is 2

89. $x^2 + y^2 = 3 + 2 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right) = 2$

90. Equation is true only when $\sin^2 y = 1$ $\tan^4 x = 1$

$$\Rightarrow x = \pm \frac{\pi}{4}, y = \pm \frac{\pi}{2}$$



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant
ICON Central Office – Madhapur – Hyderabad

Sec: **Sr.ICON-Batch-1 & 2**
Time: 09.00 Am to 12.00 Pm

Jee-Main
PTM-10

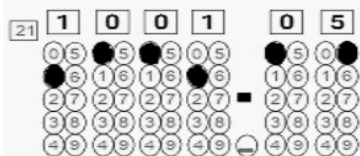
Date: 29-04-2021
Max.Marks:300

IMPORTANT INSTRUCTIONS:

- 1. Immediately fill in the Admission number on this page of the Test Booklet with Blue/Black Ball Point Pen only.
- 2. The candidates should not write their Admission Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
- 3. The test is of **3 hours** duration.
- 4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
- 5. There are **three** parts in the question paper 1, 2, 3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **two sections**.

(I) Section-I contains 20 **multiple choice** questions with **only one correct** option.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

(II) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only. First 5 attempted questions will be considered if more than 5 questions attempted.
To cancel any attempted question bubble on the question number box.
For example: To cancel attempted question 21, bubble on 21 as shown below.



Question Answered for marking



Question Cancelled for marking

Marking scheme : +4 for correct answer and 0 in all other cases.

- 6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
- 7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall.
- 8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
- 9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
- 10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capitals): _____

Admission Number:

Candidate's Signature: _____ Invigilator's Signature: _____

29-04-2021_Sr.ICON-Batch-1 & 2_Jee-Main_PT M-10_Test Syllabus

PHYSICS	: GRAVITATION, STATIC ELECTRICITY, STATIC ELECTRICITY, CURRENT ELECTRICITY, CAPACITORS
CHEMISTRY	: SOLID STATE, SOLUTIONS & COLLIGATIVE PROPERTIES, CHEMICAL KINETICS, IONIC & CHEMICAL EQUI, ATOMIC STRUCTURE, S-BLOCK & HYDEROGEN AND ITS COMPOUNDS
MATHEMATICS	: QUADRATIC EQUATIONS & EXPRESSIONS, SEQUENCES & SERIES, SEQUENCES & SERIES, MATRICES & DETERMINANTS, SYSTEM OF LINEAR EQUATIONS, STATISTICS & MATHEMATICAL REASONING, HEIGHTS & DISTANCES

PHYSICS

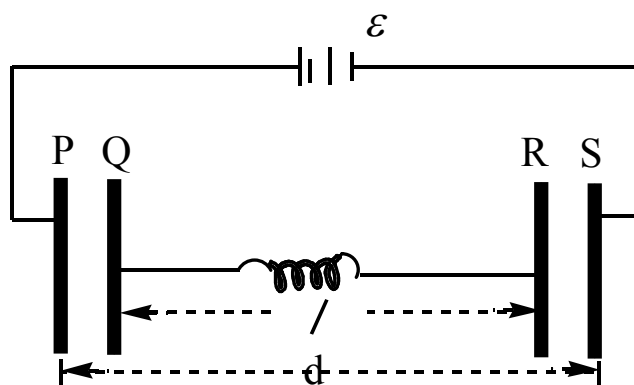
Max Marks: 100

(SINGLE CORRECT ANSWER TYPE)

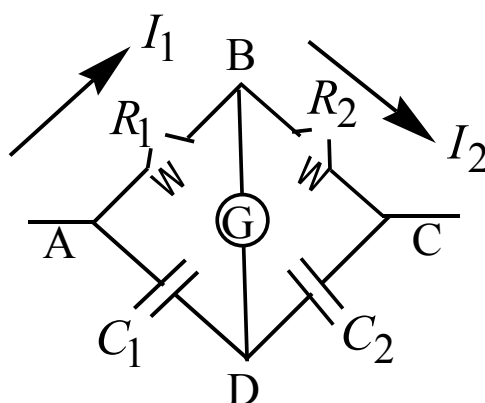
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

1. Two parallel plate capacitors with area A are connected through a conducting spring of natural length l in series as shown in Fig. Plates P and S have fixed positions at separation d . Now the plates are connected by a battery of emf ε as shown. If the extension in the spring in equilibrium is equal to the separation between the plates, find the spring constant k .

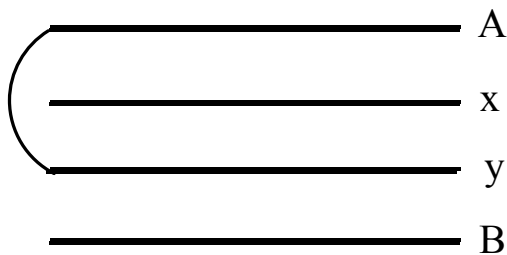


- 1) $k = \frac{27}{8} \frac{A\varepsilon_0\varepsilon^2}{(d-l)^3}$ 2) $k = \frac{27}{8} \frac{A\varepsilon_0\varepsilon}{(d-l)^3}$
- 3) $k = \frac{28}{8} \frac{A\varepsilon_0\varepsilon}{(d-l)^3}$ 4) $k = \frac{27}{8} \frac{A\varepsilon_0\varepsilon}{(d-l)^2}$
2. In the circuit in Fig. if no current flows through the galvanometer when the key k is closed, the bridge is balanced. The balancing condition for bridge is:

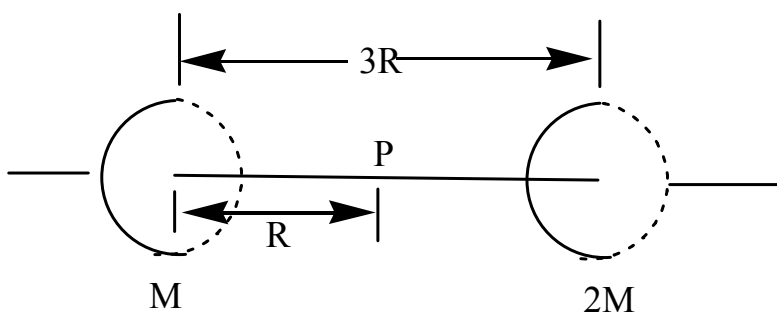


- 1) $\frac{C_1}{C_2} = \frac{R_1}{R_2}$ 2) $\frac{C_1}{C_2} = \frac{R_2}{R_1}$ 3) $\frac{C_1^2}{C_2^2} = \frac{R_1^2}{R_2^2}$ 4) $\frac{C_2^1}{C_2^2} = \frac{R_2}{R_1}$

3. In Figure, shows four identical metal plates, each of area A . The plates are kept at equal distance ' d ' successively. A source of emf 12 V is connected between x and B . Potential difference x and y is :

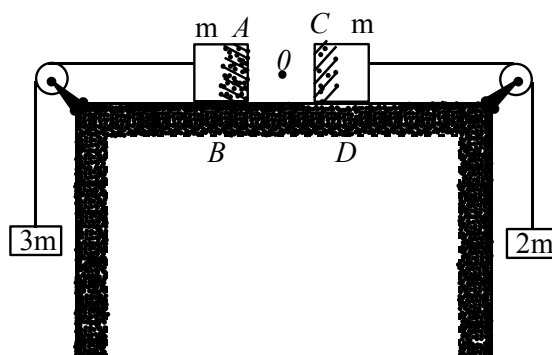


- 1) 10 V 2) 8 V 3) 6 V 4) 4 V
4. Two rings having masses M and $2M$, respectively, having the same radii are placed coaxially as shown in the figure. If the mass distribution on both the rings is non-uniform, then the gravitational potential at point P is

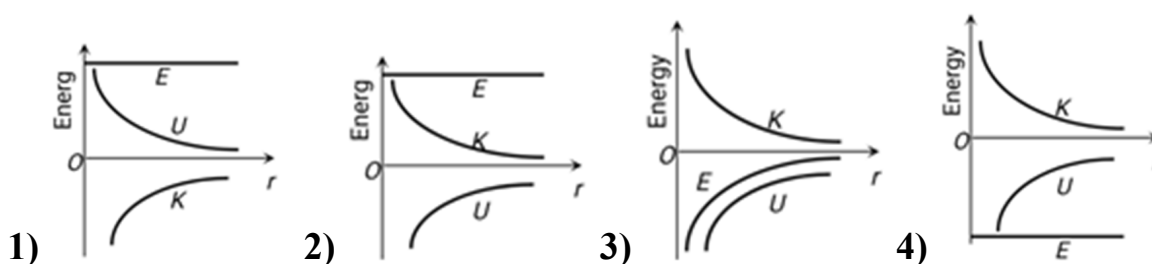


- 1) $-\frac{GM}{R} \left[\frac{1}{\sqrt{2}} + \frac{2}{\sqrt{5}} \right]$
- 2) $-\frac{GM}{R} \left[1 + \frac{1}{\sqrt{2}} \right]$
- 3) Zero
- 4) cannot be determined from the given information
5. In a young's double slit experiment, slits are illuminated by a monochromatic source of wavelength $6000\text{ \AA}$ and fringes are obtained. If screen is moved by a distance of 5 cm towards slits, change in fringe width is $3 \times 10^{-5}\text{ m}$. Then separation between the slits will be :
- 1) 1 mm 2) 1.2 mm 3) 1.5 mm 4) 1.63 mm

6. Two blocks each of mass m lie on a smooth table. They are attached to two other masses as shown in the figure – 5.. The pulleys and strings are light. An object O is kept at rest on the table. The sides AB & CD of the two blocks are made reflecting. The acceleration of two images formed in those two reflecting surfaces w.r.t. each other is :



- 1) $\frac{5g}{6}$ 2) $\frac{5g}{3}$ 3) $\frac{g}{3}$ 4) $\frac{17g}{6}$
7. The correct graph representing the variation of total energy (E_t) kinetic energy (E_k) and potential energy (U) of a satellite with its distance from the centre of earth is



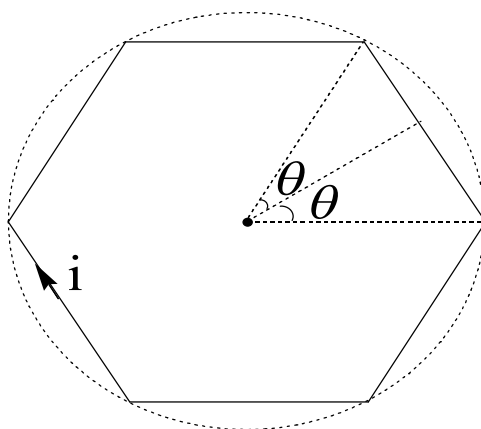
8. An electric dipole is placed along the X - axis at the origin O. A point P is at a distance of 20 cm from this origin such that OP makes an angle $\frac{\pi}{3}$ with the x-axis.

If the electric field at P makes an angle θ with the x-axis, the value of θ would be

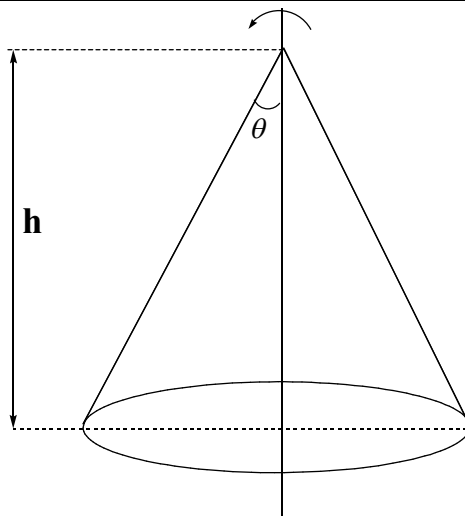
- 1) $\frac{\pi}{3}$ 2) $\frac{\pi}{3} + \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$ 3) $\frac{2\pi}{3}$ 4) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$
9. Electric charges $q, q, -2q$ are placed at the corners of an equilateral triangle ABC of side l . The magnitude of electric dipole moment of the system is

- 1) ql 2) $2ql$ 3) $\sqrt{3}ql$ 4) $4ql$

10. In an experiment for calibration of voltmeter, a standard cell of emf 1.5 V is balanced at 300 cm length of potentiometer wire. The P.D across a resistance in the circuit is balanced at 1.25 m. If a voltmeter is connected across the same resistance, it reads 0.65 V. The error in the voltmeter is:
- 1) 0.05 V 2) 0.025 V 3) 0.5 V 4) 0.25 V
11. Sun radiates energy in all direction. The average energy received at earth is $1.4 \text{ KW} / \text{m}^2$. The average distance between the earth and the sun is $1.5 \times 10^{11} \text{ m}$. If this energy is released by conservation of mass into energy, then the mass lost per day by the sun is approximately (use 1 day = 86400 sec)
- 1) $4.4 \times 10^9 \text{ kg}$ 2) $7.6 \times 10^{14} \text{ kg}$ 3) $3.8 \times 10^{12} \text{ kg}$ 4) $3.8 \times 10^{14} \text{ kg}$
12. In the following figure a wire bent in the form of a regular polygon of n sides is inscribed in a circle of radius a . Net magnetic field at centre will be :



- 1) $B = \frac{\mu_0 i}{2\pi a} \tan \frac{\pi}{n}$ 2) $B = \frac{\mu_0 n i}{2\pi a} \tan \frac{\pi}{n}$
- 3) $B = \frac{2\mu_0 n i}{\pi a} \tan \frac{\pi}{n}$ 4) $B = \frac{\mu_0 n i}{2a} \tan \frac{\pi}{n}$
13. 'Q' charge is uniformly distributed over the surface of a circular cone of semi vertex angle θ and height h . The cone is uniformly rotated about its axis at angular velocity ω . Calculate the associated magnetic dipole moment.



1) $\frac{Q\omega \tan^2 \theta h^2}{4} \hat{n}$

2) $\frac{Q\omega \tan^2 \theta h^2}{2} \hat{n}$

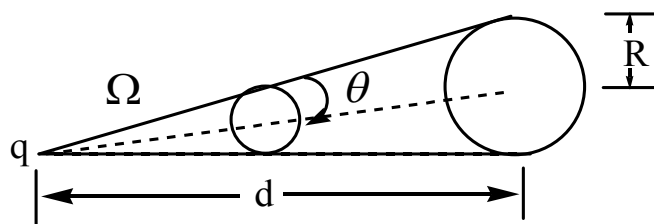
3) $\frac{Q\omega \tan^2 \theta h^2}{3} \hat{n}$

4) $Q\omega \tan^2 \theta h^2 \hat{n}$

14. A particle that carries a charge ' - q ' is placed at rest in uniform electric field $10N / C$. It experiences a force and moves. In a certain time 't' it is observed to acquire a velocity $(10\hat{i} - 10\hat{j})m / s$. The given electric field intersects a surface of area $1m^2$ in the x - z plane. Find the electric flux through the surface.

1) $2\sqrt{5}Nm^2 / c$ 2) $2\sqrt{3}Nm^2 / c$ 3) $5\sqrt{2}Nm^2 / c$ 4) $3\sqrt{2}Nm^2 / c$

15. A point charge q is placed at a distance d from the centre of a circular disc of radius R. Find electric flux through the disc due to that charge.



1) $\frac{q}{\epsilon_0} \left[1 - \frac{d}{\sqrt{d^2 + R^2}} \right]$

2) $\frac{q}{2\epsilon_0} \left[1 - \frac{d}{\sqrt{d^2 + R^2}} \right]$

3) $\frac{q}{\epsilon_0} \left[1 - \frac{R}{\sqrt{d^2 + R^2}} \right]$

4) $\frac{q}{2\epsilon_0} \left[1 - \frac{R}{\sqrt{d^2 + R^2}} \right]$

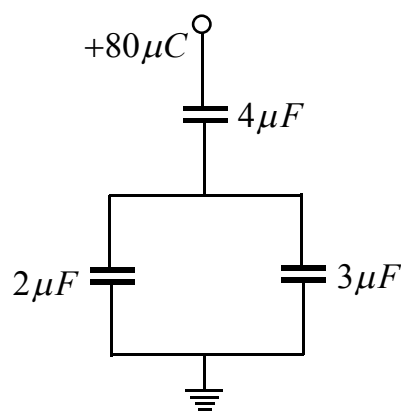
- 1) V 2) $\sqrt{2}V$ 3) $\frac{V}{\sqrt{2}}$ 4) $2V$

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30). **Attempt any five questions out of 10.**

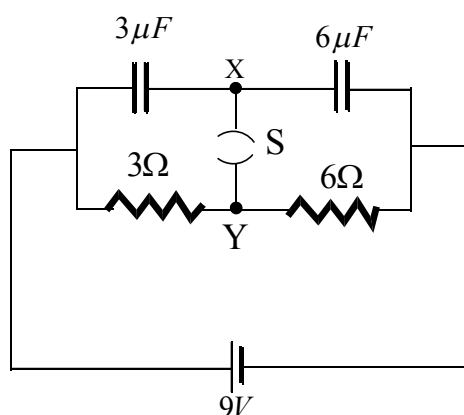
Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

21. The wave length of K_{α} X – rays produced by an X-ray tube is $0.76 \text{ \AA}$. What is the atomic number of the anode material of the tube ?
22. Light of wavelength $1800 \text{ \AA}$ ejects photoelectrons from a plate of a metal whose work function is 2eV . If a uniform magnetic field of $5 \times 10^{-5} \text{ tesla}$ is applied parallel to plate, what would be the radius of the path following by electrons ejected normally from the plate with maximum energy.
23. The current in a coil of self – inductance 2.0 henry is increasing according to $I = 2 \sin t^2 \text{ ampere}$. Find the amount of energy spent (in joule) during the period when the current changes from 0 to 2 ampere.
24. In the given circuit. A charge of $+80 \mu\text{C}$ is given to the upper plate of $4 \mu\text{F}$ capacitor. Then in the steady state, the charge on the upper plate of $3 \mu\text{F}$ capacitor in μC is.

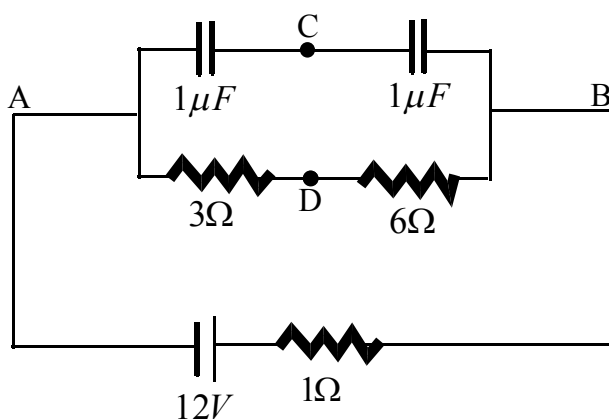


25. A $1 \mu\text{F}$ capacitor is charged to a potential difference of 1 volt. When it is allowed to discharge through an inductor having self – inductance of 1 mH , the maximum current in the circuit is found to be $\sqrt{500n} \text{ mA}$. The value of n is:
26. The bandwidth of optical fiber is 2 GHz and light of wavelength $1.3 \mu\text{m}$ is used for transmission of phone – calls. The maximum number of phone calls that could be carried by this fiber is 2.3×10^n . The value of n is.

27. A circuit is connected as shown in figure with the switch S open. When the switch is closed, the total amount of charge that flows from Y to X in μC is.



28. PD between the points C and D in the following fig. is.



29. A transistor is connected in common emitter (CE) configuration. The collector supply is 8 V and voltage drop across a resistor of 800Ω in the collector circuit is 0.5 V. If the current gain factor α is 0.96, find the base current in μA .
30. Gravitational acceleration on the surface of the planet is $\frac{\sqrt{6}}{11}g$, where g is the acceleration due to gravity on the surface of earth. The average mass density of the planet is $\frac{2}{3}$ time that of the earth. If the escape speed on the surface of the earth is taken to be $11km s^{-1}$, then find the escape speed on the surface of the planet (in $km s^{-1}$).

CHEMISTRY**(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

31. At 298K, the total pressure of an ideal solution obtained by mixing 3mol of A and 2mol of B is 184torr. Then the vapour pressure of pure B (in torr) at the same temperature is— (vapour pressure of A in pure state at 298K is 200torr)
 1) 180 2) 160 3) 16 4) 100
32. 50g of a saturated aqueous solution of KCl at 30°C is evaporated to dryness, when 13.2g of dry KCl was obtained. Then the solubility of KCl in water at 30°C is
 1) 35.87g 2) 3.587g 3) 17.95g 4) 1.795g
33. Lowering of vapour pressure due to solute in 1m aqueous solution at 100°C is
 1) 13.44mm of Hg 2) 14.12mm of Hg
 3) 31.2mm of Hg 4) 35.2mm of Hg
34. Highest freezing point was shown by
 1) 0.1m NaCl 2) 0.1m Al₂(SO₄)₃
 3) 0.1m K₄[Fe(CN)₆] 4) 0.1m Benzene
35. When one mole of solute is dissolved in 1Kg of water, the boiling point of water is determined as 100.5°C. then K_b of water is
 1) 0.5 2) 100 3) 100.5 4) 95.5
36. A first order reaction has half-life of 14.5hrs. what percent of the reactant will remain after 24hrs?
 1) 18.3% 2) 31.8% 3) 45.5% 4) 68.2%
37. A zero order reaction has $k=0.025\text{Ms}^{-1}$ for the disappearance of A what will be the concentration of A after 15sec if the initial concentration is 0.5M?
 1) 0.5M 2) 0.32M 3) 0.125M 4) 0.06M
38. What is the rate law expression for the reaction, $A + B \rightarrow C$, the following data were collected from the experiments

Exp no.	Initial [A] in M	Initial [B] in M	Rate of formation of C
1	0.1	0.1	4×10^{-4}
2	0.2	0.2	3.2×10^{-3}
3	0.1	0.2	1.6×10^{-3}

- 1) $K[A][B]^2$ 2) $K[A][B]$ 3) $K[A][B]^3$ 4) $K[A]^2[B]$

39. What % of reactant molecules will crossover the energy barrier at 325K? (heat of reaction is 0.12Kcal and activation energy of backward reaction is 0.02Kcal)
 1) 8.574 2) 84.75 3) 42.75 4) 4.275
40. Units of rate constant depends upon
 1) Temperature 2) Concentration of reactants
 3) Catalyst 4) Order of reaction
41. Equal volumes of two HCl solutions of $\text{PH}_1=3$ and $\text{PH}_2=2$ are mixed, then find the PH of final solution
 1) 3.3 2) 3.7 3) 7.0 4) 14
42. How much water must be added to 300ml of 0.2M solution of CH_3COOH ($K_a=1.8 \times 10^{-5}$) for the degree of ionization of acid to double?
 1) 600mL 2) 900mL 3) 1200mL 4) 1500ML
43. In a cubic unit cell, seven of the eight corners are occupied by atoms A and centres of faces are occupied by atoms B, then general formulae of the compound is
 1) A_7B_6 2) A_7B_{12} 3) A_7B_{24} 4) A_{24}B_7
44. Select the best indicator from the given table for titration of 20ml of CH_3COOH with 0.02M NaOH. Given PK_a of CH_3COOH is 4.74

	Indicator	PH range
I	Bromothymol blue	6-7.6
II	Thymolphthalein	9.3-10.5
III	Malachite green	11.4- 13
IV	M-Cresol purple	7.4-9.0

- 1) I 2) II 3) III 4) IV
45. Ammonium chloride and CsCl are similar in geometry. The edge length of unit cell is 387pm. If the size of Cl^- is 181pm, the size of ammonium ion is (in pm)
 1) 116pm 2) 154pm 3) 174pm 4) 206pm
46. When NaCl is doped with 10^{-5} mole % of CaCl_2 , then the number of cation vacancies are
 1) $10^{-5}N_A$ 2) $10^{-7}N_A$ 3) $2 \times 10^{-7} N_A$ 4) $2 \times 10^7 N_A$
47. K_{sp} of $\text{Zr}_3(\text{PO}_4)_4$ in terms of solubility (S) is
 1) $108S^7$ 2) $4S^3$ 3) $6912S^7$ 4) $27S^3$
48. The ionization enthalpy of hydrogen atom is $1.312 \times 10^6 \text{J/mol}$. The energy required to excite the electron in the atom from $n=1$ to $n=2$ is –(in J/mol)
 1) 9.84×10^5 2) 8.51×10^5 3) 6.56×10^5 4) 7.56×10^5

49. Not permissible one is
 1) $n=4, l=3, m=0$ 2) $n=4, l=2, m=1$ 3) $n=4, l=4, m=1$ 4) $n=4, l=0, m=0$
50. The wavelength of electron waves in two orbits is 3:5. The ratio of kinetic energy of electrons will be
 1) 25:9 2) 5:3 3) 9:25 4) 3:5

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30). **Attempt any five questions out of 10.**

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

51. Calculate the value of $X/5$, where X = Coordination number of 2D square close packing + coordination number of 2D-hcp + coordination number of 3D close packing + coordination number of 3D, ABC, ABC packing + coordination number of 3D AB,AB,AB packing
52. Identify the number of hydrogen bonded crystalline solids are in Teflon, Natural rubber, DNA, Naphthalene, sucrose, ice, Benzoic acid, H_2O_2
53. Number octahedral voids present in a crystal of radius ratio 0.8 are----
54. In zinc blende structure, the % fraction of vacancy in tetrahedral voids (per unit cell) is –
55. One molal solution of a carboxylic acid in benzene shows the elevation of boiling point of 1.518K. the degree of association (in terms of %) in the dimerization of acid in benzene is (K_b of benzene = 2.53K Kg/mol)
56. By X-ray diffraction it is found that nickel ($GAW=59$) crystallizes with ccp. The edge length of the unit cell is $3.5\text{\AA}$. Then the density of Ni crystal is (in g/cc)
57. Number of elements from following list give flame test Li, Na, K, Rb, Cs, Be, Mg, Ca, Sr, Ba
58. 100ml aqueous solution of glucose with osmotic pressure 1.2atm at 298K is mixed with 300ml aqueous solution of urea at 2.4atm at 298K. calculate the osmotic pressure of mixture
59. Normality of 11.2V H_2O_2 solution is
60. Total nodes possible for 5g sub shell are ---

MATHEMATICS**(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. Total number of words that can be formed using all letters of the word "BRIJESH" that neither begins with 'I' nor ends with 'B' is
 1) 4920 2) 4800 3) 3600 4) 3720
62. Let the term independent of X in the expansion of $\left(X^2 - \frac{1}{X}\right)^9$ has the value p and q be the sum of the coefficients of its middle terms, then $(p - q)$ equals
 1) 0 2) $2 \cdot {}^9C_4$ 3) 9C_5 4) 9C_3
63. If $1, \alpha_1, \alpha_2, \dots, \alpha_{2008}$ are $(2009)^{th}$ roots of unity, then the value of $\sum_{r=1}^{2008} r (\alpha_r + \alpha_{2009-r})$ equals
 1) 2009 2) 2008 3) 0 4) -2009
64. The value of x satisfying the equation $2 \tan^{-1}(3x) = \sin^{-1}\left(\frac{6x}{1+9x^2}\right)$ is equal to
 1) $\left[\frac{1}{3}, \infty\right)$ 2) $\left(-\infty, \frac{1}{3}\right]$ 3) $\left[\frac{-1}{3}, \frac{1}{3}\right]$ 4) $[-1, 1]$
65. $\lim_{x \rightarrow -\infty} \left(\sqrt{x^2 - \frac{4x}{3}} + 2 + x \right)$ is equal to
 1) $\frac{4}{3}$ 2) $\frac{-4}{3}$ 3) $\frac{8}{3}$ 4) $\frac{-8}{3}$
66. If $f(x) = \begin{cases} |x^5 - 4x^3 + 2| \cos(\pi x), & x \geq 0 \\ x^3 - \sin(x) - \left| (x^2 + 8x + 12)x^2 \right|, & x < 0 \end{cases}$ then number of points where $f(-|x|)$ is non derivable is
 1) 4 2) 5 3) 7 4) 8
67. If $\alpha, \beta \in \left(\frac{\pi}{2}, \pi\right)$ and $\alpha < \beta$, then which one of the following is true?
 1) $e^{\cos \alpha - \cos \beta} < \frac{\alpha}{\beta}$ 2) $e^{\cos \beta - \cos \alpha} < \frac{\beta}{\alpha}$ 3) $e^{\cos \alpha - \cos \beta} < \frac{\beta}{\alpha}$ 4) $e^{\cos \beta - \cos \alpha} < \frac{\alpha}{\beta}$

68. If F and f are different functions such that

$$F(x) = \int_0^x f(t) dt \text{ and if } F(a) = -2 \text{ and } F(b) = -2$$

where $a < b$ then which of the following must be true?

- 1) $f(x) = 0$ for some x such that $a < x < b$. 2) $f(x) < 0$ for all x such that $a < x < b$
 3) $F(x) \leq 0$ for all x such that $a < x < b$. 4) $F(x) = 0$ for some x such that $a < x < b$.

69. If $\int \frac{2 \sec x \cdot \tan x}{(\sec x - \tan x)^{10}} dx = (\sec x + \tan x)^{11} (m(\sec x - \tan x)^2 + n) + C$ then $\frac{1}{m} + \frac{1}{n}$ is equal to

- 1) 1 2) -1 3) -2 4) 2

70. The value of definite integral $\int_{-2010}^{2010} x^{2010} \cdot \cot^{-1}(2010x) dx$ is equal to

- 1) $\frac{\pi}{2011}(2011)^{2010}$ 2) $\frac{\pi}{2010}(2010)^{2011}$ 3) $\frac{\pi}{2011}(2010)^{2011}$ 4) $\frac{\pi}{2010}(2011)^{2010}$

71. Which of the following statements is a tautology?

- 1) $(\sim q \wedge p) \wedge q$ 2) $(\sim q \wedge p) \wedge (p \wedge \sim q)$
 3) $(\sim q \wedge p) \vee (p \vee \sim q)$ 4) $(p \wedge q) \wedge (\sim(p \wedge q))$

72. Let α, β are two roots of equation $x^2 + ax + b = 0$ ($a, b \in \mathbb{R}$ and $b \neq 0$). If the quadratic equation $g(x) = 0$ has two roots $\alpha + \frac{1}{\alpha}, \beta + \frac{1}{\beta}$ such that sum of roots is equal to product of roots, then the complete range of b is

- 1) $\left[\frac{1}{3}, 3\right]$ 2) $\left(\frac{1}{3}, 3\right]$ 3) $\left[\frac{1}{3}, 3\right)$ 4) $\left(-\infty, \frac{1}{3}\right) \cup (3, \infty)$

73. Let u, v, w satisfy the equations $uvw = -6, uv + vw + wu = -5, u + v + w = 2$, where $u > v > w$.

$\vec{V}_1 = u\hat{i} + v\hat{j} + w\hat{k}$ and $\vec{V}_2 = 2\hat{i} + 3\hat{j} - 5\hat{k}$ then the scalar product $\vec{V}_1 \cdot \vec{V}_2$ equals.

- 1) 9 2) 19 3) 21 4) 25

74. If the quadratic equations $3x^2 + ax + 1 = 0$ and $2x^2 + bx + 1 = 0$ have a common root, then the value of the expression $5ab - 2a^2 - 3b^2$ is

- 1) 0 2) 1 3) -1 4) 2

75. If e^λ and $e^{-\lambda}$ are the roots of equation $3x^2 - (a+b)x + 2a = 0, a, b, \lambda \in R, \lambda \neq 0$ then least integral value of b is
- 1) 4 2) 5 3) 9 4) 10
76. Consider three sequences an $A.P., a G.P. and a H.P.$ as follows
- $a, A_1, A_2, A_3, \dots, A_9, b$ ($A.P.$)
- $a, G_1, G_2, G_3, \dots, G_9, b$ ($G.P.$)
- $a, H_1, H_2, H_3, \dots, H_9, b$ ($H.P.$)
- If a, b are the roots of the equation $x^2 - (p+1)x + 16 = 0, p \in R$, then the value of $\prod_{r=1}^4 A_r H_{10-r} G_{10-2r}$ is
- 1) 16 2) 6 3) 4^9 4) 4^{12}
77. The Value of x satisfying the equation $(2011)^{\log_{10} x} = 11^{\log_{10}(1+3+5+\dots+4021)}$ lies in the interval
- 1) (95,100) 2) (100,105) 3) (110,115) 4) (120,125)
78. Let α be the solution of $16^{\sin^2 \theta} + 16^{\cos^2 \theta} = 10$ in $\left(0, \frac{\pi}{4}\right)$. If the shadow of a vertical pole is $\frac{1}{\sqrt{3}}$ of its height, then the altitude of the sun is
- 1) α 2) $\frac{\alpha}{2}$ 3) 2α 4) $\frac{\alpha}{3}$
79. The number of solutions of equation $\sin 3x - \cos x = 2$
- 1) 0 2) 1 3) 2 4) 3
80. A tower stands at the center of a circular park. A and B are two points on the boundary of the park such that $AB (= a)$ subtends an angle of 60° at the foot of the tower and the angle of elevation of the top of the tower from A or B is 30° . The height of the tower is
- 1) $\frac{2a}{\sqrt{3}}$ 2) $2a\sqrt{3}$ 3) $\frac{a}{\sqrt{3}}$ 4) $\sqrt{3}$

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30). **Attempt any five questions out of 10.**

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

81. If the system of linear equations
- $$\begin{aligned}(\cos \theta) X + (\sin \theta) Y + \cos \theta &= 0 \\ (\sin \theta) X + (\cos \theta) Y + \sin \theta &= 0 \\ (\cos \theta) X + (\sin \theta) Y - \cos \theta &= 0\end{aligned}$$
- is consistent, then the number of possible values of $\theta \in [0, 2\pi]$ is equal to
82. The total number of distinct $x \in R$ for which $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ is
83. Let A be the set of all 3×3 symmetric matrices whose diagonal elements are 1, then the number of matrices A for which the system of linear equations $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ represents three perpendicular planes is
84. A chord AB of an ellipse $\frac{x^2}{9} + \frac{y^2}{\lambda^2} = 1$ subtends a right angle at the centre of ellipse O (where O is origin), also $\frac{1}{OA^2} + \frac{1}{OB^2} = \frac{14}{45}$, then λ^2 is equal to
85. Let $A = [a_{ij}]$ be a square matrix of order two such that
- $$a_{ij} = \begin{cases} \cos \alpha & i = j \\ \cos \left(j \frac{\pi}{2} + \alpha \right) & i > j \\ \sin \left(j \frac{\pi}{2} - \alpha \right) & i < j \end{cases}, \text{ then at } \alpha = \frac{3\pi}{5}, \text{ absolute value}$$
- of sum of all the elements of matrix A^5 is
86. The sum of the first 10 terms of the series $\frac{7}{2^2 5^2} + \frac{13}{5^2 8^2} + \frac{19}{8^2 11^2} + \dots$ is $\frac{m}{n}$ then the value of $(n - 12m)$ is equal to

87. The circle lying in the first quadrant whose centre lies on the curve $y=2x^2-27$ is tangent to $4x-3y=0$ and the y -axis then the diameter of the circle is equal to.
88. If planes $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$, $\vec{r} \cdot (\hat{i} + 2\hat{j} + \hat{k}) = 2$ and $\vec{r} \cdot (a\hat{i} + a^2\hat{j} + \hat{k}) = 3$ intersect in a line, then the number of real values of a is equal to
89. Let α, β are the roots of the quadratic equation $ax^2 + bx + c = 0$ where $\beta = 4\alpha$ ($\alpha > 0$). If $3a = 2(c - b)$ and $S = \sum_{r=0}^{\infty} \beta(\alpha^r)$, then the value of $18S$ is equal to
90. If $\sqrt{3}, H, \sqrt{2}$ are in H.P., then the value of $\frac{H + \sqrt{2}}{H - \sqrt{2}} + \frac{H + \sqrt{3}}{H - \sqrt{3}}$ is equal to



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KEY SHEET

PHYSICS

1	1	2	2	3	4	4	1	5	1	6	4
7	3	8	2	9	3	10	2	11	4	12	2
13	1	14	3	15	2	16	1	17	3	18	2
19	2	20	3	21	0041.00	22	0.14 to 0.15	23	0004.00	24	0048.00
25	0002.00	26	0005.00	27	0027.00	28	0001.80	29	0026.00	30	0003.00

CHEMISTRY

31	2	32	1	33	1	34	4	35	1	36	2
37	3	38	1	39	2	40	4	41	1	42	2
43	3	44	4	45	2	46	2	47	3	48	1
49	3	50	1	51	0008.00	52	0005.00	53	0000.00	54	0050.00
55	0080.00	56	0009.00	57	0008.00	58	1.6	59	0002.00	60	0004.00

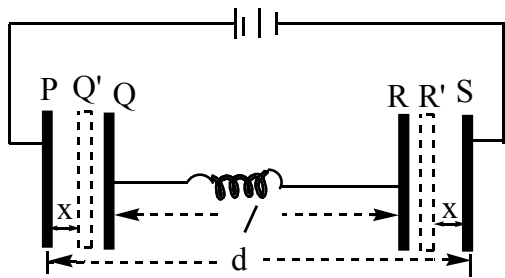
MATHEMATICS

61	4	62	4	63	4	64	3	65	3	66	2
67	2	68	1	69	4	70	3	71	3	72	1
73	2	74	2	75	2	76	4	77	4	78	3
79	1	80	3	81	0002.00	82	0002.00	83	0005.00	84	0005.00
85	0002.00	86	0004.00	87	0009.00	88	0000.00	89	0024.00	90	0002.00

SOLUTIONS

PHYSICS

1. Let charge on capacitors be q and separation between plates P & Q and R & S be x at any time. Distance between plates P & Q and R & S is same because force acting on them is same.



Capacitance of capacitor PQ, $C_1 = \frac{\epsilon_0 A}{x}$ Capacitance of capacitor RS, $C_2 = \frac{\epsilon_0 A}{x}$

From KVL, we have $\frac{q}{C_1} + \frac{q}{C_2} = \epsilon$ or $q = \frac{\epsilon_0 A \epsilon}{2x}$

At this moment extension in spring $= d - 2x - l$

Force on plate Q towards P. $F_1 = \frac{q^2}{2A\epsilon_0} = \frac{\epsilon_0^2 A^2 \epsilon^2}{8Ax^2 \epsilon_0} = \frac{A\epsilon_0 \epsilon^2}{8x^2}$ Spring force on

plate Q due to extension in spring At equilibrium,

Separation between plates = extension in spring

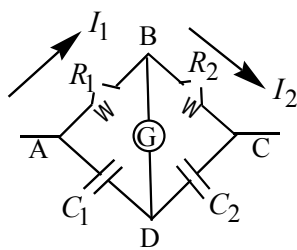
Thus, $x = y (= d - 2x - l)$ $x = \frac{d - l}{3} \dots\dots\dots(i)$

At equilibrium, $F_1 = F_2 \dots\dots\dots(ii)$

From Eqns. (i) and (ii), we have $\frac{A\epsilon_0 \epsilon^2}{8x^2} = ky = kx$ $x = \left(\frac{A\epsilon_0 \epsilon^2}{8k} \right) \dots\dots\dots(iii)$

From Eqns (i) and (iii), $\left(\frac{d - l}{3} \right)^3 = \frac{A\epsilon_0 \epsilon^2}{8k}$ $k = \frac{27}{8} \frac{A\epsilon_0 \epsilon^2}{(d - l)^3}$

2. In the steady state, no current is passing through the capacitor. Let the charge on each capacitor be q . Since the current through galvanometer between the ends of the galvanometer will be zero.

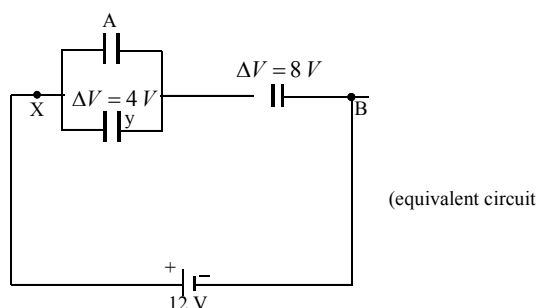


Therefore, $V_A - V_B = V_A - V_D$ $I_1 R_1 = \frac{q}{C_1} \dots\dots\dots(i)$

Similarly, $V_B - V_C = V_D - V_C$ $I_2 R_2 = \frac{q}{C_2} \dots\dots\dots(ii)$

Dividing Eq.(i) by eq.(ii), we get $\frac{I_1 R_1}{I_2 R_2} = \frac{q/C_1}{q/C_2} = \frac{C_2}{C_1}$ or $\frac{C_1}{C_2} = \frac{R_2}{R_1}$

3.



4. Potential at a pt on the ring at a distance x is

$$V = -\frac{GM}{\sqrt{R^2 + X^2}} \quad \text{So, } V_{net} = -\frac{GM}{\sqrt{2}R} - \frac{G(2M)}{\sqrt{5}R}$$

5. Fringe widths β and β' in a YDSE setup is given as

$$\beta = \frac{\lambda D}{d} \text{ and } \beta' = \frac{\lambda D'}{d} \Rightarrow \beta - \beta' = \frac{\lambda(D - D')}{d}$$

$$\Rightarrow d = \frac{\lambda(D - D')}{(\beta - \beta')} \Rightarrow d = \frac{6000 \times 10^{-10} \times 5 \times 10^{-2}}{3 \times 10^{-5}} = 10^{-3} \text{ m} = 1 \text{ mm.}$$

6. Acceleration of block $AB = \frac{3mg}{3m + m} = \frac{3}{4}g$;

Acceleration of block $CD = \frac{2mg}{2m + m} = \frac{2}{3}g$

Acceleration of image in mirror AB = 2 acceleration of mirror $= 2\left(\frac{-3g}{4}\right) = \frac{-3}{2}g$

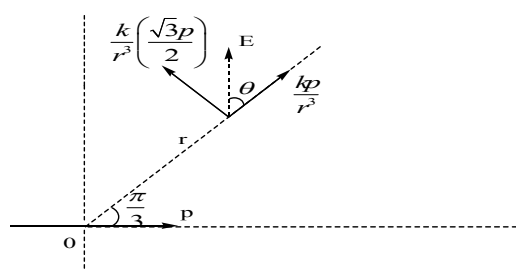
Acceleration of image in mirror $CD = 2\left(\frac{2g}{3}\right) = \frac{4g}{3}$

Acceleration of the two image w.r.t each other is given as

$$a_{rd} = \frac{4g}{3} - \left(\frac{-3g}{2}\right) = \frac{17g}{6}.$$

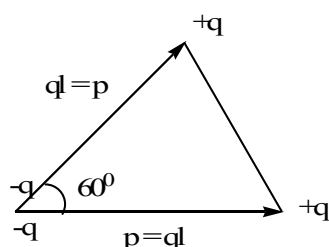
7. $PE = -\frac{GMm}{R} \quad KE = \frac{GMm}{R^2}, \quad TE = -\frac{GMm}{R^2}$

8.



$$\theta = \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

9.



Net momentum is $= \sqrt{3}p$

$$= \sqrt{3}(ql)$$

10. $300 \times 10^{-3} m \Rightarrow 1.5 V; 1.25 m \rightarrow ?$ $V = 0.625$

Error in ammeter reading = $0.625 - 0.65 = 0.025 V$

11. Power of Sun is $P_s = 1.4 \times 10^3 \times 4\pi (1.5 \times 10^{11})^2$

Energy released by Sun per day is $E_{\text{sun/day}} = P_s \times 86400$

Mass lost by sun per day is $m = \frac{E_{\text{sun/day}}}{C^2}$

$$m = \frac{1.4 \times 10^3 \times 4 \times 3.14 \times (1.5 \times 10^{11})^2 \times 86400}{(3 \times 10^8)^2} \quad m = 3.79 \times 10^{14} \text{ kg}$$

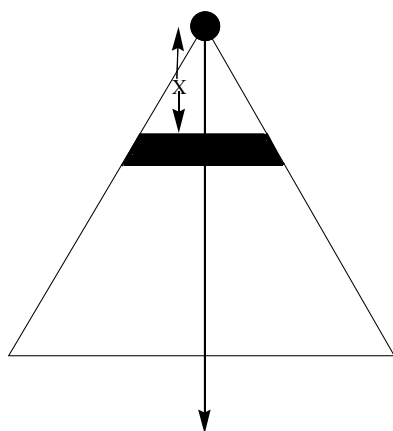
12. Magnetic field at the centre due to one side

$$B_1 = \frac{\mu_0 i}{2\pi r} \sin \theta, \text{ where } r = a \cos \theta$$

$$B_1 = \frac{\mu_0 i}{2\pi a \cos \theta} \sin \theta = \frac{\mu_0 i}{2\pi a} \tan \theta$$

$$\therefore B_{\text{net}} = nB_1$$

13. Surface area of the cone = $\pi h^2 \tan \theta \sec \theta$



Hence, charge per unit surface area = $\frac{Q}{\pi h^2 \tan \theta \sec \theta}$

$$= \frac{Q}{\pi h^2 \tan \theta \sec \theta} (2\pi)(x \tan \theta) \frac{dx}{(\cos \theta)} = \left(\frac{2Q}{h^2} \right) x dx$$

The current dl due to the rotation this charge is given by, $dl = \left(\frac{\omega}{2\pi} \right) \left(\frac{2Q}{h^2} \right) x dx$

Hence, the magnetic moment ($d\mu$) associated with this current would be,

$$d\vec{\mu} = \left(\frac{Q\omega}{\pi h^2} \right) (x)(\pi)(x \tan \theta)^2 (dx) \hat{n} = \frac{(Q\omega \tan^2 \theta)}{h^2} x^3 dx \hat{n} \quad (\hat{n} = \text{unit vector along the axis})$$

$$\vec{\mu} = \left(\frac{Q\omega \tan^2 \theta}{h^2} \right) \left(\int_0^h x^3 dx \right) \hat{n} = \frac{Q\omega \tan^2 \theta h^2}{4} \hat{n}$$

14. Force on charge $\vec{F} = q\vec{E}$ Particle moves opposite to $\vec{E}$ with $\vec{V}$

Unit vector in the direction of $\vec{V}$ is $\frac{\hat{i}}{\sqrt{2}} - \frac{\hat{j}}{\sqrt{2}}$

Unit vector in the direction of $\vec{E}$ is $\frac{\hat{i}}{\sqrt{2}} - \frac{\hat{j}}{\sqrt{2}}$

$$\vec{E} = 10 \left[-\frac{\hat{i}}{\sqrt{2}} + \frac{\hat{j}}{\sqrt{2}} \right], \text{ i.e., } \vec{A} = 1 \times \hat{j} \quad \text{Electric flux } \phi = \vec{E} \cdot \vec{A} = 5\sqrt{2} \text{ Nm}^2 / \text{C}$$

15. We know that total flux originated from a point charge q in all directions is $\frac{q}{\epsilon_0}$. This flux is originated in a solid angle 4π . In the given case solid angle subtended by the cone subtended by the disc at the point charge is $\Omega = 2\pi(1 - \cos\theta)$. So the flux of q which is passing through the surface of the disc is $\phi = \frac{q}{\epsilon_0} \frac{\Omega}{4\pi} = \frac{q}{2\epsilon_0}(1 - \cos\theta)$

From the figure, $\cos\theta = \frac{d}{\sqrt{d^2 + R^2}}$ so $\phi = \frac{q}{2\epsilon_0} \left\{ 1 - \frac{d}{\sqrt{d^2 + R^2}} \right\}$

16. $\lambda_A = \frac{12420}{E_{eV}} \text{ \AA} = \frac{12420}{14.4} = 862.5 \text{ \AA}$

Which lies in ultraviolet region $(\approx 100 \text{ \AA} \text{ to } 3800 \text{ \AA})$

17. $l = l_1 + l_2 + l_3$

$$\frac{k}{n} = \frac{k}{n_1} + \frac{k}{n_2} + \frac{k}{n_3} \quad \frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$$

18. Applying KVL in the loop $+4 + i \times 1 + i \times 2 + 2 + i \times 1 + i \times 2 + i \times 2 + 3 + i \times 1 - 3 + i \times 1 + i \times 2 = 0$
 $6 + 12i = 0 \quad i = -\frac{1}{2} \text{ A (Anticlockwise)}$

$$0 - V_A = +4 - \frac{1}{2} \times 3 + 2 - \frac{1}{2} \times 3 = 6 - \frac{6}{2} = 3V \quad \Rightarrow V_A = -3V$$

19. The potential due to dq at P is $dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$

The total potential at P due to the rod is

$$V = \int dV = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}$$

For uniform distribution of charge $dq = \lambda dr$ from $r_1 = l$ to $r_2 = x + l$.

Then, $V = \frac{1}{4\pi\epsilon_0} \int_x^{x+l} \frac{\lambda dr}{r} = \frac{\lambda}{4\pi\epsilon_0} \ln \frac{l+x}{x}$

20. $T \propto \Delta l \quad V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{k\Delta l}{m}} = \sqrt{\frac{k(\Delta l)l}{m}}$

$$\therefore V \propto \sqrt{(\Delta l)l} \quad \therefore \frac{V_2}{V_1} = \sqrt{\frac{(\Delta l_2) \times l_2}{(\Delta l_1) \times l_1}} = \sqrt{\frac{(0.05) \times 1.05}{0.1 \times 1.10}}$$

$$\therefore \frac{V_2}{V_1} = \frac{1}{\sqrt{2}} \quad \therefore V_2 = \frac{V}{\sqrt{2}}$$

21. K_α X-rays are produced when an electron makes a transition from $n = 2$ to $n = 1$ to fill a vacancy in K – shell. The wavelength of X-ray lines is given by

$$\frac{1}{\lambda_{k\alpha}} = (Z-1)^2 \left(\frac{1}{I^2} - \frac{1}{2^2} \right) \Rightarrow \frac{1}{\lambda_{k\alpha}} = \frac{3}{4} R (Z-1)^2$$

$$\Rightarrow (Z-1)^2 = \frac{4}{3R\lambda_{k\alpha}} \Rightarrow (Z-1)^2 = \frac{4}{3 \times (1.097 \times 10^7) \times (0.76 \times 10^{-10})} = 1599.25$$

$$\Rightarrow (Z-1)^2 \approx 1600 \Rightarrow Z-1 = 40 \Rightarrow Z = 41.$$

22. Energy of incident photons in eV is given as $E = \frac{12431}{1800} eV$

As work function of metal is $2eV$, the maximum kinetic energy of ejected electrons is

$$KE_{\max} = E - \phi \Rightarrow KE_{\max} = 6.9 - 2eV \Rightarrow KE_{\max} = 4.9 eV$$

If $v_{\max}$ be the speed of fastest electrons then we have $\frac{1}{2}mv_{\max}^2 = 4.9 \times 1.6 \times 10^{-19} \text{ joule}$

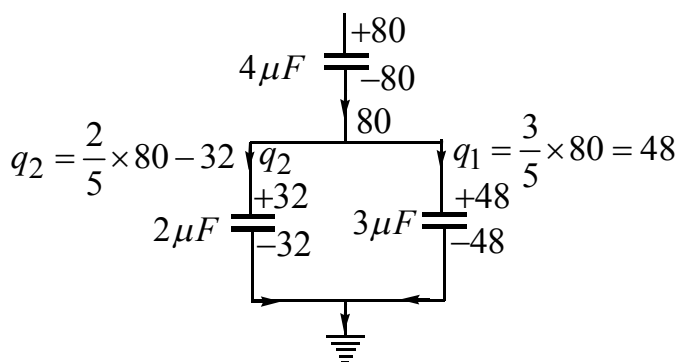
$$\Rightarrow v_{\max} = \sqrt{\frac{2 \times 4.9 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}} \Rightarrow v_{\max} = 1.31 \times 10^6 \text{ m/s}$$

When an electron with this speed enters a uniform magnetic field normally it follows a circular path whose radius can be given by

$$r = \frac{mv}{qB} \quad \left[\text{As } qvB = \frac{mv^2}{r} \right] \quad r = \frac{9.1 \times 10^{-31} \times 1.31 \times 10^6}{1.6 \times 10^{-19} \times 5 \times 10^{-5}} \quad r = 0.149 \text{ m}$$

23. $W = \int_0^t eI dt = \int_0^t \left(\frac{LdI}{dt} \right) I dt = \int_0^2 LI dI = \frac{1}{2} L I^2 \Big|_0^2 = \frac{1}{2} \times 2 \times 2(4-0) = 4 \text{ joule}$

24.



$2 \mu F$ and $3 \mu F$ are in parallel combination

25. When a capacitor charged to voltage V_0 discharged through an inductor, then the maximum current flowing in the circuit is I_0 (say). Then $\frac{1}{2} LI_0^2 = \frac{1}{2} CV_0^2$

$$I_0 = \sqrt{\frac{C}{L}} V = \sqrt{\frac{10^{-6}}{10^{-3}}} \times 1 = \sqrt{10^{-3}} \text{ A} = \sqrt{1000} \text{ mA} = \sqrt{500n} \text{ mA} \quad n = 2$$

26. $v = \frac{c}{\lambda} = \frac{3 \times 10^8}{1.3 \times 10^6} = 2.3 \times 10^{14} \text{ Hz}$

$$\text{Max. bit rate} = 2v = 2 \times 2.3 \times 10^{14} = 4.6 \times 10^{14}$$

$$\text{No. of phone calls} = \frac{\text{Max. bit rate}}{\text{bandwidth}} = \frac{4.6 \times 10^{14}}{2 \times 10^9} = 2.3 \times 10^5$$

$$n = 5$$

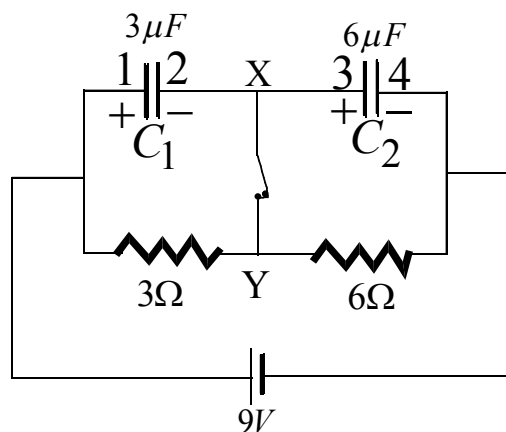
27. When Switch S is open: Same charge will exist on both the capacitors and is given by

$$\text{Charge} = \left(\frac{C_1 C_2}{C_1 + C_2} \right) V = \left(\frac{3 \times 6}{3 + 6} \right) \times 9 = 18 \mu C$$

When Switch S is closed: In steady state, current in the lower branch is

$$I = \frac{9}{3+6} = 1A. \text{ Now, } 3 \mu F \text{ and } 3 \Omega \text{ are in parallel (also } 6 \mu F \text{ and } 6 \Omega \text{ are in parallel),}$$

therefore,



$$V_{C_1} = V_{3\Omega} = 1 \times 3 = 3V$$

$$V_{C_2} = V_{6\Omega} = 1 \times 6 = 6V$$

$\therefore$ Final charge across capacitors

$$q_{C_1} = C_1 V_{C_1} = 3 \times 3 = 9 \mu C$$

$$q_{C_2} = C_2 V_{C_2} = 6 \times 6 = 36 \mu C$$

Consider plates 2 and 3,

$$(\Delta q)_{y \rightarrow x} = (+36 - 9) - (18 - 18) = 27 \mu C$$

28. Current will flow through lower branch only and is given by

$$I = \frac{12}{3+6+1} = 1.2A$$

$\therefore$ P.D. across the cell is

$$V = E - Ir = 12 - 1.2 \times 1 = 10.8V$$

This P.D. is divided equally among the two capacitors, therefore, potential at C is

$$V_C = \frac{10.8}{2} = 5.4V$$

$$\text{The P.D. at D is given by } V_D = I \times 3 = 1.2 \times 3 = 3.6V$$

$$\therefore V_C - V_D = 5.4 - 3.6 = 1.8V$$

29. We know that $\beta = \frac{\alpha}{1-\alpha} = \frac{0.96}{1-0.96} = 24$

The collector current I_C is given by

$$I_C = \frac{V_C}{R} = \frac{0.5V}{800\Omega} = 0.625 \times 10^{-3} A$$

$$\text{Further } I_B = \frac{I_C}{\beta} = \frac{0.625 \times 10^{-3}}{24}$$

$$= 26 \times 10^{-6} A = 26 \mu A$$

30. We know that, $g = \frac{GM}{R^2} = \frac{4}{3} \pi G M \rho$

$$\therefore \frac{g_p}{g_e} = \frac{R_p \rho_p}{R_e \rho_e} \text{ also } v_e = \sqrt{2gR}$$

$$\therefore \frac{(v_e)_p}{(v_e)_e} = \sqrt{\frac{g_p R_p}{g_e R_e}} = \sqrt{\frac{g_p}{g_e} \cdot \frac{g_p \rho_e}{g_e \rho_p}} = \frac{g_p}{g_e} \sqrt{\frac{\rho_e}{\rho_p}}$$

$$= \frac{\sqrt{6}}{11} \sqrt{\frac{3}{2}} = \frac{3}{11}$$

$$\therefore (v_r)_p = \frac{3}{11} (v_e)_e = \frac{3}{11} (11) = 3 km s^{-1}$$

CHEMISTRY

31. $P_t = P_A^0 X_A + P_B^0 X_B$
32. Solubility = (mass of solute/mass of solvent)100
33. $P^0 - P^s/P^0 = m/(m+55.55)$
34. Freezing point of solvent $\propto 1/\text{no of solute particles}$
35. $\Delta T_b = K_b m$
36. Use first order equation, $K = \frac{2.3031}{t} \log \frac{a}{(a-x)}$
37. $x = kt = 0.025 \times 15 = 0.375$
38. Conceptual
39. $\Delta H = (E_a)_f - (E_a)_b$;
 $\ln \ln \frac{K}{A} = -\frac{E}{RT}$
40. Conceptual
41. When volume is doubled concentration becomes half
42. Since α is neglecting wrt 1, $K_a = C_1 \alpha_1^2 = C_1 \alpha_2^2$
 $C_1 V_1 = C_2 V_2$
43. Conceptual
44. At equivalent point, $\text{pH} = 1/2[\text{pK}_w + \text{pK}_a + \log C]$, $[\text{Salt}] = m \text{ mol of acid/total volume}$.
45. $2(r^{+} + r^{-}) = (3)^{1/2} a$
46. Each Ca^{+2} can create one cation vacancy.
47. Conceptual
48. $\Delta E = E_2 - E_1$
49. l never be equal or greater than n .
50. $\lambda = h/\sqrt{2mE}$
51. Conceptual
52. Conceptual
53. Radius ratio = 0.8 indicate the geometry of crystal is bcc
54. Only 4 among 8 TH voids are occupied by Zn^{+2}
55. $\Delta T_b = k_b \cdot m \cdot i$; $DOA = \frac{1-i}{1-(\frac{1}{n})}$
56. $d = \frac{Z_{\text{eff}} \cdot A}{N_0 \cdot a^3}$
57. Conceptual
58. $\pi_{\text{mix}} = \pi_1 V_1 + \pi_2 V_2 / V_1 + V_2$
59. For 22.4V, 4N
60. Total nodes = $n-1$

MATHEMATICS

61) Total words, without any restriction = $7!$

Total words beginning with $I = 6!$

Total words ending with $B = 6!$

Total words with I and ending with $B = 5!$

Thus, total number of required words = $7! - (6! + 6! - 5!)$
 $= 7! - 2(6!) + 5!$

$$62) T_{r+1} \text{ in } \left(x^2 - \frac{1}{x}\right)^9 \text{ is } {}^9C_{1X} 2(9-r) \left(-\frac{1}{x}\right)^r = {}^9C_r \cdot x^{18-3r} \cdot (-1)^r$$

for term independent of x , $18 - 3r = 0 \Rightarrow r = 6$

$\therefore 7^{\text{th}}$ term is independent of x and equals ${}^9C_r = {}^9C_3 = 84$.

Also there are 10 terms, hence 5^{th} term and 6^{th} are the two middle term

$T_5 = {}^9C_4 \cdot x^6$; $T_6 = -{}^9C_5 \cdot x^3$ $\therefore q =$ coefficient of 5^{th} + coefficient of 6^{th} are the two middle term
 ${}^9C_4 - {}^9C_5 = 0$

Hence $p = 84, q = 0$ $\therefore p - q = {}^9C_3$

$$63) \text{ Let } S = 1(\alpha_1 + \alpha_{2008}) + 2(\alpha_2 + \alpha_{2007}) + 3(\alpha_3 + \alpha_{2006}) + \dots + 2008(\alpha_{2008} + \alpha_1) \dots (1)$$

$$\text{Also } S = 2008(\alpha_{2008} + \alpha_1) + 2007(\alpha_2 + \alpha_{2007}) + \dots + 2(\alpha_2 + \alpha_{2007}) + 1(\alpha_1 + \alpha_{2008}) \dots (2)$$

(written in reverse order) $\therefore$ On adding (1) and (2), we get

$$2S = 2009 [2(\alpha_1 + \alpha_2 + \alpha_3 + \dots + \alpha_{2008})]$$

$$2S = 2009 \left[2 \underbrace{(1 + \alpha_1 + \alpha_2 + \alpha_3 + \dots + \alpha_{2008} - 1)}_{\text{zero}} \right] \quad \text{Hence } S = -2009$$

$$64) \text{ Given, } 2 \tan^{-1}(3x) = \sin^{-1} \left(\frac{2(3x)}{1 + (3x)^2} \right)$$

$$\text{Put } 3x = \tan \theta \Rightarrow \theta = \tan^{-1}(3x)$$

$$\therefore \sin^{-1} \left(\frac{2(3x)}{1 + (3x)^2} \right) = \sin^{-1} \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right)$$

$$= \sin^{-1}(\sin 2\theta) = 2\theta, \frac{-\pi}{2} < 2\theta < \frac{\pi}{2}$$

$$2 \tan^{-1}(3x), \frac{-1}{3} \leq x \leq \frac{1}{3}$$

$$65) \lim_{x \rightarrow \infty} \left(\sqrt{x^2 - \frac{4x}{3}} + 2 + x \right) \text{ Put } x = -x \quad = \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + \frac{4x}{3}} + 2 - x \right) = \left(\sqrt{1 + \frac{4}{3x}} + 2 - x \right)$$

$$= \lim_{x \rightarrow \infty} \left(x \left(1 + \frac{4}{3x} \right)^{1/2} + 2 - x \right) = \lim_{x \rightarrow \infty} \left\{ x \left(1 + \frac{1}{2} \left(\frac{4}{3x} \right) + \frac{1}{2!} \left(\frac{1}{2} \right) \left(\frac{1}{2} - 1 \right) \left(\frac{4}{3x} \right)^2 + \dots - x \right) \right\}$$

$$\Rightarrow \frac{9}{45} = \frac{1}{\lambda^2} \Rightarrow \lambda^2 = 5$$

66) $f(x)$ is not differentiable at $x = -2, -6$ for $x < 0$

$\therefore f(-|x|)$ is not differentiable at $x = \pm 2$ and ± 6 .

Also $f(0)$ is not defined for $x < 0$ and $f(x)$ is discontinuous at $x = 0$

$\therefore f(-|x|)$ is not differentiable at $x = 0$

$\therefore f(-|x|)$ is not differentiable at 5 points.

67) Consider $f(x)=\frac{e^{\cos x}}{x}$

Now, $f'(x)=\frac{-e^{\cos x}(1+x\sin x)}{x^2}$

As $f'(x)<0\Rightarrow f(\alpha)>f(\beta)$

So, $\frac{e^{\cos \alpha}}{\alpha}>\frac{e^{\cos \beta}}{\beta}$.

68) Use Rolle's Theorem for $F(x)$ in $[a,b]$; $F(a)=F(b)$, also f is differentiable.

Hence , we have $F'(x)=f(x)=0\Rightarrow$ continuous for some $x\in(a,b)$.

69) Let $\sec x-\tan x=t\Rightarrow \sec x+\tan x=\frac{1}{t}\Rightarrow 2\tan x=\frac{1}{t}-t$

also $\sec x+(\tan x-\sec x)dx=dt$

$$\Rightarrow \int \frac{2 \sec x \tan x}{(\sec x-\tan x)^{10}} dx = \int \frac{\left(t-\frac{1}{t}\right)}{t^{11}} dt = \frac{t^{-9}}{-9} + \frac{t^{-11}}{11} + C$$
$$= (\sec x-\tan x)^{-11} \left(\frac{(\sec x-\tan x)^2}{-9} + \frac{1}{11} \right) + C$$
$$= (\sec x+\tan x)^{11} \left(\frac{(\sec x-\tan x)^2}{-9} + \frac{1}{11} \right) + C \Rightarrow m=-\frac{1}{9} \ \& \ n=\frac{1}{11}$$

70) Put $a=2010$

$$l = \int_{-a}^a x^a \cdot \cot^{-1}(ax) dx$$
$$l = \int_{-a}^a (-x)^a \cdot \cot^{-1}(ax) dx \qquad l = \int_{-a}^a (x)^a \cdot (\pi - \cot^{-1} ax) dx$$
$$2l = \pi \int_{-a}^a x^a dx \qquad 2l = \pi \int_{-a}^a x^a dx$$
$$l = \pi \int_{-a}^a x^a dx = \pi \cdot \frac{a^{a+1}}{a+1} \bigg|_0^a = \pi \cdot \frac{a^{a+1}}{a+1} = \pi \frac{(2010)^{2011}}{2011}$$

71)

q	$\sim q$	p	$\sim p$	$\sim q \wedge q$	$p \vee \sim p$	$(\sim q \wedge p) \vee (p \vee \sim p)$
T	F	T	F	F	T	T
T	F	F	T	F	T	T
F	T	T	F	T	T	T
F	T	F	T	F	T	T

72) $x^2+ax+b=0<\frac{\alpha}{\beta}$

$$\alpha+\beta=-a;\alpha\beta=b \qquad \dots\dots(1)$$
$$g(x)=0<\frac{\alpha+1/\alpha}{\beta+1/\beta}$$

sum of roots =products of roots

$$\alpha+\beta+\frac{1}{\alpha}+\frac{1}{\beta}=\left(\alpha+\frac{1}{\alpha}\right)\left(\beta+\frac{1}{\beta}\right) \Rightarrow \alpha+\beta+\frac{\alpha+\beta}{\alpha\beta}=\alpha\beta+\frac{\alpha}{\beta}+\frac{\beta}{\alpha}+\frac{1}{\alpha\beta}$$

Using equation (1)

$$\Rightarrow -a-\frac{a}{b}=b+\frac{a^2-2b}{b}+\frac{1}{b} \Rightarrow a^2a(b+1)+(b^2-2b+1)=0$$

$D \geq 0$ (Quadratic in a)

$$(b+1)^2 - 4(b^2 - 2b + 1) \geq 0 \Rightarrow (3b-1)(b-3) \leq b \in \left[\frac{1}{3}, 3\right]$$

73) The equation whose roots are

$$u, v, w \text{ is } x^3 - (u+v+w)x^2 + (uv+vw+wu)x - uvw = 0$$

$$\text{or } x^3 - 2x^2 + 5x + 6 = 0$$

by trial method, $x=1$ satisfies it.

$$\therefore x^3 - 2x^2 - 5x + 6 = (x-1)(x^2 - x - 6)$$

$$\text{Now } x^2 - x - 6 = 0 \Rightarrow (x-3)(x+2) = 0 \quad \therefore x = -2, 3$$

$$\therefore u = 3, v = 1, w = -2 \text{ (As } u > v > w)$$

$$\vec{V}_1 = 3\hat{i} + \hat{j} - 2\hat{k} \text{ and } \vec{V}_2 = 2\hat{i} + 3\hat{j} - 5\hat{k}$$

$$\vec{V}_1 \cdot \vec{V}_2 = 6 + 3 + 10 = 19$$

74) $6x^2 + 2ax + 2 = 0$ and $6x^2 + 3bx + 3 = 0$

subtracting $x(2a-3b) - 1 = 0$

$$\Rightarrow x = \frac{1}{2a-3b} \text{ (put in any equation)}$$

$$\therefore 2 \frac{1}{(2a-3b)^2} + \frac{b}{2a-3b} + 1 = 0$$

$$2 + b(2a-3b) + (2a-3b)^2 = 0$$

$$4a^2 + 9b^2 - 12ab + 2ab - 3b^2 + 2 = 0 \Rightarrow 10ab + 6b^2 + 4a^2 + 1 = 0$$

$$\Rightarrow 5ab - 3b^2 - 2a^2 = 1 \Rightarrow B$$

75) $f(x) = 3x^2 - (a+b)x + 2a = 0 \left\langle \begin{matrix} e^{-\lambda} \\ e^{\lambda} \end{matrix} \right\rangle$

$$\text{Product of the roots} \Rightarrow \frac{2a}{3} = 1 \Rightarrow a = \frac{3}{2}$$

$$\text{Now, sum of the roots} = e^{-\lambda} + e^{\lambda} = \frac{a+b}{3} > 2$$

$$\Rightarrow b > 6 - a \Rightarrow b > 6 - \frac{3}{2} \Rightarrow b > \frac{9}{2} \quad \therefore \text{Least value of } b \text{ is } 5$$

76) $\underbrace{A_1 H_9 G_8}_{16} \underbrace{A_2 H_8 G_6}_{16} \underbrace{A_3 H_7 G_4}_{16} \underbrace{A_4 H_6 G_2}_{16}$

$$\text{Now } A_1 H_9 = A_2 H_8 = A_3 H_7 = ab = 16$$

$$\therefore \text{Expression} = 16^4 \times G_2 G_4 G_6 G_8 \quad \therefore 16^4 \times 16^2 = 4^{12}$$

77) $(2011)^{\log_{10} X} = 11^{\log_{10} (1+3+5+\dots+4021)}$

$$\text{Now } \underbrace{1+3+5+\dots+4021}_{\substack{\text{sum of the first 2011} \\ \text{odd natural number}}} = (2011)^2 \quad \therefore X^{\log_{10} 2011} = 11^{\log_{10} (2011)^2}$$

$$\therefore 11^{2 \log_{10} 2011} = (121)^{\log_{10} 2011}$$

$$\text{Hence } x = 121$$

78) We have $16^{\sin^2 \theta} + 16^{\cos^2 \theta} = 10 \Rightarrow 16^{\sin^2 \theta} + 16^{1-\sin^2 \theta} = 10$

$$\Rightarrow x + \frac{16}{x} = 10, \text{ where } x = 16^{\sin^2 \theta} \Rightarrow x = 2, 8 \Rightarrow 16^{\sin^2 \theta} = 2, 8$$

$$\Rightarrow 2^{4 \sin^2 \theta} = 2, 2^3 \Rightarrow 4 \sin^2 \theta = 1, 3$$

$$\Rightarrow \sin^2 \theta = \left(\frac{1}{2}\right)^2, \left(\frac{\sqrt{3}}{2}\right) \Rightarrow \theta = \frac{\pi}{6}, \frac{\pi}{3} \quad \therefore \alpha = \frac{\pi}{6}$$

Let the altitude of the sum be θ . Then,

$$\tan \theta = \frac{h}{\frac{h}{\sqrt{3}}} \Rightarrow \tan \theta = \sqrt{3} \quad \Rightarrow \theta = \frac{\pi}{3} \quad \Rightarrow \theta = 2\alpha$$

79) Given, $\sin 3x - \cos x = 2 \quad \Rightarrow \sin 3x = 2 + \cos x$

$$\therefore L.H.S \leq 1 \text{ \& } R.H.S \geq 1 \quad \Rightarrow L.H.S = R.H.S = 1 \text{ (each)}$$

$$\Rightarrow \cos x = -1 \quad \Rightarrow x = (2n+1)\pi, n \in I$$

But, for these values of x , $\sin 3x = \sin 3(2n+1)\pi = 0$. So, the above equation is not possible. Hence, the given equation has no real solution.

80) Let h be the height of a tower. Since $\angle AOB = 60^\circ$,

$\therefore \triangle OAB$ is an equilateral triangle.

$$\therefore OA = OB = AB = a$$

$$\text{In } \triangle ACD, \tan 30^\circ = \frac{CD}{AC} \Rightarrow h = \frac{a}{\sqrt{3}}$$

81) Put $\Delta = 0$

$$\Rightarrow \begin{vmatrix} \cos \theta & \sin \theta & \cos \theta \\ \sin \theta & \cos \theta & \sin \theta \\ \cos \theta & \sin \theta & -\cos \theta \end{vmatrix} = 0 \Rightarrow \cos \theta \cdot \cos 2\theta = 0$$

$$\therefore \theta = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}$$

$$\text{But } \therefore \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \text{ (rejected)}$$

As lines are parallel. So, system is inconsistent.

$$82) \quad x^3 \begin{vmatrix} 1 & 1 & 1+x^3 \\ 2 & 4 & 1+8x^3 \\ 3 & 9 & 1+27x^3 \end{vmatrix} = 10 \quad \Rightarrow x^3 \begin{vmatrix} 1 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 3 & 9 \end{vmatrix} + x^3 \cdot x^3 \begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & 8 \\ 3 & 9 & 27 \end{vmatrix} = 0$$

$$\Rightarrow x^3 (25 - 23) + 6x^6 \cdot 2 = 10$$

$$\Rightarrow 6x^6 + x^3 - 5 = 0 \quad \Rightarrow x^3 = \frac{5}{6}, -1$$

$\therefore$ Two distinct real solutions

$$83) \text{ Let } A \begin{bmatrix} 1 & a & b \\ +a & 1 & c \\ +b & +c & 1 \end{bmatrix} \quad \therefore A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \Rightarrow x + ay + bz = 1$$

84) Let $OA = r_1$ and $OB = r_2 \quad \Rightarrow A = (r_1 \cos \theta, r_1 \sin \theta)$

$$\text{A lies on ellipse } \frac{x^2}{9} + \frac{y^2}{\lambda^2} = 1 \quad \Rightarrow \frac{r_1^2 \cos^2 \theta}{9} + \frac{r_1^2 \sin^2 \theta}{\lambda^2} = 1$$

$$\frac{1}{r_1^2} = \frac{\cos^2 \theta}{9} + \frac{\sin^2 \theta}{\lambda^2} \quad \dots\dots\dots(1)$$

Now replace θ by $90^\circ + \theta_1$ we get

$$\Rightarrow \frac{1}{r_1^2} = \frac{\sin^2 \theta}{9} + \frac{\cos^2 \theta}{\lambda^2} \quad \dots\dots(2)$$

$$(1) + (2) \Rightarrow \frac{14}{45} = \frac{1}{9} + \frac{1}{\lambda^2} \quad \Rightarrow \frac{9}{45} = \frac{1}{\lambda^2} \quad \Rightarrow \lambda^2 = 5$$

$$85) \quad a_{11} = \cos \alpha = a_{22} \quad i < j \Rightarrow a_{12} = \sin(\pi - \alpha) = \sin \alpha$$

$$i > j \Rightarrow a_{21} = \sin(\pi + \alpha) = -\sin \alpha$$

$$\Rightarrow A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix}$$

$$\text{Similarly } A^5 = \begin{bmatrix} \cos 5\alpha & \sin 5\alpha \\ -\sin 5\alpha & \cos 5\alpha \end{bmatrix}$$

$$\text{Put } \alpha = \frac{3\pi}{5} \Rightarrow A^5 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$86) \quad S = \frac{7}{2^2 5^2} + \frac{13}{5^2 8^2} + \frac{19}{8^2 11^2} + \dots\dots$$

$$3S = \frac{21}{2^2 5^2} + \frac{39}{5^2 8^2} + \frac{57}{8^2 11^2} + \dots\dots$$

$$3S = \sum_{r=1}^{10} \frac{(3r+2)^2 - (3r-1)^2}{(3r-1)^2 (3r+2)^2}$$

$$3S = \sum_{r=1}^{10} \left(\frac{1}{(3r-1)^2} - \frac{1}{(3r+2)^2} \right)$$

$$3S = \frac{1}{2^2} - \frac{1}{2^{10}} \quad 3S = \frac{2^8 - 1}{2^{10}}$$

$$S = \frac{85}{1024} = \frac{m}{n} \text{ Hence, } (n - 12m) = 4$$

$$87) \quad a = \left| \frac{4a - 3(2a^2 - 27)}{5} \right|$$

$\therefore (0, 2)$ and centre lies on same side of line and on putting $(0, 2)$ we get -ve sign

$$5a = (+6a^2 - 4a - 81)$$

$\therefore$ On putting $(a, 2a^2 - 27)$ in the line we get -ve sign

$$\Rightarrow 6a^2 - 9a - 81 = 0 \quad \Rightarrow 2a^2 - 3a - 27 = 0$$

$$\Rightarrow 2a^2 - 9a - 6a - 27 = 0 \quad \Rightarrow 2a(a+3) - 9(a+3) = 0$$

$$\Rightarrow a = -3, a = \frac{9}{2} \quad \therefore \text{diameter} = 9$$

$$88) \quad \text{Given equation of planes are}$$

$$P_1 : x + y + z = 1$$

$$P_2 : x + 2ay + z = 2$$

$$P_3 : ax + a^2y + z = 3$$

If 3 planes intersect in a line, then

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2a & 1 \\ a & a^2 & 1 \end{vmatrix} = 0$$

Applying $C_1 \rightarrow C_1 - C_2$ and $C_2 \rightarrow C_2 - C_3$

$$\begin{vmatrix} 0 & 0 & 1 \\ 1-2a & 2a-1 & 1 \\ a-a^2 & a^2-1 & a \end{vmatrix} = 0$$

∴ On expanding along C_1 , we get

$$\Rightarrow (1-2a)(a^2-1) - (a-a^2)(2a-1) = 0 \Rightarrow -1+3a+2a^2 = 0 \Rightarrow a = \frac{1}{2}, 1$$

But, for $a=1$, planes P_1 and P_3 are parallel and for $a=\frac{1}{2}$, planes P_1 and P_3 are parallel.

Hence, we conclude that 3 planes will never intersect in a line for any real value of a .

89) $ax^2 + bx + c = 0$ α $5\alpha = \frac{-b}{a}$ and $4\alpha^2 = \frac{c}{a}$

$$\therefore b = -5a\alpha \text{ and } c = 4a\alpha^2$$

Now $3a = 2(c-b)$

$$\therefore 8\alpha^2 + 10\alpha - 3 = 0$$

$$\alpha = \frac{1}{4} \text{ or } \alpha = \frac{-3}{2} \text{ (rejected)}$$

hence $\beta = 1$

Now $S = \beta(1 + \alpha + \dots + \infty)$

$$S = \frac{\beta}{1-\alpha} = \frac{1}{1-\frac{1}{4}} = \frac{4}{3}$$

$$18S = 24.$$

90) $\sqrt{3}, H, \sqrt{2}$ in H.P. $\Rightarrow H = \frac{2\sqrt{6}}{\sqrt{2} + \sqrt{3}} \Rightarrow \frac{H}{\sqrt{2}} = \frac{2\sqrt{3}}{\sqrt{2} + \sqrt{3}}, \frac{H}{\sqrt{3}} = \frac{2\sqrt{2}}{\sqrt{2} + \sqrt{3}}$

Using componendo diviendo

$$\Rightarrow \frac{H + \sqrt{2}}{H - \sqrt{2}} = + \frac{H + \sqrt{3}}{H - \sqrt{3}} = 2$$



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant
ICON Central Office – Madhapur – Hyderabad

Sec: **Sr.ICON-Batch-1 & 2**

Jee-Main

Date: 06-05-2021

Time: 09.00 Am to 12.00 Pm

PTM-11

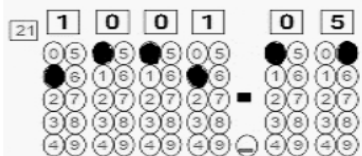
Max.Marks:300

IMPORTANT INSTRUCTIONS:

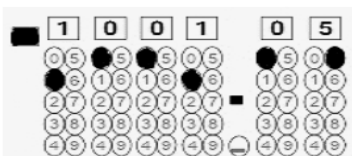
1. Immediately fill in the Admission number on this page of the Test Booklet with Blue/Black Ball Point Pen only.
2. The candidates should not write their Admission Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1, 2, 3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **two sections**.

(I) Section-I contains 20 **multiple choice** questions with **only one correct** option.
Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

(II) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only. First 5 attempted questions will be considered if more than 5 questions attempted. To cancel any attempted question bubble on the question number box. For example: To cancel attempted question 21, bubble on 21 as shown below.



Question Answered for marking



Question Cancelled for marking

Marking scheme : +4 for correct answer and 0 in all other cases.

6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capitals): _____

Admission Number:

Candidate's Signature: _____ Invigilator's Signature: _____

06-05-2021 **Sr.ICON-Batch-1 & 2** _Jee-Main_ **PTM-11** _Test Syllabus

- PHYSICS** : GRAVITATION, STATIC ELECTRICITY, STATIC ELECTRICITY, CURRENT ELECTRICITY, CAPACITORS, **EMW & COMMUNICATION SYSTEM, CLASSICAL MAGNETISM, EMI & AC, ELECTRO MAGNETISM**
- CHEMISTRY** : SOLID STATE, SOLUTIONS & COLLIGATIVE PROPERTIES, CHEMICAL KINETICS, IONIC & CHEMICAL EQUI, **ATOMIC STRUCTURE, S-BLOCK & HYDEROGEN AND ITS COMPOUNDS. ELECTRO CHEMISTRY, THERMO CHEMISTRY & THERMODYNAMICS, 15 th & 16 th GROUP ELEMENTS, CHEMICAL BONDING & PERIODIC TABLE**
- MATHEMATICS** : QUADRATIC EQUATIONS & EXPRESSIONS, SEQUENCES & SERIES, SEQUENCES & SERIES, MATRICES & DETERMINANTS, SYSTEM OF LINEAR EQUATIONS, STATISTICS & MATHEMATICAL REASONING, **HEIGHTS & DISTENCES, BINOMIAL THEOREM, PARABOLA, ELLIPSE, HYPERBOLA, P&C,**

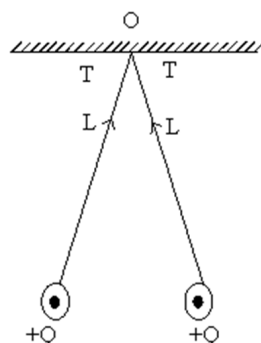
PHYSICS**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

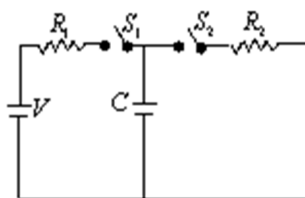
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- 1 Two planets of radii in the ratio 2:3 are made from the materials of density in the ratio 3:2. Then the ratio of acceleration due to gravity g_1 / g_2 at the surface of two planets will be
 1. 1 2. 2.25 3. $\frac{4}{9}$ 4. 0.12
- 2 Two spherical bodies of mass M and 5M and radii R and 2R respectively are released in free space with initial separation between their centres equal to 12R. If they attract each other due to gravitational force only, then the distance covered by the smaller body just before collision is
 1. 2.5 R 2. 4.5 R 3. 7.5 R 4. 1.5 R
- 3 Two insulating plates are both uniformly charged in such a way that the potential difference between them is $(V_2 - V_1) = 20V$. (i.e., plate 2 is at a higher potential). The plates are separated by $d=0.1$ m and can be treated infinitely large. An electron is released from rest on the inner surface of plate 1. Its speed when it hits plate 2 is
 1. $2.65 \times 10^6 \text{ m/s}$ 2. $7.02 \times 10^{12} \text{ m/s}$ 3. $1.87 \times 10^6 \text{ m/s}$ 4. $32 \times 10^{-19} \text{ m/s}$
- 4 Two identical rings of radii 0.1 m are placed co-axially at a distance 0.5 m apart. The charges on the rings are $2\mu C$ and $4\mu C$ respectively. The work done in transferring $5\mu C$ charge from the centre of one ring to that of the other will nearest to
 1. 0.50J 2. 0.75 J 3. 1.00 J 4. 1.50 J
- 5 An electric dipole is placed along the x-axis at the origin O. A point P is at a distance of 20cm from this origin such that OP makes an angle $\pi/3$ with the x-axis. If the electric field at P makes an angle θ with the x-axis, the value of θ would be
 1. $\frac{\pi}{3}$ 2. $\frac{\pi}{3} + \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$ 3. $\frac{2\pi}{3}$ 4. $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

- 6 Two small spheres each having equal positive charge Q (Coulomb) on each are suspended by two insulating strings of equal length L (meter) from a rigid hook (shown in Fig.). The whole set up is taken into satellite where there is no gravity. The two balls are now held by electrostatic forces in horizontal position, the tension in each string is then

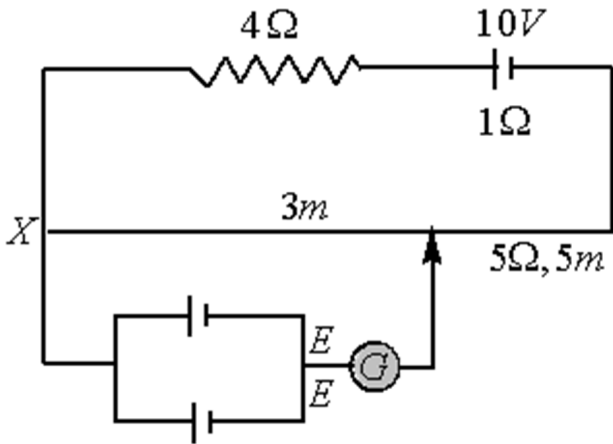


1. $\frac{Q^2}{16\pi\epsilon_0 L^2}$ 2. $\frac{Q^2}{8\pi\epsilon_0 L^2}$ 3. $\frac{Q^2}{4\pi\epsilon_0 L^2}$ 4. $\frac{Q^2}{2\pi\epsilon_0 L^2}$
- 7 If 8 drops, each of capacitance $3\mu F$, coalesce to form a single big drop, then the ratio of the energy stored in the big drop to that in each small drop will be
1. 8 : 1 2. 2 : 1 3. 32 : 1 4. 64 : 1
- 8 A battery of e.m.f. V volt, resistors R_1 and R_2 , a condenser C and switches S_1 and S_2 are connected in a circuit shown. The condenser will get fully charged to V volt when



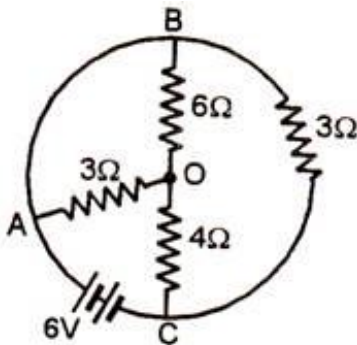
1. S_1 and S_2 are both closed 2. S_1 and S_2 are both open
3. S_1 is open and S_2 is closed 4. S_1 is closed and S_2 is open
- 9 A potential difference of $5V$ is applied across a conductor of length $0.1m$. If drift velocity of electrons is $2.5 \times 10^{-4} ms^{-1}$, calculate the electron mobility in $___ \times 10^{-6} m^2 V^{-1} s^{-1}$
1. 5 2. 4 3. 0.5 4. 15

10 A resistance of 4Ω and a wire of length 5 m and resistance 5Ω are joined in series and connected to a cell of emf 10 V and internal resistance 1Ω . A parallel combination of two identical cells is balanced across 3 m of the wire. The emf E of each cell is :



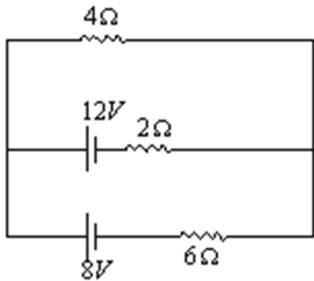
1. 1.5 V
2. 3.0 V
3. 0.67 V
4. 1.33 V

11 In the circuit the total current supplied by the battery is



1. 1 A
2. 2 A
3. 3 A
4. 4 A

12 In the given circuit, current through 6Ω is



1. 4 A
2. 3.3 A
3. 1.2 A
4. 0

13 **STATEMENT_I** :The power drawn by a resistor connected across a cell of zero internal resistance will not undergo any change even if another cell of non – zero internal resistance is also connected parallel with it whatever may be its *emf* and internal resistance. And

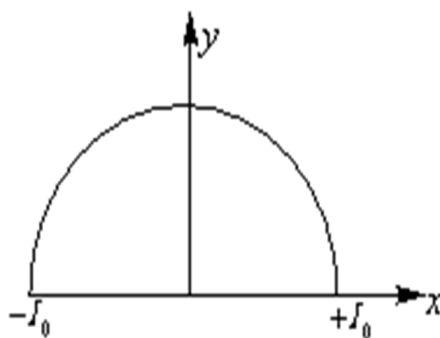
STATEMENT_II :The effective *emf* and the effective internal resistance do not change by the addition of the second cell

1. Statement I is True, Statement II is True and Statement II is correct explanation of Statement I
2. Statement I is True, Statement II is True but Statement II is not correct explanation of Statement I
3. Statement I is True, Statement II is False
4. Statement I is False, Statement II is True

14 When a known resistance 10Ω and a conductor are connected in right and left gaps and conductor is kept at 0°C , The balancing length is 50cm. If the temperature of the conductor in the left gap is increased to 100°C , the balancing length shifts to 60cm. The temperature coefficient of resistance of the material of the conductor is

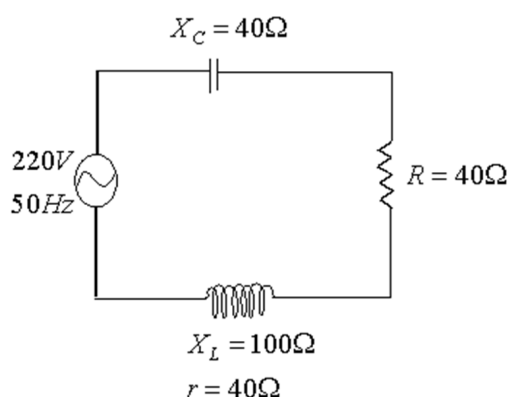
1. $5 \times 10^{-3} / ^\circ\text{C}$
2. $0.0025 / ^\circ\text{C}$
3. $0.05 / ^\circ\text{C}$
4. $0.025 / ^\circ\text{C}$

15 The rms value of a semicircular wave which has a maximum value I_0 is



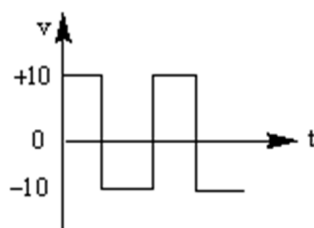
1. $\sqrt{\frac{2}{3}} I_0$
2. $\sqrt{\frac{3}{2}} I_0$
3. $\sqrt{\frac{3}{5}} I_0$
4. $\sqrt{\frac{5}{3}} I_0$

- 16 The power factor of the circuit shown in fig is



1. 0.2 2. 0.4 3. 0.8 4. 0.6
- 17 The magnetic field of a plane electromagnetic wave is given by:
 $\vec{B} = B_0 \hat{i} [\cos(kz - \omega t)] + B_1 \hat{j} \cos(kz + \omega t)$ Where $B_0 = 3 \times 10^{-5} T$ and $B_1 = 2 \times 10^{-6} T$. The rms value of the force experienced by a stationary charge $Q = 10^{-4} C$ at $z = 0$ is
1. 0.6 N 2. 0.1 N 3. $3 \times 10^{-2} N$ 4. 0.9 N
- 18 ASSERTION: A parallel plate capacitor is charged by connecting it across a battery. The battery is then disconnected and separation between the plates is decreased as a result of which the energy stored in the capacitor decreases.
 REASONING: Energy stored in the capacitor is equal to the work done in charging the capacitor.
1. Assertion is True, Reason is True and Reason is correct explanation of Assertion
 2. Assertion is True, Reason is True but Reason is not correct explanation of Assertion
 3. Assertion is True, Reason is False
 4. Assertion is False, Reason is True
- 19 A flood light is covered with a filter that transmits red light. The electric field of the emerging beam is represented by a sinusoidal plane wave
 $E_x = 36 \sin(1.20 \times 10^7 z - 3.6 \times 10^{15} t) \frac{V}{m}$. The average intensity of beam in $\frac{\text{watt}}{(\text{metre})^2}$ will be:
1. 6.88 2. 3.44 3. 1.72 4. 0.86

20 The r.m.s, voltage of the wave form shown is



1. 10 V

2. 7V

3. 6.37 V

4. None of these

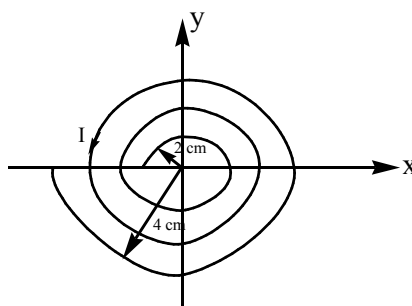
(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30). **Attempt any five questions out of 10.**

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

21 Two amplifiers are connected one after the other in series. The first amplifier has a voltage gain of 10 and the second has a voltage gain of 20. If the input signal is 0.012 volt then output signal (in V) is ____

22 A long insulated copper wire is closely wound as a spiral of N turns. The spiral has inner radius 2 cm and outer radius 4 cm. the spiral lies in x-y plane and a steady current I flows through the wire. The z-component of magnetic field at the center of spiral in SI units is

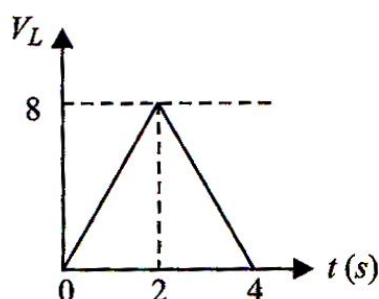


$$\frac{\mu_0 N I}{T \times 10^{-2}} \log 2 \text{ Find } T$$

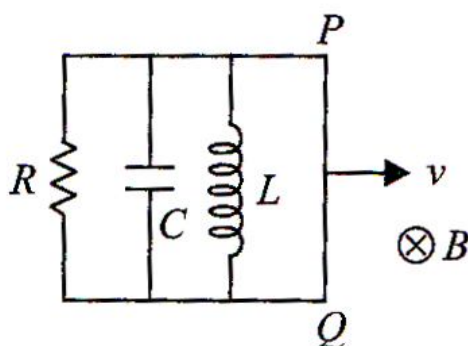
23 A 60 pF capacitor is fully charged by a 20 V supply. It is then disconnected from the supply and is connected to another uncharged 60 pF capacitor in parallel. The electrostatic energy that is lost in this process by the time the charge is redistributed between them is (in nJ)

24 The area of each plate of a parallel plate capacitor is 36cm^2 . The electric field in the gap between the plates changes at the rate of 10^{12} V/ms . The displacement current between the plates is $\frac{1}{x}\text{ A}$, where $x = \underline{\hspace{2cm}}$

- 25 The coercivity of a bar magnet is 120 A/m . It is to be demagnetized by placing it inside a solenoid of length 120 cm and number of turns 72 . The current flowing through the solenoid is
- 26 Two rings of radii 2 cm and 5 cm , charged with charge density -5 C/m are placed co-axially. Centered at the same point. At what points along the axes of the two rings will the electric field be zero?
- 27 The potential difference across a 4 H inductor varies with time as shown. The current is zero at time $t=0$. Find the current (in A) at $t=2 \text{ sec}$.



- 28 In the figure as shown, a conducting wire PQ of length 1 m is moving in a uniform magnetic field $B=3 \text{ T}$ with constant velocity $v=2 \text{ m/sec}$ towards right. $R=2 \Omega$, $C=1 \text{ F}$, and $L=2 \text{ H}$. The currents through the resistance, capacitor, and inductor at any time t are I_1, I_2 and I_3 respectively. Find the value of I_3 (in A) at time $t=2 \text{ sec}$.



- 29 A person brings a mass of 2 kg from A to B. The increase in kinetic energy of the mass is 4 and the work done by the person on the mass is -10 J , then $V_A - V_B = \underline{\hspace{2cm}}$
Where V_A and V_B are gravitational potentials at A and B.
- 30 A 100 W light bulb is placed at the center of spherical chamber of radius 20 cm . Assume that 60% of the energy supplied to the bulb is converted into light and that the surface of the chamber is perfectly absorbing. The pressure exerted by the light on the surface is $\times 10^{-7} \text{ Pa}$

CHEMISTRY**(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- 31 A metal has bcc structure and the edge length of its unit cell is $3.04 \text{ \AA}$. The volume of the unit cell in cm^3 will be
 1. $1.6 \times 10^{-21} \text{ cm}^3$ 2. $2.81 \times 10^{-23} \text{ cm}^3$ 3. $6.02 \times 10^{-23} \text{ cm}^3$ 4. $6.6 \times 10^{-24} \text{ cm}^3$
- 32 200ml of 1M H_2SO_4 , 300ml of 3M HCl and 100ml of 2M HCl are mixed and made up to 1 litre. The proton concentration in the resulting solution is
 1. 1.25M 2. 1.5 N 3. 2.5 M 4. 0.75 M
- 33 If BaCl_2 ionizes to an extent of 80% in aqueous solution, the value of Van't Hoff factor is
 1. 2.6 2. 0.4 3. 0.8 4. 2.4
- 34 For the reaction $2\text{A} + \text{B} \rightarrow \text{Products}$, it is found that doubling the concentration of both reactants increases the rate by a factor of 8. But doubling the concentration of B alone only doubles the rate. What is the order of the reaction w.r.t to A?
 1. 2 2. 3 3. 0 4. 1
- 35 The p^H of a solution is 3.0. This solution is diluted by 100 times. Then the p^H of the resulting solution is
 1. 5 2. 7 3. 1 4. 11
- 36 Solubility of AgCl will be minimum in
 1. 0.01M Na_2SO_4 2. 0.01M CaCl_2 3. Pure water 4. 0.001M AgNO_3
- 37 The kinetic energy of photo electron do not depends on
 A) Frequency of incident radiation
 B) Velocity of incident radiation
 C) Intensity of incident radiation
 D) Wavelength of incident radiation
 1. Both A and B 2. Both A and D 3. Both B and C 4. Both C and D

- 38 When CO_2 is bubbled into an aqueous solution of Na_2CO_3 the following is formed :
1. $NaHCO_3$
 2. H_2O
 3. $NaOH$
 4. OH^-
- 39 The boiling point of H_2O is less than D_2O it is because
1. H_2O has a higher ionic product
 2. D_2O has a lower dielectric constant
 3. The molecular weight of H_2O is lesser than D_2O
 4. D_2O is an associated liquid
- 40 Most reactive form of phosphorus is
1. White
 2. Red
 3. Black
 4. Yellow
- 41 The oxidation states of sulphur in the anions SO_3^{2-} , $S_2O_4^{2-}$ and $S_2O_6^{2-}$ follow the order
1. $S_2O_4^{2-} < SO_3^{2-} < S_2O_6^{2-}$
 2. $SO_3^{2-} < S_2O_4^{2-} < S_2O_6^{2-}$
 3. $S_2O_4^{2-} < S_2O_6^{2-} < SO_3^{2-}$
 4. $S_2O_6^{2-} < S_2O_4^{2-} < SO_3^{2-}$
- 42 Which of the following species is paramagnetic ?
1. CO
 2. NO
 3. O_2^{2-}
 4. CN^-
- 43 The bond energies of $H-H$, $X-X$ and $H-X$ are 104 K.Cal, 38 K.Cal and 135 K.Cal respectively the electronegativity of 'X' is
1. 3.0
 2. 3.2
 3. 3.76
 4. 2.76
- 44 **Assertion:** $KCl / NaCl / NH_4Cl$ etc. cannot be used in the salt bridge of a cell containing silver
Reason: A salt bridge contains concentrated solution of an inert electrolyte like KCl , KNO_3 , NH_4NO_3 etc. or solidified solution of such an electrolyte in agar-agar and gelatin.
1. Assertion is True, Reason is True and Reason is correct explanation of Assertion
 2. Assertion is True, Reason is True but Reason is not correct explanation of Assertion
 3. Assertion is True, Reason is False
 4. Assertion is False, Reason is True

- 45 Molar conductances of $BaCl_2$, H_2SO_4 and HCl at infinite dilutions are x_1, x_2 and x_3 respectively. Equivalent conductance of $BaSO_4$ at infinite dilution will be :
1. $\frac{(x_1 + x_2 - x_3)}{2}$ 2. $x_1 + x_2 - 2x_3$ 3. $\frac{(x_1 - x_2 - x_3)}{2}$ 4. $(x_1 + x_2 - 2x_3)$
- 46 What is the electrode potential (in V) of the following electrode at $25^\circ C$? $Ni^{2+}(0.1M) | Ni_{(s)}$
(Standard reduction potential of $Ni^{2+} | Ni$ is $-0.25 V$; $2.303 \frac{RT}{F} = 0.06$)
1. -0.25 2. -0.28 3. $+0.25$ 4. -0.31
- 47 In a hydrogen-oxygen fuel cell, combustion hydrogen occurs to
1. produce high purity water
2. create potential difference between two electrodes
3. generate heat
4. remove adsorbed oxygen from electron surfaces
- 48 Which among the following is not a state function
1. Internal energy 2. Free energy 3. Work 4. Enthalpy
- 49 At constant pressure 200 g of water is heated from $10^\circ C$ to $20^\circ C$. Thus, increase in entropy is (molar heat capacity of water at constant pressure is $75.3 J K^{-1} mol^{-1}$)
1. $29.0 J K^{-1}$ 2. $227.0 J K^{-1}$ 3. $-227.0 J K^{-1}$ 4. $-29.0 J K^{-1}$
- 50 ΔS for vapourization of 900 g water (in KJ/K) is $[\Delta H_{vap} = 40 KJ / mol]$
1. (900×40) 2. $\frac{50 \times 40}{373}$ 3. $\frac{900 \times 40}{373}$ 4. $\frac{18 \times 40}{373}$

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30). **Attempt any five questions out of 10.**

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

- 51 The dissociation energy of CH_4 is $400 K.Cal mol^{-1}$ and that of ethane is $670 K.Cal mol^{-1}$.
The C–C bond energy is:
- 52 Density of lithium atom is $0.53 g / cm^3$. The edge length of Li is $3.5 \text{ \AA}$. The number of lithium atoms in a unit cell will be _____ (Atomic mass of lithium is 6.94)
- 53 A certain substance A is mixed with an equal amount of a substance B. At the end of 1.0 h, A is 70% reacted. How much will it be left unreacted at the end of 2.5 h, if the reaction with respect to A is of first order?

- 54 How many statements amongst the following are correct?
- (i) one of the most important outcomes of the Heisenberg's principle is that the Bohr's statement in which the electrons are considered revolving in some definite orbits does not appear to hold good
 - (ii) the variable quantity that characterizes the de-Broglie wave is called the wave function.
 - (iii) the wave function has no direct physical significance
 - (iv) ionic bond is directional
 - (v) the scattering experiment of Rutherford did not provide information regarding the exact charge within the nucleus.
- 55 When 6×10^{22} electrons are used in the electrolysis of a metallic salt, 1.9 gm of the metal is deposited at the cathode. The atomic weight of that metal is 57. So oxidation state of the metal in the salt is
- 56 The standard emf of a cell involving one electron change is found to be 0.591V at 25°C . The equilibrium constant of the reaction is $x \times 10^{10}$. The value of x is ($F = 96,500 \text{ C mol}^{-1}$, $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)
- 57 $\text{Pt} : \text{H}_2 (1 \text{ atm}, 25^\circ\text{C}) \parallel \text{HCOOK} (0.01\text{M}) \parallel \text{NH}_4\text{Cl} (0.01\text{M}) \parallel \text{H}_2 : \text{Pt} (1 \text{ atm}, 25^\circ\text{C})$ K_b values of HCOO^- and NH_3 are 10^{-6} and 10^{-8} respectively. The potential of the given cell is "0.0591 X volts". What is the value of "X"?
- 58 10 grams of argon gas is compressed isothermally and reversibly at a temperature of 27°C from 10 L to 5 L. Calculate the work done for this process in multiple of 10^2 calories.
- 59 Enthalpies of combustion of Rhombic sulphur and monoclinic sulphur are 70.96 and 71.03 K cal/mol respectively. Then the heat of conversion of Rhombic sulphur to monoclinic sulphur is $10x \text{ cal/mol}$. The value of x is _____
- 60 For a reversible process, the total change in entropy of the universe is

MATHEMATICS**(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- 61** If $a, b, c \in \{1, 2, 3, 4, 5\}$, the number of equation of the form $ax^2 + bx + c = 0$ which have real roots is
1. 25 2. 26 3. 27 4. 24
- 62** The values of 'a' for which one root of the equation $x^2 - (a+1)x + a^2 + a - 8 = 0$ exceeds 2 and the other is lesser than 2, are given by:
1. $3 < a < 10$ 2. $a \geq 10$ 3. $-2 < a < 3$ 4. $a \leq -2$
- 63** A person is to count 4500 currency notes. Let a_n denote the number of notes he counts in the n^{th} minute. If $a_1 = a_2 = \dots = a_{10} = 150$ and $a_{10}, a_{11}, \dots$ are in an AP with common difference -2, then the time taken by him to count all notes is
1. 135 minutes 2. 24 minutes 3. 34 minutes 4. 125 minutes
- 64** The sum of the series $1 + 3 + 7 + 15 + 31 + \dots$ to 51 terms is
1. $2^{50} - 50$ 2. $2^{51} - 52$ 3. $2^{52} - 53$ 4. $2^{52} - 51$
- 65** $[p \wedge (\sim q)] \wedge [(\sim p) \vee q]$ is
1. a tautology
2. a contradiction
3. a tautology and a contradiction
4. neither a tautology nor a contradiction
- 66** The statement $(\sim(p \vee q)) \vee (\sim p \wedge q)$ is logically equivalent to
1. $\sim p$ 2. p 3. q 4. $\sim q$
- 67** Let $x_1, x_2, \dots, x_n$ be observations such that $\sum x_i^2 = 400$ and $\sum x_i = 80$. Then a possible value of n among the following is
1. 18 2. 15 3. 12 4. 9

- 68 In an experiment with 15 observations on x , the following results were available :
 $\sum x^2 = 2830, \sum x = 170$. One observation 20 was found to be wrong and was replaced by the correct value 30. Then the corrected variance is
 1. 78.00 2. 188.66 3. 177.33 4. 8.33
- 69 A man is walking towards a vertical pillar in a straight path, at a uniform speed. At a certain point A on the path, he observes that the angle of elevation of the top of the pillar is 30° . After walking for 10 minutes from A in the same direction, at a point B, he observes that the angle of elevation of the top of the pillar is 60° . Then the time taken (in minutes) by him, from B to reach the pillar, is :
 1. 6 2. 10 3. 20 4. 5
- 70 A ladder of length 'l' leans against a wall making an angle 30° with horizontal. Its foot is pulled towards the wall through a distance 'a' metres then the ladder moves up the wall through a distance of 'b' metres. If the ladder now makes an angle of 60° with horizontal then.
 1. $a = 2b$ 2. $a = b$ 3. $2a = b$ 4. $2a = 3b$
- 71 Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$ and U_1, U_2, U_3 be column matrices satisfying
 $AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, AU_2 = \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix}, AU_3 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ if U is 3×3 matrix whose columns are U_1, U_2, U_3 , then $|U| =$
 1. 3 2. -3 3. $3/2$ 4. 2
- 72 The number of 4 letter words can be formed from the letters of the word 'INFINITY' so that 2 are same 2 are different is
 1. 180 2. 120 3. 144 4. 96
- 73 The coefficient of $a^8 b^5 c^7$ in the expansion of $(ab + bc + ca)^{10}$ is _____
 1. 5220 2. 2250 3. 2520 4. 2025

- 74 $(1+x+x^2)^8 = a_0 + a_1x + \dots + a_{16}x^{16}$ then $a_1 - a_3 + a_5 - a_7 + \dots - a_{15} =$
1. 1 2. 2 3. 3 4. 0
- 75 A water jet from a fountain reaches its maximum height of 4m at a distance 0.5 m from the vertical passing through the point O of water outlet. The height of the jet above the horizontal OX at a distance of 0.75 m from the point O is
1. 5m 2. 6m 3. 3m 4. 7m
- 76 Minimum distance between the parabolas $y^2 - 4x - 8y + 40 = 0$ and $x^2 - 8x - 4y + 40 = 0$, is
1. 0 2. $\sqrt{3}$ 3. $2\sqrt{2}$ 4. $\sqrt{2}$
- 77 If the ellipse $\frac{x^2}{4} + y^2 = 1$ meets the ellipse $x^2 + \frac{y^2}{a^2} = 1$ in four distinct points and $a = b^2 - 5b + 7$, then b does not lie in
1. $[4, 5]$ 2. $(-\infty, 2) \cup (3, \infty)$ 3. $(-\infty, 0)$ 4. $[2, 3]$
- 78 If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points $P(x_1, y_1), Q(x_2, y_2), R(x_3, y_3), S(x_4, y_4)$ then
1. $x_1 + x_2 + x_3 + x_4 = 0$ 2. $y_1 + y_2 + y_3 + y_4 = 2$
3. $x_1x_2x_3x_4 = 2c^4$ 4. $y_1y_2y_3y_4 = 2c^4$
- 79 If a ray of light incident along the line $3x + (5 - 4\sqrt{2})y = 15$, gets reflected from the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ then its reflected ray goes along the line
1. $x\sqrt{2} - y + 5 = 0$ 2. $\sqrt{2}y - x + 5 = 0$ 3. $\sqrt{2}y - x - 5 = 0$ 4. None of these
- 80 If PQR is an equilateral triangle inscribed in the auxiliary circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a > b)$ and $P'Q'R'$ is corresponding triangle inscribed within the ellipse then centroid of the triangle $P'Q'R'$ lies at
1. Centre of ellipse
2. Focus of ellipse
3. Between focus and centre on major axis
4. None of these

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30). **Attempt any five questions out of 10.**

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

- 81 If the two equations $x^2 - Kx - 21 = 0$ and $x^2 - 3Kx + 35 = 0$ have a common root, and $K > 0$ then the value of K is
- 82 If 11 Arithmetic Means are inserted between 28 and 10 then number of integral Arithmetic Means are?
- 83 ABC is a triangular park with $AB = AC = 100$ m. A vertical tower is situated at the midpoint of BC. If the angles of elevation of the top of the tower at A and B are $\cot^{-1}(3\sqrt{2})$ and $\operatorname{cosec}^{-1}(2\sqrt{2})$ respectively, then the height of the tower (in metres) is
- 84 The system of equations $3x - 2y + z = 0$, $\lambda x - 14y + 15z = 0$, $x + 2y - 3z = 0$ has non-zero solution then $\lambda =$
- 85 Let A be a symmetric matrix of order '2' with integer entries. If the sum of the diagonal elements of A^2 is 1, then the possible number of such matrices is _____
- 86 $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{pmatrix}$ and if $A^{-1} = \frac{1}{6}[\alpha A^2 + \beta A + \gamma I]$ when I is unit matrix of order 3 then the value of $\alpha + \beta + \gamma$ is
- 87 If $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$ then
- $$\operatorname{tr}(A) + \operatorname{tr}\left(\frac{ABC}{2}\right) + \operatorname{tr}\left(\frac{A(BC)^2}{4}\right) + \operatorname{tr}\left(\frac{A(BC)^3}{8}\right) + \dots =$$
- 88 If A is order 3 square matrix such that $|A| = 2$, then $|\operatorname{adj}(\operatorname{adj}(\operatorname{adj}A))|$ is
- 89 The least value of the expression $x^2 + 4y^2 + 3z^2 - 2x - 12y - 6z + 14$ is
- 90 Let $a_1, a_2, \dots, a_{10}$ be in A.P and $h_1, h_2, \dots, h_{10}$ be in H.P if $a_1 = h_1 = 2$ and $a_{10} = h_{10} = 3$ then $a_4 h_7$ is _____



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KEY SHEET

PHYSICS

1	1	2	3	3	1	4	2	5	2	6	1
7	3	8	4	9	1	10	2	11	3	12	4
13	1	14	1	15	1	16	3	17	1	18	2
19	3	20	1	21	2.4	22	4.0	23	2.14	24	31.42
25	2.0	26	3.0	27	2.0	28	6.0	29	7.0	30	4.0

CHEMISTRY

31	2	32	2	33	1	34	1	35	1	36	2
37	3	38	1	39	3	40	1	41	1	42	2
43	3	44	2	45	4	46	2	47	2	48	3
49	1	50	2	51	70.0	52	2.0	53	5.0	54	4.0
55	3.0	56	1.0	57	6.0	58	1.0	59	7.0	60	0.0

MATHEMATICS

61	2	62	3	63	3	64	3	65	2	66	1
67	1	68	1	69	4	70	2	71	1	72	3
73	3	74	4	75	3	76	4	77	4	78	1
79	3	80	1	81	4.0	82	5.0	83	20.0	84	5.0
85	4.0	86	6.0	87	6.0	88	256.0	89	1.0	90	6.0

SOLUTIONS

PHYSICS

1) $\frac{2}{3} \times \frac{3}{2}$
 $g \propto R$

- 2) Distance between the surface of the spherical bodies = $12R - R - 2R = 9R$
 Force $\propto$ Mass
 Acceleration $\propto$ Mass

$$\text{Distance} \propto \text{Acceleration} \Rightarrow \frac{a_1}{a_2} = \frac{M}{5M} = \frac{1}{5} \Rightarrow \frac{S_1}{S_2} = \frac{1}{5} \Rightarrow S_2 = 5S_1$$

$$S_1 + S_2 = 9 \Rightarrow 6S_1 = 9 \Rightarrow S_1 = \frac{9}{6} = 1.5 \quad S_2 = 1.5 \times 5 = 7.5$$

Note : Maximum distance will be travelled by smaller bodies due to the greater acceleration caused by the same gravitational force

3) $v^2 - u^2 = 2a \Rightarrow v^2 = 2 \frac{Eq}{m} d$

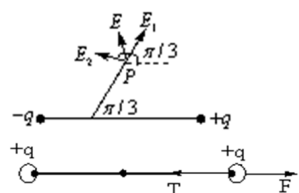
4) 0.75 J

5) From figure, $\theta = \frac{\pi}{3} + \alpha$, where

$$\tan \alpha = \frac{E_2}{E_1} \left(\frac{p \sin \theta}{4\pi \epsilon_0 r^3} \right) \left(\frac{4\pi \epsilon_0 r^3}{2p \cos \theta} \right) = \frac{1}{2} \tan \theta$$

$$\tan \left(\theta - \frac{\pi}{3} \right) = \frac{1}{2} \tan \frac{\pi}{3} = \frac{\sqrt{3}}{2} \quad \text{Or } \theta - \frac{\pi}{3} = \tan^{-1} \left(\frac{\sqrt{3}}{2} \right)$$

$$\text{Or } \alpha = \tan^{-1} \left(\frac{\sqrt{3}}{2} \right) \text{ or } \theta = \frac{\pi}{3} + \tan^{-1} \left(\frac{\sqrt{3}}{2} \right)$$



6) $T = F = \frac{1}{4\pi \epsilon_0} = \frac{q^2}{(2L)^2} = \frac{Q^2}{16\pi \epsilon_0 L^2}$

7) $V_1 : V_2 = n^{\frac{5}{3}} : 1 = 8^{\frac{5}{3}} : 1 = 32 : 1$

- 8) When S_1 is closed and S_2 is opened, the capacitor will get charged to a potential difference of V volts

9) $\mu = \frac{v}{E} \text{ \& } E = \frac{V}{l}$

- 10) For potentiometer, emf
 $E \propto l$ On $E = kl$, Where k is a constant

$$\text{Also } E = \frac{V}{l} = \frac{iR}{L} \times l \therefore E = \frac{E'}{(R_1 + R_2 + r)} \times \frac{R}{L} \times l \therefore E = \frac{10}{(5+4+1)} \times \frac{5}{5} \times 3 = 3V$$

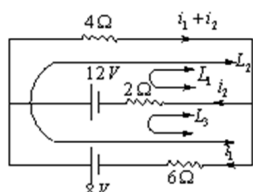
- 11) Here the resistance of arm AO and OB are in parallel. They together are in series with 4Ω , which together are in parallel to 3Ω .

Total resistance of resistances $3\ \Omega$ (arm AO), $6\ \Omega$ (arm OB) and $4\ \Omega$ (arm OC) is

$$R' = \frac{3 \times 6}{3 + 6} + 4 = 6\ \Omega \quad \text{Effective resistance, } R_{\text{eff}} = \frac{6 \times 3}{6 + 3} = 2\ \Omega$$

Current from battery, $I = \frac{6}{2} = 3\text{ A}$.

12)



Loop equations:

$$L_1: 12 - 4(i_1 + i_2) - 2i_2 = 0 \quad (1)$$

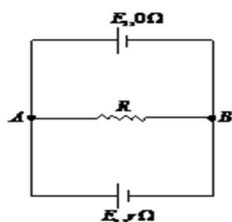
$$L_3: 8 - 12 + 2i_2 - 6i_1 = 0 \quad (2)$$

Simplify (1) $\rightarrow 4i_1 + 6i_2 = 12 \Rightarrow 2i_1 + 3i_2 = 6$

Simplify (2) $\rightarrow i_2 - 3i_1 = 2$

Solving the above equations: $i_1 = 0; i_2 = 2\text{ A}$

13)



Before connecting II cell power $= \frac{E_1^2}{R}$

After connecting II cell $V_{AB} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} = \frac{E_1 r_2 + 0}{r_2} = E_1$

Effective internal resistance $r = \frac{r_1 r_2}{r_1 + r_2} = 0$

Power in this case is $\frac{E_1^2}{R}$

14)

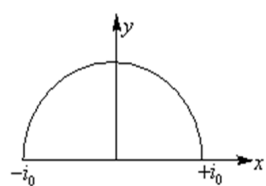
$$1) \frac{R_0}{10} = \frac{50}{50} \Rightarrow R_0 = 10\ \Omega$$

$$2) \frac{R_{100}}{10} = \frac{60}{40} = 3/2 \Rightarrow R_{100} = 15\ \Omega$$

$$3) \alpha = \frac{15 - 10}{10(100)} = 5 \times 10^{-3} / ^\circ\text{C}$$

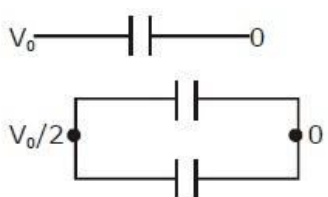
15)

The equation of a semicircular wave is or $x^2 + y^2 = I_0^2$ $y_2 = I_0^2 - x^2$

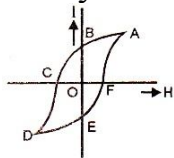


$$\therefore i_{\text{rms}}^2 = \frac{1}{2I_0} \int_{-I_0}^{I_0} y^2 dx = \frac{1}{2I_0} \int_{-I_0}^{I_0} (I_0^2 - x^2) dx$$

$$= \frac{1}{2I_0} \left[I_0^2 x - \frac{x^3}{3} \right]_{-I_0}^{I_0} = \frac{1}{2I_0} \left[\left(I_0^3 - \frac{I_0^3}{3} \right) - \left(-I_0^3 + \frac{I_0^3}{3} \right) \right] = \frac{2I_0^2}{3} I_{\text{rms}} = \sqrt{\frac{2I_0^2}{3}} = \sqrt{\frac{2}{3}} I_0$$

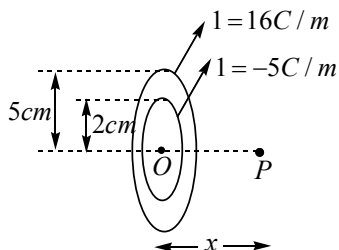
- 16) $R = 40 + 40 = 80\Omega$
 $X_L - X_C = 100 - 40 = 60\Omega$
 $Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{80^2 + 60^2} = 100$
Power factor, $\cos\phi = \frac{R}{Z} = \frac{80}{100} = 0.8$.
- 17) $\frac{\sqrt{7}-1}{\sqrt{7}+1} = 10 \sqrt{\left(\frac{3 \times 10^{-5}}{\sqrt{2}}\right)^2 + \left(\frac{2 \times 10^{-6}}{\sqrt{2}}\right)^2} \times 3 \times 10^8 = 0.63$
- 18) Both assertion and reason are correct. This follows from the relation $U = \frac{1}{2} \frac{q_0^2}{C}$. Also work done in charging the capacitor is stored as potential energy.
- 19) $I_{\text{aver}} = \frac{1}{2} \epsilon_0 E_0^2 \times C = 1.719 \frac{\text{W}}{\text{m}^2}$
- 20) $i_{\text{rms}} = \left[\frac{\int_0^T i^2 dt}{\int_0^T dt} \right]^{1/2}$
- 21) Voltage gain $= \frac{V_0}{V_i} \therefore 10 = \frac{V_0}{0.012}$ & $20 = \frac{V_0^1}{V_0}$
 $200 = \frac{V_0^1}{0.012}$ $V_0^1 = 0.012 \times 200 = 2.4\text{V}$
- 22) Magnetic field (z component) at the centre of spiral is
 $B = \frac{\mu_0 NI}{2(b-a)} \log \frac{b}{a}$ $B = \frac{\mu_0 NI}{2(4-2) \times 10^{-2}} \log \left(\frac{4 \times 10^{-2}}{2 \times 10^{-2}} \right)$ $B = \frac{\mu_0 NI}{4 \times 10^{-2}} \log 2$
- 23) 
 $V_0 = 20\text{V}$ Heat loss $= U_i = U_f = \frac{1}{2} C V_0^2 = \left[\frac{1}{2} C \left(\frac{V_0}{2} \right) \right] = \frac{C V_0^2}{4}$
 $= \frac{(60 \times 10^{-12})(20)^2}{4} = 6 \times 10^{-9} \text{J} = 6 \text{nJ}$
- 24) Given, The area of each plate $A = 36 \text{cm}^2$
 $A = 36 \times 10^{-4} \text{m}^2$
Rate of electric field $\frac{dE}{dt} = 10^2 \text{V/ms}$ $i_d = \epsilon_0 \frac{d\phi_E}{dt}$
 $\epsilon_0 A \frac{dE}{dt} = \frac{1}{4\pi \times 10^9 \times 9} \times 36 \times 10^{-4} \times 10^{12} = \frac{1}{10\pi} = \frac{1}{31.42} \text{A}$

- 25) Coercivity = $OC = H = ni$ where n number of turns per unit length.



Here, $n = \frac{72}{120} \times 100 = 60$ and $H = 120 \text{ A/m} \therefore i = \frac{H}{n} = \frac{120}{60} = 2 \text{ A}$.

- 26) Suppose that the field is zero at the point P, which is at a distance of x centimeters from the centre O. Potential at P



$$V_p = \frac{1}{4\pi\epsilon_0} \left[\frac{16 \times 2\pi r_1}{(r_1^2 + x^2)^{1/2}} - \frac{5 \times 2\pi r_2}{(r_2^2 + x^2)^{1/2}} \right]$$

$$V_p = \frac{1}{2\epsilon_0} \left[\frac{16r_1}{(r_1^2 + x^2)^{1/2}} - \frac{5r_2}{(r_2^2 + x^2)^{1/2}} \right] \text{ where } r_1 = 5\text{ cm, where } r_2 = 2\text{ cm}$$

Since the electric field at P is zero. $E = \frac{dV}{dX} = 0 \quad \left(\frac{dV}{dX} \right) = 0 = \frac{-x}{2\epsilon_0} \left[\frac{16r_1}{(r_1^2 + x^2)^{1/2}} - \frac{5r_2}{(r_2^2 + x^2)^{1/2}} \right]$

Which leads to $x = 0$ Or $\left(\frac{r_2^2 + x^2}{r_1^2 + x^2} \right)^{3/2} = \frac{5r_2}{16r_1} = \frac{5}{16} \times \frac{2}{5} = \frac{1}{8}$ Or $\frac{4 + x^2}{25 + x^2} = \frac{1}{4}$

Solving, we get $x = \pm\sqrt{3}\text{ cm}$

So we conclude that the potential has a minimum at $x = 0$ and maximum at $x = \pm\sqrt{3}\text{ cm}$ or field is zero at these points.

27) $V_L = L \frac{di}{dt} \quad Ldi = V_L dt$

$L(i_f - i_0) = \text{area under } (V_L - t) \text{ graph}$

$$4(i_f - 0) = \frac{1}{2}(8)(2) \quad i_f = 2 \text{ A}$$

28) $V = BVL = (3)(2)(1) = 6V \quad V_B = V_C = V_L = 6V$ (all are in parallel)

$$V_L = L \frac{di}{dt} \quad 6 = (2) \frac{di}{dt} \quad 6 = (2) \frac{di}{dt} \quad \frac{di}{dt} = 3 \text{ A/sec}$$

After 2 sec, current will be 6A.

29) $W_{A \rightarrow B} + W_{\text{Person}} = 4$

$$W_{AB} - 10 = 4$$

$$W_{AB} = 14 \text{ J}$$

$$m[V_A - V_B] = 14$$

$$2[V_A - V_B] = 14$$

$$\therefore V_A - V_B = 7 \text{ joule/kg}$$

30) $P = IC = \frac{60\% \text{ of } P}{4\pi r^2} C = \frac{60 \times 100 \times 3 \times 10^8}{100 \times 4 \times \pi \times r^2} = 4 \times 10^{-7} \text{ Pa}$

CHEMISTRY

31) Given $a = 3.04 \text{ \AA}$

Volume of unit cell $= a^3$

$$= (3.04 \times 10^{-8})^3 = 2.81 \times 10^{-23} \text{ cm}^3$$

32) $[H^+] = N_{mx} = \frac{V_1 N_1 + V_2 N_2 + V_3 N_3}{\text{Total volume}} \quad (\because N = \text{Normality})$

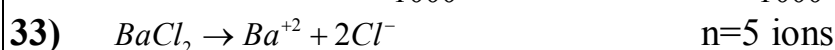
$V_1 = 200 \text{ ml}, N_1 = 1M \text{ H}_2\text{SO}_4 = 2N \text{ H}_2\text{SO}_4$

$V_2 = 300 \text{ ml}, N_2 = 3M \text{ HCl} = 3N \text{ HCl}$

$V_3 = 100 \text{ ml}, N_3 = 2M \text{ HCl} = 2N \text{ HCl}$

Total Volume $= 1000 \text{ ml}$

$$\therefore N_{mx} = \frac{200 \times 2 + 300 \times 3 + 100 \times 2}{1000} = \frac{400 + 900 + 200}{1000} = \frac{15}{10} = 1.5 N$$



given $\alpha = \text{Degree of dissociation} \Rightarrow 80\%$

$i = \text{Vanthoff factor}$

$$\alpha = \frac{i-1}{n-1} \quad \frac{80}{100} = \frac{i-1}{3-1} \quad 0.8 = \frac{i-1}{2} \quad 1.6 = i-1 \quad i = 2.6$$

34) $\text{Rate} = K[A]^x[B]^y$ ——— (1)

$8r = K[2A]^x[2B]^y$ ——— (2)

$2r = K[A]^x[2B]^y$ ——— (3)

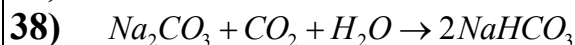
$$\frac{(2)}{(3)} \Rightarrow 4 = 2^x \Rightarrow x = 2$$

$\therefore$ order w.r.t A is 2

35) $3 + \log 100 = 3 + 2 = 5$

36) CaCl_2 is strong electrolyte and common ion effect.

37) Both B and C



39) Conceptual

40) White phosphorus contain discrete P_4 units

41) Conceptual

42) See energy level diagram

43) $\Delta = E_{A-B}^1 - \frac{1}{2}(E_{A-A} + E_{B-B}) = 135 - \frac{1}{2}(104 + 38) = 64$

$$X_A - X_B = 0.208\sqrt{\Delta} \quad X_A - 2.1 = 0.208\sqrt{64} \quad X_A = 3.76$$

44) $\text{KCl} / \text{NaCl} / \text{NH}_4\text{Cl}$ etc. cannot be used as salt bridge in a cell containing silver as one of the electrodes because they react to form a precipitate of AgCl .

45) $\lambda_m^\infty(\text{BaSO}_4) = \lambda_{\text{Ba}^{2+}}^\infty + \lambda_{\text{SO}_4^{2-}}^\infty = \lambda_m^\infty(\text{BaCl}_2) + \lambda_m^\infty(\text{H}_2\text{SO}_4) - 2\lambda_{(\text{HCl})}^\infty$

$$= x_1 + x_2 - 2x_3 ; \lambda_e^0 = \frac{1}{2} \lambda_m^0(\text{BaSO}_4)$$

$$46) E_{red} = E_{red}^0 + \frac{0.06}{n} \log C = -0.25 + \frac{0.06}{2} \log 10^{-1} = -0.28V$$

47) In $H_2 - O_2$ full cell, the combustion of H_2 occurs to create potential difference between the two electrodes

48) Work

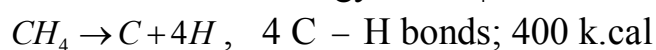
$$49) \text{ The number of moles of } H_2O = \frac{200}{18} \text{ mol}$$

$$\Delta S = 2.303 n C_p \log \frac{T_2}{T_1} = 2.303 \times \frac{200}{18} \times 75.3 \log \frac{293}{283} = 29.06 \text{ J}$$

$$50) 18g \quad H_2O = 1 \text{ mole} \quad 900g \quad H_2O = 50 \text{ mole} \quad \Delta H_{vap} = 40 \text{ KJ / mole}$$

$$\text{For 50 mole} = 50 \times 40 \quad \Delta S = \frac{\Delta H}{T} = \frac{40 \times 50}{373} \text{ KJ / K.}$$

51) The dissociation energy of CH_4 is 400K.cal/mole



$$1 \text{ C - H bond; } \frac{400}{4} = 100 \text{ k.cal}, \quad C_2H_6 \longrightarrow 2C + 6H$$

$$1 \text{ C - C} + 6 \text{ C - H} = 670 \text{ k.cal}$$

$$1 \text{ C - C} + 600 \text{ k.cal} = 670$$

$$1 \text{ C - C bond dissociation energy} = 670 - 600 = 70 \text{ k.cal}$$

$$52) d = \frac{Zn}{a^3 N_0} \Rightarrow 0.53 = \frac{x \times 6.94}{(3.5 \times 10^{-8})^3 \times (6.023 \times 10^{23})} \quad Z = 2$$

$$53) k = \frac{1.303}{t} = \log \frac{[A]_0}{[A]_t} = \frac{2.303}{1.0} \log \frac{100}{x}$$

$$\text{Again, } K = \frac{2.303}{2.5} \log \frac{100}{x} \quad \text{Hence, } \frac{2.303}{1.0} \log \frac{100}{30} = \frac{2.303}{2.5} \log \frac{100}{x} \text{ or } \log \frac{100}{x} = 2.5 \log \frac{100}{30}$$

$$\log \frac{100}{x} = 2.5 (\log 100 - \log 30) = 2.5 \times 0.5229 = 1.307 \text{ or } \log \frac{100}{x} = \text{anti log } 1.307 \text{ or } \frac{100}{x} = 20.28$$

$$\text{or } x = \frac{100}{20.28} = 4.93\% = 5\%$$

$$54) \lambda = \frac{h}{mV} \Rightarrow \frac{\lambda_e}{\lambda_p} = \frac{m_p V_p}{m_e V_e} = \frac{1.67 \times 10^{-27}}{9.11 \times 10^{-31}} \times \frac{\frac{1}{3} V}{V} = \frac{1.67 \times 10^{-27}}{9.11 \times 10^{-31}} \times \frac{1}{3}$$

$$= 0.0611 \times 10^4 = 6.11 \times 10^2 \approx 611$$

$$55) 1.9 \text{ g of metal } -6 \times 10^{22} e^- \quad -6 \times 10^{23} e^-$$

$$56) E_{cell} = \frac{0.0591}{4} \log K_c \quad 0.591 = \frac{0.0591}{1} \log K_c \text{ or } \log K_c = \frac{0.591}{0.0591} = 10$$

$$\therefore K_c = \text{Antilog } 10 = 10^{10}$$

$$57) H_2 | H^+ || H^+ | H_2$$

$$58) W_{\text{Isothermal compression}} = +nRT \ln \frac{V_1}{V_2} = 2.303 \times \frac{10}{40} \times 2 \times 300 \times \log \frac{10}{5} = 1.03 \times 10^2$$

$$59) S_R + O_2 \rightarrow SO_2, \quad \Delta H = -70.96, \quad S_H + O_2 \rightarrow SO_2, \quad \Delta H = -71.03$$

$$S_R \rightarrow S_M, \quad \Delta H = -70.96 + 71.03 = 0.07 \text{ k.cal} = 70 \text{ cal}$$

60) Conceptual

MATHEMATICS

61) For real roots $ac \leq \frac{b^2}{4}$

b	$b^2 / 4$	possible values of ac such that $ac \leq \frac{b^2}{4}$	No. of possible pairs (a, c)
2	1	1	1
3	2.25	1, 2	3
4	4	1, 2, 3, 4	8
5	6.25	1, 2, 3, 4, 5, 6	14
total			26

values of ac	possible pairs (a, c)
1	(1,1)
2	(1,2), (2,1)
3	(1,3), (3,1)
4	(1,4), (4,1), (2,2)
5	(1,5), (5,1)
6	(1,6), (6,1), (2,3), (3,2)

62) $x^2 - (a + 1)x + a^2 + a - 8 = 0$. Since roots are different, therefore $D > 0$

$\Rightarrow (a + 1)^2 - 4(a^2 + a - 8) > 0 \Rightarrow (a - 3)(3a + 1) < 0$

There are two cases arises. Case I. $a - 3 > 0$ and $3a + 1 < 0 \Rightarrow a > 3$ and $a < -\frac{11}{3}$

Hence, no solution in this case. Case II: $a - 3 < 0$ and $3a + 1 > 0 \Rightarrow a < 3$ and $a > -\frac{11}{3}$

$\therefore -\frac{11}{3} < a < 3 \Rightarrow -2 < a < 3$

63) Till 10th minute number of counted notes = 1500

$3000 = \frac{n}{2} [2 \times 148 + (n - 1)(-2)] = n [148 - n + 1]$

$\Rightarrow n^2 - 149n + 3000 = 0 \Rightarrow n = 125, 24 \Rightarrow n = 125$ is not possible

$\Rightarrow$ Total time = 24 + 10 = 34 min

64) $T_1 = 2^1 - 1, T_2 = 2^2 - 1, T_3 = 2^3 - 1, \dots$ and $T_n = 2^n - 1$

$S_{51} = \sum_{n=1}^{51} (2^n - 1) = (2 + 2^2 + 2^3 + \dots + 2^{51}) - 51 = 2(2^{51} - 1) - 51 = 2^{52} - 53$

65) Conceptual

66) $p \quad q \quad \sim p \quad p \vee q \quad \sim (p \vee q) \quad (\sim p \wedge q) \quad \sim (p \vee q) \vee (\sim p \wedge q)$

T	T	F	T	F	F	F
T	F	F	T	F	F	F
F	T	T	T	F	T	T
F	F	T	F	T	F	T

$\therefore (\sim (p \vee q)) \vee (\sim p \wedge q)$

67) Using well known fact that root mean square of number $\geq$ A.M of the numbers

$\Rightarrow \sqrt{\frac{400}{n}} \geq \frac{80}{n} \Rightarrow \frac{20}{\sqrt{n}} \geq \frac{80}{n} \Rightarrow \sqrt{n} \geq 4 \Rightarrow n \geq 16$

N = 17 but not given in choice

So n = 18 is correct number

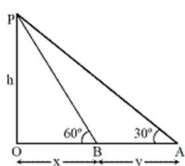
68) $\sum x = 170, \sum x^2 = 2830$

The increase in $\sum x$ is 10, then $\sum x^1 = 170 + 10 = 180$

The increase in $\sum x^2$ is $900-400=500$, then $\sum x^2 = 2830+500=3330$

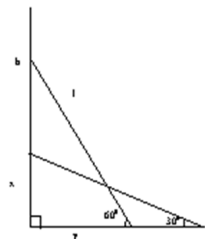
$$\text{Variance} = \frac{1}{n} \sum x^2 - \left(\frac{1}{n} \sum x \right)^2 \Rightarrow \frac{1}{15} \times 3330 - \left(\frac{1}{15} \times 180 \right)^2 \Rightarrow 222 - 144 = 78$$

69) $\tan 60^\circ = \frac{h}{x} \Rightarrow h = \sqrt{3}x$ $3x = x + y \Rightarrow 2x = y$



Time taken from A to B is 10 min So time taken from B to pillar is 5 min

70)



$$\sin 60^\circ = \frac{b+x}{l} \quad \sin 30^\circ = \frac{x}{l} \Rightarrow \frac{b}{l} = \sin 60^\circ - \sin 30^\circ$$

$$\cos 60^\circ = \frac{y}{l}, \quad \cos 30^\circ = \frac{a}{l} \Rightarrow \frac{a}{l} = \cos 30^\circ - \cos 60^\circ$$

$$\therefore \frac{a}{b} = \frac{2 \sin 45^\circ}{2 \cos 45^\circ} \Rightarrow a = b$$

71)

Let $U_1 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$, $U_2 = \begin{bmatrix} p \\ q \\ r \end{bmatrix}$ and $U_3 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Then,

$$AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a \\ 2a+b \\ 3a+2b+c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow a=1, b=-2, c=1$$

$$AU_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} p \\ 2p+q \\ 3p+2q+r \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \Rightarrow x=2, y=-1, r=-$$

$$\text{And, } AU_3 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ 2x+y \\ 3x+2y+z \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \Rightarrow x=2, y=-1, z=-3$$

$$\therefore U = [U_1 U_2 U_3] = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -1 \\ 1 & -4 & -3 \end{bmatrix} \Rightarrow |U| = \begin{vmatrix} 1 & 2 & 2 \\ -2 & -1 & -1 \\ 1 & -4 & -3 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 0 \\ -2 & -1 & 0 \\ 1 & -4 & 1 \end{vmatrix} \xrightarrow{C_3 \rightarrow C_3 - C_2} |U| = 3$$

72) That repeated letter is either I or N this can be done in 2 ways remaining 2 letters can be selected in 4C_2 ways from the remaining $2 \times {}^4C_2 \times \frac{4!}{2!} = 144$

73) Let $a^8 b^5 c^7 = (ab)^x \cdot (bc)^y \cdot (ca)^z = a^{x+z} b^{x+y} c^{y+z}$ and

$$x + y + z = 10 \Rightarrow x + y = 5, y + z = 7, z + x = 8 \Rightarrow x - y = 1 \Rightarrow x = 3, y = 2, z = 5$$

$$\therefore \text{The coefficient of } a^8 b^5 c^7 \text{ in the expansion of } (ab + bc + ca)^{10} \text{ is } \frac{10!}{3!2!5!} = 2520$$

74) $(1+x+x^2)^8 = a_0 + a_1 x + a_2 x^2 + \dots + a_{16} x^{16}$

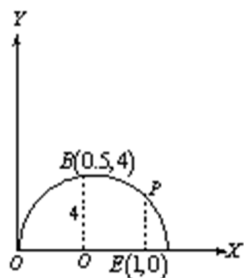
Put $x = i$

$$(1+i-1)^8 = (a_0 - a_2 + a_4 \dots + a_{16}) + i(a_1 - a_3 + a_5 \dots)$$

Comparing 'i' on both sides

$$a_1 - a_3 + a_5 - a_7 + \dots - a_{15} = 0$$

75)



The path of the water jet is a parabola

Let its equation be $y = ax^2 + bx + c$ It should pass through $(0,0), (0.5,4), (1,0)$

$$\Rightarrow c = 0, a = -16, b = 16$$

$$\Rightarrow y = -16x^2 + 16x$$

If $x = 0.75$ we get $y = 3$.

76) The equations of the parabolas are

$$y^2 - 4x - 8y + 40 = 0$$

$$x^2 - 8x - 4y + 40 = 0$$

We observe that if we interchange x and y in equation (i), we obtain equation(ii). So, the two parabolas are symmetric about $y = x$ If the two parabolas intersect on $y = x$, then the minimum distance between them is zero.On solving $y = x$ and $y^2 - 4x - 8y + 40 = 0$ we get

So, the parabolas do not intersect. Consequently, the distance between them is not zero.

Minimum distance between the two parabolas is the distance between tangents to the two parabolas which are parallel to $y = x$ Differentiating (ii), w. r. to x , we get

$$2x - 8 - 4 \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{x-4}{2}$$

If the tangent is parallel to $y = x$, then $\frac{x-4}{2} = 1 \Rightarrow x = 6$ If the tangent is parallel to $y = 7$ Thus, the coordinates of a point on parabola (ii) are $P(6, 7)$. The corresponding point on parabola (i) is $Q(7, 6)$ Hence, required distance $= PQ = \sqrt{1+1} = \sqrt{2}$ 77) For the two ellipses to intersect at four distinct points, $a > 1$

$$\Rightarrow b^2 - 5b + 7 > 1 \Rightarrow b^2 - 5b + 6 > 0 \Rightarrow b \in (-\infty, 2) \cup (3, \infty)$$

$$\Rightarrow b \text{ does not lie in } [2, 3]$$

78)

$$(x_i, y_i), i = 1, 2, 3, 4 \text{ lies on } xy = c^2 \Rightarrow y_i = \frac{c^2}{x_i}$$

Now the point (x_i, y_i) lies on $x^2 + y^2 = a^2 \Rightarrow x_i^2 + \frac{c^4}{x_i^2} = a^2$

$$\Rightarrow x_i^4 - a^2 x_i^2 + c^4 = 0 \text{ its roots are } x_1, x_2, x_3, x_4$$

$$\therefore x_1 + x_2 + x_3 + x_4 = 0$$

$$x_1 x_2 + x_1 x_3 + x_1 x_4 + x_2 x_3 + x_2 x_4 + x_3 x_4 = a^2$$

$$x_1 x_2 x_3 + x_1 x_2 x_4 + x_1 x_3 x_4 + x_2 x_3 x_4 = 0$$

$$x_1 x_2 x_3 x_4 = c^4$$

Clearly © is not correct

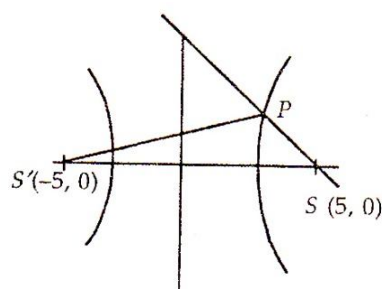
$$\text{Now } y_1 y_2 y_3 y_4 = \frac{c^2}{x_1} \cdot \frac{c^2}{x_2} \cdot \frac{c^2}{x_3} \cdot \frac{c^2}{x_4} = 0$$

$$\text{And } y_1 + y_2 + y_3 + y_4 = \frac{c^2 (\sum x_1 x_2 x_3)}{x_1 x_2 x_3 x_4} = 0$$

$$79) \quad 9 = 16(e^2 - 1)$$

$$e = 5/4$$

Since $(5, 0)$ satisfying the equation of the line $3x + (5 - 4\sqrt{2})y = 15$, so the reflected ray must pass through $(-5, 0)$ and $P = (4\sqrt{2}, 3)$



$\therefore$ Equation of $S'P$ is $\sqrt{2}y = x + 5$

$$80) \quad \text{Let } P(\theta), Q\left(\theta + \frac{2\pi}{3}\right), R\left(\theta + \frac{4\pi}{3}\right)$$

$$\text{then } P' \equiv (a \cos \theta, b \sin \theta), \quad Q' \equiv \left(a \cos \theta \left(\theta + \frac{2\pi}{3}\right), b \sin \left(\theta + \frac{2\pi}{3}\right)\right)$$

$$R' \equiv \left(a \cos \left(\theta + \frac{4\pi}{3}\right), b \sin \left(\theta + \frac{4\pi}{3}\right)\right)$$

$$\begin{aligned} \text{Let centroid of } \Delta P'Q'R' &\equiv (x', y') = a \left[\frac{\cos \theta + \cos \left(\theta + \frac{2\pi}{3}\right) + \cos \left(\theta + \frac{4\pi}{3}\right)}{3} \right] \\ y' &= \frac{a}{3} \left[\sin \theta + \sin \left(\theta + \frac{2\pi}{3}\right) + \sin \left(\theta + \frac{4\pi}{3}\right) \right] = \frac{a}{3} \left[\sin \theta + 2 \sin \left(\theta + \pi\right) + \sin \frac{\pi}{3} \right] = 0 \\ &= 0 \end{aligned}$$

$$\begin{aligned} 81) \quad \text{Two equations } a_1 x^2 + b_1 x + c_1 = 0, a_2 x^2 + b_2 x + c_2 = 0, \text{ where } a_1 b_2 - a_2 b_1 \neq 0 \text{ have common root, if} \\ (c_1 a_2 - c_2 a_1)^2 = (a_1 b_2 - a_2 b_1)(b_1 c_2 - b_2 c_1) \\ \Rightarrow (-21 - 35)^2 = (-3K + K)(-35K - 63K) \\ \Rightarrow K^2 = 16 \end{aligned}$$

$$\Rightarrow K=4$$

82) $28, A_1, A_2, \dots, A_{11}, 10$

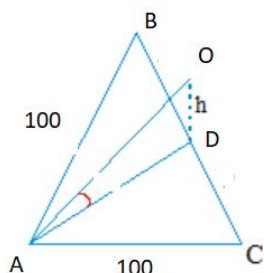
$$d = \frac{b-a}{n+1} = \frac{10-28}{12} = \frac{-18}{12} = -3/2$$

$$A_1 = 28 + d, \quad A_2 = 28 + 2d$$

$\therefore$ All even AM's are integral numbers

$\therefore A_2, A_4, A_6, A_8, A_{10}$ are integers H

83)



Given $\frac{OD}{AB} = \frac{1}{3\sqrt{2}}$ and $\frac{OB}{OD} = 2\sqrt{2}$

$$(OB)^2 = (OD)^2 + (BD)^2 \text{ and } 100^2 = (AD)^2 + (BD)^2$$

$$\therefore OD=20$$

84)

$$\begin{vmatrix} 3 & -2 & 1 \\ \lambda & -14 & 15 \\ 1 & 2 & -3 \end{vmatrix} = 0$$

$$\Rightarrow 36 - 6\lambda - 30 + 2\lambda + 14 = 0$$

$$\Rightarrow \lambda = 5$$

85)

Let $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ $A^2 = \begin{bmatrix} a^2 + b^2 & b(a+c) \\ b(a+c) & b^2 + c^2 \end{bmatrix}$

$$a^2 + b^2 + b^2 + c^2 = 1 \Rightarrow a^2 + 2b^2 + c^2 = 1$$

$$a=0, b=0, c=1, \quad a=1, b=0, c=0$$

$$a=0, b=0, c=-1 \quad a=-1, b=0, c=0$$

Total no 'of matrices = 4

86)

Using caley hamilton then $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & -2 & 4-\lambda \end{vmatrix} = 0 \quad (1-\lambda)[(1-\lambda)(4-\lambda)+2] = 0$$

$$\text{i.e } (1-\lambda)[4-\lambda-4\lambda+\lambda^2+2] = 0$$

$$= (\lambda-1)(\lambda^2-5\lambda+6) = 0$$

$$= \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0 \Rightarrow A^3 - 6A^2 + 11A - 6I = 0$$

$$6A^{-1} = A^2 - 6A + 11I \Rightarrow A^{-1} = \frac{1}{6}[A^2 - 6A + 11I] \quad \alpha=1, \beta=-, \gamma=11 \quad \alpha + \beta + \gamma = 6$$

87)

$$BC=1$$

$$\Rightarrow \text{tr}(A) + \text{tr}\left(\frac{A}{2}\right) + \text{tr}\left(\frac{A}{2^2}\right) + \dots$$

$$= \text{tr}(A) \left[1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \dots \right] = 3 \left[\frac{1}{1 - \frac{1}{2}} \right] = 6$$

88) We know that $|\text{adj}(\text{adj}A)| = |A|^{(n-1)^2}$

$$|\text{adj}(\text{adj}(\text{adj}A))| = |\text{adj}A|^{(n-1)^2}$$

$$= |A|^{(n-1)^3}$$

$$= 2^8 = 256$$

89) Let $f(x, y, z) = x^2 + 4y^2 + 3z^2 - 2x - 12y - 6z + 14$

$$= (x-1)^2 + (2y-3)^2 + 3(z-1)^2 + 1$$

For least value of $f(x, y, z)$

$$x-1=0; 2y-3=0 \text{ and } z-1=0$$

$$\therefore x=1; y=3/2; z=1$$

Hence least value of $f(x, y, z)$ is $f(1, 3/2, 1) = 1$.

90) d be the common difference of A.P

$$a_{10} = 2 + 9d = 3 \quad a_1 = n_1 = 2$$

$$d = \frac{1}{9} \quad a_{10} = h_{10} = 3$$

$$\therefore a_4 = 2 + 3d = \frac{7}{3}$$

D be the common difference of $\frac{1}{h_1}, \frac{1}{h_2} - \frac{1}{h_{10}}$

$$\frac{1}{3} = \frac{1}{h_{10}} = \frac{1}{2} + \frac{9}{D}$$

$$D = \frac{1}{54}$$

$$\frac{1}{h_7} = \frac{1}{h_1} + 6D = \frac{7}{18} \Rightarrow h_7 = \frac{18}{7}$$

$$a_4 h_7 = \frac{7}{3} \cdot \frac{18}{7} = 6$$



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

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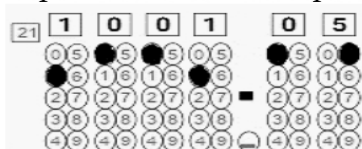
Max.Marks:300

IMPORTANT INSTRUCTIONS:

1. Immediately fill in the Admission number on this page of the Test Booklet with Blue/Black Ball Point Pen only.
2. The candidates should not write their Admission Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1, 2, 3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **two sections**.

(I) Section-I contains 20 **multiple choice** questions with **only one correct** option.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

(II) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only. First 5 attempted questions will be considered if more than 5 questions attempted.
To cancel any attempted question bubble on the question number box.
For example: To cancel attempted question 21, bubble on 21 as shown below.



Question Answered for marking



Question Cancelled for marking

Marking scheme : +4 for correct answer and 0 in all other cases.

6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
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Name of the Candidate (in Capitals): _____

Admission Number:

Candidate's Signature: _____ Invigilator's Signature: _____

13-05-2021_Sr.ICON-Batch-1 & 2_Jee-Main_PT-12_Test Syllabus

PHYSICS

:GRAVITATION, STATIC ELECTRICITY, STATIC ELECTRICITY, CURRENT ELECTRICITY, CAPACITORS, EMW & COMMUNICATION SYSTEM, CLASSICAL MAGNETISM, EMI & AC, **ELECTRO MAGNETISM , RAY & WAVE OPTICS, SHM & SOUND WAVES**

CHEMISTRY

: SOLID STATE, SOLUTIONS & COLLIGATIVE PROPERTIES,CHEMICAL KINETICS, IONIC & CHEMICAL EQUI,**ATOMIC STRUCTURE, S-BLOCK & HYDEROGEN AND ITS COMPOUNDS.** ELECTRO CHEMISTRY, THERMO CHEMISTRY & THERMODYNAMICS, **15 th & 16 th GROUP ELEMENTS, CHEMICAL BONDING & PERIODIC TABLE, GOC, HALOGEN COMPOUNDS, HYDRO CARBONS INCLUDING BENZENE, SURFACE CHEMISTRY, STATES OF MATTER**

MATHEMATICS:

QUADRATIC EQUATIONS & EXPRESSIONS, SEQUENCES & SERIES, SEQUENCES & SERIES, MATRICES & DETERMINANTS, SYSTEM OF LINEAR EQUATIONS, STATISTICS & MATHEMATICAL REASONING, **HEIGHTS & DISTENCES**, BINOMIAL THEOREM, PARABOLA, ELLIPSE, HYPERBOLA, **P&C, RANDOM VARIABLES & DIRSTIBUTIONS, IN DEFFINITE & DEFFINITE INTEGRALS, PROBABILITY, TANGENTS & NORMALS AND MVT**

PHYSICS

Max Marks: 100

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

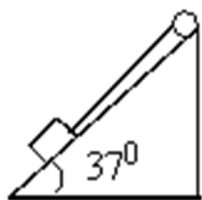
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- 1 A particle moves in the X-Y plane under the influence of a force $\vec{F}$ such that its instantaneous momentum is $\vec{p} = \hat{i}2\cos t + \hat{j}2\sin t$. What is the angle between the force and instantaneous momentum?

1. 0° 2. 45° 3. 90° 4. 180°

- 2 A shell of mass $2m$ fired with a speed u at an angle θ to the horizontal explodes at the highest point of its trajectory into two fragments of mass m each. If one fragment falls vertically, the distance at which the other fragment falls from the gun is given by

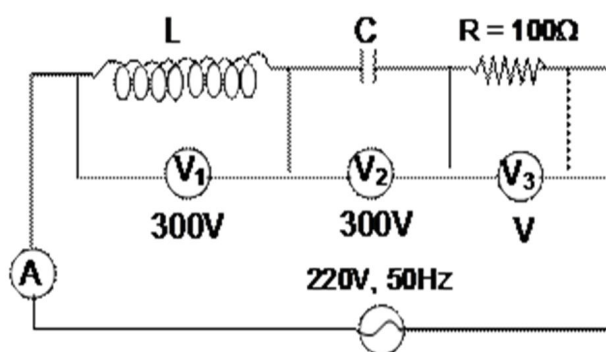
1. $\frac{u^2 \sin 2\theta}{g}$ 2. $\frac{3u^2 \sin 2\theta}{2g}$ 3. $\frac{2u^2 \sin 2\theta}{g}$ 4. $\frac{3u^2 \sin 2\theta}{g}$



- 3 A block of mass 6 kg is attached to a string wrapped around a wheel of radius 20 cm . The acceleration of the block moving down the inclined plane is 4 ms^{-2} . The tension in the string is (I of wheel $= 0.2\text{ kgm}_2$)

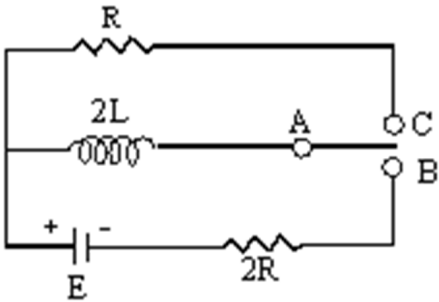
1. 24.5 N 2. 23.4 N 3. 20 N 4. 46.8

- 4 In the circuit shown below, what will be the reading of the voltmeter V_3 and ammeter A?



1. $800\text{V}, 2\text{A}$ 2. $300\text{V}, 2\text{A}$ 3. $220\text{V}, 2.2\text{A}$ 4. $100\text{V}, 2\text{A}$

5 In the circuit shown, A is joined to B for a long time, and then A is joined to C. The total heat produced in R is

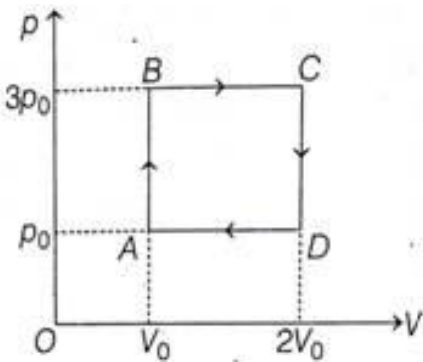


1. $\frac{LE^2}{R^2}$

2. $\frac{LE^2}{2R^2}$

3. $\frac{LE^2}{4R^2}$

4. $\frac{LE^2}{8R^2}$
- 6 An ideal monoatomic gas is carried along the cycle ABCDA as shown in the figure. The total heat absorbed during this process is

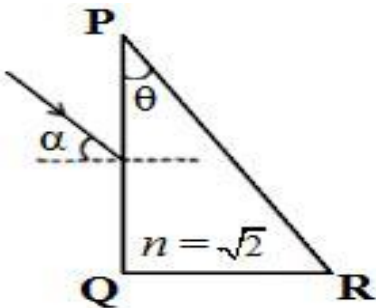


1. $10.5 \rho_0 V_0$

2. $7.5 \rho_0 V_0$

3. $2.5 \rho_0 V_0$

4. $1.5 \rho_0 V_0$
- 7 A parallel beam of light is incident from air at an angle α on the side PQ of a right angled triangular prism of refractive index $n = \sqrt{2}$. Light undergoes total internal reflection in the prism at the face PR when α has a minimum value of 45° . The angle θ of the prism is



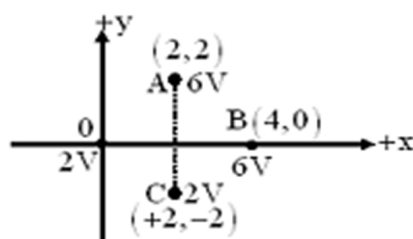
1. 15°

2. 22.5°

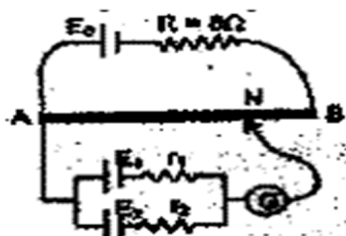
3. 30°

4. 45°

- 8 The coordinates in metre and potentials in volt at different points in a region are as shown. Find the electric field intensity in the region in Vm^{-1} ?



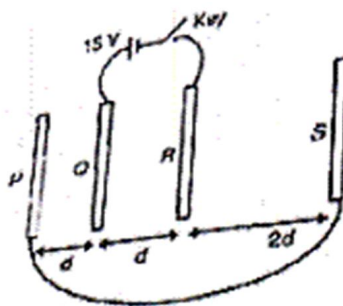
1. $2\hat{i} + 4\hat{j}$ 2. $-(\hat{i} + \hat{j})$ 3. $-2\hat{i} + 4\hat{j}$ 4. $\sqrt{2}\hat{i} - \sqrt{2}\hat{j}$
- 9 A battery of emf $E_0 = 12 \text{ V}$ is connected across a 4m long uniform wire having resistance $4\Omega/m$. The cells of small emf $E_1 = 2V$ and $E_2 = 4V$ having internal resistance 2Ω and 6Ω respectively, are connected as shown in figure if galvanometer shows no deflection at the point N, the distance of point N from the point A is equal to



1. 1/6 cm 2. 80 cm 3. 125 cm 4. 250 cm
- 10 An infinitely long cylindrical wire carries a current 'i' distributed uniformly over its cross section. The magnetic energy stored in a cylindrical shell of height 'h', inner & outer radii R & 2R is :
1. $\frac{\mu_0 i^2 h \ln 2}{4\pi}$ 2. $\frac{\mu_0 i^2 R^2 \ln 3}{4\pi h}$ 3. $\frac{\mu_0 i^2 \ln\left(\frac{2+R}{R}\right)}{\pi}$ 4. None
- 11 If wavelength of photon emitted due to transition of an electron from third orbit to first orbit in a hydrogen atom is λ , then the wavelength of photon emitted due to transition of electron from fourth orbit to second orbit will be

1. $\frac{128}{27}\lambda$ 2. $\frac{25}{9}\lambda$ 3. $\frac{36}{7}\lambda$ 4. $\frac{125}{11}\lambda$

- 12 Consider a system of four identical uncharged infinite metal plates marked P, Q, R and S arranged as shown. The key is now closed. What is the potential difference (in volt) between plates Q and S in steady state?

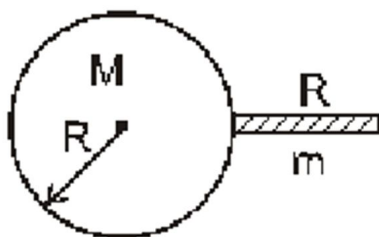


1. 6 2. 3 3. 4 4. 5
- 13 An alternating current $I = I_0 \sin \omega t$ is flowing in a circuit, the ratio of r.m.s current in the interval of 0 to T and the average current in the circuit for the time interval from T/8 to 3T/8 (Where T is time period)
1. $\frac{\pi}{2\sqrt{2}}$ 2. $\frac{\pi}{2}$ 3. $\frac{\pi}{4}$ 4. $\frac{\sqrt{2}\pi}{4}$
- 14 Beryllium nucleus (${}^9_4\text{Be}$) when bombarded by alpha particle undergoes a nuclear reaction producing ${}^{12}_6\text{C}$. Binding energy per nucleon for different species are as under.
 $\text{BE}_{\text{Be}} : 6.4763\text{MeV}; \text{BE}_{\text{He}} : 7.0819\text{MeV}; \text{BE}_{\text{C}} : 7.6885\text{MeV}$
1. The other product that is formed apart from ${}^{12}\text{C}$ is Proton
 2. The other product that is formed apart from ${}^{10}\text{C}$ is Neutron
 3. The energy released in the reaction is 5.65 MeV
 4. The energy released in the reaction is 6.65 MeV
- 15 The magnetic field existing in a region is given by, $B = B_0 \left[1 + \frac{x}{l} \right] \hat{k}$. A square loop of edge 'l' and carrying current 'I' is placed with its edges parallel to the 'x' and 'y' axes. The magnitude of the net magnetic force experienced by the loop is
1. $2B_0Il$ 2. Zero 3. B_0Il 4. $4B_0Il$
- 16 Three coherent waves having amplitudes 12 mm, 6 mm and 4 mm arrived at a given point with successive phase difference of $\frac{\pi}{2}$, if the amplitude of the resultant wave is (x+3)mm find x
1. 7 2. 6 3. 2 4. 4

- 17 A steel wire of length 1m, mass 0.1 kg and uniform cross – sectional area $10^{-6} m^2$ is fixed at both ends. The temperature of the wire is lowered by $20^\circ C$. If transverse waves are set up by plucking the string in the middle, the fundamental mode of vibration is
 $(\gamma = 2 \times 10^{11} Nm^{-2}, \alpha = 1.21 \times 10^{-5} / C^\circ)$

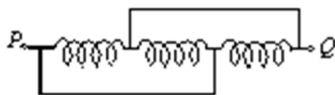
1. 22 Hz 2. 11 Hz 3. 100 Hz 4. 5 Hz

- 18 A uniform thin rod of mass m and length R is placed normally on surface of earth as shown. The mass of earth is M and its radius is R . Then the magnitude of gravitational force exerted by earth on the rod is



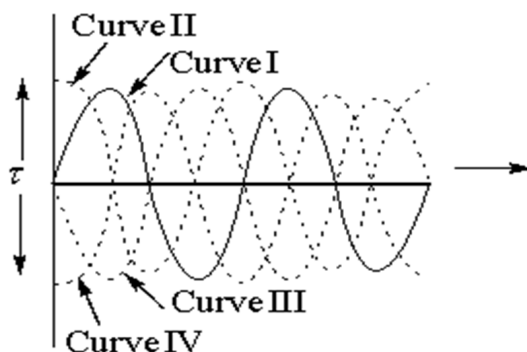
1. $\frac{GMm}{2R^2}$ 2. $\frac{GMm}{6R^2}$ 3. $\frac{4GMm}{9R^2}$ 4. $\frac{GMm}{8R^2}$

- 19 Pure inductors each of inductance 3 H are connected as shown. The equivalent induction of the circuit is



1. 1 H 2. 2 H 3. 3 H 4. 9 H

- 20 The electric dipole is situated in an electric field as shown in fig. both the dipole and the electric field are in the plane of the paper. The dipole is rotated about an axis perpendicular to the plane of the paper and passing through a point C. the direction of rotation is anti-clockwise. If the angle of rotation θ is measured with reference to a line which is perpendicular to the electric field, then the variation of τ with θ is best represented by curve



1. I 2. II 3. III 4. IV

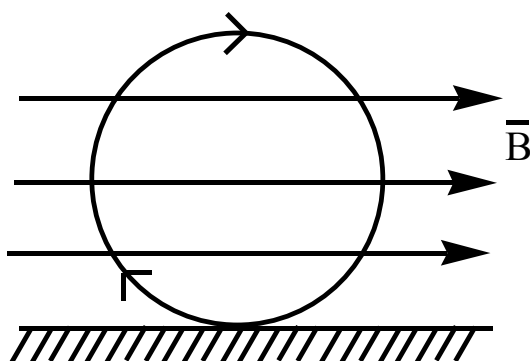
(NUMERICAL VALUE TYPE)

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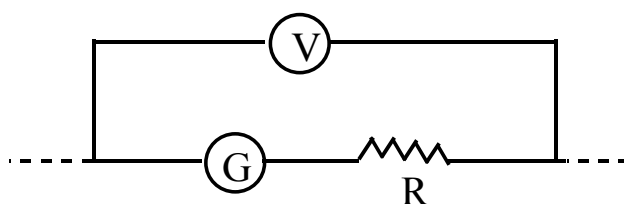
Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

- 21 A thin rod has a total charge Q uniformly distributed along its length L . If the rod rotates with angular velocity ω about an axis passing through center and parallel to length, then its magnetic moment is $\frac{KQ\omega L^2}{24}$. The value of K is...

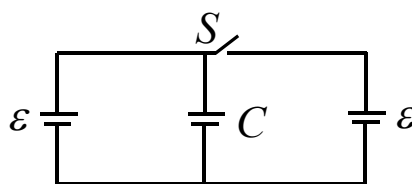
- 22 A conducting ring of mass 2 kg and radius 0.5 m is placed on a smooth horizontal plane. The ring carries a current $i = 4A$. A horizontal magnetic field $B = 10T$ is switched on at time $t = 0$ as shown in figure. The initial angular acceleration of the ring is $n\pi \text{ rad} / s^2$, then $n = \text{-----}$



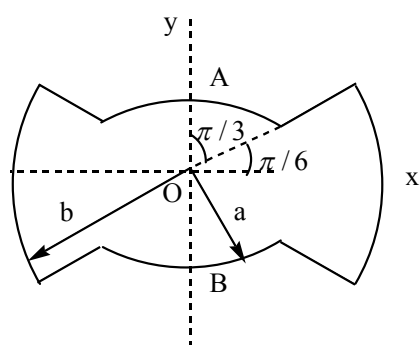
- 23 The galvanometer shown in the figure reads 3A, while the ideal voltmeter reads 24 volt. The value of $R = 7\Omega$. The galvanometer resistance in (ohm) is



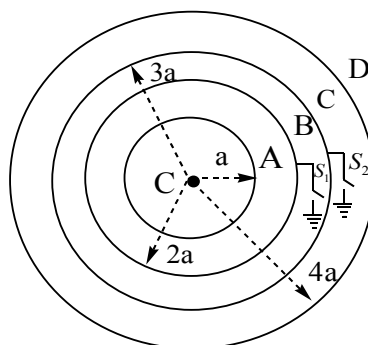
- 24 Initially the circuit is in steady state. When the switch S is closed, the heat generated in the circuit will be :



- 25 A capacitor and a coil in Series are connected to a 6Volt a. c. source by varying the frequency of the source; maximum current of 600mA is observed. If the same coil is now connected to a cell of Emf 6volt and internal Resistance of 2 ohms. The current through it in steady state in $p \times 10^{-1} \text{ Amp}$ the value of P is _____
- 26 There exists a long conductor along z-axis carrying current of 5 A along +ve z-direction. Find the total torque τ on the loop (symmetrical about x and y axes) in xy-plane as shown in the diagram. Resistance per unit length of the wire forming loop equals $2\Omega/m$. Potential difference $V_{BA} = 28\text{volt}$ is applied by using a battery. Radii 'a' = 0.3m and 'b' = 0.9m as shown. If $\tau = \frac{15}{n} \times 10^{-7} \text{ Nm}$ then $n =$



- 27 In the figure, A, B, C and D are four concentric spheres of radius a, 2a, 3a and 4a and contain, charges +3q, +6q, -9q and +12q respectively. In the beginning switches S_1 and S_2 are open. After earthing the sphere C, Δq_1 amount of charge flows from the sphere C to the earth. If the sphere B is also earthed, then Δq_2 amount of charge flows from the sphere B to the earth, find the value of $\frac{|\Delta q_1|}{|\Delta q_2|}$



- 28 A thin lens has focal length ' f ' and its aperture has diameter ' d '. It forms an image of intensity I . Now, the central part of the aperture upto diameter $d/2$ is blocked by an opaque paper. The focal length becomes $n \times f$. The value of ' n ' is
- 29 Equation of a stationary wave and a travelling waves are as follows $y_1 = a \sin kx \cos \omega t$ and $y_2 = a \sin(\omega t - kx)$. The phase difference between two points $x_1 = \frac{\pi}{3k}$ and $x_2 = \frac{3\pi}{2k}$ is ϕ_1 in the standing wave (y_1) and is ϕ_2 in travelling wave (y_2) then ratio $\frac{\phi_1}{\phi_2}$ is $\frac{2x}{7}$ then value of x is
- 30 A particle of mass ' m ' and charge ' q ' is projected with a velocity v_0 in a viscous medium, Where a uniform and constant magnetic field of induction B exists everywhere in a direction perpendicular to the direction of projection of the particle. The force of viscous drag on the particle is given by the law $\vec{f} = -b\vec{v}$, where ' b ' is a positive constant and $\vec{v}$ is velocity of the particle. The distance travelled by the particle during a time interval from the instant of projection until velocity vector turns by 2π radians is $\frac{mv_0}{b} \left(1 - e^{-\frac{n\pi b}{qB}} \right)$. Find ' n '?

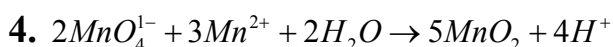
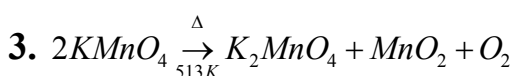
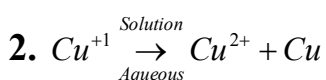
CHEMISTRY**(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

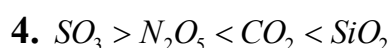
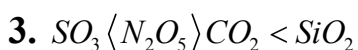
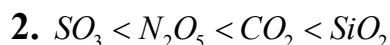
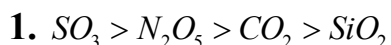
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- 31 For $2H_2O_2 \rightarrow 2H_2O + O_2$, $t_{0.5} = 0.301 \text{ hr}$. When $[H_2O_2]$ at $t = 0$ is 0.5 M , initial rate is
1. 2.303 M.h^{-1} 2. 1.151 M.h^{-1} 3. 4.606 M.h^{-1} 4. 0.301 M.h^{-1}
- 32 Among the following which is not a Lewis acid?
1. $SnCl_2$ 2. $AlCl_3$ 3. CCl_4 4. $SiCl_4$
- 33 In an AB type crystal having rock salt structure, all the atoms along a body diagonal plane are removed. What is the new formula of crystal.
1. AB_3 2. A_3B 3. AB 4. A_2B_3
- 34 Kinetic energy and pressure of a gas are related as
1. $P = \frac{2}{3}E$ 2. $P = \frac{3}{2}E$ 3. $P = \frac{3}{2}E$ 4. $P = 2E$
- 35 In a reaction, KCl is converted into $KClO_4$. Change in oxidation number is
1. 3 2. 5 3. 7 4. 8
- 36 Sodium sulphate is soluble in water but barium sulphate is sparingly soluble because
1. The hydration energy of Na_2SO_4 is more than its lattice energy while the lattice energy of $BaSO_4$ is more than its hydration energy
2. The lattice energy has no role to play in solubility
3. The lattice energy of Na_2SO_4 is more than its hydration energy
4. None of these
- 37 $2Al + 6OH^- + 6H_2O \rightarrow X + 3H_2$
 $2Al + 2OH^- + 6H_2O \rightarrow Y + 3H_2$
 Here, X and Y are
1. $[Al(OH)_5H_2O]^{-2} [Al(OH)_6]^{-3}$ 2. $[Al(OH)_6]^{-3}, [Al(OH)_4(H_2O)_2]^{-}$
3. $[Al(OH)_4]^{-} [Al(OH)_6]^{-3}$ 4. $[Al(OH)_6]^{-3} [Al(OH)_3(H_2O)_3]^{-}$

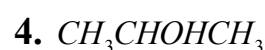
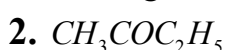
38 Which of the following is a comproportionation reaction



39 Increasing order of acidic character is



40 Haloform reaction is not given by



41 Each carbon in carbon suboxide (C_3O_2) is

1. sp^2 - hybridized

2. sp^3 - hybridized

3. sp - hybridized

4. sp^2 - hybridized but linked with one co-ordinate bond

42 Which of the following isoelectronic ions has the lowest ionization energy?



43 The reaction between an alcohol and carboxylic acid leads to the formation of

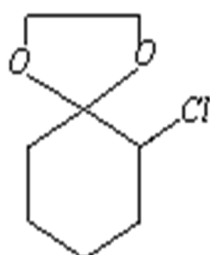
1. Aldehyde

2. Ester

3. Ketone

4. Paraffins

44 Acid catalysed hydrolysis of the acetal gives



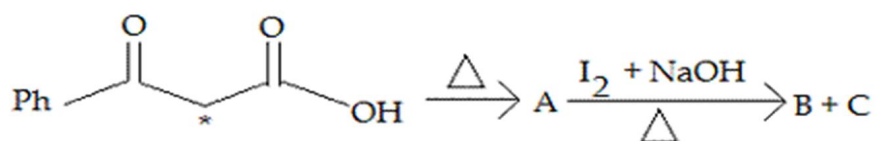
1. Ethanal and 2-chloro cyclohexanol

2. Ethanol and 2-chloro cyclohexanol

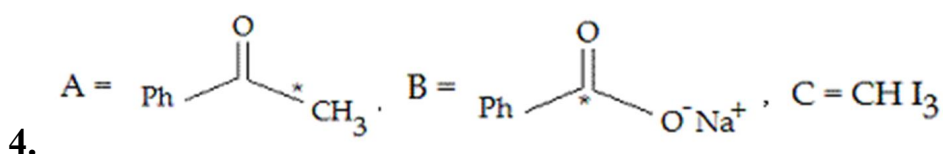
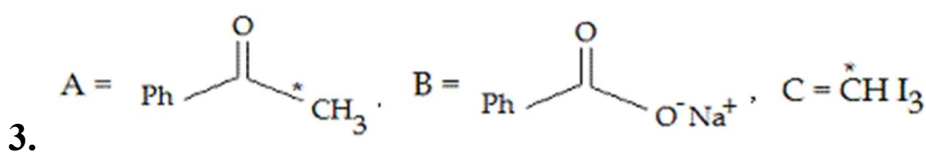
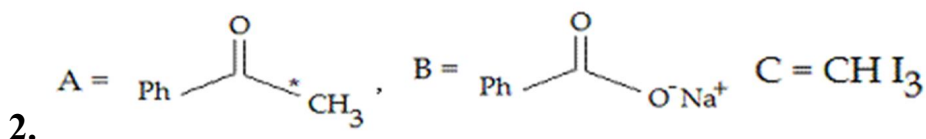
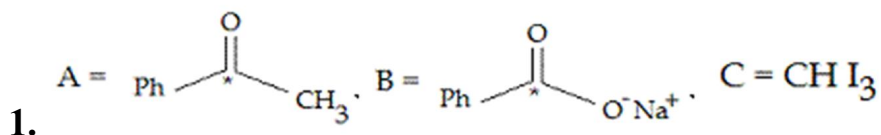
3. 1,2-ethanediol and 2-chlorocyclohexanone

4. 1,2-ethanediol 2-chloro cyclohexanol

45 In the following reaction sequence



The correct structures of A, B and C are



46 Which of the following is not an example of heterogeneous catalytic reaction?

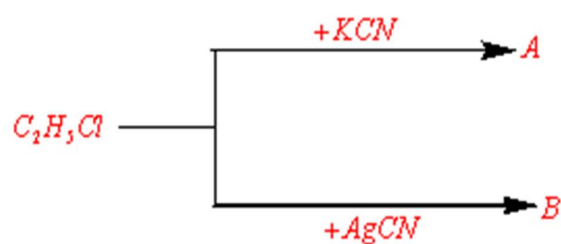
1. Ostwald's process
2. Haber's process
3. Combustion of coal
4. Hydrogenation of vegetable oils

47 4.6×10^{22} atoms of an element weigh 13.8 g. the atomic weight of that element is

1. 290
2. 180
3. 34.4
4. 10.4

48 The radius ratio of first, second and third orbits in hydrogen atom

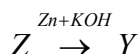
1. 1:4:9
2. 1:1:1
3. 1:3:2
4. 1:2:4



49 A and B are

1. Functional isomers
2. Chain isomers
3. Positional isomers
4. Metamers

50 $X \xrightarrow{KOH} Y$ (gas turns red litmus to blue) + Z



$X \xrightarrow{\Delta}$ gas (supports combustion at higher temperature) Identify (X) to (Z)

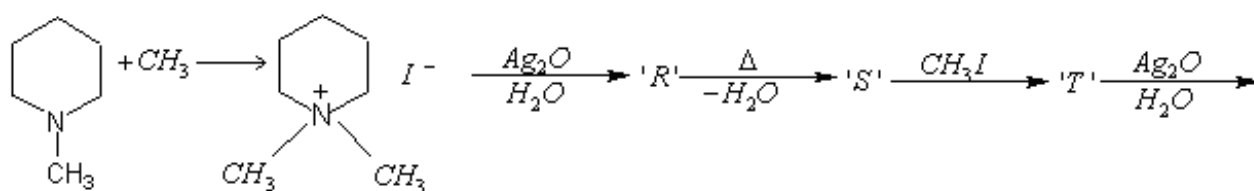
1. $X = NH_4NO_2, Y = NH_3, Z = KNO_2$
2. $X = (NH_4)_2Cr_2O_7, Y = NH_3, Z = Cr_2O_3$
3. $X = (NH_4)_2SO_4, Y = NH_3, Z = K_2SO_4$
4. $X = NH_4NO_3, Y = NH_3, Z = KNO_3$

(NUMERICAL VALUE TYPE)

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51 The photo electric emission requires a threshold wavelength, λ_0 . For a certain metal $\lambda_1 = 2200 \text{ \AA}$ and $\lambda_2 = 1900 \text{ \AA}$ produce electrons with a maximum kinetic energy KE_1 and KE_2 . If $KE_2 = 2KE_1$, calculate corresponding λ_0 .



52 $'U' \xrightarrow[-H_2O]{\Delta} 'V' \xrightarrow{\Delta} W + H_2O + (CH_3)_3N$

The number of π bonds present in W is

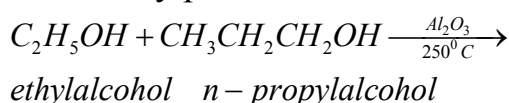
53 The ratio of $\left(\frac{\Lambda_m}{\Lambda_e}\right)$ for $Ca_3(PO_4)_2$ will be equal to _____.

54 The number of greenhouse gases in the following is _____
 $CO_2, O_2, N_2O, CH_4, CFC, CO, H_2O(g), O_3$

55 A gaseous hydrocarbon on combustion produces four times its volume of CO_2 by consuming six times its volume of oxygen. What is the ratio of gram atoms of hydrogen and carbon in that hydrocarbon.

56 What volume of methane is formed at NTP by fusing 7.32 g of sodium acetate with soda lime?

57 How many products are formed for the following reaction



58 $2CH_3COCH_3 \xrightarrow{Mg/benzene} \text{Pinacole}$. The number of '-OH' groups in pinacole are _____

59 The number of possible dipeptides formed with two amino acids A and B are

60 The number of moles of conc. HNO_3 reacts with one mole of copper metal

MATHEMATICS**(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- 61 A letter is known to have come either from 'JALALABAD' or 'ADILABAD'. On the envelope. Just two consecutive letters 'LA' are visible. The probability that the letter comes from 'ADILABAD' is
1. $\frac{4}{11}$ 2. $\frac{7}{11}$ 3. $\frac{5}{11}$ 4. $\frac{6}{11}$
- 62 A man alternately tosses a coin and throws a die continuously. The probability of his getting a head on the coin before he gets 4 on the die is
1. $\frac{1}{2}$ 2. $\frac{6}{7}$ 3. $\frac{3}{4}$ 4. $\frac{2}{3}$
- 63 If x follows a binomial distribution with parameters $n=8$ and $p=1/2$, then $P(|x-4| \leq 2)$ is:
1. 119/128 2. 119/228 3. 19/128 4. 18/28
- 64 $\int \frac{\sqrt{x^2+1} [\log(x^2+1) - 2 \log x]}{x^4} dx$ is equal to :
1. $\frac{1}{3} \left(1 + \frac{1}{x^2}\right)^{1/2} \left[\log \left(1 + \frac{1}{x^2}\right) + \frac{2}{3} \right] + c$ 2. $-\frac{1}{3} \left(1 + \frac{1}{x^2}\right)^{3/2} \left[\log \left(1 + \frac{1}{x^2}\right) - \frac{2}{3} \right] + c$
3. $\frac{2}{3} \left(1 + \frac{1}{x^2}\right)^{3/2} \left[\log \left(1 + \frac{1}{x^2}\right) + \frac{2}{3} \right] + c$ 4. None of these
- 65 $\int e^{\tan^{-1}x} (1+x+x^2) d(\cot^{-1}x)$ is equal to
1. $-e^{\tan^{-1}x} + c$ 2. $e^{\tan^{-1}x} + c$ 3. $-x e^{\tan^{-1}x} + c$ 4. $x e^{\tan^{-1}x} + c$
- 66 If $I_1 = \int_0^{\frac{\pi}{2}} \frac{\sin(\sin x)}{\sin x} dx$, $I_2 = \int_0^{\frac{\pi}{2}} \frac{\sin x}{x} dx$ and $I_3 = \int_0^{\frac{\pi}{2}} \frac{\sin(\tan x)}{\tan x} dx$
- then which of the following is true
1. $I_1 > I_3$ 2. $I_2 < I_3$ 3. $I_1 < I_2$ 4. $I_1 < I_3$
- 67 $f(x) = \min\{2 \sin x, 1 - \cos x, 1\}$ then $\int_0^{\pi} f(x) dx$ is equal to
1. $\frac{\pi}{3} + 1 - \sqrt{3}$ 2. $\frac{2\pi}{3} - 1 + \sqrt{3}$ 3. $\frac{2\pi}{3} - 1 - \sqrt{3}$ 4. $\frac{5\pi}{6} + 1 - \sqrt{3}$

- 68 The tangent at any point on the curve $x^4 + y^4 = a^4$ cuts intercepts p and q on the axes then $p^{-4/3} + q^{-4/3} =$
1. a 2. $\sqrt{a}$ 3. $\frac{1}{\sqrt{a}}$ 4. $\frac{1}{\sqrt[3]{a^4}}$
- 69 Let $f(x)$ and $g(x)$ be differentiable functions for $0 \leq x \leq 1$, such that $f(0) = 2, g(0) = 0, f(1) = 6$. Let there exist a real number c in $(0,1)$ such that $f'(c) = 2g'(c)$, then the value of $g(1)$ must be
1. 1 2. 2 3. -2 4. -1
- 70 In $[0,1]$ Lagrange's mean value theorem is not applicable to :
1. $f(x) = \begin{cases} \frac{1}{2} - x, & x < 1/2 \\ \left(\frac{1}{2} - x^2\right), & x \geq 1/2 \end{cases}$ 2. $f(x) = \begin{cases} \sin x / x, & x \neq 0 \\ 1, & x = 0 \end{cases}$
3. $f(x) = x|x|$ 4. $f(x) = |x|$
- 71 If $2a + 3b + 6c = 0$ ($a, b, c \in R$) then the quadratic equation $ax^2 + bx + c = 0$ has
1. at least one in $[0,1]$ 2. at least one root in $[2,3]$
3. at least one root in $[4,5]$ 4. None of these
- 72 The value of 'a' for which the system of equations $a^3x + (a+1)^3y + (a+2)^3z = 0$, $ax + (a+1)y + (a+2)z = 0$ and $x + y + z = 0$ has a non-zero solution is
1. 1 2. 0 3. -1 4. 2
- 73 The proposition $\sim(p \rightarrow q) \rightarrow (\sim p \vee \sim q)$ is
1. a tautology 2. a contradiction
3. either (A) or (B) 4. neither (A) nor (B)
- 74 The coefficient of x^4 in the expansion of $(2 - x + 3x^2)^6$
1. 3000 2. 3500 3. 3660 4. 4000
- 75 If $f(x+y) = f(x)f(y)$ and $f(x) = 1 + \tan 2x g(x)$ where $g(x)$ is continuous then $f'(x) =$
1. $f(x)g(0)$ 2. $\frac{1}{2}f(x)g(0)$ 3. $2f(x)g(0)$ 4. $2f(x)g'(0)$
- 76 The common tangents to the circles $x^2 + y^2 + 2x = 0$ and $x^2 + y^2 - 6x = 0$ form a triangle which is:
1. Equilateral 2. Isosceles 3. right angled 4. none of these

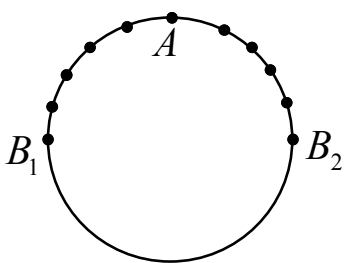
- 77 The normal $y = mx - 2am - am^3$ to the parabola $y^2 = 4ax$ subtends a right angle at the vertex if
1. $m = 1$ 2. $m = \sqrt{2} + 1$ 3. $m = \pm\sqrt{2}$ 4. $m = \sqrt{2} - 1$
- 78 The plane $4x + 7y + 4z + 81 = 0$ is rotated through a right angle about its line intersection with the plane $5x + 3y + 10z = 25$. The equation of the plane in its new position is
1. $x - 4y + 6z = 106$ 2. $x - 8y + 13z = 103$ 3. $x - 4y + 6z = 110$ 4. $x - 8y + 13z = 105$
- 79 Area inside the parabola $y = 5x^2$ and outside the parabola is $2x^2 - y + 9 = 0$ is ____ sq. units.
1. $12\sqrt{3}$ 2. $6\sqrt{3}$ 3. $9\sqrt{3}$ 4. $3\sqrt{3}$
- 80 Let $y = y(x)$ be the solution of the differential equation
- $$\sin x \frac{dy}{dx} + y \cos x = 4x, \quad x \in (0, \pi).$$
- If $y\left(\frac{\pi}{2}\right) = 0$, then $y\left(\frac{\pi}{6}\right)$ is equal to:
1. $\frac{4}{9\sqrt{3}}\pi^2$ 2. $\frac{-8}{9\sqrt{3}}\pi^2$ 3. $-\frac{8}{9}\pi^2$ 4.

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30). **Attempt any five questions out of 10.**

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

- 81 Fifteen persons, among whom are A and B , sit down at random at a round table. If p is the probability that there are exactly 4 persons between A and B find $7p$



- 82 If $\int \sin(101x) \cdot \sin^{99} x \, dx = \frac{\sin kx (\sin x)^k}{k} + c$ then $\frac{k}{20}$ is equal to k then $|k|$ is ____

- 83 The value of $\frac{29 \int_0^1 (1-x^4)^7 \, dx}{4 \int_0^1 (1-x^4)^6 \, dx}$ is

- 84 If the curves $x^3 - 3xy^2 + 2 = 0$ and $3x^2y - y^3 - 2 = 0$ intersect at an angle θ , then the value of $\sin^2 \theta$ is _____.
- 85 If the tangent at any point $P(4m^2, 8m^3)$ of $x^3 - y^2 = 0$ is a normal also to the curve $x^3 - y^2 = 0$, then $9m^2$ is equal to _____.
- 86 The integers x, y and z each are perfect squares and $x > y > z > 0$. If x, y, z form an A.P. then the smallest possible value of x is _____.
- 87 If $f: \{1, 2, 3, \dots\} \rightarrow \{0, 1, 2, \dots\}$ is defined by $f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ -\left(\frac{n-1}{2}\right) & \text{if } n \text{ is odd} \end{cases}$, then $f^{-1}(-100)$ is _____.
- 88 The number of all pairs (x, y) of real numbers $\left(x \in \left(0, \frac{\pi}{2}\right)\right)$ such that
$$\frac{(\sin x)^{2y}}{(\cos x)^{\frac{y^2}{2}}} + \frac{(\cos x)^{2y}}{(\sin x)^{\frac{y^2}{2}}} = \sin 2x.$$
- 89 The edges of a parallelepiped are of unit length and are parallel to non-coplanar unit vectors $\vec{a}, \vec{b}, \vec{c}$ such that $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = \frac{1}{2}$. Then the volume of the parallelepiped is _____.
- 90 In $\triangle ABC$, orthocentre is $(6, 0)$ and Circumcentre is $(2, 2)$. Equation of side BC is $2x + y = 17$. The radius of Circumcircle of $\triangle ABC$ is _____.



Sri Chaitanya IIT Academy., India.

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KEY SHEET

PHYSICS

1	3	2	2	3	3	4	3	5	3
6	1	7	1	8	2	9	3	10	4
11	1	12	4	13	3	14	3	15	3
16	1	17	2	18	1	19	1	20	2
21	0.0	22	40.0	23	1.0	24	0.0	25	5.0
26	2.0	27	3.0	28	1.0	29	3.0	30	2.0

CHEMISTRY

31	2	32	3	33	3	34	1	35	4
36	1	37	2	38	4	39	1	40	3
41	3	42	4	43	2	44	3	45	3
46	3	47	2	48	1	49	1	50	4
51	2612.5	52	2.0	53	6.0	54	6.0	55	2.0
56	2.0	57	3.0	58	2.0	59	4.0	60	4.0

MATHEMATICS

61	1	62	2	63	1	64	2	65	3
66	1	67	4	68	1	69	2	70	1
71	1	72	3	73	1	74	3	75	3
76	1	77	3	78	1	79	1	80	3
81	1.0	82	5.0	83	7.0	84	1.0	85	2.0
86	49.0	87	201.0	88	1.0	89	0.5	90	8.0

SOLUTIONS

PHYSICS

1) $\vec{F} = \frac{d\vec{p}}{dt} = -\hat{i}2\sin t + \hat{j}2\cot t$

Hence $\vec{F} \cdot \vec{p} = 0$, hence angle between $\vec{F}$ and $\vec{p}$ is 90°

2) $X = u \cos \theta \times \frac{u \sin \theta}{g} + 2u \cos \theta \times \frac{u \sin \theta}{g} \Rightarrow \frac{3u^2 \sin 2\theta}{2g}$

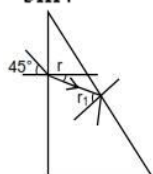
3) $\tau = T.r = I \propto \quad T = \frac{Ia}{r^2} = \frac{0.2 \times 4}{(20 \times 10^{-2})^2} \quad T = 20 \text{ N}$

4) In resonating LCR circuit, $V_L = V_C \Rightarrow V_R = \varepsilon_{rms} \quad i_{rms} = \frac{e_{rms}}{R}$

5) Conceptual

6) Conceptual

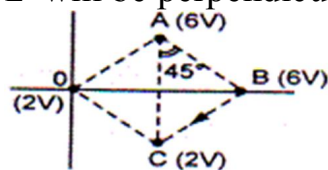
7) $\frac{\sin 45^\circ}{\sin r} = \sqrt{2} \Rightarrow \quad r = 30^\circ$



Also $r_1 \geq 45^\circ$ for internal reflection. $90 - \theta = r + 90 - r_1$

$r_1 > 30 + \theta \geq 45^\circ \quad \theta \geq 15^\circ$

8) $\vec{E}$ will be perpendicular to AB in the direction of decreasing potential



$|\vec{E}| = \frac{(6-2)}{AC \sin 45^\circ} = \frac{4}{4/\sqrt{2}} = \sqrt{2}$

$\vec{E} = -\sqrt{2} \times \cos 45^\circ \hat{i} - \sqrt{2} \sin 45^\circ \hat{j} = -(\hat{i} + \hat{j})$

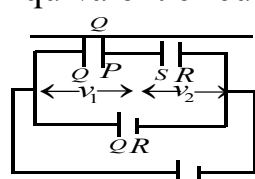
9) $V = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} = \frac{2 \times 6 + 4 \times 2}{8} = \frac{5}{2} \quad V = ipl, \quad \frac{5}{2} = \frac{12}{8+16} \times 4 \times l \quad l = 125 \text{ cm}$

10) Energy density $u = \frac{B^2}{2\mu_0}$ Energy = $\int u dv \quad \int u dv$

11) $\frac{1}{\lambda} = R \left[\frac{1}{1^2} - \frac{1}{3^2} \right] \Rightarrow \lambda = \frac{9}{8R}$ Again, $\frac{1}{\lambda'} = R \left[\frac{1}{2^2} - \frac{1}{4^2} \right] \Rightarrow \lambda' = \frac{16}{3R}$

Now, $\Rightarrow \frac{\lambda'}{\lambda} = \frac{16}{3R} \times \frac{8R}{9} \text{ or } \lambda' = \frac{128}{27} \lambda$

12) Equivalent circuits



$$C_{Pa} = C_{\theta R} = \frac{\epsilon_0 A}{d} = C \quad C_{SR} = \frac{\epsilon_0 A}{2d} = \frac{C}{2} \quad V_{GS} = V_{GR} \Rightarrow V_1 = \frac{C_{CR} V}{C_{SR} + C_{GP}} = \frac{\left(\frac{C}{2}\right) 15}{\frac{C}{2} + C} \Rightarrow \frac{15}{3} = 5V$$

13)

$$I_{rms} = \frac{I_0}{\sqrt{2}} \quad i_{avg} = \frac{\int I dt}{\int dt} = \frac{\int_0^{3T/8} I_0 \sin \omega t dt}{3T/8 - T/8} \quad i_{avg} = \frac{2\sqrt{2}i_0}{\pi}$$

14)

$${}^4_2\text{He} + {}^9_4\text{Be} \rightarrow {}^{12}_6\text{C} + n \quad \text{Energy released} = \text{Final binding energy} - \text{initial binding energy}$$

$$= (12 + 7.6885 - 9 \times 6.4763 - 4 \times 7.6819) = 5.65 \text{ MeV}$$

15)

$$\mathbf{F}_x + \mathbf{F}_{da} = 0$$

$$\mathbf{F}_{ab} = I B_0 \left(1 + \frac{x}{l}\right)$$

$$= I B_0 l + I B_0 x$$

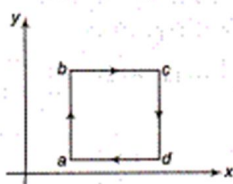
$$\mathbf{F}_{cd} = I B_0 \left[1 + \frac{x+l}{l}\right]$$

$$= 2I B_0 l + I B_0 x$$

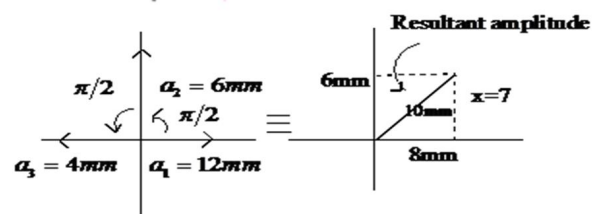
(towards left)

$$\mathbf{F}_{net} = \mathbf{F}_{cd} - \mathbf{F}_{ab} = I B_0 l$$

(towards left)



16)



17)

Conceptual

18)

$$E \text{ due to rod at centre} = \frac{Gm}{2R^2}$$

19)

$$\text{All three are in parallel. Hence } L_p = \frac{L}{3} = 1H$$

20)

$$\vec{\tau} = \vec{p} \times \vec{E} = pE \sin \phi, \text{ Where } \phi \text{ is the angle between } \vec{p} \text{ and } \vec{E}.$$

$$\text{Here, } \phi = 90 - \theta \Rightarrow \tau = pE \sin(90 - \theta)$$

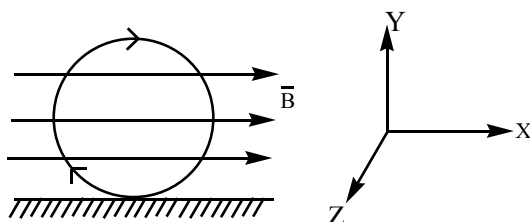
$$\tau = pE \cos \theta \quad \tau \propto \cos \theta.$$

21)

$$\frac{M}{L} = \frac{q}{2m} \text{ and, } l=0$$

22)

$$\vec{M} = iA(-K) = -(4)(\pi \times 0.5^2) \hat{k} \quad \vec{M} = -\pi \hat{k}, \quad \vec{T} = \vec{M} \times \vec{B} = (-\pi \hat{k}) \times (10 \hat{i}) = -10\pi \hat{j}$$



Axis of rotation is along T i.e., axis of rotation is the y -axis Moment of inertia of ring about y -axis is

$$I = \frac{1}{2} m R^2 = \frac{1}{2} (2) (0.5)^2 = 0.25 \text{ kgm}^2 \quad \therefore \quad \alpha = \frac{|\bar{T}|}{I} = \frac{10\pi}{0.25} = 40\pi \text{ and } s^{-2}$$

23) $V = I(G + R)$ or $24 = 3(G+7)$ or $G = 1\Omega$

24) Same p.d exists $\therefore$ heat generated = 0

25) at resonance $X_C = X_L$ and $I_C = \frac{V}{R}$

$$R = 10\Omega$$

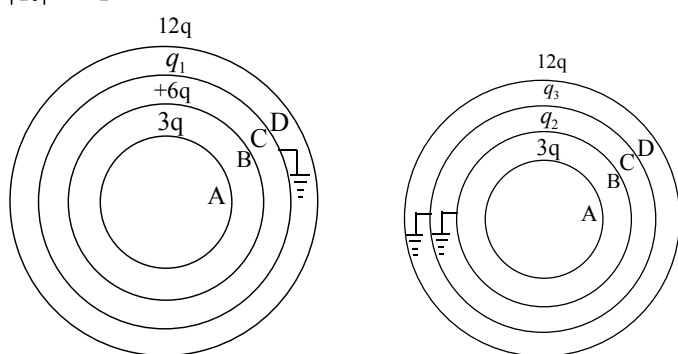
$$\therefore \text{2nd case of d.c source } i = \frac{E}{R+r} = \frac{6}{10+2} = \frac{6}{12} = 0.5$$

26) $-7.5 \times 10^{-7} \hat{i} \text{ N} - m$

27) $\frac{1}{4\pi\epsilon_0} \left[\frac{3q}{3a} + \frac{6q}{3a} + \frac{q_1}{3a} + \frac{12q}{4a} \right] = 0$

$$\Rightarrow q_1 = -18q$$

$$|q_1| = 9q$$



$$\frac{1}{4\pi\epsilon_0} \left[\frac{3q}{3a} + \frac{q_2}{3a} + \frac{q_3}{3a} + \frac{12q}{4a} \right] = 0 \quad \frac{1}{4\pi\epsilon_0} \left[\frac{3q}{2a} + \frac{q_2}{2a} + \frac{q_3}{3a} + \frac{12q}{4a} \right] = 0$$

$$\text{Solving } q_2 = -3q \quad |\Delta q_2| = 3q \quad \frac{|\Delta q_2|}{|\Delta q_1|} = 3$$

28) Focal length does not depend on aperture diameter for paraxial rays

29) $\phi_1 = \pi \quad \phi_2 = k \quad \Delta x = k \left(\frac{3\pi}{2k} - \frac{\pi}{3k} \right) = \left(\frac{7\pi}{6} \right) \frac{\phi_1}{\phi_2} = \frac{6}{7}$

30) $m \frac{dv}{dt} = -bv$; $\frac{dv}{dt}$ is tangential acceleration

$$m(dv)v = -bvRd\theta$$

$$dv = -\frac{b}{m} \left(\frac{mv}{qB} \right) d\theta \quad \int_{v_0}^v \frac{dv}{v} = -\frac{b}{2B} \int_0^{2\pi} d\theta$$

$$v = v_0 e^{\frac{-b2\pi}{qB}}$$

But distance covered $S = \frac{m}{b} (v_0 - v)$ till it attains a speed 'v'

CHEMISTRY

$$31) \quad K = \frac{0.693}{t_{\frac{1}{2}}} = \frac{0.693}{0.301} \quad \text{Rate} = K [H_2O_2]^1 = \frac{0.693}{0.301} \times 0.5 = 1.151 M.h^{-1}$$

32) Conceptual

33) Body diagonal plane consists of four corners, 2 edges, 2 Face centres and Body centre

$$34) \quad KE = \frac{3}{2} RT = \frac{3}{2} PV \quad \frac{2}{3} E = P$$

35) Conceptual

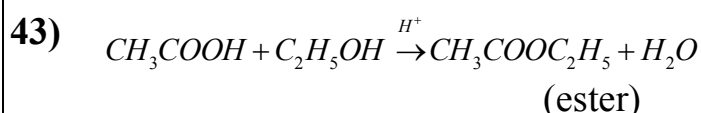
36) Conceptual

37) Conceptual

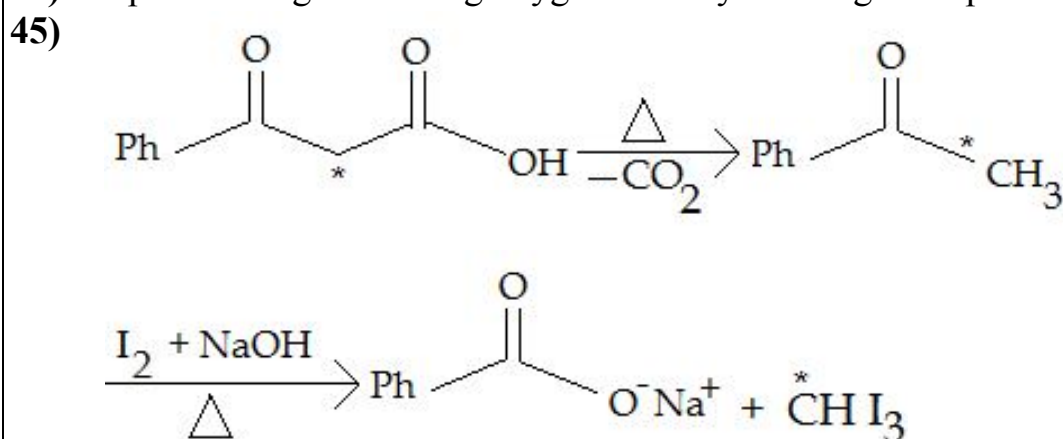
38) Conceptual

39) Non – metallic oxides are acidic

40) Conceptual

41) $O=C=C=C=O$ is sp hybridised42) Amongst the isoelectronic species, the size of S^{2-} is largest hence its IE will be the lowest.

44) Open the ring containing oxygen atoms you will get the products.



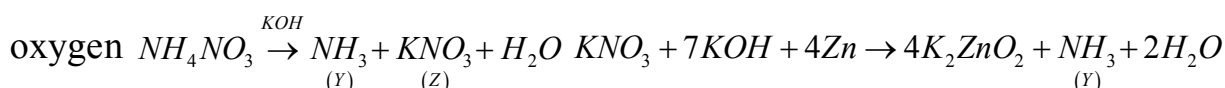
46) Then is no catalyst is required for combustion of coal

47) 4.6×10^{22} atoms weigh = 13.8 g

$$\text{Hence, } 6.02 \times 10^{23} \text{ atoms will weight} = \frac{13.8 \times 6.02 \times 10^{23}}{4.6 \times 10^{22}} = 180.6 \text{ g (molar mass)}$$

48) Conceptual

49) Conceptual

50) N_2O supports combustion at higher temperature because on decomposition, it releases

51) Conceptual

52) $CH_2 = CH - CH_2CH = CH_2$ (W)

$$53) \quad \Lambda_m = \Lambda_e \times n$$

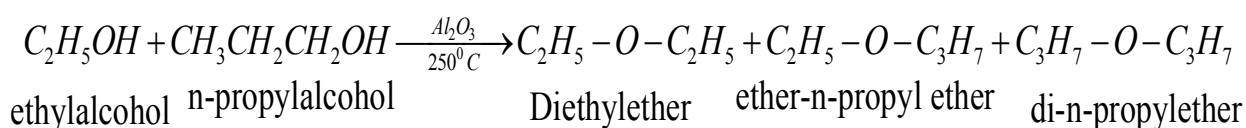
54) Greenhouse gases that absorb and emit radiant energy within a thermal infrared range

which causes an increase in atmospheric temperature. The list of gases are water H_2O , CO_2 , CH_4 , N_2O , O_3 & CFC_s

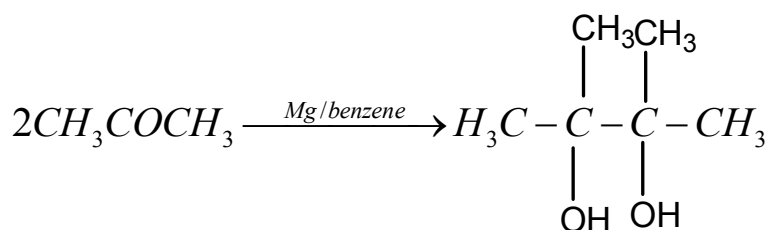
55) $C_xH_y + \left(x + \frac{y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$ ratio of hydro carbon and CO_2 gives no of carbon atom

56) Conceptual

57) If mixture of two alcohols are subjected to dehydration with Al_2O_3 at $250^\circ C$ or conc. H_2SO_4 at $140^\circ C$, we get a mixture of ethers



58)



pinacol

59) Conceptual

60) $Cu + 4HNO_3 \rightarrow 2NO_2 + Cu(NO_3)_2 + 2H_2O$

MATHEMATICS

61) Using Baye's theorem

$$p(E_1) = p(E_2) = \frac{1}{2}$$

$$p\left(\frac{A}{E_1}\right) = \frac{2}{8} = \frac{1}{4}, p\left(\frac{A}{E_2}\right) = \frac{1}{7}$$

$$p\left(\frac{E_2}{A}\right) = \frac{\frac{1}{7}}{\frac{1}{4} + \frac{1}{7}} = \frac{4}{11}$$

$$62) P(E) = \frac{1}{2} + \left(\frac{1}{2} \cdot \frac{5}{6}\right) \frac{1}{2} + \left(\frac{1}{2} \cdot \frac{5}{6}\right)^2 \frac{1}{2} + \dots = \frac{6}{7}$$

$$63) |x-4| \leq 2 \Rightarrow -2 \leq x-4 \leq 2 \Rightarrow 2 \leq x \leq 6. \quad p=1/2, n=8, \text{ then } P(n=8, p=1/2, 2 \leq x \leq 6) \\ = \frac{1}{2^8} \{ {}^8C_2 + {}^8C_3 + \dots + {}^8C_6 \} = \frac{2^8 - (1+8+8+1)}{2^8} = \frac{119}{128}.$$

$$64) I = \int \frac{x\sqrt{1+1/x^2} \log(1/x^2+1) dx}{x^4}$$

Putting $1 + \frac{1}{x^2} = t, \frac{-2}{x^3} dx = dt$, we get

$$I = -\frac{1}{2} \int \sqrt{t} \log t dt = -\frac{1}{2} \left\{ \frac{2}{3} t^{3/2} \log t - \frac{2}{3} \int t^{3/2} \cdot \frac{1}{t} dt \right\} \\ = -\frac{1}{2} \left\{ \frac{2}{3} t^{3/2} \log t - \frac{2}{3} \cdot \frac{2}{3} t^{3/2} \right\} + c = -\frac{t^{3/2}}{3} \left\{ \log t - \frac{2}{3} \right\} + c = -\frac{1}{3} \left(1 + \frac{1}{x^2} \right)^{3/2} \left\{ \log \left(1 + \frac{1}{x^2} \right) - \frac{2}{3} \right\} + c.$$

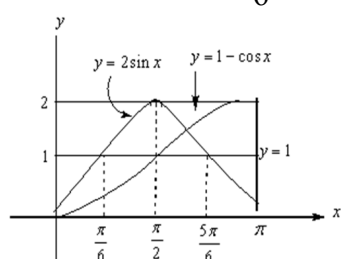
$$65) I = \int e^{\tan^{-1} x} (1+x+x^2) - \left(\frac{1}{1+x^2} \right) dx = -\int e^{\tan^{-1} x} \left(1 + \frac{x}{1+x^2} \right) dx = -\int e^{\tan^{-1} x} dx - \int x \frac{e^{\tan^{-1} x}}{1+x^2} dx \\ = -\int e^{\tan^{-1} x} dx - x e^{\tan^{-1} x} + \int e^{\tan^{-1} x} dx = -x e^{\tan^{-1} x} + c$$

$$66) 0 < x < \frac{\pi}{2} \Rightarrow \frac{\sin x}{x} \text{ is decreasing and } \sin x < x < \tan x$$

$$\Rightarrow \frac{\sin(\sin x)}{\sin x} > \frac{\sin x}{x} > \frac{\sin(\tan x)}{\tan x} \Rightarrow I_1 > I_2 > I_3$$

67) From the figure, we have $f(x) = \min \{2 \sin x, 1 - \cos x, 1\}$

$$1 - \cos x, \quad 0 \leq x \leq \frac{\pi}{2} \\ = \{1, \quad \frac{\pi}{2} \leq x \leq \frac{5\pi}{6} \\ 2 \sin x, \quad \frac{5\pi}{6} \leq x \leq \pi$$



So, $\int_0^\pi f(x) dx = \int_0^{\pi/2} (1 - \cos x) dx + \int_{\pi/2}^{5\pi/6} dx + \int_{5\pi/6}^\pi 2 \sin x dx$

$$= [x - \sin x]_0^{\pi/2} + [x]_{\pi/2}^{5\pi/6} - 2[\cos x]_{5\pi/6}^\pi = \frac{\pi}{2} - 1 + \frac{5\pi}{6} - \frac{\pi}{2} - 2\left[-1 + \frac{\sqrt{3}}{2}\right]$$
$$= \frac{5\pi}{6} - 1 + 2 - \sqrt{3} = \frac{5\pi}{6} + 1 - \sqrt{3}$$

68) Given curve $x^4 + y^4 = a^4$
Let $P(x_1, y_1)$ be any point on the curve

$$\frac{dy}{dx} = \frac{-4x^3}{4y^3} = \frac{-x^3}{y^3} \left(\frac{dy}{dx} \right)_p = \frac{-x_1^3}{y_1^3} = m$$

Equation of tangent at $y - y_1 = \frac{-x_1^3}{y_1^3} (x - x_1)$

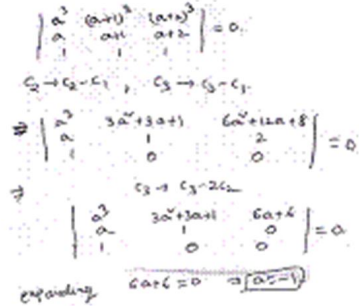
$$xx_1^3 + yy_1^3 = a^4 \frac{x}{a^4} + \frac{y}{a^4} = 1$$
$$\frac{x}{x_1^3} + \frac{y}{y_1^3} = 1$$
$$p = \frac{a^4}{x_1^3} \quad q = \frac{a^4}{y_1^3} \quad p^{\frac{-4}{3}} + q^{\frac{-4}{3}} = \frac{x_1^4}{a^3} + \frac{y_1^4}{a^3} = \frac{a^4}{a^3} = a$$

69) $\frac{f'(c)}{g'(c)} = \frac{f(1) - f(0)}{g(1) - g(0)} \Rightarrow 2 = \frac{6 - 2}{g(1) - 0}$
(Take $\phi(x) = f(x) + kg(x)$, k to be determined such that $\phi(0) = \phi(1)$ and use Rolle's theorem to the function $\phi(x)$ in $[0, 1]$ assuming that $g'(x)$ does not vanish in $(0, 1)$).

70) The function $f(x)$ defined in (A) is not differentiable at $x = 1/2 \in (0, 1)$
Hence, Lagrange's mean value theorem is not applicable to $f(x)$ given in (A).

71) Conceptual

72)



expanding $6a + 6 = 0 \Rightarrow a = -1$

73) Write truth table (a tautology)

P	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg(p \rightarrow q)$	$\neg p \vee \neg q$	$\neg(p \rightarrow q) \rightarrow (\neg p \vee \neg q)$
T	T	F	F	T	F	F	T
T	F	F	T	F	T	T	T
F	T	T	F	T	F	T	T
F	F	T	T	T	F	T	T

74) By definition of Multinomial Theorem, we have

$$(x + y + z)^n = \sum \frac{n!}{p!q!r!} x^p y^q z^r \dots\dots\dots (1)$$

Where, $p + q + r = n \therefore (2 - x + 3x^2)^6 = \sum \frac{6!}{p!q!r!} 2^p (-x)^q (3x^2)^r$

Also, $p + q + r = 6$ (2)

Term contains coefficient of x^4 , we must have

$$q + 2r = 4 \quad \dots (3)$$

Now we have to solve (2) and (3) and find nonnegative integral values of p, q, r .

p	2	3	4
q	4	2	0
r	0	1	2

Putting the above values in (2), the coefficient of x^4 is

$$\frac{6!}{2!4!0!} \cdot 2^2 \cdot 1 \cdot 1 + \frac{6!}{3!2!1!} \cdot 2^3 \cdot 1 \cdot 3 + \frac{6!}{4!0!2!} \cdot 2^4 \cdot 1 \cdot 3^2$$

$$(15 \times 4) + (60 \times 24) + (15 \times 144)$$

$$15(4 + 96 + 144) = 15 \times 244 \quad \therefore \boxed{\text{Coefficient of } x^4 \text{ is } 3660}$$

75) $f'(x) \text{ if } \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x) \text{ if } \lim_{h \rightarrow 0} \frac{g(h) \cdot \tan(2h)}{h} = 2f'(x)g'(0)$

76) Conceptual

77) Equation of the lines joining the origin, vertex of the parabola $y^2 = 4ax \frac{(mx-y)}{2am+am^3}$ or

$$4amx^2 - 4axy = (2am + am^3) y^2 = 0$$

These are at right angles of $4am - 2am - am^3 = 0 \Rightarrow m = \pm\sqrt{2}$

78) The equation of the plane through the line of intersection of the planes

$$4x + 7y + 4z + 81 = 0 \text{ and } 5x + 3y + 10z = 25 \text{ is}$$

$$(4x + 7y + 4z + 81) + \lambda(5x + 3y + 10z - 25) = 0$$

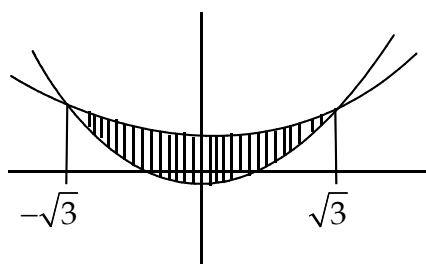
$$\text{or } (4 + 5\lambda)x + (7 + 3\lambda)y + (4 + 10\lambda)z + 81 - 25\lambda = 0$$

which is perpendicular to $4x + 7y + 4z + 81 = 0$

$$\Rightarrow 4(4 + 5\lambda) + 7(7 + 3\lambda) + 4(4 + 10\lambda) = 0 \text{ or } 81\lambda + 81 = 0 \text{ or } \lambda = -1$$

Hence, the plane is $x - 4y + 6z = 106$

79) $\text{Area} = \int_{-\sqrt{3}}^{\sqrt{3}} (2x^2 + 9 - 5x^2) dx$



$$= 2 \int_0^{\sqrt{3}} (9 - 3x^2) dx = 12\sqrt{3}$$

80) $\frac{d}{dx}(y \sin x) = 4x \Rightarrow y \sin x = 4 \frac{x^2}{2} + c$ now $y(\pi/6) = -\frac{8}{9}\pi^2$

81) Let A occupy any seat at the round table.

Then there are 14 seats available for B. If there are to be four persons between A and B, then B has only two ways to sit, as Thus, the probability of the required event is $2/14 = 1/7$.

$$\begin{aligned}
 82) \quad I &= \int \sin(101x) \sin^{99} x dx \\
 \sin(101x) &= \sin(100x + x) \\
 &= \sin 100x \cos x + \cos 100x \sin x \\
 I &= \int \sin 100x \cos x \sin^{99} x dx + \int \cos 100x \sin^{100} x dx \\
 &= \sin 100x \cdot \frac{\sin^{100} x}{100} - \int 100 \cos 100x \cdot \frac{\sin^{100} x}{100} dx + \int \cos 100x \sin^{100} x dx \\
 &= \sin 100x \cdot \frac{\sin^{100} x}{100} + c \therefore k = 100 \Rightarrow \frac{k}{20} = 5
 \end{aligned}$$

$$\begin{aligned}
 83) \quad I &= \int_0^1 (1-x^4)^7 dx \Rightarrow I = \left(x(1-x^4)^7 \right)_0^1 + 7 \times 4 \int_0^1 x^4 (1-x^4)^6 dx \\
 \Rightarrow I &= 28 \int_0^1 (1-(1-x^4)) (1-x^4)^6 dx \Rightarrow I = 28I + 28 \int_0^1 (1-x^4)^6 dx \\
 \Rightarrow 29I &= 28 \int_0^1 (1-x^4)^6 dx \Rightarrow \frac{29 \int_0^1 (1-x^4)^7 dx}{4 \int_0^1 (1-x^4)^6 dx}
 \end{aligned}$$

$$\begin{aligned}
 84) \quad \text{We have } x^3 - 3xy^2 + 2 &= 0 \dots\dots(1) \\
 \text{And } 3x^2y - y^3 - 2 &= 0 \dots\dots(2) \\
 \text{Differentiating (1) and (2) w.r.t. } x, &\text{ we get} \\
 \left(\frac{dy}{dx} \right)_{c_1} = \frac{x^2 - y^2}{2xy} &\text{ and } \left(\frac{dy}{dx} \right)_{c_2} = \frac{-2xy}{x^2 - y^2}.
 \end{aligned}$$

$$\text{Therefore, } \left(\frac{dy}{dx} \right)_{c_1} \times \left(\frac{dy}{dx} \right)_{c_2} = -1.$$

Hence, the two curves intersect at right angles

$$85) \quad y^2 = x^3 \dots\dots\dots(1)$$

$$\Rightarrow 2y \times \frac{dy}{dx} = 3x^2$$

$$\text{Slope of tangent at } P = \left. \frac{dy}{dx} \right|_P = \left. \frac{3x^2}{2y} \right|_{(4m^2, 8m^3)} = 3m$$

$$\therefore \text{Equation of tangent at } P: y - 8m^3 = 3m(x - 4m^2)$$

$$y = 3mx - 4m^3$$

It cuts curve again at point Q, solve (1) and (2) we get $x = 4m^2, m^2$

Put $x = m^2$ in equation (2)

$$\Rightarrow y = 3m(m^2) - 4m^3 = -m^3 \therefore Q \text{ is } (m^2, -m^3)$$

$$\text{Slope of tangent at } Q = \left. \frac{dy}{dx} \right|_{m^2, -m^3} = \frac{3(m^4)}{2 \times (-m^3)} = -\frac{3}{2}m$$

Slope of normal at Q = $\frac{1}{(-3/2)m} = \frac{2}{3m}$. Since tangent at P is normal at Q.

$$\therefore \frac{2}{3m} = 3m \Rightarrow 9m^2 = 2$$

86) Let $x = a^2, y = b^2, z = c^2$. Then $2b^2 = a^2 + c^2$ since b is at least 2, consider the values of b from 2 onwards. The first such value occurs when $b = 5$ (and then $a = 7, c = 1$) so, the least possible value of $x = a^2 = 49$.

87) $f^{-1}(-100) = x \Rightarrow f(x) = -100$

$$\Rightarrow -\left(\frac{x-1}{2}\right) = -100$$

$$x = 201$$

88) Applying the A.M-G.M inequality on the LHS.

$$LHS \geq 2(\sin x \cos x)^{2\left(y - \frac{y^2}{4}\right)}$$

Since $\sin x \cos x < 1$, we must have $y - \frac{y^2}{4} \geq 1 \Rightarrow \left(1 - \frac{y}{2}\right)^2 \leq 0 \Rightarrow y = 2$

Using $y = 2$ in the given equality yields $x = \frac{\pi}{4}$

The required pair is $\left(\frac{\pi}{4}, 2\right)$

89)

$$\Rightarrow [\bar{a} \ \bar{b} \ \bar{c}]^2 = \begin{vmatrix} \bar{a}.\bar{a} & \bar{a}.\bar{b} & \bar{a}.\bar{c} \\ \bar{b}.\bar{a} & \bar{b}.\bar{b} & \bar{b}.\bar{c} \\ \bar{c}.\bar{a} & \bar{c}.\bar{b} & \bar{c}.\bar{c} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix} = 1\left(1 - \frac{1}{4}\right) - \frac{1}{2}\left(\frac{1}{2} - \frac{1}{4}\right) + \frac{1}{2}\left(\frac{1}{4} - \frac{1}{2}\right) = \frac{1}{4}$$

$$\therefore \text{Volume of parallelepiped} = [\bar{a} \ \bar{b} \ \bar{c}] = \frac{1}{2} = 0.5$$

90) Reflection of orthocenter with respect to the side lies on circumcircle

$\therefore$ Circum radius = 8



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant
ICON Central Office – Madhapur – Hyderabad

Sec: **Sr.ICON-Batch-1 & 2**

Jee-Main

Date: 20-05-2021

Time: 09.00 Am to 12.00 Pm

PTM-13

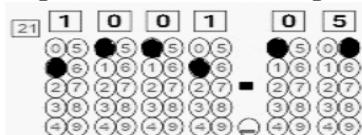
Max.Marks:300

IMPORTANT INSTRUCTIONS:

1. Immediately fill in the Admission number on this page of the Test Booklet with Blue/Black Ball Point Pen only.
2. The candidates should not write their Admission Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1, 2, 3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **two sections**.

(I) Section-I contains 20 **multiple choice** questions with **only one correct** option.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

(II) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only. First 5 attempted questions will be considered if more than 5 questions attempted. To cancel any attempted question bubble on the question number box. For example: To cancel attempted question 21, bubble on 21 as shown below.



Question Answered for marking



Question Cancelled for marking

Marking scheme : +4 for correct answer and 0 in all other cases.

6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capitals): _____

Admission Number:

Candidate's Signature: _____ Invigilator's Signature: _____

20-05-2021_Sr.ICON-Batch-1 & 2_Jee-Main_PT-13_Test Syllabus

PHYSICS

:GRAVITATION, STATIC ELECTRICITY, CURRENT ELECTRICITY, CAPACITORS, EMW & COMMUNICATION SYSTEM, CLASSICAL MAGNETISM, EMI & AC, **ELECTRO MAGNETISM, RAY & WAVE OPTICS, SHM & SOUND WAVES, ERROR ANALYSIS**ATOMIC & NUCLEAR PHYSICS, **MOTION IN A LINE & PLANE, PROPERTIES OF MATTER**

CHEMISTRY

: SOLID STATE, SOLUTIONS & COLLIGATIVE PROPERTIES,CHEMICAL KINETICS, IONIC & CHEMICAL EQUI, **ATOMIC STRUCTURE, S-BLOCK & HYDEROGEN AND ITS COMPOUNDS.** ELECTRO CHEMISTRY, THERMO CHEMISTRY & THERMODYNAMICS, **15 th & 16 th GROUP ELEMENTS, CHEMICAL BONDING & PERIODIC TABLE, GOC, HALOGEN COMPOUNDS, HYDRO CARBONS INCLUDING BENZENE, SURFACE CHEMISTRY, STATES OF MATTER, ALCOHOLS PHENOLS AND ETHERS, ALDEHYDES & KETONES, 15 th & 16 th GROUP ELEMENTS, 17th & ZERO GROUP**

MATHEMATICS

: QUADRATIC EQUATIONS & EXPRESSIONS, SEQUENCES & SERIES, SEQUENCES & SERIES, MATRICES & DETERMINANTS, SYSTEM OF LINEAR EQUATIONS, STATISTICS & MATHEMATICAL REASONING, **HEIGHTS & DISTENCES**, BINOMIAL THEOREM, PARABOLA, ELLIPSE, HYPERBOLA, **P&C**, RANDOM VARIABLES & DIRSTIBUTIONS, IN DEFFINITE & DEFFINITE INTEGRALS, **PROBABILITY, TANGENTS & NORMALS AND MVT, DIFFERENTIAL EQUATIONS, VECTORS & 3D, AREAS**

PHYSICS

Max Marks: 100

(SINGLE CORRECT ANSWER TYPE)

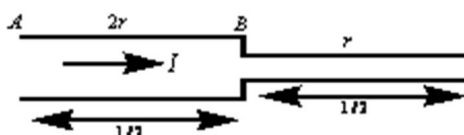
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

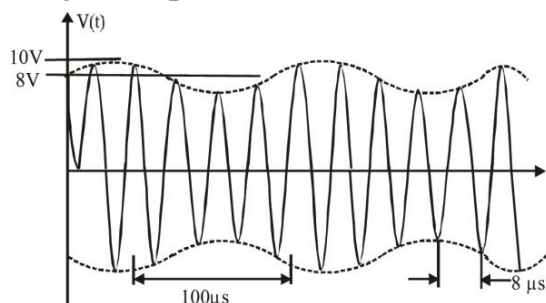
- 1 Find the gravitational potential at a distance half the radius of Earth from the centre of Earth, if the potential on the surface of earth is V_0 ?

1. $\frac{11}{4}V_0$ 2. $\frac{V_0}{2}$ 3. $\frac{11}{8}V_0$ 4. $2V_0$

- 2 If a steady current I is flowing through a cylindrical element ABC, Choose the correct relationship



1. $V_{AB} = 2V_{BC}$
2. Power across BC is 4 times the power across AB
3. Current densities in AB and BC are equal
4. Electric field due to current inside AB and BC are equal
- 3 A parallel plate capacitor with air between the plates has a capacitance of $9pF$. The separation between its plates is ' d '. The space between the plates is now filled with two dielectrics. One of the dielectrics has dielectric constant $k_1 = 3$ and thickness $\frac{d}{3}$ while the other one has dielectric constant $k_2 = 6$ and thickness $\frac{2d}{3}$. Capacitance of the capacitor is now
1. $1.8pF$ 2. $45pF$ 3. $40.5pF$ 4. $20.25pF$
- 4 An amplitude modulated signal is plotted below:-



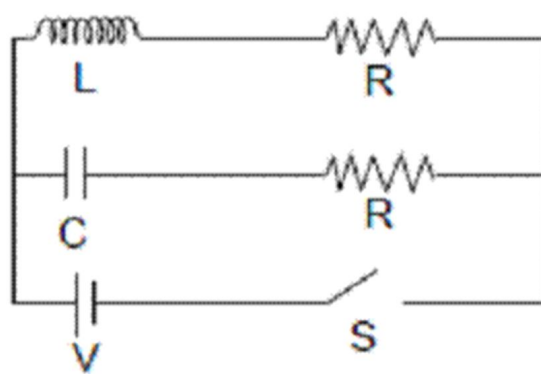
Which one of the following best describes the above signal?

1. $(9 + \sin(2.5\pi \times 10^5 t)) \sin(2\pi \times 10^4 t) V$ 2. $(9 + \sin(4\pi \times 10^4 t)) \sin(5\pi \times 10^5 t) V$
 3. $(1 + 9 \sin(2\pi \times 10^4 t)) \sin(2.5\pi \times 10^5 t) V$ 4. $(9 + \sin(2\pi \times 10^4 t)) \sin(2.5\pi \times 10^5 t) V$

- 5 A short magnet of moment $6.75 A - m^2$ produces a neutral point on its axis. If the horizontal component of earth's field is $5 \times 10^{-5} Wb/m^2$, the distance of the neutral point should be:

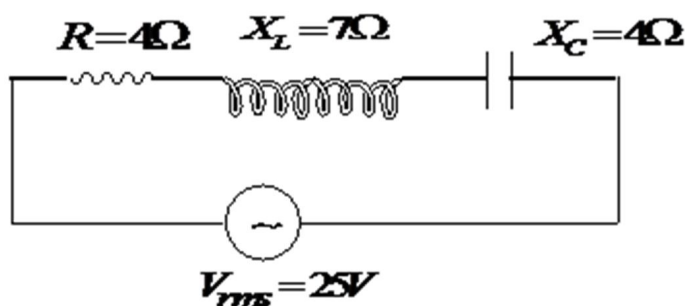
1. 10 cm 2. 20 cm 3. 30 cm 4. 40 cm

- 6 In the circuit shown in the figure, $R = \sqrt{\frac{L}{C}}$. Switch S is closed at time $t = 0$. The current through C and L would be equal after a time t equal to



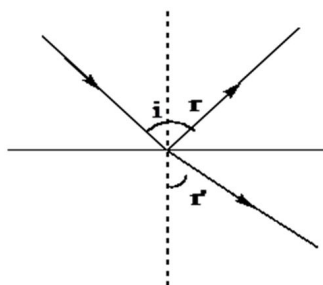
1. CR 2. $CR \ln(2)$ 3. $\frac{L}{R \ln(2)}$ 4. LR

- 7 For the L-C-R series circuit as shown in figure the heat developed in 80sec and amplitude of wattless current i is



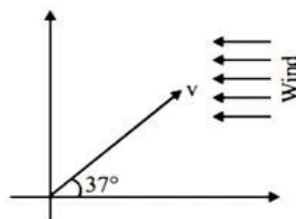
1. 4000J, 3A 2. 8000J, 3A 3. 4000J, 4A 4. 8000J, 5A

- 8 A ray of light is incident at an angle i from denser to rare medium. The reflected and refracted rays are perpendicular to each other. The angle of reflection and the angle of refraction are respectively r and r' , then the critical angle will be



1. $\sin^{-1}(\sin r)$ 2. $\sin^{-1}(\tan r')$ 3. $\sin^{-1}(\tan i)$ 4. $\tan^{-1}(\sin i)$
- 9 **List-I** **List-II**
 I) Diverging lens P) Focal length does not change on dipping in water
 II) Converging lens Q) Always forms a virtual, erect and diminished image of a real object
 III) Concave mirror R) Can form virtual, erect and magnified image of a real object
 IV) Convex mirror S) Can form real, inverted and diminished image of a real object
 T) Focal length changes on dipping in water
- Q. Which of the following options has the correct combinations considering List – I and List – II ?
1. I – Q, T 2. II – P, R, S 3. I – R, S, T 4. II – P, Q
- 10 A physical quantity z depends on four observables a, b, c and d , as $z = \frac{a^2 b^3}{\sqrt{c} d^3}$. The percentages of error in the measurement of a, b, c and d are 2%, 1.5%, 4% and 2.5% respectively. The percentage of error in z is:
1. 12.25% 2. 16.5% 3. 14% 4. 14.5%
- 11 The wavelength of the first spectral line in the Balmer series of hydrogen atom is $6561 \text{ \AA}$. The wavelength of the second spectral line in the Balmer series of singly ionized helium atom is:
1. $1215 \text{ \AA}$ 2. $1640 \text{ \AA}$ 3. $2430 \text{ \AA}$ 4. $4687 \text{ \AA}$
- 12 Consider the nuclear reaction: $X^{200} \rightarrow A^{110} + B^{90}$
 If the binding energy per nucleon for X, A and B is 7.4 eV, 8.2 MeV and 8.2 MeV respectively, what is the energy released?
1. 200 MeV 2. 160 MeV 3. 110 MeV 4. 90 MeV

- 13 A proton and an α - particle are accelerated through the same potential difference. The ratio of their de Broglie wavelengths is:
 1. $\sqrt{2}:1$ 2. $1/\sqrt{2}:1$ 3. $2\sqrt{2}:1$ 4. $2:1$
- 14 The relation between time t and distance x is $t = ax^2 + bx$, where a and b are constants. If v is velocity of particle the acceleration of particle will be
 1. $-2abv^2$ 2. $2bv^3$ 3. $-2av^3$ 4. $2av^2$
- 15 A ball of mass m is thrown at angle 37° with horizontal with a speed of 20 m/s without air resistance. When ball reaches at highest point, the wind starts blowing horizontally which imparts a constant horizontal acceleration ' a ' to the ball in opposite direction.



Q: Mark the correct statement

1. If air does not blow then the time of flight is T and due to blowing of wind the time of flight becomes T' then $T' < T$
2. If air does not blow then the time of flight is T and due to blowing of wind the time of flight becomes T' then $T' > T$
3. If air does not blow then the maximum height attained is H and due to blowing of wind the maximum height attained is H' then $H' = H$
4. If air does not blow then the maximum height attained is H and due to blowing of wind the maximum height attained is H' then $H' > H$
- 16 Starting from the origin at time $t = 0$, with initial velocity $5\hat{j} \text{ ms}^{-1}$, a particle moves in the x - y plane with a constant acceleration of $(10\hat{i} + 4\hat{j}) \text{ ms}^{-2}$. At time t , its coordinates are $(20m, y_0 m)$. The values of t and y_0 are, respectively:
 1. $2s$ and $18m$ 2. $4s$ and $52m$ 3. $2s$ and $24m$ 4. $5s$ and $25m$
- 17 The pressure at the bottom of a tank of water is $3P$ where P is the atmosphere pressure. If the water is drawn out till the level of water is lowered by one fifth, the pressure at the bottom of the tank will now be
 1. $2P$ 2. $(13/5)P$ 3. $(8/5)P$ 4. $(4/5)P$

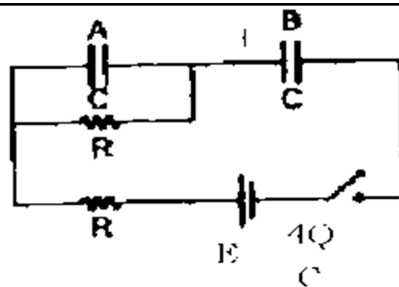
- 18 A liquid is kept in a cylindrical vessel which is being rotated about a vertical axis through the centre of the circular base. If the radius of the vessel is r and angular velocity of rotation is ω , then the difference in the heights of the liquid at the centre of the vessel and the edge is
1. $\frac{r\omega}{2g}$ 2. $\frac{r^2\omega^2}{2g}$ 3. $\sqrt{2gr\omega}$ 4. $\frac{\omega^2}{2gr^2}$
- 19 To what depth must a rubber ball be taken in deep sea so that its volume is decreased by 0.1%. (The bulk modulus of rubber is $9.8 \times 10^8 \text{ N/m}^2$; and the density of sea water is 10^3 kg/m^3)
1. 0.1m 2. 1m 3. 10 m 4. 100 m
- 20 Two soap bubbles coalesce to form a single bubble. If V is the subsequent change in volume of contained air and S change in total surface area, T is the surface tension and P atmospheric pressure, then which of the following relation is correct? (Temperature is constant)
1. $4PV + 3ST = 0$ 2. $3PV + 4ST = 0$ 3. $2PV + 3ST = 0$ 4. $3PV + 2ST = 0$

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30). **Attempt any five questions out of 10.**

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

- 21 The time period of revolution of a planet A around the sun is 27 times that of another planet B. The distance of planet A from sun is how many times greater than that of the planet B from the sun
- 22 A sphere of radius R carries charge such that its volume charge density is proportional to the square of the distance from the centre. What is the ratio of the magnitude of the electric field at a distance $2R$ from the centre to the magnitude of the electric field at a distance of $R/2$ from the centre (i.e. $E_{r=2R} / E_{r=R/2}$)?
- 23 Consider the circuit shown in the figure. Capacitors A and B, each have capacitance $C=2\text{F}$. The plates of capacitor A are shorted using a wire of resistance $R = 1\Omega$ while the capacitor B is given an initial charge $Q = 4 \text{ coulomb}$ with the polarity as shown. The switch is closed at time $t=0$. What will be the initial current (in amp) drawn from the battery immediately after the switch is closed?



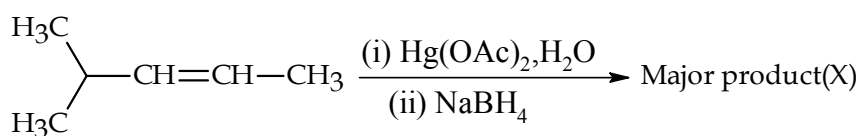
- 24 The electric field (in NC^{-1}) in an electromagnetic wave is given by $E = 50 \sin \omega(t - x/c)$ The energy stored in a cylinder of cross-section $10cm^2$ and length 100 cm along the x-axis will be $n \times 10^{-11} J$. The value of n is ($\epsilon_0 = 8.85 \times 10^{-12} C^2 / Nm^2$)
- 25 In a series RC circuit with an AC source, $R = 300\Omega$, $C = 25\mu F$, $\epsilon_0 = 50V$ and $\nu = 50 / \pi Hz$. Find the peak current in the circuit ($in \times 10^{-1} A$).
- 26 In a YDSE bi-chromatic light of wavelengths 400 nm and 560 nm are used. The distance between the slits is 0.1 mm and the distance between the plane of the slits and the screen is 1 m. The minimum distance between two successive regions of complete darkness is $4(2k+1)$ mm. The value of k is
- 27 If the length and breadth of a thin rectangular sheet are measured using a meter scale as 16.2 cm and 10.1 cm respectively, its surface area is $164 \pm x cm^2$. Then value of 'x' is.
- 28 An electron in an excited state of Li^{2+} ion has angular momentum $3h / 2\pi$. The de Broglie wavelength of the electron in this state is $p \pi a_0$ (where a_0 is the Bohr radius). The value of p is
- 29 A freshly prepared sample of a radioisotope of half-life 1386s has activity 10^3 disintegrations per second. Given that $\log_e 2 = 0.693$, find the fraction of the initial number of nuclei (expressed in nearest, integer percentage) that will decay in the first 80 s after preparation of the sample.
- 30 Two stream of photons, possessing energies equal to thrice and nine times the work function of metal are incident on the metal surface successively. The value of ratio of maximum velocities of the photoelectrons emitted in the two respective cases is $x : y$. The value of y is

CHEMISTRY**(SINGLE CORRECT ANSWER TYPE)**

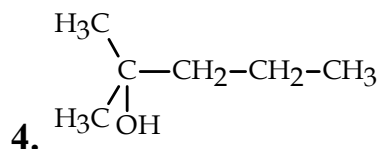
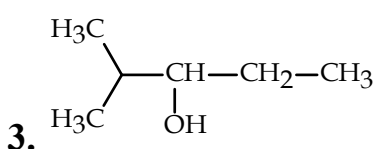
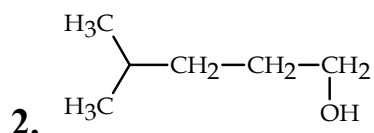
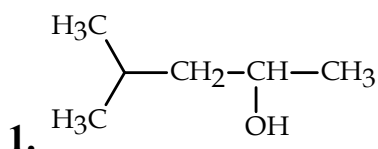
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

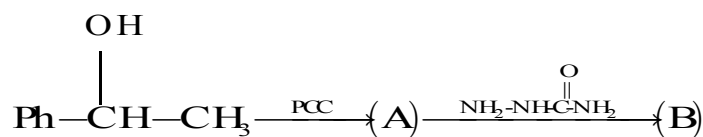
31 In the reaction



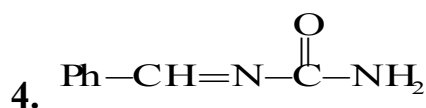
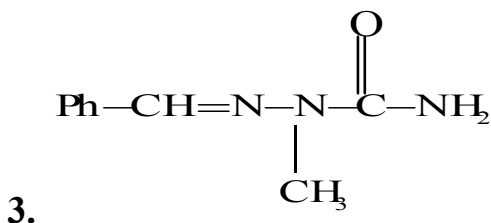
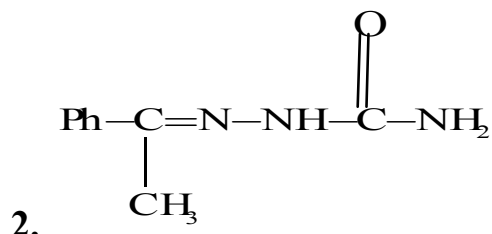
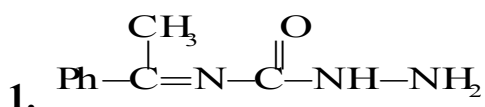
X will be



32



product (B) is



33 Which of the following compound is resistant to nucleophilic attack by hydroxyl ions?

1. Acetonitrile 2. Diethyl ether 3. Acetamide 4. Methyl acetate

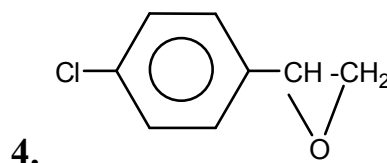
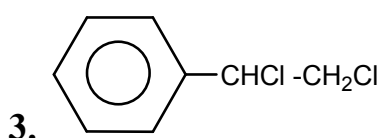
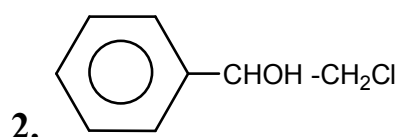
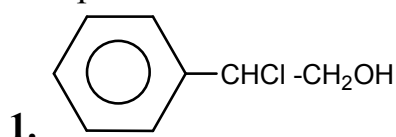
34 **Assertion:** Anhydrous HI in THF of $(CH_3)_3COCH_3$ (aqueous) gives two different products when treated with S_N1 .

Reason: Polar solvent favours S_N1 to give $3^\circ R-X$ and polar aprotic solvent favour S_N2 to give primary alkyl halide.

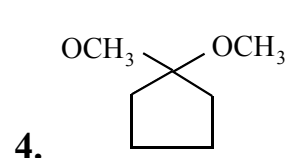
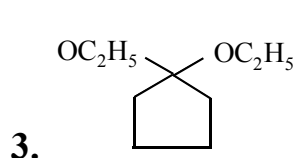
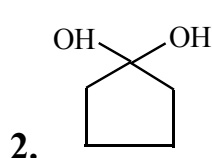
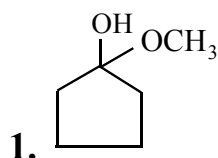
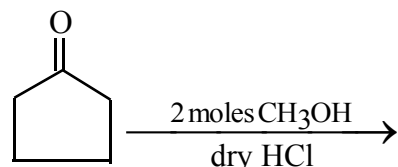
1. Assertion is True, Reason is True and Reason is correct explanation of Assertion
2. Assertion is True, Reason is True but Reason is not correct explanation of Assertion
3. Assertion is True, Reason is False
4. Assertion is False, Reason is True



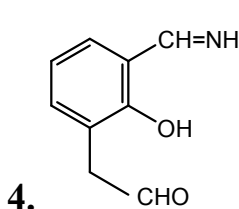
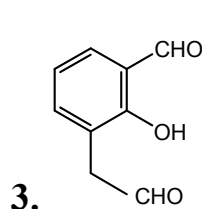
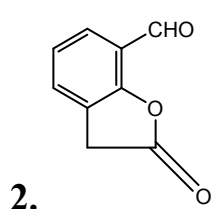
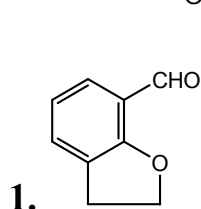
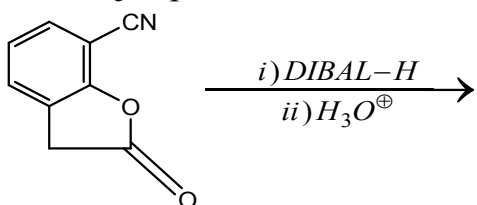
35 The product of the above reaction is

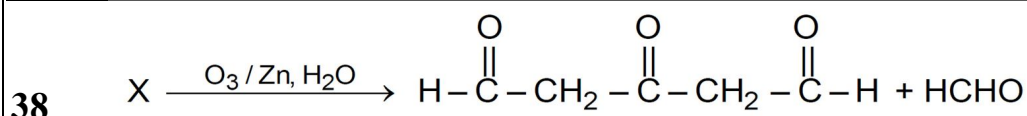


36 The product of this reaction is

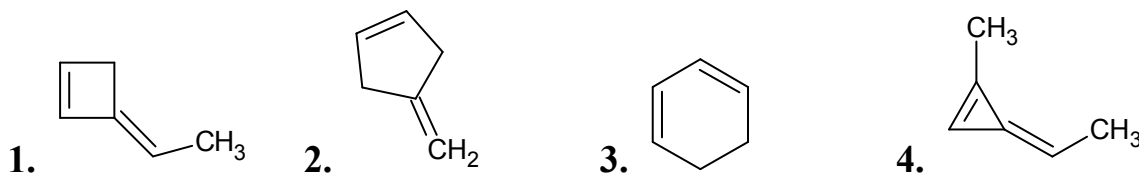


37 The major product of the following reaction is :





The structure of X will be :



39 Nitrogen differ from the rest of the members of this group due to

- I) Its smaller size
- II) High electro negativity
- III) Ionization enthalpy
- IV) non-availability of d-orbitals

1. I, II only 2. I, III, IV only 3. II, III, IV only 4. All

40 The true statement acids of phosphorus H_3PO_2 , H_3PO_3 and H_3PO_4 is

- 1. The order of their acidity is $H_3PO_4 > H_3PO_3 > H_3PO_2$
- 2. All of them are reducing in nature
- 3. All of them are tribasic acids
- 4. The geometry of phosphorus is tetrahedral in all the three

41 Which of the following is wrong with respect to ozone?

- 1. It is a dry bleach 2. It can reduce potassium permanganate
- 3. It is a bent molecule 4. It is a resonance hybrid

42 The compound in which the number of $d\pi - p\pi$ bonds are equal to those present in ClO_4^-

- 1. XeF_4 2. XeO_3 3. XeO_4 4. XeF_6

43 Regarding XeF_2 incorrect statement is

- 1. It is a linear molecule with sp^3d hybridized xenon atom
- 2. It has 3 lone pairs of electrons on central xenon atom
- 3. It's hydrolysis product is Xenon
- 4. It is obtained when xenon and excess fluorine react with each other at 70 bar pressure

- 44 A ball of mass 200gm is moving with velocity of 10ms^{-1} . If the error in measurement of velocity is 0.1%, the uncertainty in its position is
1. $3.3 \times 10^{-31}\text{m}$ 2. $3.3 \times 10^{-27}\text{m}$ 3. $5.3 \times 10^{-25}\text{m}$ 4. $2.64 \times 10^{-32}\text{m}$
- 45 Following are the events taking place to explain adsorption theory
- I: desorption
 II: diffusion of the reactants along the surface
 III: adsorption of the reactants
 IV: formation of the activated surface complex
- These events are taking place in the following order
1. I, II, III, IV 2. III, II, IV, I 3. III, IV, I, II 4. IV, III, II, I
- 46 Which of the following statement is not correct regarding sucrose is
1. Sucrose is made of β -D glucose and D(+) fructose
 2. Hydrolysis of sucrose gives dextrorotatory glucose and laevorotatory fructose
 3. Whole molecule is dextrorotatory before hydrolysis
 4. Sucrose on hydrolysis gives equimolar mixture of D(-) glucose and pp_b fructose
- 47 The concentration of fluoride, lead, nitrate and iron in a water sample from an underground lake was found to be 1000 pp_b , 40 pp_b , 100 ppm and 0.2 ppm, respectively. This water is unsuitable for drinking due to high concentration of
1. Fluoride 2. Lead 3. Nitrate 4. Iron
- 48 The compound having maximum dipole moment is
1. NH_3 2. NF_3 3. NCl_3 4. NI_3
- 49 Which one of the following compounds has sp^2 hybridization?
1. CO_2 2. SO_2 3. N_2O 4. CO
- 50 The rate constant of a reaction is $5 \times 10^{-8} \text{ mole litre}^{-1} \text{ sec}^{-1}$. How long it would take to change concentration for $4 \times 10^{-2} \text{ M}$ to $2 \times 10^{-2} \text{ M}$?
1. $4 \times 10^6 \text{ sec}$ 2. $4 \times 10^5 \text{ sec}$ 3. $4 \times 10^3 \text{ sec}$ 4. 40 sec

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30). **Attempt any five questions out of 10.**

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

- 51 The number of unshared pair of electrons in the electrophile involved in conversion of phenol to salicylaldehyde using CHCl_3/KOH is
- 52 $\text{C}_6\text{H}_5 - \text{O} - \text{CH}_2 - \text{C}_6\text{H}_5 \xrightarrow{\text{HI}}$, How many products are formed
- 53 In CsCl structure, the number of Cs^+ ions that occupy second nearest neighbour locations of a Cs^+ ion is _____.
- 54 Two weak monobasic organic acids HA and HB have dissociation constants as 1.6×10^{-5} and 0.4×10^{-5} respectively at 25°C . If 500 mL of 1M solution of each of these two acids are mixed to produce 1 L of mixed solution, what is the pH of the resulting solution?
- 55 The equilibrium constant for the following reaction is 10^{2x} . The value of 'x' is _____.
 $\text{In}^{2+} + \text{Cu}^{2+} \longrightarrow \text{In}^{3+} + \text{Cu}^+$ (at 298K)
 [Given $E_{\text{Cu}^{2+}/\text{Cu}^+}^0 = +0.15\text{V}$; $E_{\text{In}^{3+}/\text{In}^+}^0 = -0.42\text{V}$; $E_{\text{In}^{2+}/\text{In}^+}^0 = -0.40\text{V}$]
- 56 500 gr of water contains 6×10^{-3} grs of dissolved MgSO_4 in lit. Calculate the hardness of water in ppm of CaCO_3 ?
- 57 A metal crystallizes in f.c.c. Then, the ratio of number of its Second nearest neighbours to the nearest neighbours is
- 58 Number of halides that undergo partial hydrolysis- BF_3 , CaF_2 , NF_3 , SiF_4 , SbCl_3 , SF_6 , CH_4 , BiCl_3
- 59 The mass of glucose that would be dissolved in 50 g of water in order to produce the same lowering of vapour pressure as is produced by dissolving 1 g of urea in the same quantity of water is :
- 60 The equilibrium constant for the mutarotation, $\alpha - D - \text{glucose} \rightleftharpoons \beta - D - \text{glucose}$ is 1.8. What percent of the α - form remains under equilibrium?

MATHEMATICS**(SINGLE CORRECT ANSWER TYPE)**

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61 Let $f(x) = 1 + 3^x \log 3$ and $F(x)$ be its anti-derivative which assume the value 7 for $x = 2$.

The value of x for which the curve $F(x)$ cuts the abscissa axis is:

1. 0 2. 1 3. 2 4. 3

62 The integral $\int \frac{\sec^2 x}{(\sec x + \tan x)^{\frac{11}{2}}} dx$ equals (for some arbitrary constant K)

1. $\frac{-1}{(\sec x + \tan x)^{\frac{11}{2}}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

2. $\frac{1}{(\sec x + \tan x)^{\frac{11}{2}}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

3. $\frac{-1}{(\sec x + \tan x)^{\frac{11}{2}}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

4. $\frac{1}{(\sec x + \tan x)^{\frac{11}{2}}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

63 $\lim_{n \rightarrow \infty} \left(\frac{n!}{n^n} \right)^{\frac{1}{n}} =$

1. e 2. $2e$ 3. $\frac{1}{2e}$ 4. $\frac{1}{e}$

64 If $f(x) = \frac{e^x}{1+e^x}$, $I_1 = \int_{f(-a)}^{f(a)} xg(x(1-x))dx$ and $I_2 = \int_{f(-a)}^{f(a)} g(x(1-x))dx$ then the value of $\frac{I_2}{I_1}$

is

1. -1 2. -2 3. 1 4. 2

- 65 The area bounded by the x-axis and the curve $y = f(x)$ and the lines $x = 1, x = b$ is $\sqrt{b^2 + 1} - \sqrt{2}$ for all $b > 1$ then $f(x) =$
1. $\sqrt{x-1}$ 2. $\sqrt{x+1}$ 3. $\sqrt{x^2+1}$ 4. $\frac{x}{\sqrt{x^2+1}}$
- 66 The area bounded by the curves $y = \ln x$, $y = \ln|x|$, $y = |\ln x|$ and $y = |\ln|x||$ is
1. 4 sq units 2. 6 sq units 3. 10 sq units 4. 8 sq units
- 67 If $y = y(x)$ is the solution of the differential equation $\left(\frac{2 + \sin x}{y+1}\right) \frac{dy}{dx} + \cos x = 0$ with $y(0) = 1$, then $y\left(\frac{\pi}{2}\right) =$
1. $\frac{1}{3}$ 2. $\frac{2}{3}$ 3. 1 4. $\frac{4}{3}$
- 68 If x-intercept of any tangent is 3 times x-coordinate of the point of tangency, then the equation of the curve lying in the first quadrant, given that it passes through (1,1) is
1. $y = \frac{1}{x}$ 2. $y = \frac{1}{x^2}$ 3. $y = \frac{1}{\sqrt{x}}$ 4. $y = \sqrt{x}$
- 69 If the relation between subnormal SN and sub tangent ST at any point S on the curve $by^2 = (x+a)^3$ is $p(SN) = q(ST)^2$, then $\frac{p}{q}$ is equal to
1. $\frac{a}{27b}$ 2. $\frac{8a}{27b}$ 3. $\frac{8b}{27b}$ 4. $\frac{8b}{27}$
- 70 Let $f(x)$ and $g(x)$ be differentiable for $0 \leq x \leq 2$ such that $f(0) = 2, g(0) = 1$ and $f(2) = 8$. let there exist a real number c in $[0, 2]$ such that $f'(c) = 3g'(c)$, then the value of $g(2)$ is _____.
1. 2 2. 4 3. 3 4. 1

- 71 $\vec{a} = 2\vec{i} + \vec{j} - 2\vec{k}$ and $\vec{b} = \vec{i} + \vec{j}$ if $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|$, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ is 30° , then $|(\vec{a} \times \vec{b}) \times \vec{c}| =$
1. $2/3$ 2. $3/2$ 3. 2 4. 3
- 72 A line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube, then $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta =$
1. $\frac{1}{3}$ 2. $\frac{2}{3}$ 3. $\frac{1}{5}$ 4. $\frac{4}{3}$
- 73 A line makes the same angle ' θ ' with each of the x and z axis. If the angle β , which it makes with y -axis, is such that $\sin^2 \beta = 3 \sin^2 \theta$ then $\cos^2 \theta =$
1. $\frac{2}{3}$ 2. $\frac{1}{5}$ 3. $\frac{3}{5}$ 4. $\frac{2}{5}$
- 74 The relation R in the set of natural number N is defined by $xRy \Leftrightarrow x^2 - 4xy + 3y^2 = 0; x, y \in N$, R is
1. Reflexive but not symmetric and transitive
2. Symmetric but not reflexive and transitive
3. Transitive but not reflexive and symmetric
4. Equivalence relation
- 75 Out of the two roots of $x^2 + (1 - 2\lambda)x + (\lambda^2 - \lambda - 2) = 0$, one root is greater than 3 and the other root is less than 3, then the limits of λ are
1. $\lambda < 2$ 2. $2 < \lambda < 5$ 3. $\lambda < 5$ 4. $\lambda = \frac{5}{2}$
- 76 Suppose a, b, c are in A.P and a^2, b^2, c^2 are in G.P if $a < b < c$ and $a + b + c = \frac{3}{2}$, then the value of a is
1. $\frac{1}{2\sqrt{2}}$ 2. $\frac{1}{2\sqrt{3}}$ 3. $\frac{1}{2} - \frac{1}{\sqrt{3}}$ 4. $\frac{1}{2} - \frac{1}{\sqrt{2}}$

- 77 The number of ordered pairs (m, n) , $m, n \in \{1, 2, \dots, 100\}$ such that $7^m + 7^n$ is divisible by 5 is
1. 1250 2. 2000 3. 2500 4. 5000
- 78 The coefficient of $a^8b^5c^7$ in the expansion of $(ab + bc + ca)^{10}$ is _____
1. 5220 2. 2250 3. 2520 4. 2025
- 79 If A is order 3 square matrix such that $|A| = 2$, then $|\text{adj}(\text{adj}(\text{adj}A))|$ is
1. 512 2. 256 3. 64 4. None of these
- 80 The statement $(\sim(p \vee q)) \vee (\sim p \wedge q)$ is logically equivalent to
1. $\sim p$ 2. p 3. q 4. $\sim q$

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30). **Attempt any five questions out of 10.**

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

- 81 $\bar{a}, \bar{b}$ are such that $|\bar{a}| = \sqrt{3}$, $|\bar{b}| = 2$ and $(\bar{a}, \bar{b}) = \frac{\pi}{3}$. Then the area of the triangle with adjacent sides $\bar{a} + 2\bar{b}$ and $2\bar{a} + \bar{b} =$ _____
- 82 The equation of the plane passing through the intersection of the planes $2x - 5y + z = 3$, $x + y + 4z = 5$ and parallel to the plane $x + 3y + 6z = 1$ is $x + 3y + 6z = k$, where k is
- 83 Five different digits from the set of numbers $\{1, 2, 3, 4, 5, 6, 7\}$ are written in random order. If the probability that 5 digit numbers thus formed is divisible by 9 is $\frac{x}{y}$ (x and y are co-prime numbers), then the value of $(y - 10x)$ is
- 84 The normal at any point $P(t^2, 2t)$ on the parabola $y^2 = 4x$ meets the curve again at Q , the area of ΔPOQ , O being the origin is $\frac{k}{|t|}(1+t^2)(2+t^2)$, then $k =$
- 85 If $\ell x + my + n = 0$ is an equation of the line joining the extremities of a pair of semi-conjugate diameter of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$, then $\frac{9\ell^2 + 4m^2}{n^2} =$

- 86 If the product of the lengths of perpendiculars drawn from any point on the hyperbola $x^2 - 2y^2 - 2 = 0$ to its asymptotes is P , then the value of $[P]$, where $[.]$ represents the greatest integer function, is
- 87 If $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}; B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$ then $tr(A) + tr\left(\frac{A(BC)}{2}\right) + tr\left(\frac{A(BC)^2}{4}\right) + \dots \infty =$
- 88 If $A = \begin{bmatrix} 5 & 2004 \\ 1 & 401 \end{bmatrix}$, then the absolute value of $\det\begin{pmatrix} A^{2005} & A^{2004} \\ A & -5A \end{pmatrix}$ equal to
- 89 If $\sum_{i=1}^{18} (x_i - 8) = 9$ and $\sum_{i=1}^{18} (x_i - 8)^2 = 45$ then the standard deviation of the observations $x_1, x_2, \dots, x_{18}$ is
- 90 The angular depressions of the top and the foot of a chimney as seen from the top of a second chimney, which is 150m high and standing on the same level plane as the first are θ & ϕ respectively. If $\tan \theta = \frac{4}{3}$ & $\tan \phi = \frac{5}{2}$, then the distance between their tops is



Sri Chaitanya IIT Academy., India.

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PTM-13

Max.Marks:300

KEY SHEET

PHYSICS

1	3	2	2	3	3	4	4	5	3
6	2	7	1	8	3	9	1	10	3
11	1	12	2	13	3	14	3	15	3
16	1	17	2	18	2	19	4	20	2
21	9.0	22	2.0	23	6.0	24	1.11	25	0.1
26	3.0	27	3.0	28	2.0	29	4.0	30	2.0

CHEMISTRY

31	3	32	2	33	2	34	1	35	1
36	4	37	3	38	2	39	4	40	4
41	2	42	2	43	4	44	4	45	2
46	1	47	3	48	1	49	2	50	2
51	7.0	52	2.0	53	6.0	54	2.5	55	5.0
56	10.0	57	0.5	58	4.0	59	3.0	60	35.71

MATHEMATICS

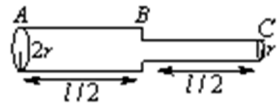
61	2	62	3	63	4	64	4	65	4
66	1	67	1	68	3	69	4	70	3
71	2	72	4	73	3	74	1	75	2
76	4	77	3	78	3	79	2	80	1
81	4.5	82	7.0	83	1.0	84	2.0	85	2.0
86	0.0	87	6.0	88	2004.0	89	1.5	90	100.0

SOLUTIONS

PHYSICS

1) For $r = \frac{R}{2}$, $V = \frac{-GM}{2R^3} \left[3R^2 - \left(\frac{R}{2} \right)^2 \right] = \frac{-11}{8} \frac{GM}{R}$

At $\Rightarrow V = \frac{11}{8} V_0 \left[\because V_0 = \frac{-GM}{R} \right]$ (a) $\frac{V_{AB}}{V_{BC}} = \frac{I_{AB} R_{AB}}{I_{BC} R_{BC}} = \frac{R_{AB}}{R_{BC}} = \frac{\rho \frac{\ell}{2} [\pi \times 4r^2]}{\rho [\pi r^2]} = \frac{1}{4}$

2)  $I_{AB} = I_{BC}$

[Since, (b) $\frac{P_{BC}}{P_{AB}} = \frac{I^2 R_{BC}}{I^2 R_{AB}} = \frac{\rho \frac{\ell}{2} [\pi \times 4r^2]}{\rho \frac{\ell}{2} [\pi r^2]}$ wire is of same material] Therefore, option (a) is

wrong. (c) $\frac{J_{AB}}{J_{BC}} = \frac{\frac{I}{\pi \times 4r^2}}{\frac{I}{\pi \times r^2}} = \frac{1}{4}$;

Therefore, (b) is correct

(d) $\frac{E_{AB}}{E_{BC}} = \frac{\left[\frac{V_{AB}}{\ell/2} \right]}{\frac{V_{BC}}{\ell/2}} = \frac{1}{4}$; Therefore, (c) is incorrect

$C_0 = \frac{\epsilon_0 A}{d} = 9PF$ Therefore, (d) is incorrect

3) $C_1 = 3K_1 C_0 = 3 \times 3 \times 9 = 81PF$

$C_2 = \frac{3}{2} K_2 C_0 = \frac{3}{2} \times 6 \times 9 = 81PF$

$C_1, C_2 \therefore \frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$ are in series $C_s = 40.5PF$ 9 ± 1

4) Analysis of graph says

(1) Amplitude varies as 8-10 V or μs (2) Two time period as

100 $8\mu s$ (signal wave) & $\left[9 \pm 1 \sin \left(\frac{2\pi t}{T_1} \right) \right] \sin \left(\frac{2\pi t}{T_2} \right)$ (carrier wave)

Hence signal is $= 9 \pm 1 \sin (2\pi \times 10^4 t) \sin 2.5\pi \times 10^5 t \frac{\mu_0}{4\pi} \frac{2M}{r^3} = B_H$

5) $\frac{10^{-7} \times 2 \times 6.75}{r^3} = 5 \times 10^{-5} \Rightarrow r = 30cm$ $i_c = \frac{V}{R} e^{-t/RC} = \frac{V}{R} e^{-t/\sqrt{LC}}$

6) $i_c = \frac{V}{R} e^{-t/RC} = \frac{V}{R} e^{-t/\sqrt{LC}}$
 $i_L = \frac{V}{R} \left(1 - e^{-\frac{Rt}{L}} \right) = \frac{V}{R} \left(1 - e^{-t/\sqrt{LC}} \right)$

$$i_C = i_L \text{ will be at } e^{-t/\sqrt{LC}} = \frac{1}{2} \Rightarrow t = \sqrt{LC} \ln(2) = RC \ln 2$$

$$7) Z = 5\Omega \quad H = i_{rms}^2 R t = \frac{25}{2 \times 5} = \left(\frac{25}{5}\right)(4)(80) = 4000J \quad i = i_0 \sin \phi = 5 \times \frac{3}{5} = 3A$$

$$\text{Wattles current } r + r' = 90^\circ$$

$$8) \text{ We have, } r' = 90^\circ - r = 90^\circ - i \text{ Or } \frac{\sin i}{\sin r'} = \frac{\sin i}{\sin(90^\circ - i)}$$

$$\text{Now } \tan i = \frac{1}{r \mu_d} = \mu_r$$

$$\text{Critical angle, } \sin C = \tan i \quad = \therefore C = \sin^{-1}(\tan i) \quad Z = \frac{a^2 b^{1/3}}{\sqrt{cd^3}}$$

9) Conceptual

10) Given: Z

$$\text{Percentage error in } = \frac{\Delta Z}{Z} = \frac{2\Delta a}{a} + \frac{1}{3} \frac{\Delta b}{b} + \frac{1}{2} \frac{\Delta c}{c} + \frac{3\Delta d}{d},$$

$$= 2 \times 2 + \frac{1}{3} \times 1.5 + \frac{1}{2} \times 4 + 3 \times 2.5 = 14\%. \quad \frac{1}{\lambda_{H_2}} = R Z_1^2 \left[\frac{1}{4} - \frac{1}{9} \right], z_1 = 1$$

$$11) \frac{1}{\lambda_{H_2}} = R Z_1^2 \left[\frac{1}{4} - \frac{1}{9} \right], z_1 = 1 \quad \frac{1}{\lambda_{H_e}} = R Z_2^2 \left[\frac{1}{4} - \frac{1}{16} \right], z_2 = 4 \text{ On solving } \lambda_{H_e} = 1215 \text{ A}^0$$

$$12) \frac{\lambda P}{\lambda \alpha} = \sqrt{\frac{m_\alpha q_\alpha}{m_p q_p}}$$

$$13) \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{8}{1}} = 2\sqrt{2} : 1 \quad \frac{dt}{dx} = 2ax \frac{dx}{dt} + b \frac{dx}{dt}$$

$$14) (3) \Rightarrow v = \frac{dx}{dt} = \frac{1}{(2ax+b)} \Rightarrow \frac{d^2x}{dt^2} = \frac{-2a}{(2ax+b)^3} = -2av^3 \quad \vec{u} = 5 \hat{j} \text{ m/s}$$

15) Conceptual

$$16) \text{ Given : } \vec{a} = 10 \hat{i} + 4 \hat{j}$$

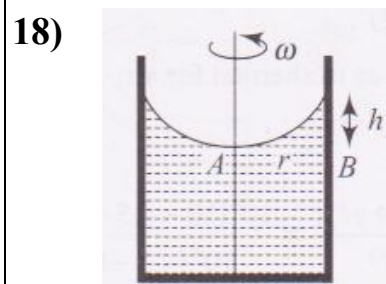
$$\text{Acceleration, } (20, y_0) \text{ and final coordinate } t \text{ in time } S_x = u_x t + \frac{1}{2} a_x t^2.$$

$$\Rightarrow 20 = 0 + \frac{1}{2} \times 10 \times t^2 \Rightarrow t = 2s \quad S_y = u_y t + \frac{1}{2} a_y t^2 \quad y_0 = 5 \times 2 + \frac{1}{2} \times 4 \times 2^2 = 18m$$

$$P^1 = \frac{4}{5} \times 2P + P = \frac{13P}{5}$$

17) The pressure of liquid = $3P - P = 2P$

$$\text{Thus, in the next case pressure, } P_A + \frac{1}{2} dv_A^2 + dgh_A = P_B + \frac{1}{2} dv_B^2 + dgh_B$$



From Bernoulli's theorem,

$$P_A + \frac{1}{2}dv_A^2 + dgh_A = P_B + \frac{1}{2}dv_B^2 + dgh_B$$

$$\text{Here, } h_A = h_B \therefore P_A + \frac{1}{2}dv_A^2 = P_B + \frac{1}{2}dv_B^2 \Rightarrow P_A - P_B = \frac{1}{2}d[V_B^2 - v_A^2]$$

$$\text{Now, } v_A = 0, v_B = r\omega \text{ and } P_A - P_B = hdg \therefore hdg = \frac{1}{2}dr^2\omega^2 \text{ or } h = \frac{r^2\omega^2}{2g}$$

$$19) B = \frac{\Delta P}{\frac{\Delta v}{v}} \Rightarrow \Delta P = B \frac{\Delta v}{v} \quad \Delta P = 9.8 \times 10^8 \times 10^{-3} \text{ N/m}^2$$

$$h = \frac{\Delta P}{dg} = \frac{9.8 \times 10^5}{10^3 \times 9.8} \quad h = 10^2 \text{ m} = 100 \text{ m}$$

$$20) = \left(P + \frac{4T}{r} \right)$$

$$\text{Take pressure} = \frac{4}{3}\pi r^3$$

Volume OH^-

And solve it

$$21) T^{2/3} \propto R \Rightarrow \frac{R_A}{R_B} = \left(\frac{T_A}{T_B} \right)^{2/3} = \left(\frac{27T_B}{T_B} \right)^{2/3} = 9$$

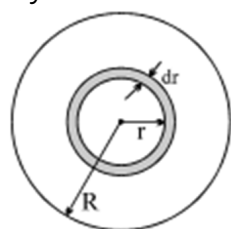
$$22) \rho = \alpha r^2 \text{ [where } \alpha \text{ is a constant]}$$

Charge in the shell (element)

$$\delta\theta = \rho(4\pi r^2 \delta r) = \alpha(4\pi)r^4 \delta r$$

$$\text{Charge enclosed in sphere of radius } r, q = 4 \int_0^r r^4 dr = \frac{4\pi\alpha r^5}{5}$$

By Gauss's theorem,



at $r = R/2$

$$\text{Total charge enclosed, } Q = \frac{4\pi\alpha R^5}{5}$$

$$23) \text{ P.D across } B = \frac{Q}{C} = 2 \text{ volt \& Battery emf} = 8V$$

$\therefore$ The capacitor 'A' acts as short circuit at $t = 0$

$\therefore$ Hence current at $t=0$, is

$$I = \frac{8-2}{1} = 6A$$

$$24) \text{ Energy contained in a cylinder}$$

$$U = \text{average energy density} \times \text{Volume} = \frac{1}{2} \epsilon_0 E_0^2 \times Al$$

$$= \frac{1}{2} \times (8.85 \times 10^{-12}) \times (50)^2 \times (10 \times 10^{-4}) \times 1 = 1.10625 \times 10^{-11} \text{ J}$$

25) Given $R = 300\Omega, C = 25\mu F = 25 \times 10^{-6} F$

$$E_0 = 50V$$

$$f_x = \frac{50}{\pi} \text{ Hz}$$

$$X_C = \frac{1}{\omega C} = \frac{1}{50 / \pi \times 2\pi \times 25 \times 10^{-6}} = \left(\frac{10^4}{25} \right)$$

$$Z = \sqrt{R^2 + X_0^2} = \sqrt{(300)^2 + (10^4 / 25)^2}$$

$$\sqrt{(300)^2 + (400)^2} = 500$$

Peak current $\frac{E_0}{Z} = \frac{50}{500} = 0.1A$

26) $\beta = \frac{D\lambda}{d}$

27) $\ell = 162 \pm 0.1 \text{ cm}$

$$b = 10.1 \pm 0.1 \text{ cm}$$

$$\ell b = 163.62 \pm 2.6 \text{ cm}^2$$

using proper rules

$$\ell b = 164 \pm 3 \text{ cm}^2$$

28) $L = \frac{nh}{2\pi} = \frac{3h}{2\pi} \quad n=3 \quad \lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h2\pi r}{3h} = \frac{2\pi r}{3}$

$$r = a_0 \frac{n^2}{Z} \quad \lambda = \frac{2\pi}{3} a_0 \frac{n^2}{Z} = \frac{2\pi}{3} a_0 \frac{3^2}{3} = 2\pi a_0$$

29) Since, $N = N_0 e^{-\lambda t}$

$$\Rightarrow \frac{N}{N_0} = e^{-\lambda t} \Rightarrow \frac{N}{N_0} = e^{-\frac{\log_e 2}{1386} \times 80}$$

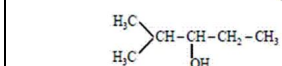
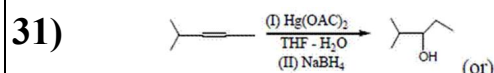
$$\Rightarrow \frac{N}{N_0} = e^{-\frac{0.693 \times 80}{1386}} \Rightarrow \frac{N}{N_0} = e^{-0.04} \Rightarrow \frac{N}{N_0} = \left(\frac{1}{e} \right)^{0.04}$$

$$\text{Fraction of nuclei decayed} = 1 - \frac{N}{N_0} = 1 - \left(\frac{1}{e} \right)^{0.04} = 0.04 = 4\%$$

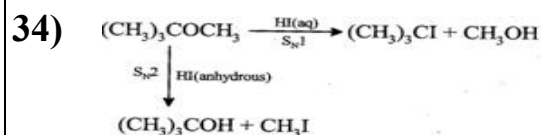
30) $KE_{\max} = h\nu - \phi \quad \frac{1}{2}mv^2 = h\nu - \phi$

$$V = \sqrt{\frac{2(h\nu - \phi)}{m}}$$

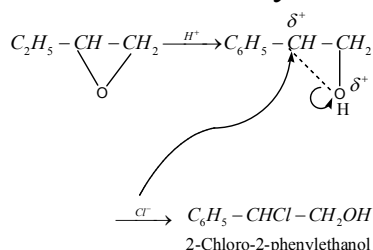
$$\text{Given } h\nu_1 = 3\phi \quad h\nu_2 = 9\phi \therefore \frac{V_1}{V_2} = \sqrt{\frac{h\nu_1 - \phi}{h\nu_2 - \phi}} \quad \frac{V_1}{V_2} = \sqrt{\frac{3\phi - \phi}{9\phi - \phi}} = \frac{1}{2}$$

CHEMISTRY

33) Diethyl ether itself being a Lewis base, is not attacked by nucleophiles like, $C_6H_5^-$ ions, And all others contains an electrophile carbon and are readily attacked by nucleophiles.

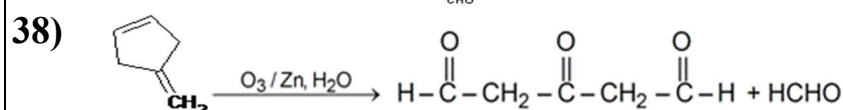
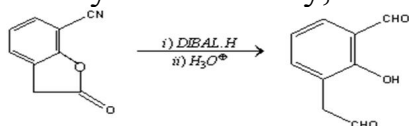


35) The $H_3PO_2 > H_3PO_3 > H_3PO_4$ group stabilizes the incipient carbocation thereby favouring the attack at benzylic carbon to give 2-chloro-2-phenylethanol, i.e.,



36) Conceptual

37) Nitriles are selectively reduced by DIBAL-H to imines, which on hydrolysis yield aldehydes. Similarly, esters are also reduced to aldehydes



39) Conceptual

40) Order of acidity is $H_3PO_2 > H_3PO_3 > H_3PO_4$

H_3PO_4 is not a reducing agent

Only H_3PO_4 is a tribasic acid

In all three phosphorus is tetrahedral geometry

41) Conceptual

42) Conceptual

43) Conceptual

44) $mass = 200g = 2 \times 10^{-3} kg$ $v = 10ms^{-1}$ $v = \frac{10 \times 0.1}{100} = 10^{-2} ms^{-1}$

$$\Delta x \cdot \Delta v = \frac{h}{4\pi m} \quad \Delta x = \frac{6.626 \times 10^{-34}}{4 \times 3.14 \times 2 \times 10^{-3} \times 10^{-2}}$$

$$= 2.64 \times 10^{-32} m$$

45) Conceptual

46) Sucrose is made of β -D-glucose and $t = \frac{10^{-1}}{k} = \frac{4 \times 10^{-2} - 2 \times 10^{-2}}{5 \times 10^{-8}} = 4 \times 10^5$ fructose

47) Conceptual

48) CONCEPTUAL

49) Conceptual

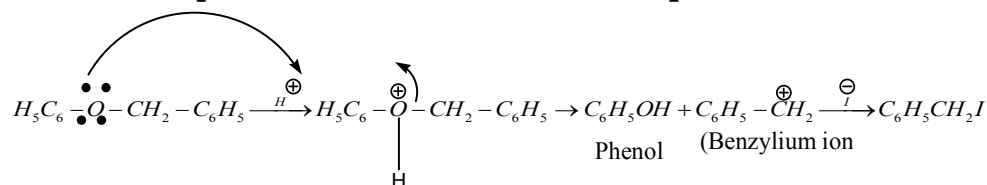
50) The unit of k suggest it to be zero

$$Co - c = kt$$

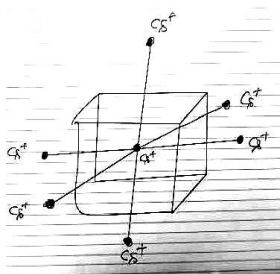
$$t = \frac{10^{-1}}{k} = \frac{4 \times 10^{-2} - 2 \times 10^{-2}}{5 \times 10^{-8}} = 4 \times 10^5$$

51) The electrophile is :CCl₂ which has 1 lp on carbon and 3 lp on each Cl.

52)



53)



$$54) (H^+) = \sqrt{C_1 k a_1 + c_2 k a_2}$$

$$55) In^{3+} + 2e^- \rightarrow In^+; E^0 = -0.42V$$

$$In^{2+} + e^- \rightarrow In^+; E^0 = -0.40V \therefore \text{For } In^{3+} + e^- \rightarrow In^{2+}; E^0 = -0.44V$$

$$E_{cell}^0 = 0.15 - (-0.44) = 0.59V$$

$$E_{cell}^0 = \frac{0.059}{n} \log k \quad 0.59 = \frac{0.059}{1} \log k \Rightarrow k = 10^{10}$$

56)

$$10^6 \text{ gr of water} = \frac{10^6}{500} \times 6 \times 10^{-3} = 12 \text{ gr}$$

$$120 \text{ gr of } MgSO_4 = 100 \text{ gr of } CaCO_3$$

$$12 \text{ gr of } MgSO_4 = 10 \text{ gr of } CaCO_3$$

$$\text{Hardness of water} = 10 \text{ ppm.}$$

57) nearest neighbours is 12

Second nearest neighbours is 6

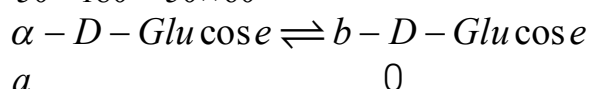
58) CONCEPTUAL

59)

$$(\Delta p)_{glucose} = (\Delta p)_{urea} (x_B)_{glucose} = (x_B)_{urea} \left(\frac{n_B}{n_A} \right)_{glucose} = \left(\frac{n_A}{n_B} \right)_{urea}$$

$$\frac{w_B}{50} \times \frac{18}{180} = \frac{1 \times 18}{50 \times 60} w_B = 3g$$

60)



$$\text{Equilibrium } a - x \quad x$$

$$K = \frac{x}{a - x} = 1.8 \Rightarrow x = \frac{9}{14} a$$

$$\therefore \% \text{ of } \alpha - D - Glucose \text{ remained} = \frac{a - x}{a} \times 100 = 35.7\%$$

MATHEMATICS

61) $F(x) = \int f(x) dx + C = x + 3^x + C$ but $F(2) = 7$ so $7 = 2 + 9 + C \Rightarrow C = -4$

Hence $F(x) = x + 3^x - 4$ Now, $F(x) = 0 \Rightarrow x = 1$

62) $I = \int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ Let $\sec x + \tan x = t \Rightarrow \sec x - \tan x = \frac{1}{t}$

$\Rightarrow \sec x = \frac{1}{2} \left(t + \frac{1}{t} \right)$ Also $\sec x (\sec x + \tan x) dx = dt$

$\Rightarrow \sec x dx = \frac{dt}{t} \therefore I = \frac{1}{2} \int \frac{\left(t + \frac{1}{t} \right) dt}{t^{9/2} \cdot t} = \frac{1}{2} \int (t^{-9/2} + t^{-13/2}) dt$

$= \frac{1}{2} \left[\frac{t^{-9/2+1}}{-\frac{9}{2}+1} + \frac{t^{-13/2+1}}{-\frac{13}{2}+1} \right] + K = \frac{-1}{7} t^{-7/2} - \frac{1}{11} t^{-11/2} + K$

$= -\frac{1}{7t^{7/2}} - \frac{1}{11t^{11/2}} + K = -\frac{1}{t^{11/2}} \left(\frac{1}{11} + \frac{t^2}{7} \right) + K$

$= \frac{-1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

63) Let $K = \lim_{n \rightarrow \infty} \left(\frac{n!}{n^n} \right)^{\frac{1}{n}}$ $\log K = \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \log \left(\frac{1.23 \dots n}{n.n.n \dots n} \right)$

$= \lim_{n \rightarrow \infty} \frac{1}{n} \left[\log \left(\frac{1}{n} \right) + \log \left(\frac{2}{n} \right) + \dots + \log \left(\frac{n}{n} \right) \right] = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \log \left(\frac{r}{n} \right)$

$\log k = \int_0^1 \log x dx$ $\log k = [\log x \cdot x]_0^1 - \int_0^1 \frac{1}{x} dx = (0 - 0) - (x)_0^1 = -1$

$\log k = -1 \Rightarrow k = e^{-1} \Rightarrow k = \frac{1}{e}$

64) $f(x) = \frac{e^x}{1+e^x} \therefore f(a) = \frac{e^a}{1+e^a}$ And $f(-a) = \frac{e^{-a}}{1+\frac{1}{e^a}} = \frac{e^{-a}}{1+\frac{1}{e^a}} = \frac{1}{1+e^a}$

$\Rightarrow f(a) + f(-a) = \frac{e^a + 1}{1+e^a} = 1$ Let $f(-a) = \alpha \therefore f(a) = 1 - \alpha$ Now, $I_1 = \int_{\alpha}^{1-\alpha} xg(x(1-x)) dx$

$= \int_{\alpha}^{1-\alpha} (1-x)g((1-x)(1-(1-x))) dx = \int_{\alpha}^{1-\alpha} (1-x)g(x(1-x)) dx \therefore 2I_1 = \int_{\alpha}^{1-\alpha} g(x(1-x)) dx = I_2 \therefore \frac{I_2}{I_1} = 2$

65) Area $= \int_1^b f(x) dx = \sqrt{b^2 + 1} - \sqrt{2} \Rightarrow \sqrt{b^2 + 1} - \sqrt{1+1} = \left| \sqrt{x^2 + 1} \right|_1^b$

$f(x) = \frac{d}{dx} (\sqrt{x^2 + 1}) = \frac{1}{2} \frac{2x}{\sqrt{x^2 + 1}} = \frac{x}{\sqrt{x^2 + 1}}$

66) Required area $= 4 \int_0^1 -\log x dx$

67) CONCEPTUAL

68) Equation of tangent is $Y = y = \frac{dy}{dx}(X - x)$

$$\text{For } Y = 0, X = 3x, \text{ we get } \frac{dy}{dx} = -\frac{y}{2x} \Rightarrow \frac{dy}{y} = -\frac{1}{2} \frac{dx}{x}$$

$$\Rightarrow \ln|y| = -\frac{1}{2} \ln|x| + \ln c \Rightarrow y = \frac{c}{\sqrt{x}}$$

$$\Rightarrow y = \frac{1}{\sqrt{x}} \quad (c = 1, \text{ as the curve passes through } (1, 1))$$

69) Here $by^2 = (x + a)^3$, differentiating both sides, we get

$$2by \frac{dy}{dx} = 3(x + a)^2 \cdot 1 \Rightarrow \frac{dy}{dx} = \frac{3(x + a)^2}{2by}$$

$$\therefore \text{Length of subnormal} \Rightarrow SN = y \frac{dy}{dx} = \frac{3(x + a)^2}{2b} \quad \dots (1)$$

$$\text{And length of sub tangent} \Rightarrow ST = y \frac{dx}{dy} = \frac{2by^2}{3(x + a)^2} \quad \dots (2)$$

$$\frac{p}{q} = \frac{(ST)^2}{(SN)^2} \quad (\text{given}) \Rightarrow \frac{p}{q} = \frac{(2by^2) \cdot 2b}{\{3(x + a)^2\}^2 \cdot 3(x + a)^2} \quad \{\text{using (i) and (ii)}\}$$

$$= \frac{8b \{(x + a)^3\}^2}{27 (x + a)^6} \quad \{\text{using, } by^2 = (x + a)^3\} = \frac{8b}{27} \therefore \frac{p}{q} = \frac{8b}{27}$$

70) As $f(x)$ and $g(x)$ are continuous and differentiable in $[0, 2]$, then there exist at least one

$$c \text{ such that } \frac{f'(c)}{g'(c)} = \frac{f(2) - f(0)}{g(2) - g(0)} \Rightarrow \frac{8 - 2}{g(2) - 1} = 3 \Rightarrow g(2) - 1 = 2 \Rightarrow g(2) = 3$$

$$71) |\vec{a}| = 3, |\vec{a} \times \vec{b}| = |2\vec{i} + 2\vec{j} + \vec{k}| = 3$$

$$|\vec{c} - \vec{a}| = 2\sqrt{2} \Rightarrow |\vec{c} - \vec{a}|^2 = 8 \Rightarrow |\vec{c}|^2 + |\vec{a}|^2 - 2(\vec{a} \cdot \vec{c}) = 8$$

$$\Rightarrow c^2 + 9 - 2c = 8 \quad \because \vec{a} \cdot \vec{c} = |\vec{c}|$$

$$\Rightarrow c^2 - 2c + 1 = 0 \Rightarrow (c - 1)^2 = 0 \therefore c = 1 \therefore |\vec{c}| = 1$$

$$|(\vec{a} \times \vec{b}) \times \vec{c}| = |\vec{a} \times \vec{b}| |\vec{c}| \sin(30) = 3 \times 1 \times \frac{1}{2} = \frac{3}{2}$$

$$72) \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3}$$

$$73) l = \cos \theta, n = \cos \theta, m = \cos B$$

$$2 \cos^2 \theta + \cos^2 \beta = 1 \quad 2 \cos^2 \theta = 3 \sin^2 \theta = 3 - 3 \cos^2 \theta \quad \cos^2 \theta = \frac{3}{5}$$

$$74) \text{ Given } xRy \text{ if } x^2 - 4xy + 3y^2 = 0$$

If $y = x \Rightarrow x^2 - 4xy + 3y^2 = 0 \Rightarrow R$ is reflexive R is not symmetric $3R1 = 0$ but $1R3 \neq 0$ R is not transitive $\therefore$ for example $9R3 = 0$; $3R1 = 0$ but $9R1 \neq 0$

$$75) f(3) > 0 \text{ and } \Delta > 0$$

76) Since $a + c = 2b, b = \frac{1}{2}$. Also $b^2 = \pm ac \Rightarrow ac = \pm \frac{1}{4}$

And $a + c = 1$ hence a, c are the roots of the equation

$$x^2 - x \pm \frac{1}{4} = 0 \Rightarrow a = \frac{1}{2} - \frac{1}{\sqrt{2}}$$

77) Note that $7^r (r \in \mathbb{N})$ ends in 7, 9, 3 or 1 (corresponding to $r = 1, 2, 3$ and 4 respectively).

Thus, $7^m + 7^n$ cannot end in 5 for any values of $m, n \in \mathbb{N}$.

In other words, for $7^m + 7^n$ to be divisible by 5, it should end in 0.

For $7^m + 7^n$ to end in 0, the forms of m and n should be as follows

	m	n
1	$4r$	$4s + 2$
2	$4r + 1$	$4s + 3$
3	$4r + 2$	$4s$
4	$4r + 3$	$4s + 1$

Thus, for a given value of m there are just 25 values of n for which $7^m + 7^n$ ends in 0.

[For instance, if $m = 4r$, then $n = 2, 6, 10, \dots, 981$

There are $100 \times 25 = 2500$ ordered pairs (m, n) for which $7^m + 7^n$ is divisible by 5.

78) Let $a^8 b^5 c^7 = (ab)^x \cdot (bc)^y \cdot (ca)^z = a^{x+z} \cdot b^{x+y} \cdot c^{y+z}$ and

$$x + y + z = 10 \Rightarrow x + y = 5, y + z = 7, z + x = 8 \Rightarrow x - y = 1 \Rightarrow x = 3, y = 2, z = 5$$

$$\therefore \text{The coefficient of } a^8 b^5 c^7 \text{ in the expansion of } (ab + bc + ca)^{10} \text{ is } = \frac{10!}{3! \cdot 2! \cdot 5!} = 2520$$

79) We know that $|adj(adjA)| = |A|^{(n-1)^2}$

$$|adj(adj(adjA))| = |adjA|^{(n-1)^2} = |A|^{(n-1)^3} = 2^8 = 256$$

80)

p	q	$\sim p$	$p \vee q$	$\sim(p \vee q)$	$(\sim p \wedge q)$	$\sim(p \vee q) \vee (\sim p \wedge q)$
T	T	F	T	F	F	F
T	F	F	T	F	F	F
F	T	T	T	F	T	T
F	F	T	F	T	F	T

$$\therefore (\sim(p \vee q)) \vee (\sim p \wedge q)$$

81) Area of triangle = $\frac{1}{2} |(\bar{a} + 2\bar{b}) \times (2\bar{a} + \bar{b})|$

$$= \frac{1}{2} |2(\bar{a} \times \bar{a}) + (\bar{a} \times \bar{b}) + 4(\bar{b} \times \bar{a}) + 2(\bar{b} \times \bar{b})|$$

$$= \frac{1}{2} |\bar{0} + (\bar{a} \times \bar{b}) - 4(\bar{a} \times \bar{b}) + \bar{0}| = \frac{1}{2} |-3(\bar{a} \times \bar{b})| = \frac{3}{2} |(\bar{a} \times \bar{b})|$$

$$= \frac{3}{2} |\bar{a}| |\bar{b}| \sin(\bar{a}, \bar{b}) = \frac{3}{2} \sqrt{3} (2) \sin \frac{\pi}{3} = 3\sqrt{3} \frac{\sqrt{3}}{2} = \frac{9}{2} = 4.5$$

82) Equation of plane passing through the intersection of the planes $2x - 5y + z = 3$ and $x + y + 4z = 5$ is

$$(2x - 5y + z - 3) + \lambda(x + y + 4z - 5) = 0 \Rightarrow (2 + \lambda)x + (-5 + \lambda)y + (1 + 4\lambda)z - 3 - 5\lambda = 0 \quad \dots(i)$$

which is parallel to the plane $x + 3y + 6z = 1$.

$$\text{Then } \frac{2 + \lambda}{1} = \frac{-5 + \lambda}{3} = \frac{1 + 4\lambda}{6}$$

$$\text{Then, } \frac{2 + \lambda}{1} = \frac{-5 + \lambda}{3} = \frac{1 + 4\lambda}{6} \therefore \lambda = \frac{-11}{2}$$

from eq. (i),

$$-\frac{7}{2}x - \frac{21}{2}y - 21z + \frac{49}{2} = 0 \therefore x + 3y + 6z = 7$$

Hence, $k = 7$

83) If the sum of digits in a number is divisible by 9, then number itself is divisible by 9.

Either number comprises $\{1, 2, 4, 5, 6\}$ or $\{1, 2, 3, 5, 7\}$

$$\therefore P = \frac{2.5!}{{}^7P_5} = \frac{2}{21} = \frac{x}{y} \Rightarrow y - 10x = 1$$

84) Given $P \equiv (t^2, 2t)$

Given parabola is $y^2 = 4x \dots\dots\dots(1)$

Here $a = 1$ Let $Q = (t_1^2, 2t_1)$

Since normal at P meets the curve again at Q.

$$\therefore t_1 = -t - \frac{2}{t} = \frac{-(t^2 + 2)}{t} \dots(2)$$

Now $O \equiv (0, 0), P \equiv (t^2, 2t), Q \equiv (t_1^2, 2t_1)$

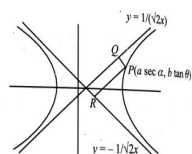
$$\begin{aligned} \text{Given, } \frac{k}{|t|}(1 + t^2)(2 + t^2) &= \text{area of } \Delta POQ = \frac{1}{2}|t^2 \cdot 2t_1 - 2t \cdot t_1^2| \\ &= |t^2 t_1 - t t_1^2| \\ &= \left| t^2 \left(\frac{t^2 + 2}{t} \right) - t \frac{(t^2 + 2)^2}{t^2} \right| = (t^2 + 2) \left| t + \frac{t^2 + 2}{t} \right| = (t^2 + 2) 2 \frac{(1 + t^2)}{|t|} \therefore k = 2 \end{aligned}$$

85) Homogenising, $\frac{x^2}{9} + \frac{y^2}{4} - 1 \left(\frac{\ell x + my}{-n} \right)^2 = 0$

$$\left(\frac{1}{9} - \frac{l^2}{n^2} \right) x^2 - \frac{2lm}{n^2} xy + \left(\frac{1}{4} - \frac{m^2}{n^2} \right) y^2 = 0$$

$$\text{Product of the slopes} = \frac{-b^2}{a^2} = \frac{-4}{9} \Rightarrow 9 \left(\frac{1}{9} - \frac{l^2}{n^2} \right) = -4 \left(\frac{1}{4} - \frac{m^2}{n^2} \right) \Rightarrow \frac{9l^2 + 4m^2}{n^2} = 2$$

86)



$$PQ \times PR = \frac{|a \sec \alpha - \sqrt{2} b \tan \alpha|}{\sqrt{3}} \times \frac{|a \sec \alpha + \sqrt{2} b \tan \alpha|}{\sqrt{3}} = \frac{|a^2 \sec^2 \alpha - 2b^2 \tan^2 \alpha|}{3}$$

$$= \frac{2(\sec^2 \alpha - \tan^2 \alpha)}{3} [As a = \sqrt{2}, b = 1] = \frac{2}{3} \therefore [P] = 0 \text{ or we can use } P = \frac{a^2 b^2}{a^2 + b^2}$$

$$87) \quad BC = I = \frac{\text{tr}(A)}{1 - (1/2)} = 6.$$

$$88) \quad |A| = 1$$

$$|A^{2005} - 5A^{2004}| = |A^{2004}(A - 5I)|$$

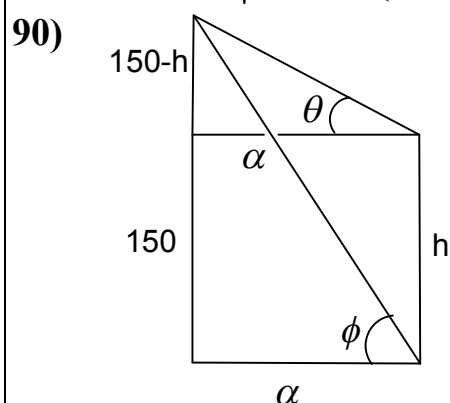
$$= |A|^{2004} |A - 5I|$$

$$= |A - 5I|$$

$$89) \quad \text{Let } x_i - 8 = x$$

$$\sum_{i=1}^{18} x = 9, \quad \sum_{i=1}^{18} x^2 = 45$$

$$\therefore S.D = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2} = \sqrt{\frac{9}{4}} = \frac{3}{2}$$



$$\tan \theta = \frac{4}{3}, \quad \tan \phi = \frac{5}{2}$$

$$\tan \phi = \frac{150}{d} \Rightarrow \frac{5}{2} = \frac{150}{d} \Rightarrow d = 60$$

$$\tan \theta = \frac{150-h}{d} \Rightarrow \frac{4}{3} = \frac{150-h}{60} \Rightarrow 150-h = 80$$

$\therefore$ Dis tan ce between the tops

$$\sqrt{(60)^2 + (80)^2} = 100$$